

BREAST CANCER MANAGEMENT

Exhaustive Preparation Notes

A Slide-by-Slide Elaborated Study Guide, With Line-by-Line Explanation in Context to the Referenced Source Texts and Guidelines

Companion Document to: BreastCancer_Management_Enhanced.pptx (74 slides, 19 sections)

Primary Reference Sources

Schwartz's Principles of Surgery, 11th Edition -- Chapter 17: Diseases of the Breast (Parker CC, Damodaran S, Bland KI, Hunt KK)

DeVita, Hellman, and Rosenberg's Cancer: Principles and Practice of Oncology, 10th Edition -- Chapter 106: Cancer of the Breast (Burstein HJ, Harris JR, Morrow M)

ESMO Clinical Practice Guideline -- Metastatic Breast Cancer, 2026 Update

NCCN Clinical Practice Guidelines in Oncology -- Breast Cancer, Version 6.2026

JNCCN -- NCCN Guidelines Insights: Genetic/Familial High-Risk Assessment, Version 2.2026

Federal Ministry of Health, Ethiopia -- National Breast Cancer Guideline, 2024-2028

Compiled for GSR-3 Surgical Oncology Residency -- September 2026

Dr. Muluamlak -- EBS Unit

How This Document Is Organized

This preparation note walks through the enhanced 74-slide teaching deck, "Breast Cancer Management," section by section and slide by slide. For every content slide, the original bullet points are first recapped in condensed form so the mapping to the source slide is unambiguous, followed by a "Detailed Explanation and Source Context" passage that expands on the underlying mechanism, the supporting trial or epidemiologic evidence, and the clinical reasoning connecting the point to surgical oncology practice.

Explanations are written in direct correspondence to five reference bodies, cited by name at the point of use: Schwartz's Principles of Surgery Chapter 17 (surgical anatomy, operative technique, classic paradigms), DeVita's Cancer Chapter 106 (tumor biology, systemic therapy, epidemiology), the ESMO 2026 Metastatic Breast Cancer guideline (new-agent algorithms, ESCAT/MCBS scoring), NCCN Breast Cancer v6.2026 and the companion JNCCN Genetic/Familial High-Risk Assessment v2.2026 (risk-adapted screening and hereditary-risk thresholds), and the Ethiopian FMOH National Breast Cancer Guideline 2024-2028 (resource-adapted practice in a low-and-middle-income setting).

Two visual conventions are preserved from the presentation. Passages shaded green and marked [ETHIOPIA] discuss the Ethiopian guideline's approach and explicitly state where and why it diverges from the international standard -- the "gap" is named, not merely implied. Passages shaded red and marked [UPDATE] discuss material that is new in the 2026 guideline cycle (ESMO 2026, NCCN v6.2026, JNCCN v2.2026) and would not have appeared in an edition of Schwartz or DeVita predating those updates.

Numeric symbols have been rendered in plain text for portability across PDF readers: ">=" for "at least," "<=" for "at most," "->" for "leads to / progresses to," "+/-" for "with or without," and "increased/decreased" written out in place of arrows. This is a fidelity choice, not a content change -- every quantitative threshold, hazard ratio, and percentage from the source slide is preserved. A clickable outline (PDF bookmarks) is provided in the reader's sidebar, structured as Section > Slide, allowing direct navigation to any of the 74 slides' notes. The in-document Table of Contents on the following page lists the same structure with page numbers.

Table of Contents

SECTION 01 -- History of Breast Cancer Therapy **7**

SLIDE 3 -- Section Divider -- History of Breast Cancer Therapy 7

SLIDE 4 -- History of Breast Cancer Therapy 7

SECTION 02 -- Anatomy and Physiology (including Embryology) **10**

SLIDE 5 -- Section Divider -- Anatomy and Physiology 10

SLIDE 6 -- Embryology of the Breast 10

SLIDE 7 -- Breast Anatomy 12

SLIDE 8 -- Lymphatic Drainage 14

SLIDE 9 -- Physiology and Gynecomastia 15

SECTION 03 -- Epidemiology and Natural History **18**

SLIDE 10 -- Section Divider -- Epidemiology and Natural History 18

SLIDE 11 -- Epidemiology of Breast Cancer 18

SLIDE 12 -- Natural History of Breast Cancer 20

SECTION 04 -- Risk Factors for Breast Cancer **23**

SLIDE 13 -- Section Divider -- Risk Factors for Breast Cancer 23

SLIDE 14 -- Risk Factors -- Overview 23

SLIDE 15 -- BRCA1 and BRCA2 Mutations 24

SLIDE 16 -- Hormonal and Environmental Risk Factors 26

SLIDE 17 -- Risk Assessment Models 28

SECTION 05 -- Hereditary Breast Cancer **31**

SLIDE 18 -- Section Divider -- Hereditary Breast Cancer 31

SLIDE 19 -- BRCA Carrier Management Strategies 31

SLIDE 20 -- NCCN 2026 -- Genetic Risk Assessment Updates 33

SECTION 06 -- Screening and Prevention **36**

SLIDE 21 -- Section Divider -- Screening and Prevention 36

SLIDE 22 -- Screening Mammography 36

SLIDE 23 -- MRI Screening Indications and Chemoprevention 38

SLIDE 24 -- Ethiopian Guideline -- Screening and Early Detection 40

SECTION 07 -- Diagnosis: Imaging and Biopsy	42
SLIDE 25 -- Section Divider -- Diagnosis: Imaging and Biopsy	42
SLIDE 26 -- Clinical Presentation and Examination	42
SLIDE 27 -- Imaging Modalities	44
SLIDE 28 -- Biopsy Techniques	46
SLIDE 29 -- Ethiopian Guideline -- Timely Diagnosis (Pillar 2)	47
 SECTION 08 -- Pathology -- In Situ Carcinomas	 50
SLIDE 30 -- Section Divider -- Pathology, In Situ Carcinomas	50
SLIDE 31 -- LCIS -- Features and Natural History	50
SLIDE 32 -- DCIS -- Features and Management Algorithm	52
 SECTION 09 -- Pathology -- Invasive Carcinomas	 55
SLIDE 33 -- Section Divider -- Pathology, Invasive Carcinomas	55
SLIDE 34 -- Invasive Carcinoma Types	55
SLIDE 35 -- Special Histologic Types and Presentations	56
 SECTION 10 -- AJCC Staging (TNM), 8th Edition	 60
SLIDE 36 -- Section Divider -- AJCC Staging (TNM)	60
SLIDE 37 -- AJCC 8th Edition -- T and N Categories	60
SLIDE 38 -- AJCC Staging Groups and Workup by Stage	61
 SECTION 11 -- Prognostic Factors and Molecular Subtypes	 64
SLIDE 39 -- Section Divider -- Prognostic Factors and Molecular Subtypes	64
SLIDE 40 -- Traditional Prognostic and Predictive Factors	64
SLIDE 41 -- Multigene Assays -- Oncotype DX and MammaPrint	67
SLIDE 42 -- Molecular Subtypes (Intrinsic/Sorlie-Perou Classification)	69
 SECTION 12 -- Local-Regional Therapy	 71
SLIDE 43 -- Section Divider -- Local-Regional Therapy	71
SLIDE 44 -- Breast-Conserving Surgery (BCS) versus Mastectomy	71
SLIDE 45 -- Axillary Surgery -- SLNB and the Z0011 Trial	73
SLIDE 46 -- Radiation Therapy in Breast Cancer	75
SLIDE 47 -- Mastectomy Types and Reconstruction	77
 SECTION 13 -- Systemic Therapy (Adjuvant)	 80

SLIDE 48 -- Section Divider -- Systemic Therapy (Adjuvant)	80
SLIDE 49 -- Adjuvant Endocrine Therapy	80
SLIDE 50 -- Adjuvant Chemotherapy Regimens	81
SLIDE 51 -- HER2-Targeted Therapy	83
SLIDE 52 -- Neoadjuvant (Preoperative) Systemic Therapy	85
SECTION 14 -- Management by Stage	88
SLIDE 53 -- Section Divider -- Management by Stage	88
SLIDE 54 -- Stage 0 -- In Situ Disease Management	88
SLIDE 55 -- Stage I-II B -- Curative Intent Management	90
SLIDE 56 -- Stage III -- Locally Advanced Breast Cancer (LABC)	92
SLIDE 57 -- Stage IV -- Metastatic Breast Cancer (MBC)	94
SECTION 15 -- ESMO 2026 -- Metastatic Breast Cancer Updates	97
SLIDE 58 -- Section Divider -- ESMO 2026 MBC Updates	97
SLIDE 59 -- ESMO 2026 -- HR+/HER2- MBC: ESR1 and PI3K/AKT Updates	97
SLIDE 60 -- ESMO 2026 -- HER2+ and TNBC MBC: New Agent Algorithms	99
SECTION 16 -- Recurrence -- Types and Management	103
SLIDE 61 -- Section Divider -- Recurrence, Types and Management	103
SLIDE 62 -- Recurrence -- Classification and Definitions	103
SLIDE 63 -- Locoregional Recurrence -- Management Algorithm	105
SECTION 17 -- Special Situations	108
SLIDE 64 -- Section Divider -- Special Situations	108
SLIDE 65 -- Special Situations: Pregnancy, Male BC, and Nipple Discharge	108
SLIDE 66 -- Inflammatory BC, Local-Regional Recurrence, and Rare Cancers	110
SECTION 18 -- Ethiopian Guideline 2024 Integration	114
SLIDE 67 -- Section Divider -- Ethiopian Guideline 2024	114
SLIDE 68 -- Ethiopian BC Guideline 2024 -- Overview and Three Pillars	114
SLIDE 69 -- Ethiopia -- Treatment Scope by Healthcare Tier	116
SLIDE 70 -- Ethiopia versus International Standard -- Key Gaps and Priorities	118
SECTION 19 -- Follow-Up and Quick Reference	120
SLIDE 71 -- Section Divider -- Follow-Up and Quick Reference	120

SLIDE 72 -- Follow-Up Guidelines (ASCO Recommendations)120

SLIDE 73 -- Quick Reference -- Subtypes, Key Numbers, and High-Yield Distinctions121

SLIDE 74 -- Summary -- Key Take-Home Messages123

SECTION 01 -- History of Breast Cancer Therapy

From the earliest recorded description of breast cancer through the paradigm shift from radical extirpative surgery to biology-driven, multidisciplinary precision oncology. This history matters clinically because nearly every modern controversy in breast surgery -- how much tissue to remove, whether to operate on the axilla, when to irradiate -- is a direct descendant of the Halsted-versus-Fisher debate introduced here and revisited in Section 03.

SLIDE 3 -- Section Divider -- History of Breast Cancer Therapy

Detailed Explanation and Source Context

This divider slide frames the historical arc that both Schwartz Chapter 17 and DeVita Chapter 106 open with, because the surgical management of breast cancer cannot be understood ahistorically: each successive era's dominant operation was a direct expression of the prevailing theory of how the disease spreads. The subtitle, "From Halsted radical mastectomy to precision oncology," previews the two poles of that theory -- Halsted's anatomic, contiguous-spread model, and the modern molecular, systemic-disease model -- which are treated in full in Section 03's discussion of the Halsted-versus-Fisher paradigm (Slide 12).

SLIDE 4 -- History of Breast Cancer Therapy

Slide content recap:

- ~3000 BC: Edwin Smith Papyrus -- earliest recorded case (male); "there is no treatment"
- 1700s: Le Dran and Moore -- local spread via lymphatics; first node removal
- 1894: Halsted and Meyer -- radical mastectomy: en-bloc skin, pec major/minor, axilla levels I-III
- 1943: Haagensen and Stout -- defined "grave signs" of inoperability
- 1948: Patey and Dyson -- modified radical: preserve pectoralis major
- 1976: Veronesi and Fisher -- randomized trials proving BCS = mastectomy for survival (NSABP B-06)
- 1980s-90s: sentinel lymph node concept; tamoxifen era
- 2000s: HERA, NSABP B-31 -- trastuzumab for HER2+ BC; survival benefit proven
- 2010s: SOFT/TEXT, Z0011 -- ovarian suppression; SLNB alone safe in T1-2N1 BCS
- 2020s: DESTINY, MONARCH-3 -- T-DXd for HER2-low/+; CDK4/6 inhibitors standard 1st line MBC

Detailed Explanation and Source Context

The Edwin Smith Papyrus (circa 3000 BC, translated by James Henry Breasted in 1930) is the earliest surviving medical document to describe breast cancer -- eight cases of "bulging tumors" of the breast are catalogued, with the physician-author's stark conclusion for the most advanced case: "there is no treatment." Schwartz Chapter 17 opens with this case specifically because it establishes that breast cancer's basic clinical phenotype (a hard, cool, bulging mass) was recognized four millennia before any effective therapy existed, framing the subsequent 5,000-year narrative as a search for that therapy.

Le Dran (1757) and, later, Sir Benjamin Bell were among the first to observe that disease which had spread to the axillary lymph nodes carried a substantially worse prognosis than disease confined to the breast -- an empirical observation, made without any understanding of lymphatic anatomy or physiology, that nonetheless launched the practice of removing palpable axillary nodes together with the breast. This is the conceptual ancestor of both the sentinel lymph node biopsy and the Berg level I-III axillary dissection discussed in Section 02 (Slide 8).

William Halsted's 1894 description of the radical mastectomy (breast, pectoralis major, pectoralis minor, and axillary contents levels I-III removed en bloc) formalized a theory of "centrifugal" spread: cancer starts locally and extends in an orderly, contiguous fashion through lymphatics before becoming systemic, such that sufficiently wide local extirpation could be curative. For the next 70 years, the radical mastectomy was the standard of care worldwide, and Halsted's operation is the direct reason the pectoralis muscles and axillary lymphatic anatomy described in Section 02 remain central to surgical teaching -- even though the muscles themselves are no longer routinely resected.

Haagensen and Stout's 1943 criteria for inoperability ("grave signs": skin edema, skin ulceration, chest wall fixation, and, later, satellite skin nodules and inflammatory changes) were the first systematic attempt to identify patients for whom radical surgery alone would fail -- a direct conceptual precursor to today's T4 categories (Section 10, Slide 37) and to the mandate for neoadjuvant systemic therapy before surgery in locally advanced disease (Section 14, Slide 56).

Patey and Dyson's 1948 modified radical mastectomy -- preserving the pectoralis major (and, in most modern variants, the pectoralis minor as well) while still clearing axillary levels I-III -- reduced morbidity without a demonstrated loss of oncologic efficacy, and remains, structurally, the MRM performed today (Section 12, Slide 47).

The pivotal break from the Halsted paradigm came from Bernard Fisher's systemic-disease hypothesis and the parallel European work of Umberto Veronesi. The NSABP B-06 trial (first reported 1985, updated through the 1990s by Fisher and colleagues, with Veronesi's Milan I trial providing concordant European data) randomized women with early breast cancer to radical mastectomy, total mastectomy with radiation, or lumpectomy with radiation, and found no difference in overall survival among the three arms. This single finding -- that breast-conserving surgery plus radiotherapy is oncologically equivalent to mastectomy -- is arguably the most consequential result in the history of breast cancer treatment, and it is why BCS is presented as the preferred option "when possible" throughout Section 12 (Slide 44).

The 1980s-1990s introduced two further transformations: the sentinel lymph node concept (Giuliano and colleagues, building on Cabanas's earlier work in penile cancer), which allowed axillary staging without complete dissection in node-negative patients, and the widespread adoption of tamoxifen as adjuvant endocrine therapy following the Early Breast Cancer Trialists' Collaborative Group (EBCTCG) overviews -- both concepts that are developed fully in Sections 12 and 13.

The 2000s introduced the first validated molecularly targeted therapy in breast cancer: trastuzumab. The HERA trial and the joint NSABP B-31/NCCTG N9831 analysis (published 2005) demonstrated that adding one year of adjuvant trastuzumab to chemotherapy in HER2-positive early breast cancer reduced the risk of recurrence by roughly half -- proof of principle that a tumor's molecular subtype, not merely its anatomic stage, should dictate systemic therapy. This is the conceptual root of the entire molecular-subtyping framework covered in Section 11.

The 2010s produced two practice-changing de-escalation trials: SOFT and TEXT (ovarian function suppression combined with tamoxifen or exemestane improves outcomes in higher-risk premenopausal ER-positive disease, Section 13 Slide 49) and ACOSOG Z0011 (in appropriately selected cT1-2N0 patients undergoing breast-conserving surgery with whole-breast radiation, one or two positive sentinel nodes do not require completion axillary dissection, Section 12 Slide 45) -- both are discussed at length in their respective sections because they represent a shift toward selectively doing less surgery/treatment without sacrificing survival, the mirror image of the earlier "more is better" Halstedian era.

The 2020s, referenced by DESTINY-Breast04 (trastuzumab deruxtecan, T-DXd, extending anti-HER2 antibody-drug conjugate therapy to "HER2-low" tumors previously considered HER2-negative) and MONARCH-3 (establishing CDK4/6 inhibitors as first-line therapy for HR-positive, HER2-negative metastatic disease), mark the current era of antibody-drug conjugates and targeted small molecules, which is the subject of Section 15's dedicated ESMO 2026 update slides. Schwartz Chapter 17's historical table and DeVita Chapter 106's discussion of the evolution of systemic therapy are the primary sources synthesized in this timeline; the 2020s entries are updated here beyond the base textbooks using the ESMO 2026 and NCCN v6.2026 guideline sources that anchor Section 15.

SECTION 02 -- Anatomy and Physiology (including Embryology)

Breast embryologic development, gross and surgical anatomy, lymphatic drainage, and hormonal physiology -- the substrate on which every pathologic and operative concept in later sections depends. The embryology slide (Slide 6) was added specifically to supply background that Schwartz Chapter 17 assumes as prerequisite knowledge but does not itself re-derive in detail, concentrated here on features with direct malignant relevance.

SLIDE 5 -- Section Divider -- Anatomy and Physiology

Detailed Explanation and Source Context

Anatomy is presented before pathology and staging in this deck (as in Schwartz Chapter 17) because nearly every oncologic concept that follows is anatomically anchored: the T4a designation depends on knowing exactly which structures constitute the "chest wall"; the N-category depends on the Berg level classification of the axilla; and the rationale for breast-conserving surgery depends on understanding the retromammary and subcutaneous planes that permit an oncologically adequate resection without removing the entire breast envelope.

SLIDE 6 -- Embryology of the Breast

Slide content recap:

- Breast develops from ectoderm (milk line / galactic band) -- bilateral ridges Week 4-6
- Milk line: axilla to groin; normally involutes leaving single pectoral pair
- Persistence -> polymastia (extra breast) or polythelia (extra nipple) -- 1-2% population
- Breast buds appear Week 6; 15-20 primary epithelial buds canalize -> lactiferous ducts
- Terminal ductal-lobular units (TDLU) form in late fetal/postnatal life
- TDLU = primary site of origin for DCIS, invasive ductal carcinoma and LCIS
- Nipple-areola complex forms by areolar skin condensation; nipple elevation at puberty
- Accessory nipple along milk line: L2-T10 dermatome distribution
- Axillary tail of Spence: functional breast tissue enters axilla through deep fascia opening
- Inframammary fold: important surgical landmark; at 5th-6th rib midclavicular line
- Retromammary space: loose areolar tissue between breast and pectoralis fascia -- plane used in subglandular implant placement
- Accessory breast tissue (axilla most common) can develop cancer -- mimics lymph node
- Inverted nipple if ductal tissue fails to canalize / fibrous tethering
- Amazia (absent breast) vs amastia (absent nipple + breast) -- rare surgical anomalies
- Cooper's ligaments (fibrous septa) anchor skin -> retraction = malignancy sign
- Retroglandular fat plane -- preserved in MRM; violated in Halsted radical
- Upper outer quadrant (UOQ): 50% of breast volume -> 50% of all breast cancers
- Central zone: subareolar; Paget's disease arises here
- Retroareolar: 15-20 major lactiferous ducts converge at nipple
- Internal mammary nodes: 25-30% of medial tumors drain here -> important for staging
- TDLU undergoes cyclical proliferation/apoptosis with menstrual cycle

Detailed Explanation and Source Context

This slide was purpose-built to satisfy a specific requirement of this revision: to concentrate the embryologic, physiologic, and pathologic background that a surgical oncology text like Schwartz assumes the reader already possesses, directing it specifically toward malignant relevance rather than treating embryology as an isolated preclinical topic.

The breast is an ectodermal appendage. Between the fourth and sixth weeks of gestation, paired thickenings of ectoderm -- the "milk lines" or galactic bands -- extend from the axilla to the groin bilaterally. In humans, this ridge normally regresses everywhere except at the future pectoral (thoracic) location, where it persists and gives rise to the definitive breast. Failure of complete regression explains two clinically encountered anomalies: polymastia (a complete supernumerary breast, including glandular tissue) and polythelia (an isolated supernumerary nipple without underlying gland), together affecting an estimated 1-2% of the population. Because these ectopic remnants retain the same hormonal responsiveness and malignant potential as orthotopic breast tissue, accessory breast tissue -- most often found in the axilla -- can itself give rise to a primary breast carcinoma, and on examination a mass here may be mistaken for an enlarged lymph node; this is the direct clinical payoff of teaching the milk-line concept in an oncology context rather than a purely embryologic one.

Around week 6, primary epithelial buds (15 to 20 of them) invaginate from the epidermis into the underlying mesenchyme and progressively canalize to form the lactiferous duct system, each duct ultimately draining a separate lobe of the adult gland. The terminal ductal-lobular unit (TDLU), the functional secretory unit at the end of each ductal tree, does not fully mature until late fetal life and continues to develop postnatally and especially at puberty and during pregnancy. The TDLU is emphasized here because it is the site of origin of the overwhelming majority of breast malignancies -- ductal carcinoma in situ, invasive ductal carcinoma, and lobular carcinoma in situ all arise from TDLU epithelium (Sections 08 and 09) -- so understanding TDLU embryology directly explains why these lesions cluster anatomically and why the TDLU is the histologic reference point pathologists use when describing "usual ductal hyperplasia" versus atypia versus carcinoma in situ.

The nipple-areola complex forms by condensation of the overlying areolar skin, with the nipple itself becoming everted (elevated) around puberty under estrogen influence; failure of the underlying ductal tissue to canalize properly, or fibrous tethering, can result in a congenitally inverted nipple -- a benign developmental variant that must be distinguished on history from an acquired, new nipple inversion, which is a red-flag sign of underlying malignancy (Section 07, Slide 26) because it implies pathologic shortening of the lactiferous ducts or Cooper's ligaments by an infiltrating tumor. Rare, more severe developmental failures include amazia (absent breast tissue with a normal nipple) and amastia (absent breast and nipple), both principally of surgical/reconstructive rather than oncologic significance.

Several surgically important landmarks are embryologically derived and are reintroduced with operative significance in Section 02's anatomy slide (Slide 7) and Section 12: the axillary tail of Spence, a tongue of glandular tissue that extends through an opening in the clavipectoral (deep) fascia into the lower axilla and is the reason axillary dissection can encounter breast parenchyma rather than pure nodal tissue at its supero-medial extent; the inframammary fold, classically located at the level of the fifth to sixth rib in the midclavicular line, used as an incision site and as the lower boundary in mastectomy and implant-based reconstruction; and the retromammary space, a plane of loose areolar tissue separating the breast from the investing fascia of pectoralis major, which is the surgical plane developed for subglandular implant placement and which, notably, is preserved in a modified radical mastectomy but was deliberately violated (removed together with the pectoral fascia and muscle) in the original Halsted radical mastectomy -- tying this slide directly back to Section 01's historical narrative.

Cooper's suspensory ligaments -- fibrous septa that run from the deep fascia through the breast parenchyma to attach to the overlying dermis -- have obvious clinical importance: when a malignant process infiltrates and foreshortens these septa, the result is visible skin dimpling or retraction, one of the cardinal physical signs of breast cancer covered in Section 07 (Slide 26). The retroglandular fat plane, similarly, is a landmark whose preservation (in modified techniques) or deliberate sacrifice (in the historical Halsted operation) defines the

extent of "radical" surgery.

Quadrant distribution matters because it is not uniform: the upper outer quadrant contains roughly half of all glandular tissue by volume (owing to the axillary tail) and, correspondingly, is the site of origin of approximately half of all breast cancers -- a fact used clinically to raise the index of suspicion for any palpable abnormality in that location. The central, subareolar zone is where the major lactiferous ducts (15 to 20 of them) converge before opening onto the nipple surface, and it is specifically the site where Paget's disease of the nipple arises (Section 09, Slide 35), because Paget cells are thought to represent intraepithelial extension of an underlying ductal carcinoma along these convergent terminal ducts into the nipple epidermis.

Finally, this slide anticipates the lymphatic drainage discussion of Slide 8 by noting that 25 to 30% of tumors arising in the medial half of the breast drain preferentially to the internal mammary nodal chain rather than (or in addition to) the axilla -- a drainage pattern with direct staging consequences (internal mammary nodal involvement upstages disease to N3b, Section 10) -- and closes the physiologic loop by noting that the TDLU is not a static structure but undergoes cyclical proliferation and apoptosis across the menstrual cycle under estrogen and progesterone influence, the same hormonal cycling elaborated in Slide 9's discussion of breast physiology.

SLIDE 7 -- Breast Anatomy

Slide content recap:

- Extent: 2nd-6th rib vertically; sternal edge to mid-axillary line
- Axillary tail of Spence: breast tissue through clavipectoral fascia opening into axilla
- Cooper's suspensory ligaments: attach skin to overlying dermis and deep fascia
- Lymphatics: subdermal + parenchymal plexuses drain to axilla primarily
- 3 tissue types: glandular (15-20 lobes), stromal (fibrous), adipose
- Pectoralis major: superficial to breast; preserved in MRM (Patey)
- Pectoralis minor: removed in Halsted; divided in Patey for axillary access
- Serratus anterior: critical to preserve long thoracic nerve during ALND
- Intercostobrachial nerve: sensory arm inner surface -- often sacrificed in ALND
- Thoracodorsal nerve to latissimus: preserve in all axillary dissections
- Internal mammary (pectoral branches): 60% medial breast -> largest supply
- Lateral thoracic artery (branch of axillary): upper outer quadrant
- Intercostal perforators (2nd-4th): posterior supply
- Thoracoacromial artery: pectoralis minor region
- Venous drainage: internal mammary -> subclavian -> SVC; axillary vein
- Long thoracic nerve (of Bell): serratus anterior -> "winged scapula" if cut
- Thoracodorsal nerve: latissimus dorsi -> preserve for TRAM/LD flap reconstruction
- Medial pectoral nerve: pec minor/part pec major -> enters pec minor lateral border
- Lateral pectoral nerve: larger; enters pec major at clavicular head
- Intercostobrachial nerve: T2 -> inner arm sensation -- often sacrificed in full ALND

Detailed Explanation and Source Context

Schwartz Chapter 17 opens its anatomy section with the breast's surface boundaries -- vertically from the second to the sixth rib, horizontally from the sternal edge to the mid-axillary line, with the axillary tail of Spence extending this footprint through a gap in the clavipectoral fascia into the axilla proper. These boundaries are not decorative: they define the minimum skin envelope that must be evaluated (and, in a mastectomy, removed) to ensure the entire glandular field is addressed, and they explain why axillary surgery can encounter glandular tissue at its uppermost, medial extent rather than pure fat and nodes.

The breast parenchyma itself is composed of three tissue compartments -- glandular tissue organized into 15 to 20 discrete lobes (each drained by its own lactiferous duct, consistent with the embryologic duct-canalization pattern from Slide 6), a fibrous (stromal) framework that includes Cooper's ligaments, and a variable adipose component whose proportion increases with age and after menopause as glandular tissue involutes. This three-compartment model directly explains mammographic density (fibroglandular-dominant breasts are radiographically "dense" and harder to screen, Section 06) and is the tissue substrate oncologists manipulate during volume-displacement techniques (Section 12, Slide 47).

The muscular relationships are of primary surgical importance because they are precisely what differentiates the historical operations of Section 01 from one another: the pectoralis major lies immediately deep to the breast and is preserved in the modified radical mastectomy (Patey's 1948 modification) but was resected en bloc in Halsted's original radical mastectomy; the pectoralis minor was likewise removed by Halsted but is only divided (not removed) in the Patey technique to permit adequate exposure for a complete axillary dissection, and in many contemporary MRM variants it is simply retracted rather than divided. The serratus anterior, forming the medial wall of the axilla, must be carefully preserved during axillary lymph node dissection specifically because the long thoracic nerve of Bell runs along its superficial surface; injury to this nerve denervates serratus anterior and produces a "winged scapula" deformity, one of the classically tested complications of axillary surgery. The thoracodorsal nerve, supplying latissimus dorsi, must similarly be preserved during any axillary dissection -- both because latissimus dorsi contributes to shoulder function and, more specifically in the reconstructive context, because an intact thoracodorsal pedicle is a prerequisite if a latissimus dorsi myocutaneous flap is later used for breast reconstruction (Section 12, Slide 47). The intercostobrachial nerve, a pure sensory branch of the second intercostal nerve supplying the skin of the medial upper arm, is frequently sacrificed during a complete axillary dissection because it traverses the operative field; this is why numbness of the inner upper arm is such a common, and largely unavoidable, patient complaint after axillary surgery, and why some surgeons attempt selective preservation when oncologically feasible.

Arterial supply to the breast is multi-source, which is clinically relevant to flap viability and to understanding patterns of local recurrence: the internal mammary artery's perforating branches supply roughly 60% of the breast, concentrated medially, and are the single largest contributor; the lateral thoracic artery (a branch of the axillary artery) supplies the upper outer quadrant, correlating with that quadrant's high tumor incidence noted in Slide 6; intercostal perforators from the second through fourth intercostal spaces supply the posterior aspect; and the thoracoacromial artery supplies the region overlying pectoralis minor. Venous drainage largely mirrors arterial supply, with the internal mammary vein draining centrally to the subclavian vein and thence to the superior vena cava, and the axillary vein draining the lateral breast and axillary contents -- this is the route by which hematogenous metastasis is understood to occur, prefiguring the metastatic-cascade discussion in Section 03 (Slide 12).

Finally, the nerve anatomy of the pectoral region has direct surgical stakes beyond the long thoracic and thoracodorsal nerves already discussed: the medial pectoral nerve (despite its name, derived from the medial cord of the brachial plexus) supplies pectoralis minor and part of pectoralis major, entering along the lateral border of pectoralis minor, while the larger lateral pectoral nerve (from the lateral cord) enters pectoralis major near its clavicular head. Preserving both pectoral nerves during a modified radical mastectomy is what allows the pectoralis major to retain innervation and bulk despite the muscle's superficial surface being exposed during the dissection -- a technical detail Schwartz emphasizes because inadvertent injury can cause avoidable pectoral muscle atrophy even when the muscle itself is anatomically "preserved."

SLIDE 8 -- Lymphatic Drainage

Slide content recap:

- Level I (low axilla): lateral to pectoralis minor -- first echelon; 60% breast drainage
- Level II (mid-axilla): posterior to / behind pec minor -- includes Rotter's nodes (interpectoral)
- Level III (apical): medial to pec minor / subclavicular -- "complete dissection"
- Skip metastasis to Level III without Level I/II: ~4% -- clinically significant
- Internal mammary nodes lie in intercostal spaces 1-6, within 3 cm of sternal edge (usually spaces 1-3)
- 25-30% of medial/central tumors drain to IMN; 15% of outer quadrant tumors
- IMN involvement: independent predictor of poor prognosis; included in Stage N3b
- Extended radical mastectomy (Urban): included IMN; no survival benefit over MRM
- Sentinel node = first draining node(s) from tumor site -- best reflects nodal status
- False-negative rate (FNR): ~5-8% with dual tracer (blue dye + Tc-99m sulfur colloid)
- Z0011 trial: 1-2 positive SLNs in cT1-2, BCS + WBRT -> ALND not needed (non-inferior OS/DFS)

Detailed Explanation and Source Context

The Berg classification divides the axillary lymph nodes into three anatomic levels defined entirely relative to the pectoralis minor muscle, which is why the detailed muscular relationships taught in Slide 7 are a direct prerequisite for understanding this slide. Level I nodes lie lateral (inferior and lateral) to the lateral border of pectoralis minor and constitute the first echelon of drainage, receiving roughly 60% of total breast lymphatic outflow; Level II nodes lie directly posterior to (behind) pectoralis minor and include the interpectoral, or Rotter's, nodes situated between pectoralis major and minor -- these were specifically targeted in the Patey modified radical mastectomy but were not separately addressed in Halsted's original description; and Level III (apical) nodes lie medial to the medial border of pectoralis minor, extending toward the subclavian region, and their clearance was what Halsted considered a "complete" axillary dissection. Because lymphatic flow in the axilla is not perfectly sequential, disease can occasionally bypass Level I and II to involve Level III directly -- so-called skip metastasis -- at a rate of approximately 4%, a fact of direct relevance when a surgeon must decide whether to extend a dissection to Level III based on intraoperative findings at Level I/II.

Internal mammary nodes constitute an entirely separate drainage basin from the axilla, situated within the intercostal spaces (most often the first through third, though anatomically the chain extends through the sixth) within about 3 centimeters of the sternal edge. Because the medial and central portions of the breast drain preferentially along the internal mammary perforating vessels (recall Slide 7's arterial anatomy), tumors arising in these locations metastasize to the internal mammary chain in 25 to 30% of cases, compared with roughly 15% for outer-quadrant tumors that still drain here to a lesser degree via lymphatic crossover. Involvement of the internal mammary nodes is an independent adverse prognostic factor and is formally incorporated into the AJCC nodal staging system as N3b when combined with axillary involvement (Section 10, Slide 37) -- a direct link between this slide's anatomy and the staging system two sections later. Historically, this drainage pattern motivated Jerome Urban's "extended radical mastectomy," which added deliberate resection of the internal mammary chain to the standard radical mastectomy; despite the anatomic rationale, no survival benefit over the standard (non-extended) operation was ever demonstrated, and the extended procedure -- with its added morbidity from entering the pleural space -- was abandoned, a case study in how anatomic completeness does not automatically translate into oncologic benefit, foreshadowing the same lesson that Z0011 would later teach about the axilla itself.

The sentinel lymph node is defined operationally, not anatomically: it is simply the first node (or nodes) that a given tumor's lymphatics drain to, and the biological premise of sentinel node biopsy is that its histologic status is the single most reliable, least invasive predictor of the entire nodal basin's status. Localization uses a dual-tracer technique -- a blue dye (isosulfan blue) injected around the tumor or subareolarly, combined with a technetium-99m-labeled sulfur colloid radiotracer detected intraoperatively with a handheld gamma probe --

which together achieve a combined identification rate well above either tracer alone, with an accepted false-negative rate of approximately 5 to 8%, meaning that in that proportion of node-positive patients the sentinel node itself will be falsely reported negative. This false-negative rate is the numeric benchmark against which newer post-neoadjuvant techniques like targeted axillary dissection (Section 12, Slide 45) are measured and improved upon.

The clinical payoff of the sentinel node concept was realized definitively in the ACOSOG Z0011 trial, which is previewed here and detailed fully in Section 12 (Slide 45): in patients with clinical T1-2, clinically node-negative disease undergoing breast-conserving surgery with planned whole-breast radiation, finding one or two positive sentinel nodes (with only macrometastatic disease) does not require completion axillary lymph node dissection, because omitting the completion dissection produced no difference in overall or disease-free survival at ten years of follow-up. This result -- discovered decades after the sentinel node concept itself was validated -- represents the endpoint of the historical arc traced in Section 01: from Halsted's mandatory complete axillary clearance, through the intermediate step of sentinel-node-guided selective dissection, to the modern default of no further axillary surgery at all in appropriately selected, low-volume nodal disease.

SLIDE 9 -- Physiology and Gynecomastia

Slide content recap:

- Thelarche (Tanner Stage 2): estrogen-driven ductal elongation, ~age 8-13
- Progesterone: lobular development, alveolar differentiation
- Prolactin: milk production; high levels -> suppress ovulation
- Puberty: estrogen + GH + IGF-1 -> complete ductal growth
- Pregnancy: progesterone + placental lactogen -> full lobular development
- Menopause: loss of E2/P -> involution; increase adipose relative to gland
- HRT: combined E+P -> increased BC risk (WHI trial); E-alone -> no significant increase
- Gynecomastia definition: benign proliferation of breast gland in males (true glandular hypertrophy)
- Physiologic gynecomastia: neonates (maternal estrogen), puberty, elderly (relative hyperestrogenism)
- Pathologic causes: hyperthyroidism, cirrhosis, hypogonadism, Klinefelter's (XXY)
- Drug causes: spironolactone, digitalis, marijuana, anabolic steroids, anti-androgens
- Paraneoplastic: testicular/adrenal tumor (hCG-secreting), lung cancer
- Pseudogynecomastia: adipose tissue -- no glandular component
- 1% of all breast cancers are male; 6% in Sub-Saharan Africa (higher rate)
- Male BC risk factors: Klinefelter's, BRCA2 mutation, radiation, exogenous estrogen
- Male BC typically ER+ (90%), older age at diagnosis (avg 67 yrs)
- Male BC presents later: central location, nipple involvement more common
- Male BC treatment: MRM (BCS rarely used); tamoxifen preferred adjuvant endocrine
- BRCA2: most important germline mutation in male BC (lifetime risk 6-8%)
- Prolactin secreted by ant. pituitary; dopamine suppresses PRL
- Causes of hyperprolactinemia: pituitary adenoma, hypothyroidism, drugs (antipsychotics, metoclopramide)
- Bloody discharge: intraductal papilloma (most common); DCIS must be excluded
- Persistent unilateral discharge -> galactography / ductoscopy

Detailed Explanation and Source Context

Breast physiology is a hormonally sequential process that recapitulates, at each life stage, the embryologic ductal template established in Slide 6. Thelarche, the onset of breast development corresponding to Tanner Stage 2, occurs around age 8 to 13 and is driven primarily by rising estrogen, which causes ductal elongation and branching; progesterone, rising later in the menstrual cycle and dramatically in pregnancy, acts on this ductal scaffold to drive lobular (alveolar) development and differentiation -- the two hormones are therefore complementary, not redundant, with estrogen building the "pipes" and progesterone building the "factories" at their ends. Prolactin, secreted by the anterior pituitary, is the hormone actually responsible for milk synthesis

once the alveolar structures are differentiated, and physiologically high prolactin (as in lactation) suppresses gonadotropin-releasing hormone pulsatility and thereby suppresses ovulation -- the physiologic basis of lactational amenorrhea.

At puberty, estrogen acts together with growth hormone and IGF-1 to complete ductal growth; in pregnancy, the combination of progesterone and placental lactogen drives full lobular maturation in preparation for lactation, meaning the TDLU epithelium introduced in Slide 6 is at its most differentiated, and paradoxically often at its most proliferatively active, during pregnancy -- relevant to the discussion of pregnancy-associated breast cancer in Section 17. At menopause, the loss of circulating estradiol and progesterone results in glandular involution and a relative increase in the adipose component of the breast, consistent with the three-tissue-compartment model from Slide 7. Exogenous hormone replacement recapitulates this signaling: the Women's Health Initiative (WHI) trial demonstrated that combined estrogen-plus-progestin HRT modestly but significantly increases breast cancer risk, whereas estrogen-alone HRT (used only in women without a uterus, since unopposed estrogen is otherwise contraindicated for endometrial reasons) did not show a significant increase -- an important, frequently-tested distinction that is picked up again in the risk-factors discussion of Section 04 (Slide 16).

Gynecomastia is the male analog of benign ductal proliferation: true glandular hypertrophy of the male breast, to be distinguished carefully from pseudogynecomastia, which is simple fat deposition without any glandular component and requires no endocrine workup. Physiologic gynecomastia occurs in three predictable windows -- the neonatal period (from residual transplacental maternal estrogen), puberty (from a transient imbalance favoring estrogen over androgen during the pubertal hormonal surge), and old age (from relative hyperestrogenism as testicular androgen production declines while peripheral aromatization of androgens to estrogen, occurring in adipose tissue, continues or increases with age-related weight gain). Pathologic causes to screen for include hyperthyroidism (which increases sex hormone-binding globulin and shifts the free androgen-to-estrogen ratio), hepatic cirrhosis (impaired estrogen clearance), primary or secondary hypogonadism, and Klinefelter syndrome (47,XXY karyotype), which is doubly important because it is also an independent, well-established risk factor for male breast cancer itself, discussed two paragraphs below. Numerous medications are recognized causes, including spironolactone (anti-androgenic and estrogenic activity), digoxin, marijuana, exogenous anabolic steroids (via peripheral aromatization), and anti-androgen therapy used in prostate cancer treatment. A paraneoplastic mechanism should be considered when gynecomastia is of new, rapid onset in an adult male without another obvious cause: hCG-secreting testicular or adrenal tumors, and certain lung cancers, can drive gynecomastia through hCG's structural cross-reactivity with LH at the Leydig cell receptor, stimulating estrogen production.

Male breast cancer accounts for roughly 1% of all breast cancer cases globally, but this proportion is markedly higher (approximately 6%) in Sub-Saharan African populations -- a disparity flagged in this deck because it recurs as an explicit theme in the Ethiopian epidemiology discussion of Section 03 (Slide 11), and one for which the leading proposed explanations include a higher regional prevalence of conditions predisposing to relative hyperestrogenism and a possibly higher frequency of BRCA2 mutations in some African-descent populations, though the epidemiologic evidence base remains incomplete. The dominant risk factors for male breast cancer are Klinefelter syndrome, germline BRCA2 mutation (the single most important known genetic risk factor, conferring a lifetime risk on the order of 6 to 8%, compared with well under 1% in the general male population), prior radiation exposure, and conditions of exogenous or endogenous estrogen excess. Biologically, male breast cancer is overwhelmingly estrogen-receptor-positive (approximately 90% of cases) and tends to present at an older average age (around 67 years) than female breast cancer, typically as a central, retroareolar mass with nipple involvement more common at presentation than in women -- likely reflecting the smaller volume of male breast tissue, which places most tumors in proximity to the nipple-areolar complex from the outset, and also

reflecting diagnostic delay related to lower awareness. Because the breast tissue volume is small, breast-conserving surgery is rarely feasible, and modified radical mastectomy is the standard surgical approach; given the high rate of ER-positivity, tamoxifen (rather than an aromatase inhibitor, which is less effective in men because peripheral androgens continue to provide a substrate for aromatization outside the ovary, unlike the near-complete suppression achievable in postmenopausal women) is the preferred adjuvant endocrine agent. The NCCN 2026 recommendation -- also referenced explicitly in Section 05 (Slide 20) -- that male BRCA2 carriers undergo annual PSA screening from age 40 to 45 reflects the additional, and separately elevated, prostate cancer risk that BRCA2 confers in men, making male BRCA2 carriers a population requiring surveillance for two distinct hormone-driven malignancies simultaneously.

Finally, this slide closes with the physiology of nipple discharge, mechanistically tied back to the prolactin axis introduced above: prolactin secretion from the anterior pituitary is tonically suppressed by hypothalamic dopamine, so any process that either increases prolactin secretion directly (a prolactin-secreting pituitary adenoma) or removes dopaminergic suppression (hypothyroidism, via elevated TRH driving prolactin release, or dopamine-antagonist medications such as antipsychotics and metoclopramide) can produce galactorrhea. This physiologic, typically bilateral, milky discharge must be distinguished on history and examination from pathologic discharge, which is more often unilateral, from a single duct, and blood-tinged or serosanguineous; the most common cause of bloody nipple discharge is a benign intraductal papilloma, but ductal carcinoma in situ must always be actively excluded as the underlying cause before a benign etiology is accepted, and persistent unilateral single-duct discharge that is not explained by imaging warrants further evaluation with galactography or ductoscopy -- a workup pathway developed in full in Section 17 (Slide 65).

SECTION 03 -- Epidemiology and Natural History

Global and Ethiopian incidence/mortality data, and the biological timeline of tumor growth and metastasis -- establishing both why breast cancer is a public-health priority everywhere and why the specific numbers differ so sharply between a high-income and a low-resource setting such as Ethiopia.

SLIDE 10 -- Section Divider -- Epidemiology and Natural History

Detailed Explanation and Source Context

DeVita Chapter 106 opens its epidemiology discussion with the same juxtaposition used here: incidence and mortality trends in high-income countries, set against the markedly different pattern in low- and middle-income countries. This section pairs that DeVita framework directly with the Ethiopian FMOH 2024 national data, which did not exist as a single consolidated source before this guideline and is treated as a primary reference throughout, rather than as an addendum.

SLIDE 11 -- Epidemiology of Breast Cancer

Slide content recap:

- Most common cancer worldwide: 2.26 million new cases in 2020 (11.7% of all cancers)
- Leading cause of cancer death in women globally: 685,000 deaths (2020)
- Most common in high-income countries; mortality highest in LMIC due to late presentation
- 5-year survival: >90% in HIC vs 66% India, 40% South Africa
- Incidence increasing: 0.5% per year in developed countries
- Lifetime risk in US women: 12.5% (1 in 8 women)
- ~310,720 new invasive + ~56,500 DCIS cases estimated 2024 (USA)
- US mortality: ~42,000 deaths/year; declining since 1989 peak (~1.9%/year)
- Decline attributed to mammography screening + adjuvant therapy advances
- Racial disparity: White women higher incidence; Black women higher mortality (HR 1.4)
- TNBC: higher frequency in Black women (20-25% vs 10-15% White)
- Leading cancer in Ethiopia: 16,133 new cases/year (GLOBOCAN 2020)
- Ethiopia ASI rate: 41.5/100,000 -- BC accounts for 1 in 5 of ALL cancers in Ethiopia
- One third of all female cancers in Ethiopia are breast cancer
- Ethiopia age-standardized mortality: 24.1/100,000
- 65-75% of Ethiopian patients present at Stage III or IV (vs <30% in HIC)
- Male BC: 6% of all BC in Sub-Saharan Africa (vs 1% globally)
- Ethiopia: no population-based mammographic screening program -- limited infrastructure
- Ethiopia: CBE as primary early detection modality (not mammography)
- Ethiopia: majority diagnosed Stage III/IV -> limited curative options
- Ethiopia: 3 WHO GBCI 60:60:80 targets -- 60% early stage, 60-day diagnosis, 80% treatment completion
- Ethiopia target 2030: shift diagnosis to >=60% Stage I/II through CBE scale-up

Detailed Explanation and Source Context

DeVita Chapter 106 documents breast cancer as the most commonly diagnosed cancer in women worldwide, with GLOBOCAN 2020 recording 2.26 million new cases -- 11.7% of all cancer diagnoses across both sexes combined -- and 685,000 deaths, making it the leading cause of cancer death in women globally. The pattern of where breast cancer is diagnosed most often (high-income countries, reflecting both a true biological/lifestyle risk gradient and greater diagnostic capture) versus where it kills most often relative to incidence (low- and middle-income countries) is one of the central epidemiologic facts in oncology: the five-year survival exceeds 90% in high-income countries but falls to roughly 66% in India and as low as 40% in South Africa, a gap driven

overwhelmingly by stage at presentation rather than by any fundamental difference in tumor biology. Incidence itself continues to rise modestly (about 0.5% annually) in developed countries, generally attributed to a combination of increasing adoption of Western reproductive and lifestyle risk factors (Section 04) and the detection of biologically indolent disease through expanding screening.

United States-specific data, drawn from DeVita and American Cancer Society statistics, quantify a lifetime risk of 12.5% -- the familiar "1 in 8 women" figure -- with an estimated 310,720 new invasive cases and 56,500 new cases of ductal carcinoma in situ in 2024, and approximately 42,000 deaths annually. Notably, US breast cancer mortality has been declining since a peak around 1989, at a rate of roughly 1.9% per year, a trend attributed jointly to the introduction of population mammographic screening (Section 06) and to progressive improvements in adjuvant systemic therapy (Section 13) -- the two levers, early detection and better treatment, that recur throughout this entire document as the twin drivers of outcome. A persistent racial disparity is documented in the US data: white women have a modestly higher overall incidence, yet Black women have significantly higher mortality (a hazard ratio around 1.4), a gap explained by a combination of later stage at diagnosis, disparities in access to guideline-concordant treatment, and a genuinely higher biological frequency of triple-negative breast cancer in Black women (20 to 25%, compared with 10 to 15% in white women) -- TNBC being, as detailed in Section 11, the molecular subtype with the fewest targeted-therapy options and the worst prognosis stage-for-stage.

The Ethiopian data, drawn from the FMOH 2024 National Breast Cancer Guideline and GLOBOCAN 2020, establish breast cancer as the single leading cancer among Ethiopian women, with 16,133 new cases annually and an age-standardized incidence rate of 41.5 per 100,000 -- breast cancer alone accounts for one in five of all cancer diagnoses in the country and roughly one-third of all cancers diagnosed in women. [ETHIOPIA] The age-standardized mortality rate of 24.1 per 100,000 -- a strikingly high ratio of deaths to incident cases compared with the US figures above -- is the numeric signature of the central problem this guideline was written to address: 65 to 75% of Ethiopian patients present with Stage III or IV disease, versus fewer than 30% in high-income countries, meaning the majority of Ethiopian patients are, at the point of diagnosis, already beyond the stage at which surgery alone offers a realistic prospect of cure. This single statistic -- stage at presentation -- is the epidemiologic root cause that the guideline's entire three-pillar structure (Section 18) is designed to correct, and it is why "early detection" (Pillar 1) and "timely diagnosis" (Pillar 2) are given equal structural weight alongside treatment (Pillar 3) in the Ethiopian framework, in a way that a high-income-country guideline, which can generally assume early-stage presentation, does not need to emphasize as heavily. The observation that male breast cancer constitutes 6% of Sub-Saharan African breast cancer cases (versus 1% globally) is consistent with the finding already introduced in Section 02 (Slide 9) and is repeated here in its epidemiologic, population-level context.

[ETHIOPIA] The guideline note is explicit about the structural cause of Ethiopia's late-stage presentation pattern: there is no population-based mammographic screening program, reflecting the capital equipment, technician, and radiologist infrastructure that mammography requires and that is not yet available at scale; clinical breast examination (CBE) by a trained health worker is therefore adopted as the primary, resource-appropriate early-detection modality (elaborated fully in Section 06, Slide 24). Because CBE detects tumors only once they are large enough to be palpable (typically ≥ 1 cm, and often considerably larger in practice), it inherently cannot achieve the same stage-shift that mammographic screening achieves in a high-income setting -- which is precisely why the guideline sets its own locally calibrated targets rather than importing the American Cancer Society or NCCN screening targets wholesale. These are the WHO Global Breast Cancer Initiative "60:60:80" targets: at least 60% of invasive breast cancers diagnosed at Stage I or II, a maximum of 60 days from first presentation to completion of diagnostic workup, and at least 80% of diagnosed patients completing

multidisciplinary treatment. The explicit 2030 target -- shifting the Stage I/II proportion to 60% or greater primarily through CBE scale-up rather than through mammography -- is the single clearest, most quotable expression of how the Ethiopian guideline diverges from an NCCN- or ESMO-style guideline: it is not attempting to replicate high-income screening infrastructure in the near term, but to achieve stage-shift through a workforce-based (rather than equipment-based) intervention that can be deployed at the primary care level nationwide.

SLIDE 12 -- Natural History of Breast Cancer

Slide content recap:

- Average doubling time: 100-200 days (range 44-1869 days) for invasive BC
- Average tumor detectable by mammography at ~0.5-1.0 cm; clinically palpable at ~2 cm
- Estimated 6-8 years of subclinical growth before detection by mammography
- Cell kinetics: cell loss factor 50-70%; most cells die as they are produced
- S-phase fraction (SPF) and Ki-67 index reflect proliferative rate
- Halsted (centrifugal): cancer spreads orderly: local -> regional -> systemic
- -> Surgical cure possible with wider resection (radical mastectomy rationale)
- Fisher (systemic): cancer is systemic disease from early stages
- -> Surgery for local control only; systemic therapy determines survival
- Modern consensus: both apply -- local control + systemic therapy are essential
- Local invasion -> lymphovascular invasion (LVI) -> nodal spread -> hematogenous
- Key steps: EMT (epithelial-mesenchymal transition), anoikis resistance, extravasation
- Preferential metastatic sites: bone (70%), liver, lung, brain, adrenal (Batson's plexus)
- Breast cancer is the most common cause of malignant pleural effusion in women
- Brain metastasis: highest in HER2+ and TNBC subtypes (~30-50% ultimately)
- Bone-only metastasis: best prognosis among MBC; median OS 2-4 years with modern therapy
- 10-year survival by AJCC stage: I: ~85%, IIA: ~75%, IIB: ~65%, IIIA: ~55%, IIIB: ~40%, IV: ~5-25%
- Nodal status: single most powerful pathologic predictor
- 0 nodes: 70% 10-yr DFS; 1-3 nodes: 55%; 4-9 nodes: 40%; >=10 nodes: 20%
- Dominant patterns: ER+ recurrences late (years 5-20); TNBC/HER2+ early (<3 years)
- Annual hazard of recurrence: ER+ bimodal peaks; TNBC front-loaded peak at 2-3 years
- Late recurrence (>5 years): ER+ only; motivation for extended adjuvant endocrine therapy

Detailed Explanation and Source Context

Breast tumor growth kinetics, as presented in DeVita Chapter 106, are described by an average doubling time of 100 to 200 days for invasive disease, though the true range is extremely wide (44 to 1,869 days), reflecting the underlying molecular heterogeneity that Section 11 explores in depth -- a fast-doubling, high Ki-67 triple-negative tumor and a slow-doubling, low Ki-67 luminal A tumor behave, in this literal sense, like different diseases growing at different speeds. Using average doubling times, a tumor becomes detectable by mammography only once it reaches roughly 0.5 to 1.0 cm, and clinically palpable at roughly 2 cm; extrapolating backward from a single transformed cell, this implies an estimated 6 to 8 years of entirely subclinical growth before a tumor is large enough to be found by the most sensitive routine screening modality available -- a number that is the biological justification for regular screening intervals (Section 06) and that also explains why "early detection" and "primary prevention" are conceptually distinct: a screen-detected cancer has often already been present, silently, for the better part of a decade.

Not every malignant cell that is produced survives to contribute to tumor growth: the "cell loss factor" in breast cancer is estimated at 50 to 70%, meaning that the majority of cells generated by tumor cell division die (via apoptosis, necrosis, or immune clearance) as fast as they are produced, and net tumor growth reflects only the small surviving fraction. The S-phase fraction (the proportion of cells actively synthesizing DNA at a given moment, historically measured by flow cytometry) and the Ki-67 immunohistochemical index (a nuclear proliferation marker, now the more widely used clinical surrogate) both quantify this same underlying proliferative

rate, and Ki-67 in particular reappears repeatedly later in this document as a key input into molecular subtyping (Section 11, Slide 42) and into the monarchE trial's eligibility criteria for adjuvant CDK4/6 inhibition (Section 13, Slide 49).

This slide's central conceptual content is the Halsted-versus-Fisher paradigm, previewed in Section 01's historical timeline and given its full biological grounding here. Halsted's centrifugal model holds that breast cancer spreads in an orderly, predictable, anatomically contiguous sequence: first locally within the breast, then regionally to the axillary lymphatics, and only subsequently -- and, in this model, preventably -- to distant systemic sites; the direct clinical corollary is that sufficiently wide and complete local-regional surgery (the radical mastectomy) should, in principle, be curative by intercepting the disease before it can disseminate. Bernard Fisher's systemic hypothesis, developed from decades of NSABP trial data (including B-06, discussed in Section 01), instead holds that breast cancer is, in a meaningful proportion of patients, a systemic disease from a very early point in its natural history -- meaning that occult micrometastatic dissemination can occur even while the primary tumor and its regional nodes are still surgically resectable, in which case the extent of local surgery cannot, by itself, alter survival, and it is the effectiveness of systemic therapy (chemotherapy, endocrine therapy, and now targeted agents) that determines whether those occult micrometastases progress into clinically evident metastatic disease. The NSABP B-06 finding -- that breast-conserving surgery achieves equivalent survival to radical mastectomy -- was the pivotal piece of evidence supporting Fisher's model over Halsted's, because it demonstrated that more extensive local surgery does not improve survival once adequate local control and systemic therapy are provided. The modern synthesis, stated explicitly on this slide, is that both models contribute partial truth: local-regional therapy (surgery and radiation) is still necessary to achieve durable local control and to prevent the morbidity of uncontrolled local disease, while systemic therapy is what actually governs long-term survival in the majority of patients -- this dual-track logic (local AND systemic, not local OR systemic) is the organizing principle behind every stage-specific treatment algorithm in Section 14.

The metastatic cascade -- the sequence by which cancer cells escape the primary site and establish clinically evident secondary tumors -- proceeds through local invasion of the surrounding stroma, entry into lymphovascular channels (lymphovascular invasion, LVI, a histopathologic finding that independently predicts both nodal involvement and distant recurrence and is scored on every surgical pathology report, Section 11 Slide 40), and eventual hematogenous dissemination once cells achieve venous or lymphatic access to the systemic circulation. At the molecular level, this process requires epithelial-mesenchymal transition (EMT), a phenotypic switch in which epithelial tumor cells lose polarity and cell-cell adhesion and acquire a migratory, mesenchymal phenotype; resistance to anoikis (the programmed cell death that normally occurs when epithelial cells lose attachment to the extracellular matrix, which circulating tumor cells must evade to survive in the bloodstream); and extravasation, the process of exiting the vasculature to colonize a distant organ. Breast cancer shows characteristic, non-random preferential sites of distant metastasis -- bone (accounting for approximately 70% of first distant recurrences), liver, lung, and brain, with adrenal involvement also recognized -- a pattern partly explained anatomically by Batson's vertebral venous plexus, a valveless venous network connecting the pelvic and thoracic venous systems directly to the vertebral column without first passing through the pulmonary capillary bed, providing a direct hematogenous route for tumor cells to reach the axial skeleton. Breast cancer is, notably, the single most common cause of malignant pleural effusion in women, a clinically important fact for any patient presenting with a new pleural effusion and a breast cancer history. Brain metastasis shows striking subtype-selectivity, occurring in an estimated 30 to 50% of patients with HER2-positive or triple-negative disease over the full course of their illness -- far higher than in luminal (ER-positive) disease -- a pattern that directly motivates the emphasis, in Section 15's ESMO 2026 update, on selecting anti-HER2 regimens (specifically tucatinib-based combinations) with demonstrated central nervous system activity. Among the various patterns of metastatic spread, bone-only metastatic disease carries the most

favorable prognosis, with a median overall survival of 2 to 4 years achievable with modern systemic therapy, reflecting both the generally slower growth kinetics of bone metastases and the availability of effective bone-modifying agents (Section 14, Slide 57).

Survival correlates predictably, though imprecisely, with anatomic stage: approximate ten-year overall survival figures run from roughly 85% for Stage I, through 75% for Stage IIA, 65% for Stage IIB, 55% for Stage IIIA, 40% for Stage IIIB, down to a wide 5 to 25% range for Stage IV (a range reflecting the substantial heterogeneity within metastatic disease and the steadily improving outcomes achieved with the newer agents discussed in Section 15). Lymph node status remains, even in the genomic era, the single most powerful pathologic predictor of outcome, with ten-year disease-free survival falling in a clear dose-response relationship to nodal burden: approximately 70% with no involved nodes, 55% with one to three positive nodes, 40% with four to nine positive nodes, and only about 20% with ten or more positive nodes -- a relationship so robust that it anchors the entire N-category structure of AJCC staging in Section 10.

Finally, the temporal pattern of recurrence differs fundamentally by molecular subtype, a fact with direct implications for follow-up strategy (Section 19) and for the duration of adjuvant endocrine therapy (Section 13): ER-positive tumors show a distinctive bimodal hazard pattern, with recurrences occurring at a low but persistent rate for as long as fifteen to twenty years after diagnosis -- these tumors, unlike TNBC or HER2-enriched disease, can and do recur late, beyond five years, which is the direct biological rationale for extending adjuvant endocrine therapy beyond the traditional five-year duration in higher-risk patients (the ATLAS and MA.17R trials, Section 13, Slide 49). Triple-negative and HER2-enriched tumors, by contrast, show a front-loaded hazard of recurrence that peaks early -- around two to three years for TNBC and three to four years for HER2-positive disease -- and then falls off sharply, such that a TNBC patient who remains disease-free at five years has a dramatically better long-term prognosis than an ER-positive patient at the same five-year mark, whose risk of eventual late recurrence has not yet been fully retired.

SECTION 04 -- Risk Factors for Breast Cancer

Genetic, hormonal, environmental, and benign-breast-disease risk factors, and the validated quantitative models used to translate these factors into an individualized risk estimate that drives screening intensity and chemoprevention eligibility.

SLIDE 13 -- Section Divider -- Risk Factors for Breast Cancer

Detailed Explanation and Source Context

Risk factors are grouped in this section, following Schwartz Chapter 17's organization, into four broad categories -- genetic/hereditary (elevated further in Section 05's dedicated BRCA discussion), reproductive/hormonal, environmental/lifestyle, and histologic (benign breast disease) -- because each category implies a different clinical action: genetic risk triggers referral for germline testing and intensified surveillance, hormonal risk factors inform chemoprevention counseling, and histologic risk (atypical hyperplasia, LCIS) triggers a specific management algorithm covered in Section 08.

SLIDE 14 -- Risk Factors -- Overview

Slide content recap:

- BRCA1 mutation: 40-87x lifetime relative risk; predominantly TNBC phenotype
- BRCA2 mutation: 20-40x lifetime relative risk; predominantly ER+ phenotype
- Prior breast cancer (contralateral): 3-4x; annual 0.5% risk of second primary
- LCIS: 8-10x lifetime; bilateral marker; not obligate precursor
- Atypical hyperplasia (ADH/ALH): 4-5x; multiplied 10x if + family history
- First-degree relative BC: 2-3x (if 1); 3-4x if 2 or more first-degree relatives
- Prior chest radiation: 3-5x; Mantle RT for Hodgkin's lymphoma
- Mammographic density (BIRADS 4): 4-6x; dense breast -> decreased sensitivity of mammography
- Obesity (post-menopausal): 1.5-2x; increased peripheral aromatization of androgens
- Nulliparity or late first pregnancy (>35): 1.5-2x; hormonal exposure window effect

Detailed Explanation and Source Context

This overview table ranks the major recognized risk factors by their approximate relative risk, and the ordering itself teaches an important lesson: the single largest risk multipliers are all genetic (germline BRCA1/2 mutation), while the most commonly encountered risk factors in everyday practice (family history of one affected relative, nulliparity, obesity) are individually modest, in the 1.5- to 3-fold range. This distinction matters clinically because it explains why genetic counseling referral criteria (Section 05) are reserved for a specific subset of patients rather than applied universally, while modification of modifiable lifestyle factors is recommended broadly at the population level despite each individual factor's more modest effect size.

BRCA1 mutation carries the highest relative risk of any single factor on this list (40- to 87-fold, depending on the specific study and population), and its phenotype is preferentially triple-negative/basal-like, whereas BRCA2, though still markedly elevated (20- to 40-fold), is preferentially estrogen-receptor-positive -- this ER-status divergence between the two BRCA genes, despite both being involved in the same homologous-recombination DNA repair pathway, is explored mechanistically in Section 05 (Slide 15) and has direct treatment implications (BRCA1 tumors are more likely to be treated with chemotherapy and PARP inhibitors as primary systemic therapy, given the paucity of hormone-receptor targets).

A personal history of contralateral breast cancer confers a 3- to 4-fold relative risk of a second primary, translating to an ongoing absolute risk of approximately 0.5% per year for the remainder of the patient's life -- a risk that never fully returns to baseline and that underlies the recommendation for indefinite annual contralateral mammographic surveillance regardless of which side was originally treated (Section 19, Slide 72). Lobular carcinoma in situ (LCIS) is presented here as a risk factor rather than a precursor lesion -- conferring an 8- to 10-fold lifetime relative risk that is notably bilateral (LCIS in one breast signals elevated risk in both breasts equally) -- a distinction elaborated fully in Section 08 (Slide 31), where LCIS is explicitly contrasted with DCIS, which is instead an obligate, unilateral anatomic precursor.

Atypical hyperplasia -- whether atypical ductal hyperplasia (ADH) or atypical lobular hyperplasia (ALH) -- confers a 4- to 5-fold relative risk on its own, but this risk multiplies to approximately 10-fold when combined with a first-degree relative's history of breast cancer, an important gene-environment-style interaction that changes clinical management (Section 05's risk-model discussion, Slide 17, and the benign-breast-disease slide, Section 04 Slide 16) from routine surveillance to consideration of chemoprevention. Family history itself is graded: a single affected first-degree relative confers a 2- to 3-fold risk, while two or more affected first-degree relatives raises this to 3- to 4-fold, and the pattern, age of onset, and bilaterality within the family are exactly the features used by the referral criteria discussed in Section 05 (Slide 15) to decide who should undergo germline genetic testing rather than simply being counted as having a "positive family history."

Prior therapeutic chest irradiation -- classically, mantle-field radiotherapy administered for Hodgkin lymphoma, historically delivered to a field encompassing the mediastinum and, incidentally, a substantial volume of breast tissue in young patients -- confers a 3- to 5-fold relative risk, and because the radiation exposure typically occurs in adolescence or young adulthood, the resulting breast cancer risk manifests at an unusually young age, motivating a distinct, earlier-starting surveillance protocol recognized by NCCN for this specific survivor population. Mammographic breast density (specifically BIRADS category 4, the most dense category) is itself both a risk factor (4- to 6-fold) and a diagnostic confounder, because dense fibroglandular tissue reduces the sensitivity of mammography to detect an underlying malignancy -- a "double jeopardy" effect (higher risk, harder to detect) that is the specific rationale for supplemental ultrasound or MRI screening in dense-breasted women, discussed in Section 06 (Slide 23).

The remaining two factors on this table are hormonal/lifestyle and carry the most modest relative risks of the group: postmenopausal obesity (1.5- to 2-fold), mechanistically explained by increased peripheral aromatization of adrenal and ovarian androgen precursors to estrogen within adipose tissue -- a mechanism that, notably, does not apply premenopausally, where obesity is not associated with increased risk and may even be modestly protective, reflecting the different hormonal milieu before and after ovarian estrogen production ceases; and nulliparity or a late first full-term pregnancy (after age 35), also 1.5- to 2-fold, which is explained by the "hormonal exposure window" hypothesis -- an early first full-term pregnancy is thought to induce terminal differentiation of breast epithelium that renders it less susceptible to subsequent carcinogenic transformation, so delaying or foregoing this differentiating pregnancy leaves a larger population of susceptible, undifferentiated epithelial cells exposed to ongoing cyclical hormonal stimulation for a longer cumulative period.

SLIDE 15 -- BRCA1 and BRCA2 Mutations

Slide content recap:

- Tumor suppressors -- BRCA1/2 involved in HR (homologous recombination) repair
- BRCA1 (17q21) mutations -> predominantly TNBC (basal-like) phenotype; high-grade
- BRCA1 lifetime BC risk: 50-72% (by age 80)
- BRCA1 lifetime ovarian cancer risk: 40-46%

- High-penetrance: AD inheritance, variable expression
- BRCA1 associated cancers: breast, ovary, fallopian tube; lower risk pancreas/prostate
- BRCA2 (13q12.3) lifetime BC risk: 45-69%; predominantly ER+ (higher luminal A/B frequency)
- BRCA2 lifetime ovarian cancer risk: 12-20%
- BRCA2 male BC risk: 6-8% lifetime (vs <1% general population)
- BRCA2 other cancers: pancreatic (4-7%), melanoma, prostate (7-24% depending on age)
- NCCN 2026: BRCA2 carriers -> annual prostate cancer screening from age 40-45
- Biallelic BRCA2: Fanconi anemia -- embryonic lethality / severe developmental defects
- 3 common Ashkenazi founder mutations: 185delAG, 5382insC (BRCA1); 6174delT (BRCA2)
- Ashkenazi prevalence: 1 in 40 individuals (vs 1:400 general population)
- Higher prior probability -> expanded testing criteria in this population
- Testing criteria (NCCN 2026): TNBC at any age; BC diagnosed <=50 years; >=2 primary BC
- Testing criteria: ovarian/fallopian/peritoneal cancer at any age; male breast cancer
- Testing criteria: Ashkenazi ancestry; first/second degree relative meeting criteria
- BRCA1/2 in germline: opens olaparib/talazoparib eligibility in metastatic setting

Detailed Explanation and Source Context

BRCA1 (located at chromosome 17q21) and BRCA2 (13q12.3) both encode tumor suppressor proteins that function centrally in the homologous recombination pathway of double-strand DNA break repair -- a high-fidelity repair mechanism that, when intact, is one of the cell's primary defenses against the accumulation of oncogenic mutations. Loss of function in either gene (following the classic Knudson two-hit model, with one allele inactivated in the germline and the second lost somatically within the tumor) forces cells to rely on error-prone, lower-fidelity repair pathways, generating genomic instability that predisposes to malignant transformation -- and this same mechanistic vulnerability is precisely what PARP inhibitors (olaparib, talazoparib) exploit therapeutically through synthetic lethality, a concept introduced here and developed fully in Section 05's next slide (Slide 19) and in the ESMO 2026 metastatic algorithms of Section 15.

Despite sharing this common repair pathway, BRCA1 and BRCA2 carriers develop phenotypically distinct cancers, a divergence that is one of the more frequently tested and clinically consequential facts in this topic. BRCA1-associated breast cancers are predominantly high-grade and triple-negative/basal-like, conferring a lifetime breast cancer risk of 50 to 72% by age 80 and a lifetime ovarian cancer risk of 40 to 46% -- both markedly higher than BRCA2's corresponding figures. As a high-penetrance, autosomal dominant condition with variable expressivity, BRCA1 also elevates risk (to a lesser but still clinically actionable degree) for fallopian tube and primary peritoneal cancer (biologically related to ovarian cancer and often considered together), and to a smaller extent pancreatic and prostate cancer. BRCA2-associated breast cancers, by contrast, are predominantly estrogen-receptor-positive, with a higher relative frequency of luminal A and luminal B molecular subtypes (Section 11), and carry a somewhat lower lifetime breast cancer risk of 45 to 69% and ovarian cancer risk of 12 to 20%. BRCA2 is distinguished by two additional, clinically important associations not shared to the same degree by BRCA1: a substantially elevated male breast cancer risk of 6 to 8% lifetime (versus well under 1% in the general male population, as introduced in Section 02, Slide 9), and a markedly elevated and clinically aggressive prostate cancer risk in male carriers, ranging from 7 to 24% depending on age at assessment -- the [UPDATE] NCCN 2026 recommendation that BRCA2 male carriers undergo annual PSA screening beginning at age 40 to 45 is a direct response to this specific, quantified risk, and is developed in full mechanistic and statistical detail in the next slide (Slide 20). BRCA2 also elevates pancreatic cancer risk (4 to 7% lifetime) and melanoma risk more than BRCA1 does. A rare but important genetic phenomenon worth noting is biallelic BRCA2 mutation (inheriting a pathogenic variant from both parents), which does not simply double the cancer risk but instead causes Fanconi anemia complementation group D1, a severe autosomal recessive bone marrow failure and developmental disorder -- illustrating that BRCA2's role extends well beyond cancer predisposition into fundamental genomic maintenance required for normal development.

The Ashkenazi Jewish population carries a dramatically elevated prior probability of harboring a BRCA1 or BRCA2 mutation because of three specific, well-characterized founder mutations that arose in this genetically distinct population and have since achieved a carrier frequency of approximately 1 in 40 individuals -- ten times the roughly 1 in 400 background carrier frequency estimated in the general population. The three founder mutations are 185delAG and 5382insC in BRCA1, and 6174delT in BRCA2; because these specific mutations are so common in this population, genetic testing laboratories often screen for these three variants specifically as an initial, cost-effective test in patients of Ashkenazi ancestry before proceeding to full-panel sequencing if the founder-mutation panel is negative but suspicion remains high. This elevated prior probability is also why the NCCN testing criteria explicitly lower the threshold for genetic counseling referral in patients of Ashkenazi Jewish ancestry compared with the general population.

The NCCN v2.2026 germline testing criteria, summarized on this slide and expanded upon in the dedicated update slide that follows (Slide 20), identify specific clinical scenarios that should trigger genetic counseling and testing regardless of family history: triple-negative breast cancer diagnosed at any age (because of TNBC's strong biological association with BRCA1), breast cancer diagnosed at age 50 or younger, two or more primary breast cancers in the same individual (whether bilateral or two separate ipsilateral primaries), any personal history of ovarian, fallopian tube, or primary peritoneal cancer at any age, any case of male breast cancer, Ashkenazi Jewish ancestry, and having a first- or second-degree relative who independently meets any of these criteria. Confirming a germline BRCA1 or BRCA2 pathogenic or likely pathogenic variant is not merely diagnostic information for cascade family testing -- it is directly therapeutically actionable in the metastatic setting, since it establishes eligibility for PARP inhibitor monotherapy (olaparib or talazoparib), a treatment pathway detailed in both the risk-reduction context (Slide 19, next) and the ESMO 2026 metastatic treatment algorithms (Section 15, Slide 59).

SLIDE 16 -- Hormonal and Environmental Risk Factors

Slide content recap:

- Early menarche (<12): increased risk (increased lifetime estrogen exposure)
- Late menopause (>55): increased risk; nulliparity or first birth >35 -> 1.5-2x increased
- Breastfeeding: protective -- each year of breastfeeding -> 4% decreased BC risk
- HRT (combined E+P): WHI -> 1.24x increased risk for >5 yrs; absolute risk small (8 per 10,000)
- Oral contraceptives: COCP slightly increased current-use risk (RR 1.24); wanes after cessation
- DES exposure (in utero): 2x increased risk in daughters (mothers who took DES 1940-1971)
- Non-proliferative lesions (fibrocystic change, cysts, mild hyperplasia): NO increased risk
- Proliferative without atypia (moderate hyperplasia, sclerosing adenosis, papilloma): 1.5-2x increased
- Proliferative with atypia: 4-5x increased (ADH and ALH have equal predictive value)
- -> Atypical hyperplasia + first-degree relative BC: 10x increased lifetime risk
- Radial scar / complex sclerosing lesion: independent 2x increased risk; requires excision
- Radiation: ionizing radiation to chest (Hodgkin therapy, repeated fluoroscopy)
- Alcohol: dose-dependent; each drink/day -> ~7% increased risk
- Obesity (BMI >30): 1.5-2x increased post-menopausal (adipose aromatase); protective pre-menopausal
- Physical activity: 25-30% decreased risk with regular moderate exercise
- Night-shift work: possible increased risk (circadian disruption, decreased melatonin -- limited evidence)
- Diet: no single food established; Mediterranean diet pattern associated with decreased risk
- Li-Fraumeni syndrome (TP53): lifetime BC risk 40-80%; often premenopausal
- Cowden syndrome (PTEN): 25-85% lifetime BC risk; also thyroid/endometrial
- Peutz-Jeghers (STK11): 30-50% BC risk + GI hamartomas
- Hereditary diffuse gastric cancer (CDH1): 40-60% lobular BC risk
- ATM heterozygotes: 2-3x risk; homozygous -> ataxia-telangiectasia
- PALB2 mutation: 14-35% lifetime BC risk -- becoming routine in germline testing panels

Detailed Explanation and Source Context

The reproductive and hormonal risk factors on this slide all share a common unifying mechanism: cumulative lifetime exposure to unopposed cyclical estrogen (and, to a lesser degree, progesterone) signaling on breast epithelium. Early menarche (before age 12) lengthens the total duration of ovulatory cycling before any protective effect of a first pregnancy, while late menopause (after age 55) extends this same cyclical exposure at the other end of reproductive life; both increase risk through the identical mechanism of more cumulative hormone-cycle exposure, and both are frequently tested together as a pair for exactly this reason. Nulliparity, or a late first full-term pregnancy after age 35, was already introduced in Slide 14's overview and is restated here in its full hormonal-exposure-window context. Breastfeeding is the one reproductive factor on this list that is protective rather than risk-elevating: each cumulative year of lactation is associated with an approximate 4% reduction in lifetime breast cancer risk, thought to reflect both suppression of ovulatory cycling during lactation and a direct differentiating effect of lactation on breast epithelium analogous to (and additive with) the protective effect of an early full-term pregnancy.

Exogenous hormone exposure recapitulates and, in the case of combined hormone replacement therapy, adds to this endogenous risk: the Women's Health Initiative trial -- already referenced in Section 02 (Slide 9) -- found combined estrogen-plus-progestin HRT used for more than five years increased breast cancer risk by a relative 24% (hazard ratio approximately 1.24), though it is important to contextualize this figure in absolute terms, since the corresponding absolute risk increase was small, on the order of eight additional cases per 10,000 woman-years -- a distinction between relative and absolute risk that is essential for counseling patients considering HRT for menopausal symptom control. Combined oral contraceptives show a similar, modest relative risk elevation (RR approximately 1.24) confined to the period of current or very recent use, which wanes progressively after discontinuation and appears to return to baseline within roughly ten years -- meaning this is a transient, reversible risk rather than a permanent one. In utero diethylstilbestrol (DES) exposure -- a synthetic estrogen prescribed to pregnant women (primarily in the United States) between approximately 1940 and 1971 to prevent miscarriage, before its carcinogenic and teratogenic effects were recognized -- confers an approximately two-fold increased breast cancer risk specifically in the daughters who were exposed in utero, a now-historical but still clinically relevant exposure in women born during that window.

Benign breast disease is stratified into three histologic risk categories that directly determine follow-up intensity, a framework Schwartz Chapter 17 presents as a decision-tree: non-proliferative lesions (simple fibrocystic change, simple cysts, mild epithelial hyperplasia without atypia) confer no increased breast cancer risk above the general population baseline and require no additional surveillance beyond routine screening; proliferative disease without atypia (moderate or florid hyperplasia, sclerosing adenosis, intraductal papilloma without atypia) confers a modest 1.5- to 2-fold relative risk; and proliferative disease with atypia -- specifically atypical ductal hyperplasia (ADH) or atypical lobular hyperplasia (ALH) -- confers a substantially higher 4- to 5-fold relative risk, with ADH and ALH considered to carry approximately equal predictive weight despite arising from ductal versus lobular epithelium respectively. As already flagged in Slide 14, this atypia-associated risk compounds multiplicatively with family history: the combination of atypical hyperplasia and a first-degree relative with breast cancer raises lifetime risk approximately 10-fold, a combination that specifically triggers consideration of chemoprevention under the NCCN thresholds discussed in the risk-models slide that follows (Slide 17). A radial scar (also called a complex sclerosing lesion) is treated as a separate, independent risk category conferring an approximately 2-fold increased risk on its own, and -- unlike simple proliferative lesions without atypia -- radial scars identified on core needle biopsy are generally recommended for surgical excision, both because of this independent risk elevation and because of a recognized, non-trivial rate of occult malignancy or atypia found upon complete excision of lesions initially sampled only by core biopsy.

Environmental and lifestyle factors round out the modifiable side of the risk equation. Ionizing radiation to the chest -- whether from therapeutic mantle radiotherapy for Hodgkin lymphoma (already discussed in Slide 14) or, historically, from repeated diagnostic fluoroscopic chest exposures -- is a well-established carcinogen for breast tissue, with risk inversely related to age at exposure (younger exposed breast tissue, still undergoing active proliferation, is more radiosensitive). Alcohol consumption shows a clear dose-dependent relationship, with each additional standard drink per day associated with an approximate 7% relative increase in risk, mechanistically attributed to both alcohol's effect on increasing circulating estrogen levels and to direct DNA-damaging effects of its metabolite acetaldehyde. Obesity, quantified here as BMI greater than 30, again shows the postmenopausal-specific pattern already explained in Slide 14 (1.5- to 2-fold increased risk via increased peripheral aromatase activity in adipose tissue), while being neutral or even modestly protective premenopausally. Regular moderate physical activity is one of the few interventions with a genuinely protective effect, associated with a 25 to 30% risk reduction, likely mediated through multiple pathways including reduced adiposity, lower circulating insulin and insulin-like growth factor levels, and immune modulation. Night-shift work has been proposed as a risk factor through circadian disruption and consequent suppression of melatonin (itself hypothesized to have oncostatic properties), though the evidence base for this association remains limited compared with the other, better-established factors on this slide. No single dietary component has been definitively established as causal, though epidemiologic data consistently associate a Mediterranean dietary pattern (high in vegetables, fruit, whole grains, and unsaturated fats, low in processed foods) with modestly reduced risk.

The slide closes with a set of hereditary cancer syndromes beyond BRCA1/2 that all confer substantially elevated breast cancer risk through distinct molecular mechanisms, each of which is separately actionable in a genetic counseling context and, per the NCCN v2.2026 update discussed in Slide 20, increasingly captured on multi-gene panel testing rather than single-gene BRCA1/2 testing alone. Li-Fraumeni syndrome, caused by germline TP53 mutation, confers a strikingly high lifetime breast cancer risk of 40 to 80%, often presenting at unusually young, even premenopausal ages, and is associated with a broader multi-cancer phenotype (sarcoma, adrenocortical carcinoma, brain tumors, leukemia) that should prompt suspicion when breast cancer occurs alongside these other malignancies in a family. Cowden syndrome, caused by germline PTEN mutation, confers an even wider risk range (25 to 85% lifetime breast cancer risk) and is additionally associated with thyroid and endometrial cancer, plus characteristic mucocutaneous hamartomas. Peutz-Jeghers syndrome (germline STK11 mutation) confers a 30 to 50% breast cancer risk in addition to its more classically recognized gastrointestinal hamartomatous polyposis and mucocutaneous pigmentation. Hereditary diffuse gastric cancer syndrome, caused by germline CDH1 mutation (the same E-cadherin gene whose loss characterizes sporadic lobular carcinoma at the tissue level, as discussed in Section 08, Slide 31), confers a distinctively lobular-histology breast cancer risk of 40 to 60% -- linking a hereditary gastric cancer syndrome mechanistically to a specific breast cancer histologic subtype through loss of the same cell-adhesion protein in both organs. ATM heterozygote carriers face a more modest 2- to 3-fold breast cancer risk (with homozygous ATM mutation instead causing the distinct, severe pediatric syndrome ataxia-telangiectasia), and PALB2 mutation -- a gene that functions cooperatively with BRCA2 in the same homologous recombination pathway -- confers a 14 to 35% lifetime breast cancer risk and has moved, over the past several guideline cycles, from a research finding to a gene now tested routinely alongside BRCA1/2 on standard hereditary breast cancer panels, as reaffirmed in the NCCN 2026 update (Slide 20).

SLIDE 17 -- Risk Assessment Models

Slide content recap:

- Gail/BCRAT: age, menarche, first birth, family hx, biopsies, race -> 5-year risk; chemoprevention eligibility (Gail $\geq 1.7\%$)
- Tyrer-Cuzick (IBIS): reproductive hx, density, family hx, BBD, BRCA probability -> lifetime risk; most accurate for strong family hx
- BRCAPro: pedigree + BRCA1/2 test results -> probability of carrying BRCA1/2 variant
- BOADICEA/CanRisk: full pedigree, multiple susceptibility genes, PRS -> comprehensive hereditary risk incl. PALB2/ATM
- BCSC: age, race/ethnicity, density, family hx, biopsy -> population-based 5-10 year risk
- NCCN 2026 thresholds: discuss chemoprevention if Gail 5-year risk $\geq 1.7\%$; stronger recommendation if Gail $\geq 3\%$ or IBIS/Tyrer-Cuzick 10-year $\geq 5\%$

Detailed Explanation and Source Context

Because the risk factors catalogued across Slides 14 and 16 rarely occur in isolation, and because their effects are neither simply additive nor uniformly multiplicative, validated statistical risk-prediction models exist to synthesize an individual patient's specific combination of factors into a single quantitative risk estimate that can drive concrete clinical decisions -- most importantly, chemoprevention eligibility (Section 06, Slide 23) and the choice between average-risk and high-risk (MRI-supplemented) screening protocols (Section 06, Slide 23). Each model on this slide has a distinct input structure and a distinct best-use case, and recognizing which model is appropriate for which clinical question is itself a tested skill.

The Gail model (formally, the Breast Cancer Risk Assessment Tool, BCRAT) is the most widely used model in general practice, built from a relatively compact set of inputs -- current age, age at menarche, age at first live birth, number of first-degree relatives with breast cancer, number of prior breast biopsies (and whether any showed atypical hyperplasia), and race/ethnicity -- to generate a five-year and a lifetime risk estimate. Its major and well-recognized limitation is that it captures family history only crudely (counting affected first-degree relatives without incorporating the detailed multi-generational pedigree structure, age of onset, or paternal-side history that would suggest a hereditary syndrome), and consequently it systematically underestimates risk in patients from strongly hereditary (particularly BRCA-affected) kindreds. Despite this limitation, the Gail model's five-year risk threshold is the specific number embedded in the NCCN chemoprevention guideline -- a Gail five-year risk of 1.7% or greater is the threshold at which chemoprevention should be discussed with the patient (this exact 1.7% figure derives historically from the NSABP P-1 tamoxifen chemoprevention trial's own eligibility criteria), making the Gail model, despite its simplicity, the single most commonly applied tool in this specific decision.

The Tyrer-Cuzick model (also known by its associated software, IBIS) incorporates a substantially richer input set -- detailed reproductive history, mammographic breast density, a more granular multi-generational family history, personal history of benign breast disease including atypical hyperplasia and LCIS, and an estimated probability of underlying BRCA1/2 mutation status -- and is generally regarded as the most accurate model specifically for patients with a strong family history, precisely because it models pedigree structure more richly than Gail. Its recognized limitation is a tendency to overestimate risk in some populations with atypical hyperplasia or LCIS relative to observed outcomes in certain validation cohorts. BRCAPro takes a narrower, more targeted approach: using pedigree information together with any available BRCA1/2 genetic test results within the family, it calculates specifically the probability that an individual carries a BRCA1 or BRCA2 mutation -- a distinct output from the breast-cancer-risk-percentage that Gail and Tyrer-Cuzick produce, and one used primarily to help decide whether genetic testing itself is worthwhile before committing to it, or to interpret an ambiguous or negative test result in the context of residual mutation probability. BOADICEA (now more commonly accessed through its successor platform, CanRisk) is the most comprehensive model available, incorporating full pedigree data, multiple susceptibility genes beyond BRCA1/2 (including PALB2, ATM, and CHEK2, reflecting the same expanded gene panel emphasized in the NCCN 2026 update, Slide 20), and even an optional polygenic risk score (PRS) input -- though, as that same update slide notes explicitly, PRS is not yet recommended as a standard-of-care input by NCCN as of the 2026 guideline cycle, reflecting continued

uncertainty about how best to integrate common low-penetrance genetic variants into individualized risk counseling. Finally, the Breast Cancer Surveillance Consortium (BCSC) model is a population-based tool, using age, race/ethnicity, breast density, family history, and prior biopsy history to generate five- and ten-year risk estimates, and is best suited to population-level or screening-program risk stratification rather than to the detailed individualized counseling of a patient with a complex family history, which is better served by Tyrer-Cuzick or BOADICEA/CanRisk.

The NCCN 2026 thresholds that operationalize these model outputs into action are explicit and worth committing to memory precisely because they are the point at which risk assessment translates into a treatment recommendation: chemoprevention should be discussed with any patient whose Gail five-year risk reaches 1.7% or higher, and a stronger recommendation to proceed with chemoprevention applies at a Gail five-year risk of 3% or higher, or equivalently at a Tyrer-Cuzick/IBIS ten-year risk of 5% or higher -- these specific numeric thresholds are elaborated with their corresponding drug options (tamoxifen, raloxifene, exemestane, anastrozole) in Section 06 (Slide 23).

SECTION 05 -- Hereditary Breast Cancer

Management strategies once a pathogenic BRCA1/2 (or related) variant is identified -- surveillance, chemoprevention, risk-reducing surgery, and systemic-treatment implications -- plus the [UPDATE] NCCN v2.2026 and JNCCN v2.2026 revisions to genetic risk assessment that postdate the base Schwartz and DeVita chapters.

SLIDE 18 -- Section Divider -- Hereditary Breast Cancer

Detailed Explanation and Source Context

Having established in Section 04 which patients should be referred for germline testing, this section addresses the downstream question: once a pathogenic variant is confirmed, what specifically should be done. Management branches into three parallel, non-mutually-exclusive tracks -- enhanced surveillance, pharmacologic chemoprevention, and risk-reducing surgery -- and the choice among (or combination of) these tracks is a shared decision incorporating the carrier's specific gene, age, childbearing plans, and personal risk tolerance.

SLIDE 19 -- BRCA Carrier Management Strategies

Slide content recap:

- Annual breast MRI + mammography (alternating every 6 months) from age 25-75
- Clinical breast exam every 6-12 months from age 25
- BRCA1: start MRI by age 25; BRCA2: by 25-30
- Transvaginal ultrasound + CA-125: q6 months from 30-35 yrs or 5-10 yrs before earliest affected relative
- PSA screening for male BRCA1/2 carriers: annual from age 40-45 (NCCN 2026)
- Tamoxifen 20 mg/day x 5 years: reduces ER+ BC by 49% (NSABP P-1)
- -> Less established benefit in BRCA1 (predominantly TNBC); clearer benefit for BRCA2
- Raloxifene (postmenopausal): similar efficacy to tamoxifen, less uterine risk
- Aromatase inhibitors (AIs): postmenopausal only; exemestane, anastrozole
- Risk-benefit should include QoL, contraindications (VTE, endometrial cancer history)
- Risk-reducing bilateral mastectomy (RRM): decreases BC risk by ~90-95%
- Risk-reducing bilateral salpingo-oophorectomy (RRBSO): decreases ovarian Ca risk 80-96%
- RRBSO also decreases BC risk ~50% if done before menopause (BRCA1 by age 40, BRCA2 by 45)
- Nipple-sparing mastectomy: acceptable for appropriate candidates; residual risk ~5-10%
- Reconstruction counseling: implant vs autologous; body image discussion essential
- Olaparib (PARP inhibitor): FDA-approved for BRCA1/2 HER2-neg early BC (high-risk; residual dz after neoadj)
- Talazoparib: alternative PARP inhibitor; similar indication
- Neoadjuvant: BRCA1/2 -> use platinum-based chemotherapy (cisplatin active)
- Metastatic: olaparib/talazoparib preferred over single-agent chemo (OlympiAD/EMBRACA trials)
- Children: autosomal dominant -- 50% chance; genetic counseling for first-degree relatives

Detailed Explanation and Source Context

Non-surgical surveillance for confirmed BRCA1/2 carriers is more intensive than average-risk screening in three specific ways: it starts earlier, it adds MRI to mammography, and it extends to organs beyond the breast. The recommended protocol is annual breast MRI plus annual mammography, staggered so that one or the other imaging modality is performed approximately every six months, providing continuous surveillance coverage from age 25 through 75; MRI is added specifically because, as established in Section 06 (Slide 23), its sensitivity in BRCA carriers (71 to 100%) substantially exceeds mammography's sensitivity in this population (33 to 60%), reflecting both the younger age and denser breast tissue typical of BRCA carriers and the more aggressive,

faster-growing tumor biology (particularly in BRCA1 carriers) that can arise and become clinically significant between annual mammographic intervals. Clinical breast examination is recommended more frequently than in average-risk patients, every six to twelve months starting at age 25. A modest but clinically meaningful distinction exists between the two genes: BRCA1 carriers are recommended to begin MRI surveillance by age 25, while BRCA2 carriers may begin slightly later, at 25 to 30, reflecting BRCA2's somewhat later average age of onset and lower absolute risk at the youngest ages. Because BRCA1/2 mutations also substantially elevate ovarian cancer risk (Slide 15), transvaginal ultrasound combined with serum CA-125 measurement is recommended every six months starting at age 30 to 35, or five to ten years before the earliest age of ovarian cancer onset in an affected relative, whichever is earlier -- though it should be noted that this ovarian surveillance strategy has not been shown to reduce ovarian cancer mortality as effectively as risk-reducing surgery, discussed below. [UPDATE] The NCCN 2026 recommendation for annual PSA screening in male BRCA1/2 carriers, beginning at age 40 to 45, was already introduced in Section 02 (Slide 9) and Section 04 (Slide 15) and is repeated here as part of the comprehensive surveillance package, with its full evidentiary basis (the specific prostate cancer risk percentages and odds ratios) detailed in the dedicated update slide that follows (Slide 20).

Chemoprevention in BRCA carriers uses the same agents validated in the general high-risk population (Section 06, Slide 23) but with gene-specific caveats regarding expected benefit. Tamoxifen, 20 mg daily for five years, reduced the incidence of estrogen-receptor-positive breast cancer by 49% in the landmark NSABP P-1 prevention trial; because this benefit is mediated entirely through ER blockade, and because BRCA1-associated tumors are predominantly triple-negative (ER-negative, per Slide 15), tamoxifen's chemopreventive benefit is far less well established in BRCA1 carriers than in BRCA2 carriers, whose tumors are predominantly ER-positive and for whom the benefit is correspondingly clearer and more consistently observed. Raloxifene is a reasonable postmenopausal alternative to tamoxifen with broadly similar efficacy but a more favorable uterine safety profile (lower endometrial cancer and venous thromboembolism risk), and aromatase inhibitors (exemestane, anastrozole) are additional postmenopausal-only options -- none of these agents can be used premenopausally without concurrent ovarian suppression, since AIs require a postmenopausal estrogen milieu to be effective. Any chemoprevention decision must weigh quality-of-life effects and contraindications, particularly a personal history of venous thromboembolism or endometrial cancer, which are relative or absolute contraindications to tamoxifen specifically.

Risk-reducing surgery offers the largest and most durable risk reduction of any intervention available to a BRCA carrier. Risk-reducing bilateral mastectomy (RRM) reduces breast cancer incidence by approximately 90 to 95% -- not to zero, because a small amount of at-risk glandular tissue inevitably remains even after complete mastectomy, particularly at the chest wall margin and in the axillary tail; this residual risk is somewhat higher (5 to 10%) with a nipple-sparing technique, which is nonetheless considered oncologically acceptable in appropriately selected carriers who place high value on cosmetic and sensory outcome, reflecting an evolving shift toward less radical prophylactic surgery as data accumulate on its safety. Risk-reducing bilateral salpingo-oophorectomy (RRBSO) reduces ovarian cancer risk by 80 to 96%, and carries a second, distinct benefit: when performed before natural menopause, RRBSO also reduces subsequent breast cancer risk by approximately 50%, by eliminating ovarian estrogen production during what would otherwise be the remaining years of premenopausal hormonal exposure -- the recommended timing for this breast-cancer-risk-reduction benefit is age 40 for BRCA1 carriers (reflecting BRCA1's higher and earlier ovarian cancer risk, which drives the timing decision primarily) and age 45 for BRCA2 carriers. Any discussion of risk-reducing mastectomy must include detailed reconstruction counseling -- implant-based versus autologous tissue reconstruction, discussed surgically in Section 12 (Slide 47) -- as well as explicit attention to body image and sexuality, since these prophylactic operations are, unlike therapeutic mastectomy, performed on healthy tissue in an asymptomatic

patient, and the psychological calculus is correspondingly different.

Finally, confirmed germline BRCA1/2 status carries direct systemic-treatment implications that extend across the full disease spectrum, tying this slide forward to Sections 13 and 15. In the neoadjuvant setting, BRCA1/2 status favors the inclusion of platinum-based chemotherapy (cisplatin or carboplatin), since platinum agents cause DNA interstrand crosslinks that are particularly lethal to cells already deficient in homologous-recombination repair -- the same mechanistic principle underlying PARP inhibitor synthetic lethality. In the adjuvant setting for early-stage, HER2-negative disease with high-risk features or residual disease after neoadjuvant chemotherapy, olaparib is FDA-approved as adjuvant therapy for confirmed germline BRCA1/2 carriers, with talazoparib available as an alternative PARP inhibitor for a similar indication. In the metastatic setting, both the OlympiAD trial (olaparib) and the EMBRACA trial (talazoparib) demonstrated that PARP inhibitor monotherapy is preferred over standard single-agent chemotherapy in HER2-negative, BRCA1/2-mutated metastatic breast cancer, offering improved progression-free survival with a generally more favorable toxicity profile than cytotoxic chemotherapy -- this metastatic-setting recommendation is repeated and contextualized within the broader 2026 treatment algorithm in Section 14 (Slide 57). Finally, because BRCA1/2 mutations are inherited in an autosomal dominant pattern, each child of a confirmed carrier has an independent 50% chance of inheriting the same pathogenic variant, which is the basis for offering cascade genetic counseling and testing to all first-degree relatives of any newly identified carrier -- a principle that recurs in the NCCN 2026 update's discussion of cascade testing for the newly emphasized genes discussed next (Slide 20).

SLIDE 20 -- NCCN 2026 -- Genetic Risk Assessment Updates

Slide content recap:

- BRCA1 P/LP carriers: 7-26% cumulative lifetime prostate Ca risk
- BRCA2 P/LP carriers: 19-61% cumulative lifetime prostate Ca risk
- BRCA2 associated with aggressive prostate cancer (T4, Gleason ≥ 8 , metastatic)
- OR 3.19 (95% CI 1.94-5.25) for aggressive prostate Ca in BRCA2 carriers
- NEW: Annual PSA screening recommended from age 40-45 for BRCA1/2 male carriers
- Screen from age 40 with BRCA2 or positive family hx of prostate Ca
- ATM P/LP variants: 2-4x BC risk; prostate Ca risk (moderate penetrance)
- CHEK2 1100delC: 2-3x BC risk; also associated with prostate Ca risk
- PALB2: 14-35% lifetime BC risk -- now routinely tested alongside BRCA1/2
- PRS (polygenic risk score): NOT recommended as standard of care by NCCN 2026
- CanRisk/BOADICEA now includes PALB2, ATM, CHEK2 in comprehensive risk model
- HOXB13 P/LP variants found in 1.4-4.5% of prostate cancer patients
- G84E missense variant: 60% lifetime prostate Ca risk by age 80 in carriers
- HOXB13 associated with early-onset prostate cancer (age ≤ 55)
- Now included in multi-gene panel testing for hereditary cancer syndromes
- No strong association with aggressive disease phenotype (unlike BRCA2)
- BRCA1/2 carriers: annual pancreatic Ca surveillance from age 50 recommended
- Nonepithelial ovarian cancer (granulosa, Sertoli-Leydig): new testing criteria added
- BRIP1, RAD51C/D: ovarian Ca risk; relevant for counseling surgical prevention
- Germline testing cascade: all first-degree relatives of confirmed P/LP variant carrier

Detailed Explanation and Source Context

This slide is drawn directly from the JNCCN v2.2026 Genetic/Familial High-Risk Assessment guideline update, one of the seven source documents supplied for this revision, and represents content that postdates -- and in the case of the specific prostate-cancer risk quantification, meaningfully extends -- what is available in the base Schwartz and DeVita chapters. Its inclusion in a breast-cancer-focused deck is deliberate: because BRCA1/2 are shared breast-and-prostate (and, for BRCA2, pancreatic) cancer susceptibility genes, a breast oncology

teaching document is incomplete without covering the male-carrier and cascade-family implications of a BRCA diagnosis made in a female patient's workup.

The headline update is a quantified, gene-specific prostate cancer risk figure for male BRCA1/2 carriers: BRCA1 pathogenic/likely pathogenic (P/LP) carriers face a cumulative lifetime prostate cancer risk of 7 to 26%, while BRCA2 carriers face a substantially higher risk of 19 to 61% -- and, critically, BRCA2-associated prostate cancer is not merely more frequent but qualitatively more aggressive, with a demonstrated association with advanced local stage (T4), high Gleason grade (≥ 8), and metastatic presentation at diagnosis, quantified in the source literature as an odds ratio of 3.19 (95% CI 1.94-5.25) for aggressive disease phenotype in BRCA2 carriers compared with non-carriers. This combination of elevated frequency and elevated aggressiveness is precisely what justifies the new recommendation, appearing for the first time in this guideline cycle, of annual PSA screening beginning at age 40 to 45 for BRCA1/2 male carriers, with screening initiation specifically at age 40 recommended for BRCA2 carriers or for any carrier with an additional positive family history of prostate cancer -- a more aggressive and earlier-starting screening protocol than the general-population PSA screening recommendations, justified entirely by this quantified elevated and aggressive-phenotype risk.

Beyond BRCA1/2, the update formally incorporates several additional genes into the routine, multi-gene panel testing paradigm rather than treating them as second-tier or optional additions. ATM pathogenic/likely pathogenic variants confer a moderate, 2- to 4-fold breast cancer risk (as already introduced in Section 04, Slide 16) and are now also recognized to confer a moderate-penetrance prostate cancer risk in male carriers. CHEK2 1100delC, a specific and well-characterized truncating variant, confers a similar 2- to 3-fold breast cancer risk and an associated prostate cancer risk. PALB2, already discussed in Slide 16 for its 14 to 35% lifetime breast cancer risk, is explicitly described here as having transitioned from a research-context finding to a gene tested routinely alongside BRCA1/2 as standard practice -- reflecting several years of accumulated penetrance data that have matured PALB2 from investigational to actionable. Notably, and this is a specific point of continuity rather than change from prior guideline cycles, the polygenic risk score (PRS) -- which aggregates the small individual effects of many common single-nucleotide variants into a composite risk score -- is explicitly NOT recommended as a standard-of-care input by NCCN as of the 2026 cycle, reflecting persistent uncertainty about PRS calibration across different ancestral populations and about how to integrate it meaningfully alongside monogenic testing results in clinical counseling; the comprehensive risk models CanRisk and BOADICEA, discussed in Section 04 (Slide 17), have nonetheless incorporated PALB2, ATM, and CHEK2 into their standard input set, even as they continue to treat PRS as an optional, non-mandatory addition.

A genuinely new gene introduced in this guideline cycle is HOXB13, most notably its specific G84E missense variant. HOXB13 pathogenic/likely pathogenic variants are found in 1.4 to 4.5% of men diagnosed with prostate cancer, and the G84E variant specifically confers a striking 60% lifetime risk of prostate cancer by age 80 in carriers, with a characteristic association with early-onset disease (diagnosed at age 55 or younger) -- HOXB13 is therefore now included in multi-gene hereditary cancer panels. It is worth explicitly noting the contrast drawn in the source material between HOXB13 and BRCA2: despite HOXB13's high lifetime penetrance, it does not show the same strong association with an aggressive disease phenotype that characterizes BRCA2-associated prostate cancer, meaning that while HOXB13 carriers should be screened intensively because of sheer frequency of disease, the disease itself, once found, does not carry the same intrinsically higher-grade biology that BRCA2-associated prostate cancer does.

The update also addresses non-prostate, non-breast surveillance in BRCA1/2 carriers and testing-criteria expansion for BRCA-associated gynecologic malignancy. Annual pancreatic cancer surveillance beginning at age 50 is now recommended for BRCA1/2 carriers, reflecting the elevated pancreatic cancer risk (particularly in BRCA2 carriers, as noted in Slide 15) that these mutations confer. Testing criteria have been expanded to

include certain nonepithelial ovarian cancer histologies -- specifically granulosa cell tumors and Sertoli-Leydig cell tumors -- which were not previously flagged as strong indications for hereditary cancer testing but are now recognized to warrant it. Finally, BRIP1 and RAD51C/RAD51D are highlighted as additional ovarian-cancer-risk genes of direct relevance to surgical risk-reduction counseling (specifically, decisions around risk-reducing salpingo-oophorectomy), extending the risk-reducing surgery discussion of Slide 19 to a broader set of causative genes beyond BRCA1/2 alone. The update closes by reaffirming the cascade testing principle already introduced in Slide 19: once any pathogenic or likely pathogenic variant is confirmed in an index patient -- whether in BRCA1/2, PALB2, ATM, CHEK2, or HOXB13 -- all first-degree relatives should be offered targeted testing for that specific familial variant, since this is both the most efficient and the most clinically actionable way to identify at-risk relatives who would otherwise be missed by symptom- or diagnosis-triggered testing alone.

SECTION 06 -- Screening and Prevention

Mammography evidence and guideline thresholds, MRI and ultrasound supplemental screening, validated chemoprevention agents, and -- in explicit contrast -- the [ETHIOPIA] CBE-based early detection strategy that substitutes for population mammography where it is not yet available.

SLIDE 21 -- Section Divider -- Screening and Prevention

Detailed Explanation and Source Context

Screening and prevention are grouped together in this section because they represent the two available strategies for reducing breast cancer mortality before a tumor becomes symptomatic: screening detects existing but subclinical disease earlier, while chemoprevention and risk-reducing interventions (Section 05) attempt to prevent the disease from arising at all in high-risk individuals. The section closes with the Ethiopian guideline's own screening framework, deliberately placed immediately after the international mammography-based standard so the contrast is direct and unavoidable.

SLIDE 22 -- Screening Mammography

Slide content recap:

- Mammography reduces BC mortality ~15-25% (meta-analyses of RCTs)
- Sensitivity: 77-95%; lower in dense breasts (<50%) -> supplemental US or MRI
- Specificity: ~90%; recall rate 10%; biopsy rate ~1.5%
- USPSTF 2024: biennial mammography ages 40-74 (updated from 50-74)
- ACS: annual mammography from age 45 (optional 40-44); biennial option at 55+
- NCCN/ACR: annual mammography from age 40 for average risk
- Digital Breast Tomosynthesis (3D mammo, DBT): 15-30% increased cancer detection vs 2D
- 40% decreased in recall rates; 27% increased in invasive cancer detection with DBT
- DBT now standard at most screening centers in HIC
- DBT slightly higher radiation dose -- still well within safe range
- BIRADS 0: Incomplete -- needs additional imaging
- BIRADS 1: Negative -- routine screening
- BIRADS 2: Benign -- routine screening
- BIRADS 3: Probably benign -- short-interval 6-month follow-up (0.5-2% malignancy)
- BIRADS 4: Suspicious -- biopsy recommended (4A 2-10%, 4B 10-50%, 4C 50-95%)
- BIRADS 5: Highly suggestive of malignancy -- biopsy required (>95%)
- BIRADS 6: Known biopsy-proven malignancy
- Cumulative 10-year false-positive risk with annual mammography: ~61%
- Overdiagnosis estimate: 10-25% of screen-detected cancers (esp. DCIS)
- Benefits highest in age 50-69; debate continues for 40-49 age group
- Screening interval does not affect OS in most trials (annual vs biennial)
- [ETHIOPIA]: Mammography NOT yet available as population screening; CBE is primary tool

Detailed Explanation and Source Context

Mammography's mortality benefit -- an approximate 15 to 25% relative reduction in breast cancer death, derived from meta-analyses of the major randomized controlled screening trials conducted from the 1960s onward -- is the foundational evidence justifying every screening guideline discussed on this slide. Its diagnostic performance is not uniform: sensitivity ranges widely from 77 to 95% depending on breast density, falling below 50% in the densest breast tissue (BIRADS category D), which is precisely the population for whom supplemental ultrasound or MRI is recommended, as detailed on the next slide (Slide 23) and already flagged as a risk-diagnosis "double

jeopardy" in Section 04 (Slide 14). Specificity runs around 90%, translating into a recall rate (patients called back for additional imaging) of approximately 10% and an eventual biopsy rate of around 1.5% of all women screened -- numbers essential for counseling patients about the likely course of a screening program and for understanding the overdiagnosis debate discussed below.

Three major US guideline bodies differ modestly but meaningfully in their specific age and interval recommendations, and knowing the distinctions is a frequently tested point: the US Preventive Services Task Force, in its 2024 update, now recommends biennial mammography from age 40 to 74 -- a change from its prior 2016 recommendation, which had started routine screening at 50; the American Cancer Society recommends annual mammography beginning at 45, with an optional earlier start at 40 to 44 and the option to transition to biennial screening at 55 and beyond; and NCCN together with the American College of Radiology recommend annual mammography beginning at 40 for average-risk women, the most aggressive (earliest-starting, most frequent) of the three positions. These differences reflect differing weightings of the same underlying trial data -- earlier and more frequent screening detects more cancers and modestly improves mortality reduction, but at the cost of more false positives, more biopsies, and more overdiagnosis, discussed next.

Digital breast tomosynthesis (DBT, "3D mammography") acquires multiple low-dose projection images through an arc and reconstructs thin slices through the breast, largely eliminating the tissue-overlap artifact that limits conventional 2D mammography's sensitivity, particularly in dense breasts. Relative to standard 2D mammography, DBT increases cancer detection by 15 to 30%, reduces recall rates by approximately 40% (because superimposed normal tissue that would otherwise mimic a mass on 2D imaging is resolved into its true three-dimensional, benign appearance), and increases invasive cancer detection specifically by 27% -- DBT has consequently become the standard screening modality at most high-income-country centers, at the cost of a slightly higher radiation dose that nonetheless remains well within accepted safety limits.

The BIRADS (Breast Imaging Reporting and Data System) classification standardizes mammographic (and, by extension, ultrasound and MRI) reporting into a seven-category scale that directly dictates management, and each category's associated malignancy risk is worth memorizing precisely: BIRADS 0 indicates an incomplete assessment requiring additional imaging before a final category can be assigned; BIRADS 1 (negative) and BIRADS 2 (benign finding present but definitively benign) both warrant only routine screening; BIRADS 3 (probably benign) carries a malignancy risk of only 0.5 to 2% and is managed with short-interval, typically six-month, follow-up imaging rather than immediate biopsy; BIRADS 4 (suspicious) is subdivided into 4A (2 to 10% malignancy risk), 4B (10 to 50%), and 4C (50 to 95%), and biopsy is recommended across the entire 4 category regardless of sub-tier; BIRADS 5 (highly suggestive of malignancy, greater than 95% risk) mandates biopsy; and BIRADS 6 designates a lesion already confirmed malignant by prior biopsy, used when imaging that lesion again for staging or treatment-response purposes. This BIRADS framework is the direct organizing structure for the triple assessment algorithm discussed in Section 07 (Slide 26).

The costs of population mammographic screening are real and are explicitly quantified on this slide rather than glossed over: the cumulative risk of experiencing at least one false-positive result over ten years of annual screening is approximately 61% -- meaning that for a woman screened annually across a decade, a false alarm is statistically more likely than not to occur at some point -- and an estimated 10 to 25% of screen-detected cancers, particularly low-grade DCIS, represent overdiagnosis: the detection of a lesion that, left unfound, would never have progressed to cause symptoms or death within the patient's lifetime, yet which is nonetheless treated (with its attendant morbidity) once discovered, because there is currently no reliable way to distinguish prospectively which screen-detected DCIS lesions would have remained indolent. These competing considerations -- mortality benefit versus false-positive burden and overdiagnosis -- are precisely why the benefit-to-harm ratio is considered most favorable in the 50-to-69 age range and remains genuinely debated for

the 40-to-49 group, and why screening interval (annual versus biennial) has not been shown to meaningfully affect overall survival in most trial data, even though it does affect the false-positive and biopsy burden.

[ETHIOPIA] The guideline note appended to this slide states plainly what Section 03's epidemiology discussion (Slide 11) already established as the structural reality: population-based mammographic screening is not yet available in Ethiopia, and clinical breast examination therefore serves as the primary early-detection modality nationally -- the full Ethiopian screening protocol, including its specific age thresholds and the trained-health-worker CBE technique, is detailed in the dedicated Ethiopian guideline slide that closes this section (Slide 24).

SLIDE 23 -- MRI Screening Indications and Chemoprevention

Slide content recap:

- Annual MRI + mammography: $\geq 20\%$ lifetime risk (BRCA1/2, other high-risk genes)
- Sensitivity of MRI: 71-100% vs 33-60% for mammography in BRCA carriers
- Contrast-enhanced MRI: evaluate extent of disease; preop planning; occult primary
- MRI indications: BRCA carrier, prior chest RT, first-degree BRCA carrier, lifetime risk $\geq 20\%$
- MRI not recommended for average-risk women (false positive rate too high)
- MRI optional for: lifetime risk 15-20%, dense breasts with prior atypia (NCCN)
- Ultrasound recommended for dense breasts (BIRADS C/D) when MRI not available/affordable
- Whole-breast US: detects additional 2-3 cancers per 1000 screened women
- Automated whole-breast US (ABUS): reproducible; less operator-dependent
- US high FP rate: 85-90% of additional biopsies prompted by US are benign
- US cannot replace mammography: misses calcifications (DCIS) entirely
- Tamoxifen 20 mg/day x 5 yrs: 49% decreased ER+ BC (NSABP P-1); pre+postmenopausal
- Raloxifene 60 mg/day x 5 yrs: 38% decreased ER+ BC (MORE trial); postmenopausal only; decreased VTE vs Tam
- STAR trial: Tamoxifen vs Raloxifene -- similar invasive BC reduction; more uterine sarcoma with Tam
- Exemestane 25 mg/day x 5 yrs: 65% decreased risk (MAP.3 trial); postmenopausal only
- Anastrozole 1 mg/day x 5 yrs: 50% decreased (IBIS-II); postmenopausal; decreased bone density
- NCCN: discuss at Gail 5-year $\geq 1.7\%$; recommend at $\geq 3\%$ or IBIS 10-yr $\geq 5\%$
- NCCN: discuss RRM for P/LP variants in high-penetrance genes (BRCA1/2, TP53, CDH1)
- Risk reduction with RRM: ~90-95% for BC; does not eliminate risk (residual chest wall tissue)
- Survival benefit shown in BRCA1/2 carriers with intact ovaries
- Nipple-sparing RRM: acceptable in appropriate candidates; completeness is paramount
- Reconstruction: immediate implant vs tissue expander vs LD/TRAM flap -- individualized
- Counseling: body image, sexuality, future screening, genetic implications

Detailed Explanation and Source Context

Breast MRI's screening role is defined entirely by risk stratification: annual MRI in addition to annual mammography is recommended for anyone with a lifetime breast cancer risk of 20% or greater, a population dominated by confirmed BRCA1/2 (and other high-penetrance gene) carriers, individuals with a first-degree relative who is a known BRCA carrier but who have not themselves been tested, and survivors of chest radiotherapy at a young age (Section 04, Slide 14). The rationale is the sensitivity gap already introduced in Section 05 (Slide 19): in BRCA carriers, MRI sensitivity of 71 to 100% substantially exceeds mammography's sensitivity of only 33 to 60% in this same population, a gap driven by the denser breast tissue and faster-growing, often interval-appearing cancers characteristic of hereditary breast cancer. Beyond pure screening, contrast-enhanced MRI has three additional clinical uses that recur throughout this document: defining the true extent of disease in a newly diagnosed cancer (informing surgical planning, particularly the choice between breast-conserving surgery and mastectomy, Section 12), identifying an occult primary tumor in a patient presenting with an axillary metastasis and no mammographically visible lesion (Section 17, Slide 66), and assessing residual disease after neoadjuvant systemic therapy (Section 13, Slide 52). Critically, MRI is

explicitly NOT recommended as a screening tool for average-risk women, because its high false-positive rate in a population with a low pretest probability of cancer would generate an unacceptable burden of unnecessary biopsies; NCCN designates an intermediate lifetime risk band of 15 to 20% (or dense breasts combined with a personal history of atypical hyperplasia) as an "optional" MRI indication rather than a firm recommendation, reflecting a less favorable benefit-to-harm ratio at this lower risk level than at 20% and above.

Whole-breast ultrasound is positioned as the intermediate-resource supplemental option: when MRI is unavailable or unaffordable, ultrasound is recommended for women with dense breasts (BIRADS category C or D), detecting an additional 2 to 3 cancers per 1,000 women screened beyond mammography alone. Automated whole-breast ultrasound (ABUS) offers more standardized, reproducible image acquisition than traditional operator-dependent handheld ultrasound, which is valuable in a screening context where consistency across repeated examinations matters. Ultrasound's major limitation, however, is a high false-positive rate -- 85 to 90% of the additional biopsies it prompts turn out to be benign -- and, more fundamentally, ultrasound cannot detect microcalcifications, the specific finding that most commonly identifies ductal carcinoma in situ (Section 08, Slide 32); ultrasound is therefore always a supplement to, never a replacement for, mammography in a screening context.

The chemoprevention agents introduced conceptually in Section 05 (Slide 19) are given their full efficacy data here. Tamoxifen, 20 mg daily for five years, reduced ER-positive breast cancer incidence by 49% in the NSABP P-1 trial and is effective in both pre- and postmenopausal women. Raloxifene, 60 mg daily for five years, achieved a somewhat smaller 38% reduction in the MORE trial but is restricted to postmenopausal use. The head-to-head STAR trial directly compared the two agents and found broadly similar reductions in invasive breast cancer incidence, but with a more favorable safety profile for raloxifene, specifically a lower rate of uterine sarcoma than seen with tamoxifen -- a clinically important distinction when counseling a postmenopausal patient who could receive either agent. Exemestane, an aromatase inactivator, achieved the largest relative risk reduction on this slide at 65% in the MAP.3 trial, and anastrozole achieved a 50% reduction in the IBIS-II prevention trial; both aromatase inhibitors are postmenopausal-only options and carry a distinct toxicity profile from the SERMs, notably accelerated bone density loss rather than the endometrial and thromboembolic risks associated with tamoxifen. The NCCN eligibility thresholds restated here from Section 04 (Slide 17) -- discuss chemoprevention at a Gail five-year risk of 1.7% or greater, recommend it more strongly at 3% or greater (or an IBIS/Tyrer-Cuzick ten-year risk of 5% or greater) -- are the numeric gateway connecting the risk-assessment models of Section 04 to this slide's specific drug menu.

Risk-reducing mastectomy is restated here as the surgical alternative (or complement) to pharmacologic chemoprevention for the highest-risk patients, specifically those with confirmed pathogenic or likely pathogenic variants in high-penetrance genes -- BRCA1/2, TP53 (Li-Fraumeni), and CDH1 (hereditary diffuse gastric cancer, Section 04 Slide 16) are the genes NCCN specifically flags for this discussion. As detailed in Section 05 (Slide 19), risk-reducing bilateral mastectomy achieves a 90 to 95% breast cancer risk reduction without fully eliminating risk, given inevitable residual glandular tissue; a demonstrated survival benefit specifically requires that the patient's ovaries remain functional (or that risk-reducing salpingo-oophorectomy has also been performed), since ongoing systemic estrogen exposure is what the residual at-risk tissue remains sensitive to. Nipple-sparing technique is acceptable in appropriately selected candidates, with the overriding surgical principle being oncologic completeness of the gland resection rather than a compromise for cosmesis. Any risk-reducing mastectomy discussion must be accompanied by full reconstruction counseling -- weighing immediate implant placement, staged tissue-expander-to-implant reconstruction, or autologous flap options (latissimus dorsi or TRAM/DIEP, detailed surgically in Section 12, Slide 47) -- individualized to the patient's body habitus, prior radiation history, and preference, alongside frank discussion of body image, sexuality, and the psychological

adjustment specific to prophylactic (rather than therapeutic) surgery performed on an asymptomatic patient, plus the practical genetic implication that this decision does not, by itself, change the counseling needs of the patient's at-risk relatives.

SLIDE 24 -- Ethiopian Guideline -- Screening and Early Detection

Slide content recap:

- [ETHIOPIA] CBE replaces population mammography as primary screening tool
- Ethiopia primary modality: CBE (trained health worker); international: mammography (2D/3D)
- Ethiopia age to start: 30 years (CBE); international: 40 years (mammography)
- Ethiopia frequency: annual CBE; 2-yr mammogram if available; international: annual/biennial mammo
- Sensitivity for <1cm tumor: low with CBE; high (~85%) with mammography
- Infrastructure needed: minimal for CBE; significant (equipment + radiologists) for mammography
- Gap: 65-75% present Stage III/IV in Ethiopia vs <30% in HIC
- 1. Breast Health Awareness (BHA): women ≥ 20 years; monthly BSE education
- 2. Clinical Breast Examination (CBE): women ≥ 30 years; annual by trained provider
- 3. Diagnostic Imaging (US/Mammography): women ≥ 40 years every 2 years, or diagnostic
- CBE by trained health worker: identifies cancers as small as 1 cm
- CBE integrated into existing MCH services (ANC, postnatal, FP clinics)
- Physical exam: breasts and underarms in upright and lying flat positions
- Palpation looking for: lump/change in shape, fixed/hard mass, axillary lump
- Skin changes: peau d'orange, ulceration, nipple retraction, bloody discharge
- Confirmatory: if suspicious $\rightarrow$ refer for US +/- mammography and FNAC/biopsy
- Train all primary HCW; include CBE in medical and nursing curricula
- Quality assurance: proficiency assessments and standardized protocols

Detailed Explanation and Source Context

[ETHIOPIA] This slide operationalizes, in concrete protocol form, the structural gap already established in Section 03's epidemiology (Slide 11) and Section 06's mammography slide (Slide 22): because population-based mammographic screening is not available at national scale, the Ethiopian FMOH 2024 guideline builds its entire early-detection strategy around clinical breast examination performed by a trained health worker, a workforce-based rather than equipment-based intervention that can be deployed even where imaging infrastructure and radiologist staffing are limited or absent.

The direct side-by-side comparison table makes the tradeoff explicit rather than implicit: CBE requires minimal infrastructure (a trained examiner and an examination room) and can therefore be deployed at the primary care level nationwide, but its sensitivity for detecting a tumor under 1 centimeter is low, because CBE depends entirely on a lesion being large enough to be palpable; mammography, by contrast, achieves approximately 85% sensitivity for small lesions but requires the equipment, technician training, and radiologist interpretation capacity that are the very resources not yet available at scale in Ethiopia. The consequence of this infrastructure gap is stated numerically and without euphemism: 65 to 75% of Ethiopian patients present at Stage III or IV, compared with fewer than 30% in high-income countries where mammographic screening is routine -- this single comparison is the clearest, most quantified expression in the entire document of what a screening infrastructure gap costs in terms of stage at diagnosis, and it is the reason the guideline treats early detection as a distinct, resourced pillar (Pillar 1, detailed fully in Section 18, Slide 68) rather than a secondary consideration.

The Ethiopian early-detection strategy is structured as three sequential, age-stratified tiers, each matched to an appropriate level of the health system. The first tier, Breast Health Awareness (BHA), targets women aged 20 and above with monthly breast self-examination education -- a low-cost, community-deliverable intervention intended to raise general symptom awareness and prompt earlier self-referral when an abnormality is noticed,

rather than to serve as a sensitive screening test in itself. The second tier, clinical breast examination, targets women aged 30 and above for annual examination by a trained provider -- this is the core, resource-appropriate screening intervention, deliberately started at a younger age (30) than the international mammography threshold (40) precisely because CBE, being a physical rather than radiographic examination, carries none of the radiation-exposure or cost considerations that motivate mammography's later starting age, and because starting CBE at 30 casts a wider net at minimal marginal cost. The third tier, diagnostic imaging (ultrasound preferentially, with mammography where available), is reserved for women 40 and above, imaged every two years or on a diagnostic (symptom-triggered) basis -- reflecting the reality that imaging capacity remains a scarce resource to be allocated where clinical suspicion or higher baseline risk (age) most justifies it, rather than deployed as universal population screening.

A specific and important design choice is that CBE is explicitly integrated into existing maternal and child health (MCH) service touchpoints -- antenatal care, postnatal care, and family planning clinics -- rather than requiring the creation of a separate, parallel breast-screening clinical infrastructure. This integration strategy leverages contact points where reproductive-age women are already accessing care for other reasons, maximizing population reach without requiring new health-system capacity to be built from scratch, a pragmatic design principle that recurs throughout the Ethiopian guideline's broader three-pillar structure (Section 18).

The CBE protocol itself is specified in explicit, standardized examination steps, precisely because CBE's diagnostic value depends entirely on examiner technique and training consistency: the breasts and axillae are examined in both the upright and supine (lying flat) positions, since a mass may be more easily palpated in one position than the other depending on breast size and tissue distribution; palpation specifically searches for a discrete lump or a change in breast shape, a fixed or hard mass (both worrisome features, versus a soft, mobile, well-circumscribed mass more suggestive of a benign fibroadenoma), and any axillary lump suggesting nodal involvement; and the skin is inspected for the same red-flag signs detailed in the clinical presentation discussion of Section 07 (Slide 26) -- peau d'orange, ulceration, nipple retraction, and bloody nipple discharge. Any suspicious finding on CBE triggers referral for ultrasound, with or without mammography, and fine-needle aspiration cytology or biopsy -- initiating the "timely diagnosis" pathway (Pillar 2) detailed in the very next section (Section 07, Slide 29). Finally, because the entire strategy's diagnostic yield depends on consistent, high-quality examination technique across a large, geographically distributed primary-care workforce, the guideline places explicit emphasis on training every primary health care worker in CBE technique, embedding CBE instruction into the standard medical and nursing curricula so that competency is built into professional formation rather than delivered only as an ad hoc in-service training, and on ongoing quality assurance through periodic proficiency assessment and adherence to standardized examination protocols -- without which a workforce-based screening strategy would be vulnerable to exactly the kind of examiner-to-examiner variability that could undermine its population-level effectiveness.

SECTION 07 -- Diagnosis: Imaging and Biopsy

Clinical presentation and examination technique, the triple assessment framework, comparative imaging modality performance, biopsy technique selection, and the [ETHIOPIA] resource-adapted diagnostic pathway targeting a 60-day turnaround.

SLIDE 25 -- Section Divider -- Diagnosis: Imaging and Biopsy

Detailed Explanation and Source Context

This section moves from population-level screening (Section 06) to the individual diagnostic workup of a patient with a clinical finding, whether that finding arose from screening, from CBE, or from patient self-report. The organizing concept throughout is the triple assessment -- clinical examination, imaging, and pathology -- introduced on Slide 26 and referenced repeatedly through the remainder of the section, including in its resource-adapted Ethiopian form on Slide 29.

SLIDE 26 -- Clinical Presentation and Examination

Slide content recap:

- Painless lump (most common, 70%): new hard/fixed lump -> high suspicion
- Skin changes: peau d'orange (dermal lymphatic invasion), dimpling (Cooper's ligament tethering)
- Nipple changes: inversion (new), eczematous rash (Paget's), bloody discharge
- Axillary lymphadenopathy: hard, matted, fixed nodes -> suspicious
- Locally advanced: skin ulceration, chest wall fixation, erythema
- Inflammatory BC: diffuse erythema >1/3 breast, peau d'orange, rapid onset (not a skin infection)
- Arms above head + leaning forward: accentuates asymmetry, skin dimpling, nipple retraction
- Lymph nodes examined: axillary levels I-III, supraclavicular, infraclavicular
- Systematic palpation: 4 quadrants + subareolar; standing then supine
- Note: size (cm), location (clock face, distance from nipple), mobility, consistency, borders
- Skin: erythema, edema, satellite nodules, scars, visible veins
- Contralateral breast: always examine both; bilateral synchronous BC ~1-3%
- 1. Clinical Breast Examination (history + physical)
- 2. Imaging (mammogram +/- US +/- MRI)
- 3. Pathology (FNAC or core needle biopsy)
- All three concordant benign: 99.6% NPV; discordance -> biopsy mandatory
- Core needle biopsy preferred over FNAC: provides architecture, ER/PR/HER2
- Age <30 with discrete mass: US first (dense breast, low malignancy risk)
- Age ≥30 or suspicious mass: diagnostic mammogram + US +/- biopsy
- Dominant solid mass any age: biopsy if ≥BIRADS 4
- Nipple discharge: cytology of discharge + mammogram + US +/- ductoscopy
- Occult primary (axillary node + unknown primary): MRI breast before biopsy

Detailed Explanation and Source Context

A painless, new, hard, and fixed lump is the most common presenting complaint in breast cancer, accounting for roughly 70% of presentations, and its specific physical characteristics -- hardness and fixation to surrounding tissue, in contrast to the soft, mobile texture typical of a benign fibroadenoma -- are what should raise clinical suspicion even before any imaging is obtained. The cutaneous and nipple findings on this slide are all mechanistically explained by anatomy already established in Section 02: peau d'orange (an orange-peel-like skin texture) results from dermal lymphatic invasion causing localized lymphedema that makes the normal skin pores appear pitted and prominent, and skin dimpling results from tumor infiltration and consequent

foreshortening of Cooper's suspensory ligaments (Section 02, Slide 6), pulling the overlying skin inward at the point of ligament attachment. New nipple inversion (as opposed to a longstanding, congenital inversion, discussed in Section 02 Slide 6) implies pathologic shortening of the central lactiferous ducts by an infiltrating tumor, while an eczematous, non-healing nipple rash is the classic presentation of Paget's disease of the nipple, detailed fully in Section 09 (Slide 35); bloody nipple discharge was already discussed physiologically in Section 02 (Slide 9) and must always prompt exclusion of an underlying malignancy even though a benign intraductal papilloma is the more common cause. Axillary lymphadenopathy that is hard, matted (multiple nodes fused together rather than individually mobile), and fixed to surrounding tissue is a worrisome finding for nodal metastatic involvement, in contrast to soft, mobile, tender reactive nodes more typical of an infectious or inflammatory process. Locally advanced disease can present with frank skin ulceration, fixation of the tumor to the underlying chest wall (implying invasion beyond the breast parenchyma into pectoralis fascia or muscle, or beyond, per the T4 criteria detailed in Section 10, Slide 37), and diffuse erythema. Inflammatory breast cancer deserves specific emphasis because it is a clinical diagnosis, not a histologic one: diffuse erythema involving at least one-third of the breast, peau d'orange, and rapid onset (typically over weeks, not months) together define the syndrome, and the single most important diagnostic pitfall is mistaking it for mastitis or a breast abscess -- the distinguishing feature is that inflammatory breast cancer does not respond to antibiotics and is not associated with the systemic signs of true infection (fever, leukocytosis) at outset; this diagnosis is developed fully in Section 17 (Slide 66) given its unique, surgery-contraindicated-first treatment sequence.

Examination technique is standardized precisely because subtle findings can be missed without a systematic approach: having the patient raise her arms overhead and then lean forward accentuates any breast asymmetry, skin dimpling, or nipple retraction by placing tension on the suspensory ligaments and pectoral fascia, making subtle tethering far more visible than with the arms at rest. Lymph node examination should systematically assess all three axillary levels (recall the Berg classification from Section 02, Slide 8), plus the supraclavicular and infraclavicular basins, since involvement of these more distant nodal groups carries specific staging implications (N3, Section 10). Breast palpation itself should proceed systematically through all four quadrants plus the subareolar region, performed in both the standing and supine positions (since gravity redistributes breast tissue differently in each position, potentially revealing or obscuring a mass). Any palpable finding should be documented with specific descriptors that directly inform subsequent management: size in centimeters, location described by clock-face position and measured distance from the nipple (both essential for surgical planning and for correlating a palpable finding with a corresponding imaging finding), mobility (fixed versus mobile), consistency (hard versus soft), and border characteristics (well-circumscribed versus irregular/ill-defined, the latter being more suspicious). Skin should be inspected broadly for erythema, edema, satellite nodules (small separate tumor deposits in the skin, a T4b criterion), surgical scars from prior procedures, and abnormally prominent superficial veins (which can indicate an underlying mass altering local venous drainage). Finally, the contralateral breast must always be examined as a matter of routine, since synchronous bilateral breast cancer occurs in an estimated 1 to 3% of cases and would otherwise be missed if examination focused exclusively on the side of the presenting complaint.

The triple assessment -- clinical examination, imaging, and pathology, drawn together here as the organizing diagnostic framework for the entire workup -- achieves a negative predictive value of 99.6% when all three components concordantly suggest a benign process, meaning that a patient with a reassuring exam, reassuring imaging, and reassuring (or unnecessary, if imaging alone was sufficiently reassuring) pathology can be followed with a very high degree of confidence that no cancer is present; conversely, ANY discordance among the three components -- for instance, a clinically and radiographically suspicious mass with a benign-appearing core biopsy -- mandates further action (repeat biopsy, excisional biopsy) rather than acceptance of the reassuring result, because the entire premise of the triple assessment is that concordance, not any single component in

isolation, is what confers diagnostic reliability. Core needle biopsy is explicitly preferred over fine-needle aspiration as the pathology component of the triple assessment specifically because core biopsy preserves tissue architecture and yields sufficient material for the ER, PR, and HER2 biomarker testing that drives every subsequent treatment decision in this document (Section 11), whereas FNA provides only cytology (isolated cells) and cannot distinguish invasive from in situ disease or reliably support biomarker testing -- this preference, and the full biopsy technique comparison, is developed in detail on the very next slide (Slide 28).

Age-stratified imaging pathways reflect the differing pretest probability of malignancy and the differing baseline breast tissue characteristics across age groups: in a woman under 30 presenting with a discrete palpable mass, ultrasound is the appropriate first-line imaging study, both because young breast tissue is typically dense (limiting mammographic sensitivity) and because the baseline likelihood of malignancy is low enough that avoiding unnecessary mammographic radiation exposure is a reasonable priority; at age 30 or older, or in the presence of a clinically suspicious mass at any age, a full diagnostic mammogram plus ultrasound, with biopsy as indicated by BIRADS category, is the appropriate pathway. Any dominant solid mass reaching BIRADS category 4 or higher, regardless of patient age, should proceed to biopsy given the substantial malignancy risk associated with that BIRADS tier (Section 06, Slide 22). Nipple discharge, as introduced physiologically in Section 02 (Slide 9), is worked up with cytologic examination of the discharge itself together with mammography and ultrasound, with ductoscopy or galactography reserved for cases not adequately clarified by standard imaging. Finally, the specific and important scenario of occult primary breast cancer -- an axillary nodal metastasis found with no mammographically or sonographically visible breast primary -- should prompt breast MRI before proceeding to biopsy of the axillary node, since MRI identifies the occult primary tumor in a substantial majority of such cases (Section 07, Slide 27, and developed fully in Section 17, Slide 66), which then allows the biopsy and subsequent treatment planning to be directed at an identified, imageable target rather than proceeding blind.

SLIDE 27 -- Imaging Modalities

Slide content recap:

- Mammography (2D): sensitivity 75-85%, specificity 90%; screening; calcifications (DCIS)
- DBT (3D Mammo): sensitivity 85-95%, specificity 93%; screening; dense breasts; spiculated lesions
- Breast Ultrasound: sensitivity 90-95%, specificity 88%; cyst vs solid; dense breast; biopsy guidance
- Breast MRI: sensitivity 90-99%, specificity 70-80%; high-risk screening; occult primary; extent
- PET/CT: sensitivity 70-85%; stage III-IV; detect LN/distant mets; ambiguous sites
- Newly diagnosed BC: MRI evaluates extent ipsilateral + contralateral synchronous disease
- Occult primary (positive axillary node): 70% detection rate before MRI negative
- MRI evaluate response to neoadjuvant: residual disease assessment
- MRI for dense breasts or implants: supplement mammography
- MRI for BRCA/high-risk annual surveillance
- US-guided FNA/core biopsy of suspicious nodes: before decisions on surgical staging
- Mark biopsied suspicious node (clip) for targeted axillary dissection (TAD) after neoadjuvant
- cN0: SLNB standard; cN1-2: full evaluation; post-neoadjuvant cN0 -> TAD
- PET-CT: useful for detecting occult systemic disease in Stage III

Detailed Explanation and Source Context

This comparative table quantifies the sensitivity-specificity tradeoffs across the five imaging modalities used in breast cancer diagnosis and staging, and the pattern across the table is instructive: sensitivity generally rises as modalities move from mammography through DBT to ultrasound and MRI, but specificity does not rise in lockstep, and each modality's "best use" column reflects a deliberate matching of its particular strengths to a specific clinical question rather than a simple hierarchy of "better" versus "worse." Conventional 2D

mammography, with 75 to 85% sensitivity and 90% specificity, remains the foundational population-screening tool specifically because of its unique ability to detect microcalcifications -- the characteristic mammographic signature of ductal carcinoma in situ (Section 08, Slide 32) that no other modality on this table reliably identifies. Digital breast tomosynthesis improves on both sensitivity (85 to 95%) and specificity (93%) relative to 2D mammography, particularly valuable in dense breast tissue and for characterizing spiculated (irregular, radiating) mass margins, as already detailed in Section 06 (Slide 22).

Breast ultrasound achieves the highest sensitivity-specificity combination among the readily available modalities (90 to 95% sensitivity, 88% specificity) for its specific, narrower purpose: distinguishing a simple cyst from a solid mass (a distinction mammography cannot reliably make), characterizing masses in dense breast tissue where mammographic sensitivity is limited, and providing real-time image guidance for biopsy -- ultrasound's core limitations, as already noted in Slide 23, are its inability to detect calcifications and its operator dependence. Breast MRI achieves the highest sensitivity of any modality (90 to 99%) but at the cost of the lowest specificity (70 to 80%), meaning MRI will detect nearly every cancer present but will also generate a substantial number of false-positive findings requiring additional workup -- this sensitivity-specificity profile is precisely why MRI is reserved for high-risk screening and specific problem-solving indications (Slide 23) rather than deployed as a general screening tool, where its low specificity would generate an unacceptable biopsy burden in an average-risk population. PET-CT, with a comparatively modest 70 to 85% sensitivity, is not useful for detecting small primary or nodal disease (lesions under approximately 1 cm fall below its resolution), but is valuable specifically in Stage III to IV disease for detecting more distant lymph node involvement or systemic metastatic disease that would change management -- a role developed fully in the staging workup discussion of Section 10 (Slide 38).

MRI's specific indication list, restated and expanded here from Slide 23, spans the full disease timeline: at initial diagnosis, MRI evaluates the true extent of disease both within the ipsilateral breast (often revealing additional foci of disease not appreciated on mammography or ultrasound, which can change the surgical plan from breast-conserving surgery to mastectomy) and in the contralateral breast, where it can detect synchronous but occult disease in the opposite breast in a meaningful minority of patients; in the specific scenario of occult primary breast cancer presenting as an axillary metastasis, MRI identifies the primary tumor in approximately 70% of cases, allowing directed rather than blind biopsy (as already discussed in Slide 26 and developed fully in Section 17, Slide 66); during and after neoadjuvant systemic therapy, MRI is the imaging modality that best correlates with actual pathologic residual disease at surgery, informing both surgical planning and, together with targeted axillary dissection, the completeness of tumor response assessment (Section 13, Slide 52); in patients with dense breast tissue or breast implants (which can obscure mammographic assessment), MRI serves a supplemental diagnostic role; and, as already established, MRI is the surveillance imaging modality of choice in BRCA and other high-risk carriers undergoing annual screening (Section 05, Slide 19).

Finally, this slide connects imaging directly to axillary management, previewing the surgical staging discussion of Section 12 (Slide 45): any axillary lymph node that appears suspicious on ultrasound should undergo image-guided fine-needle aspiration or core biopsy before any decision is made about the extent of surgical axillary staging, since a biopsy-confirmed positive node before surgery changes management (potentially warranting neoadjuvant therapy or altering the extent of planned axillary surgery) differently than a node found positive only at the time of sentinel node biopsy itself. When such a suspicious node is biopsied and confirmed positive before neoadjuvant therapy, it should be marked with a radiographic clip specifically so that, after neoadjuvant treatment converts the node to clinically negative, that same originally-positive node can be deliberately identified and removed as part of targeted axillary dissection (TAD) -- a technique developed in full in Section 12 (Slide 45) that substantially reduces the false-negative rate of post-neoadjuvant axillary staging

compared with sentinel node biopsy alone. The overall axillary staging algorithm summarized here -- sentinel node biopsy as the standard for clinically node-negative (cN0) disease, fuller upfront evaluation for clinically node-positive (cN1-2) disease, and targeted axillary dissection when a clinically node-negative status is achieved after neoadjuvant therapy in a patient who was originally node-positive -- is the master algorithm elaborated fully in Section 12 (Slide 45).

SLIDE 28 -- Biopsy Techniques

Slide content recap:

- FNA (FNAC): cells (cytology); rapid, minimal trauma, guides treatment acutely; no architecture; cannot distinguish invasive vs in situ
- Core needle biopsy (CNB): core tissue (histology); architecture, ER/PR/HER2; standard of care; small hematoma risk
- Vacuum-assisted (Mammotome): multiple cores, 11-14G; best for calcifications; more expensive, longer procedure
- Excisional biopsy: full lesion removed; diagnostic + therapeutic if margins clear; requires OR
- Incisional biopsy: part of lesion; large lesions; prior excisional failure; disrupts tumor
- US-guided biopsy: real-time; solid masses; axillary nodes; most used technique
- Stereotactic (mammogram-guided): microcalcifications without US correlate
- MRI-guided: lesion only on MRI -- requires supine or prone coil; DCIS especially
- Clip placement mandatory: marks biopsy site for surgery or TAD after neoadjuvant
- Radiograph clip to confirm: concordance of imaging and pathology before treatment
- Discordant result: repeat biopsy -- never trust a benign core if imaging is suspicious

Detailed Explanation and Source Context

The five biopsy techniques on this table represent an escalating spectrum of tissue yield, invasiveness, and cost, and choosing among them is a matter of matching the technique to the specific clinical question and lesion characteristics. Fine-needle aspiration provides only cytologic material -- individual cells rather than intact tissue architecture -- making it fast and minimally traumatic, useful for rapidly confirming a fluid-filled cyst or providing acute guidance in a time-sensitive situation, but fundamentally limited because it cannot distinguish invasive carcinoma from carcinoma in situ (a distinction that depends entirely on visualizing an intact or breached basement membrane, which requires tissue architecture, not isolated cells) and is highly operator-dependent in its adequacy and interpretability. Core needle biopsy, by contrast, extracts an intact cylindrical core of tissue using a larger-gauge needle, preserving the architectural relationships needed to determine invasion status and providing sufficient tissue for immunohistochemical staining of ER, PR, and HER2 -- this combination of diagnostic completeness and relatively low invasiveness (an outpatient needle procedure, not a surgical incision) is exactly why core needle biopsy, not FNA, is designated the standard-of-care tissue diagnosis modality in the triple assessment (Slide 26), with only a small associated risk of hematoma. Vacuum-assisted core biopsy (commonly performed with the Mammotome device) extracts multiple contiguous cores through a single needle insertion using a larger 11-to-14-gauge needle and continuous vacuum suction, yielding a greater tissue volume with lower sampling error than standard core biopsy -- this makes it the preferred technique specifically for sampling microcalcifications, where a small, non-mass lesion visible only as scattered calcium deposits benefits from the more complete tissue retrieval that vacuum assistance provides, at the cost of a more expensive, longer procedure. Excisional biopsy removes the entire lesion surgically and is both diagnostic and, if margins are clear, potentially therapeutic in a single procedure, but requires an operating room and the associated resources and recovery; incisional biopsy removes only a representative portion of a lesion, reserved for large lesions where complete excision is impractical as an initial diagnostic step, or when a prior excisional attempt has failed to achieve diagnosis -- incisional biopsy is used only rarely because it disrupts the tumor without achieving definitive local control and generally requires the operating room like excisional biopsy.

Image guidance for percutaneous (needle) biopsy is selected based on how the target lesion is best visualized: ultrasound guidance is the most commonly used approach, providing real-time needle visualization for solid masses and for suspicious axillary lymph nodes (as introduced in Slide 27); stereotactic (mammography-guided) biopsy is reserved for lesions visible only as microcalcifications on mammography with no corresponding ultrasound correlate, since ultrasound cannot visualize calcifications at all; and MRI-guided biopsy is reserved for lesions identified only on MRI with no mammographic or sonographic correlate, requiring a dedicated supine or prone breast coil setup and being particularly relevant for detecting DCIS, which can present as a non-mass MRI enhancement pattern without a corresponding finding on other modalities.

Two procedural principles apply universally across all percutaneous biopsy techniques and are emphasized specifically because their omission can compromise subsequent treatment. First, clip placement at the biopsy site is mandatory, regardless of which image-guidance technique or needle size is used, because the biopsy itself typically removes or substantially reduces the visible/palpable abnormality, and a radiopaque clip is the only reliable way to relocate the exact biopsy site for definitive surgical excision -- this becomes especially critical in the neoadjuvant setting, where a tumor may show a complete or near-complete radiographic response to systemic therapy, leaving the clip as the only remaining marker of where the original tumor bed was located for targeted axillary dissection (Section 12, Slide 45) or for surgical planning. A post-biopsy radiograph should always confirm clip placement and, more fundamentally, imaging-pathology concordance should be explicitly assessed before any treatment decision is finalized: if a biopsy returns a benign result but the imaging appearance was suspicious (BIRADS 4 or 5), this is a discordant result that must never simply be accepted as reassuring -- the correct action is repeat biopsy (often with a different or larger-gauge technique, or excisional biopsy) rather than trusting a benign core result in the face of suspicious imaging, since sampling error is an inherent limitation of any percutaneous needle technique and a falsely reassuring benign core result in a truly malignant lesion represents one of the most consequential and avoidable diagnostic errors in breast cancer care.

SLIDE 29 -- Ethiopian Guideline -- Timely Diagnosis (Pillar 2)

Slide content recap:

- [ETHIOPIA] Target = complete workup within 60 days of referral
- Ethiopia initial imaging: US (mammogram limited); international: diagnostic mammo + US
- Ethiopia tissue diagnosis: FNAC first -> CNB; international: core needle biopsy preferred
- Ethiopia biomarker testing: ER/PR (limited); HER2 (variable); international: ER/PR/HER2/Ki-67 all sites
- Ethiopia time to diagnosis target: <=60 days; international best practice: <=2-4 weeks
- MDT available at tertiary only in Ethiopia; standard at all treating centers internationally
- Step 1: History + CBE by trained provider (history of lump, duration, changes)
- Step 2: Imaging -- US preferred (mammogram if available); assess mass characteristics
- Step 3: FNAC as initial diagnostic modality -> core needle biopsy -> incisional biopsy
- Combining all three = diagnostic concordance before treatment decisions
- FNAC first approach: practical for resource-limited settings; limitations recognized
- 85-90% of all biopsied breast lesions are ultimately benign
- Women >=30 yrs with: breast lump +/- change in shape/consistency, axillary lump, skin changes, nipple retraction/bloody discharge
- Within 60 days: coordinate US +/- mammography + FNAC/biopsy for diagnosis
- Benign lesion: treat per guideline
- Suspected/confirmed BC: refer to BC treatment facility + MDT

Detailed Explanation and Source Context

[ETHIOPIA] This slide operationalizes Pillar 2 of the Ethiopian FMOH 2024 guideline -- "Timely Diagnosis" -- and its single organizing metric, a 60-day maximum turnaround from referral to completed diagnostic workup, is

one-third of the guideline's overall 60:60:80 target framework already introduced in Section 03 (Slide 11) and detailed fully in Section 18 (Slide 68). Every element of this slide's diagnostic algorithm is a direct, resource-adapted counterpart to the international triple-assessment framework of Slide 26, and the side-by-side comparison table makes each specific point of divergence explicit rather than leaving it implicit.

The first documented gap is imaging access: where the international standard calls for diagnostic mammography combined with ultrasound (Slide 27), Ethiopian practice relies primarily on ultrasound, with mammography available only where infrastructure permits -- this mirrors precisely the screening-level gap already described in Section 06 (Slide 24), and the consequence is the same throughout: ultrasound alone cannot detect microcalcifications, meaning a subset of mammographically-detectable early ductal carcinoma in situ will be missed by an ultrasound-only diagnostic pathway. The second gap concerns the sequence of tissue diagnosis: whereas the international standard designates core needle biopsy as the preferred initial tissue-diagnosis technique specifically because it preserves architecture and yields sufficient material for full biomarker testing (Slide 28), the Ethiopian pathway begins with fine-needle aspiration cytology as a practical, lower-cost, more widely deployable first step, escalating to core needle biopsy only when FNAC is insufficient or ambiguous -- the guideline is explicit that this FNAC-first approach is a pragmatic accommodation to resource constraints, and its limitations (specifically, FNAC's inability to assess invasion status or reliably support HER2 testing, as detailed in Slide 28) are openly acknowledged rather than glossed over. The third gap concerns biomarker completeness: ER and PR testing are described as routinely available, but HER2 testing is variable and inconsistent across facilities -- a gap with direct and serious downstream treatment consequences, since HER2 status determines eligibility for trastuzumab and the other anti-HER2 agents detailed in Section 13 (Slide 51), meaning that inconsistent HER2 testing access can result in HER2-positive patients in some parts of the country never being identified as candidates for what is, internationally, considered essential targeted therapy. The fourth gap is the turnaround-time target itself: the Ethiopian guideline's own aspirational standard of 60 days is, by its own explicit acknowledgment, longer than the 2-to-4-week turnaround considered international best practice -- the 60-day figure is calibrated to what is currently achievable given existing infrastructure constraints (specifically, the referral pathway across the three-tier health system detailed in Section 18, Slide 69) rather than representing a claim that 60 days is clinically ideal. Finally, multidisciplinary team availability, essential for coordinating the treatment decisions that follow a completed diagnosis, is currently limited to tertiary centers such as Tikur Anbessa Specialized Hospital, Jimma University Specialized Hospital, and ALERT Hospital, whereas the international standard expects MDT availability at every treating center -- this specific gap is developed further in Section 18's detailed discussion of the Ethiopian MDT approach (Slide 69).

The Ethiopian diagnostic algorithm itself proceeds in three explicit steps that any primary or secondary facility should follow: first, a structured history and clinical breast examination by a trained provider, eliciting the duration of any lump and any changes in size, shape, or associated symptoms; second, imaging, with ultrasound as the preferred initial modality (mammography used where available) to characterize the mass's sonographic features; and third, tissue diagnosis beginning with fine-needle aspiration cytology as the initial modality, escalating to core needle biopsy and, if still non-diagnostic, incisional biopsy. As with the international triple assessment, the guideline emphasizes that combining all three elements -- history/examination, imaging, and cytology/histology -- to achieve diagnostic concordance, rather than relying on any single element in isolation, remains the governing principle even within this resource-adapted pathway; the guideline further notes, as a piece of context relevant to counseling patients and managing workup expectations at any resource level, that 85 to 90% of all biopsied breast lesions ultimately prove benign, underscoring that a systematic, efficient diagnostic pathway benefits the many patients who will NOT have cancer just as much as it benefits early detection in those who do.

The overall workup algorithm is triggered by a specific, explicit symptom list applicable to any woman aged 30 or older: a breast lump with or without an associated change in shape or consistency, a palpable axillary lump, skin changes (peau d'orange or ulceration, as detailed in Slide 26), or nipple retraction or bloody discharge. Any patient presenting with one or more of these findings should have ultrasound (with or without mammography) and fine-needle aspiration cytology or biopsy coordinated within the 60-day target window; a benign result is managed according to the specific benign diagnosis per standard guideline recommendations, while a suspected or confirmed cancer diagnosis triggers referral to a designated breast cancer treatment facility and initiation of multidisciplinary team evaluation -- the entry point into the treatment pathways developed across Sections 12 through 14 of this document.

SECTION 08 -- Pathology -- In Situ Carcinomas

Lobular carcinoma in situ and ductal carcinoma in situ -- their histologic features, divergent natural histories (risk marker versus obligate precursor), and their distinct management algorithms.

SLIDE 30 -- Section Divider -- Pathology, In Situ Carcinomas

Detailed Explanation and Source Context

LCIS and DCIS are presented together because they are the two entities that must be sharply distinguished from one another, despite both existing "in situ" (confined within the basement membrane, without invasion) and despite the frequent tendency of trainees to conflate them. The single most important conceptual takeaway carried through this section, and restated explicitly in the Quick Reference comparison of Section 19 (Slide 73), is that LCIS is a bilateral risk marker while DCIS is a unilateral, obligate anatomic precursor -- this distinction alone determines nearly every difference in their management.

SLIDE 31 -- LCIS -- Features and Natural History

Slide content recap:

- Origin: terminal duct-lobular unit (TDLU); lobular architecture preserved
- Cells: small, uniform, discohesive; fill and distend acini
- E-cadherin NEGATIVE (CDH1 loss) -> loss of cell-cell adhesion (also in invasive lobular)
- Classic LCIS: bilateral risk marker; NOT an obligate precursor
- Pleomorphic LCIS (PLCIS): high-grade, necrosis -> managed more like high-grade DCIS
- Florid LCIS: expansile growth -> margins warranted (controversial)
- Classic LCIS management: observation + risk stratification + chemoprevention
- Tamoxifen (NSABP P-1): 49% decreased subsequent BC risk for LCIS patients
- Excision not mandatory for classic LCIS at core biopsy -- but often done due to upgrade risk
- Pleomorphic LCIS: excision to clear margins (similar management to high-grade DCIS)
- Risk-reducing bilateral mastectomy: option for strong family history or BRCA+
- Annual mammogram + physical exam for surveillance
- Lifetime BC risk: 8-10x general population; bilateral equal risk
- ~1% per year absolute risk of developing BC after LCIS diagnosis
- Both lobular and ductal invasive carcinoma risk is elevated
- LCIS at surgical margin: no re-excision required (classic LCIS); PLCIS: excise
- ~20% of excisions -> upgrade to invasive or DCIS (reason for excision at initial CNB)
- ILC: invasive cells in single file ("Indian file") through stroma; no gland formation
- E-cadherin negative in both LCIS and ILC -- marker of lobular differentiation
- ILC: 8-14% of invasive BC; bilateral/multifocal in 20-30%
- ILC more often ER+ (90%), HER2-; often mammographically occult (MRI superior)
- ILC prognosis: similar to IDC stage-for-stage; late recurrences (bones, peritoneum, leptomeningeal)

Detailed Explanation and Source Context

Lobular carcinoma in situ arises from the terminal ductal-lobular unit introduced embryologically in Section 02 (Slide 6), but -- unlike DCIS -- it preserves the underlying lobular architecture: the malignant-appearing cells fill and distend the acini of the lobule without disrupting its basic glandular framework. The defining molecular lesion is loss of E-cadherin expression, caused by inactivation of the CDH1 gene (the same gene whose germline mutation causes hereditary diffuse gastric cancer syndrome, discussed in Section 04, Slide 16); E-cadherin is a transmembrane cell-adhesion protein, and its loss causes the characteristic discohesive growth pattern in which individual small, uniform, non-cohesive cells fail to adhere to one another -- this same E-cadherin loss and

consequent discohesive growth pattern is shared with invasive lobular carcinoma, discussed later in this slide, and is the molecular thread connecting LCIS to ILC despite the very different clinical significance of the "in situ" versus "invasive" distinction.

Three histologic variants of LCIS carry different management implications. Classic LCIS -- the variant implied whenever "LCIS" is used without qualification -- is treated fundamentally as a bilateral risk marker rather than as a cancer precursor requiring eradication: its presence signals an elevated risk of subsequently developing invasive carcinoma (in either breast, roughly equally), but classic LCIS itself is not considered obligated to progress to invasive disease if left in place. Pleomorphic LCIS (PLCIS), by contrast, shows high-grade cytologic features and often central necrosis, morphologically resembling high-grade DCIS, and is consequently managed much more aggressively -- with surgical excision to achieve clear margins, analogous to DCIS management (Slide 32) -- reflecting a more biologically aggressive, and less purely "risk marker," behavior. Florid LCIS, characterized by expansile growth filling and distending the lobular units more extensively, is managed with a more cautious, margin-conscious approach as well, though this remains an area of ongoing controversy in the literature regarding exactly how aggressively it should be treated relative to classic LCIS.

For classic LCIS, management centers on observation combined with risk stratification and consideration of chemoprevention rather than surgical eradication: annual mammography and clinical breast examination constitute the core surveillance regimen, and tamoxifen -- as established in the NSABP P-1 trial already discussed in Section 05 (Slide 19) and Section 06 (Slide 23) -- reduces subsequent breast cancer risk by 49% in patients with a history of LCIS, making it a well-supported chemoprevention option in this specific population. Surgical excision is not mandatory when classic LCIS is found incidentally on core needle biopsy, but is frequently performed in practice anyway because of a recognized upgrade risk: approximately 20% of lesions initially diagnosed as LCIS on core biopsy are found, upon complete surgical excision, to actually harbor invasive carcinoma or DCIS that the core biopsy sampling missed -- this upgrade rate is the specific clinical justification most surgeons cite for proceeding to excisional biopsy after a core-biopsy diagnosis of LCIS, even though guidelines do not universally mandate it for the classic variant. Pleomorphic LCIS, as already noted, is managed with excision to achieve clear margins given its more aggressive histologic appearance. For patients with the highest combined risk -- classic LCIS together with a strong family history or a confirmed BRCA mutation -- risk-reducing bilateral mastectomy, as detailed in Section 05 (Slide 19), remains an available option, reflecting LCIS's role as a marker that compounds with, rather than substitutes for, other risk factors in the overall risk calculus.

The natural history data underlying this management approach are specific and worth internalizing precisely: LCIS confers an 8- to 10-fold elevated lifetime breast cancer risk (as already stated in the risk-factors overview of Section 04, Slide 14), translating into an ongoing absolute risk of approximately 1% per year of developing invasive breast cancer following an LCIS diagnosis, with this elevated risk applying essentially equally to both breasts -- the "bilateral" qualifier is not a minor footnote but the central clinical fact that distinguishes LCIS from DCIS, where risk is overwhelmingly concentrated in the ipsilateral breast at the site of the lesion. Notably, the subsequent invasive cancer risk following LCIS is elevated for BOTH invasive lobular and invasive ductal histology -- LCIS is a marker of generalized elevated risk across the breast epithelium, not a lesion that specifically predicts only lobular-type invasive cancer will eventually develop. When classic LCIS happens to be present at a surgical resection margin, no re-excision is required specifically because LCIS is not being treated as a lesion requiring complete anatomic eradication (unlike DCIS, where margin status is a critical determinant of adequate treatment, Slide 32) -- though pleomorphic LCIS at a margin does warrant re-excision, consistent with its more DCIS-like management approach.

The slide closes by extending the E-cadherin-loss, discohesive-growth-pattern concept from LCIS to its invasive counterpart, invasive lobular carcinoma (ILC), because the two lesions are histogenetically linked. ILC is histologically distinctive: individual tumor cells infiltrate the stroma in a single-file ("Indian file") pattern without forming the glandular structures typical of ductal carcinoma, a direct morphologic consequence of the same E-cadherin loss that defines LCIS. ILC accounts for 8 to 14% of all invasive breast cancers, making it the second most common histologic type after invasive ductal carcinoma (Section 09, Slide 34), and shows a distinctly higher rate of bilateral or multifocal disease (20 to 30%) than ductal carcinoma -- a pattern directly explained by ILC's shared field-effect biology with LCIS, which itself confers bilateral risk. ILC is more frequently estrogen-receptor-positive (approximately 90% of cases) and HER2-negative than invasive ductal carcinoma, and it is notably prone to being mammographically occult -- its diffuse, single-file infiltrative growth pattern often fails to form a discrete, mammographically detectable mass or distortion, meaning MRI (with its superior soft-tissue contrast resolution, as discussed in Section 07, Slide 27) is frequently the more sensitive modality for both initial detection and extent-of-disease assessment in ILC. Stage-for-stage, ILC's overall prognosis is broadly similar to invasive ductal carcinoma, but its pattern of late recurrence is distinctive, showing a characteristic predilection for unusual metastatic sites -- bone, peritoneal and gastrointestinal surfaces, and leptomeningeal disease -- that differs somewhat from the more typical ductal carcinoma metastatic pattern (bone, liver, lung, brain, as detailed in Section 03, Slide 12) and that clinicians should specifically keep in mind when a patient with a remote history of ILC develops symptoms suggestive of an unusual site of recurrence.

SLIDE 32 -- DCIS -- Features and Management Algorithm

Slide content recap:

- Malignant epithelial cells within basement membrane -- no invasion
- Staged as Tis; DCIS + microinvasion (<1mm) = T1mi
- Nuclear grade: low (grade 1), intermediate (grade 2), high (grade 3)
- Architecture: comedo (necrosis), cribriform, micropapillary, solid, papillary
- Comedo DCIS (high-grade with central necrosis): most aggressive; calcifies -> mammographically visible
- Histologic extent: often underestimated on imaging; gap-like growth up to 1 cm
- If untreated: ~50% of high-grade DCIS -> invasive BC within 10 years (Lagios)
- Low-grade DCIS: longer latency; some may never progress (overdiagnosis concern)
- DCIS upgrade to invasive at excision: ~20% overall (higher for large/high-grade)
- Annual IBTR risk after BCS + RT: ~0.5-1% per year
- Without RT: IBTR ~1.5-3% per year -- halved with radiation
- Van Nuys Prognostic Index (VNPI): scoring system for DCIS: 3-9 points (size, margin, nuclear grade, age)
- VNPI 3-4: BCS alone (no RT); VNPI 5-7: BCS + RT; VNPI 8-9: mastectomy
- VNPI limitations: not universally adopted; controversial omission of RT decisions
- Current standard: margin ≥ 2 mm primary criterion for adequacy
- "No ink on tumor" insufficient for DCIS -- 2 mm margin required for RT cases
- Management pathway: BCS (lumpectomy) or Mastectomy
- After BCS: WBRT (standard Cat 1); after mastectomy: no RT needed
- Endocrine Rx (ER+ DCIS): 5 yrs tamoxifen or AI

Detailed Explanation and Source Context

Ductal carcinoma in situ is defined by malignant-appearing epithelial cells proliferating within the ductal system but remaining entirely confined within an intact basement membrane -- the presence or absence of basement membrane breach is the single defining feature that separates DCIS from invasive ductal carcinoma, and is exactly the architectural information that core needle biopsy (Section 07, Slide 28) provides but fine-needle aspiration cannot. DCIS is staged as Tis in the AJCC system (Section 10, Slide 37); when a focus of invasion is present but measures 1 millimeter or less, the lesion is upstaged specifically to T1mi (microinvasion) rather than

remaining Tis, a distinction with direct implications for whether axillary staging is required (discussed below).

DCIS is graded by nuclear features into low (grade 1), intermediate (grade 2), and high (grade 3) categories, and separately classified by architectural growth pattern into comedo, cribriform, micropapillary, solid, and papillary subtypes -- these two classification systems (grade and architecture) are related but distinct, since a given architectural pattern can occur across a range of nuclear grades. Comedo-type DCIS, characterized by high nuclear grade combined with central luminal necrosis, is specifically emphasized as the most biologically aggressive pattern; the necrotic centers frequently calcify, which is precisely why comedo DCIS -- despite its more aggressive biology -- is often more readily detected mammographically than lower-grade patterns, since the resulting microcalcifications are the classic mammographic signature of DCIS discussed in Section 06 (Slide 22) and Section 07 (Slide 27). A separate, important point about DCIS extent is that its true histologic extent is frequently underestimated by imaging, since DCIS can grow in a discontinuous, gap-containing pattern along the ductal system for up to a centimeter beyond what appears as a contiguous abnormality on mammography -- this imaging-pathology extent discordance is a major contributor to the positive-margin and re-excision rates encountered in DCIS surgical management.

The natural history of untreated DCIS differs sharply by grade, a distinction with direct implications for the overdiagnosis debate already introduced in Section 06 (Slide 22): based on classic natural-history data (notably from Lagios and colleagues, studying patients whose DCIS was initially misdiagnosed as benign and therefore left untreated), approximately 50% of high-grade DCIS progresses to invasive breast cancer within ten years if left untreated, establishing high-grade DCIS as a lesion with substantial malignant potential that clearly warrants treatment. Low-grade DCIS, by contrast, shows a longer latency to progression, and a meaningful proportion may never progress to invasive disease within a patient's natural lifetime -- this is precisely the biological basis for the overdiagnosis concern raised in Section 06, since some proportion of low-grade, screen-detected DCIS represents lesions that, if never found and never treated, would not have gone on to threaten the patient's life, yet current diagnostic technology cannot reliably distinguish prospectively which specific low-grade DCIS lesions belong to this indolent subset. Reflecting this biological range, the overall rate at which a DCIS diagnosis made on core needle biopsy is "upgraded" to invasive carcinoma upon complete surgical excision is approximately 20%, with this upgrade rate rising further for larger lesions and higher-grade disease -- this upgrade phenomenon is the specific reason surgical excision (rather than observation) remains standard even for a DCIS diagnosis made purely on core biopsy, since a meaningful minority of such diagnoses will, on complete excision, actually represent invasive disease requiring appropriately escalated management including axillary staging.

Ipsilateral breast tumor recurrence (IBTR) rates after treatment quantify the value of adjuvant radiation specifically: following breast-conserving surgery combined with whole-breast radiation, the annual IBTR risk is approximately 0.5 to 1% per year, whereas surgery alone (without radiation) carries an annual IBTR risk of 1.5 to 3% per year -- radiation therapy therefore approximately halves the annual recurrence risk, a benefit magnitude directly comparable to (and consistent with) the invasive-cancer radiation benefit discussed in Section 12 (Slide 46). The Van Nuys Prognostic Index (VNPI) was developed as a structured attempt to identify which patients might safely omit radiation therapy after breast-conserving surgery for DCIS, scoring four factors -- tumor size, margin width, nuclear grade, and patient age -- on a scale from 3 to 9 points, with a score of 3 to 4 supporting breast-conserving surgery alone (without radiation), a score of 5 to 7 supporting breast-conserving surgery plus radiation, and a score of 8 to 9 favoring mastectomy. The VNPI is presented here with an explicit caveat regarding its limitations: it has not achieved universal adoption in clinical practice, and using it to justify omitting radiation therapy in low-scoring patients remains genuinely controversial in the literature, since long-term outcome data on VNPI-guided radiation omission have shown mixed results across different validation cohorts.

The current, more widely accepted standard for adequate margin width in DCIS surgery is a minimum of 2 millimeters, a threshold that stands in explicit contrast to the "no ink on tumor" standard used for invasive carcinoma (elaborated in Section 12, Slide 44) -- DCIS requires this wider, quantitatively defined margin specifically because of its underestimated true extent (discussed above) and the correspondingly higher risk that a minimal, ink-only margin would leave residual disease behind, particularly in patients who will go on to receive radiation therapy, for whom the 2-millimeter standard is most rigorously applied. The overall management pathway proceeds along two branches depending on the extent and distribution of disease: breast-conserving surgery (lumpectomy) followed by whole-breast radiation therapy is the Category 1, guideline-preferred standard approach for most patients, while mastectomy (reserved for multicentric disease, extensive disease relative to breast size, or patient preference) does not require adjuvant radiation, since the entire at-risk breast tissue has already been removed. In either surgical pathway, adjuvant endocrine therapy -- five years of tamoxifen or, in postmenopausal patients, an aromatase inhibitor -- is added when the DCIS is estrogen-receptor-positive, reducing both ipsilateral recurrence risk and the risk of a new contralateral breast cancer, exactly analogous to (though generally considered to confer a more modest absolute benefit than) its role in invasive, ER-positive disease detailed in Section 13 (Slide 49).

SECTION 09 -- Pathology -- Invasive Carcinomas

The full spectrum of invasive histologic subtypes and their relative frequencies and prognoses, plus the special, high-acuity presentations -- inflammatory breast cancer, phyllodes tumor, Paget disease, and angiosarcoma -- that each require a distinct management approach diverging from routine invasive ductal carcinoma.

SLIDE 33 -- Section Divider -- Pathology, Invasive Carcinomas

Detailed Explanation and Source Context

Having established the in situ lesions in Section 08, this section catalogues the invasive histologic subtypes, ranked by frequency, and then singles out four special presentations -- inflammatory breast cancer, phyllodes tumor, Paget's disease, and angiosarcoma -- each of which is a recognizable, testable clinical entity with a management pathway that diverges meaningfully from standard invasive ductal carcinoma, making early recognition of the correct diagnosis clinically consequential rather than merely academic.

SLIDE 34 -- Invasive Carcinoma Types

Slide content recap:

- Invasive ductal carcinoma (NST): 70-80%; no specific features; heterogeneous; grades 1-3; most common cause BC death
- Invasive lobular carcinoma (ILC): 8-14%; Indian-file pattern; E-cadherin neg; ER+/HER2-; bilateral risk; mammogram-occult
- Tubular carcinoma: 1-4%; well-differentiated tubules; low-grade; ER+; excellent (98% 10-yr survival)
- Mucinous (colloid): 2-3%; pools of mucin; ER+; older women; good prognosis, low nodal positivity
- Medullary-like: 1-2%; pushing borders; lymphocytic infiltrate; ER-/HER2-; BRCA1; paradoxically better prognosis
- Cribriform: 0.3-0.8%; low-grade; sieves/arches; ER+; excellent, often mixed with tubular
- Papillary (invasive): 0.5-1%; papillary architecture; older women; ER+; good prognosis
- Metaplastic: 0.2-1%; squamous/spindle/chondroid/osseous metaplasia; TNBC; aggressive, chemo-resistant
- Micropapillary: 1-2%; tumor clusters in lacunar spaces; ER+; high LVI; poor prognosis, high nodal positivity

Detailed Explanation and Source Context

Invasive ductal carcinoma of no special type (also termed "not otherwise specified," NST) accounts for 70 to 80% of all invasive breast cancers, and by definition lacks the distinguishing architectural features that define the specific special-type histologies on the remainder of this table -- it is a heterogeneous category encompassing tumors graded 1 through 3 with widely varying biology, molecular subtype (Section 11), and prognosis, and, simply by virtue of its overwhelming numeric predominance, remains the single most common cause of breast cancer death, even though several of the rarer special types listed below carry a worse per-case prognosis. Invasive lobular carcinoma, already discussed in full in Slide 31 alongside its in situ counterpart, accounts for 8 to 14% of cases and is included here for completeness of the frequency ranking.

Several special histologic types carry a distinctly favorable prognosis, and recognizing them is clinically important because it can appropriately temper the intensity of adjuvant systemic therapy relative to grade-matched invasive ductal carcinoma. Tubular carcinoma (1 to 4% of cases) is composed of well-differentiated, angulated tubular structures, is essentially always low-grade and estrogen-receptor-positive, and achieves an excellent 98% ten-year survival with a low rate of nodal metastasis -- among the most favorable of all breast cancer histologies. Mucinous (colloid) carcinoma (2 to 3%) is characterized by tumor cells floating within large pools of extracellular mucin, tends to occur in older women, is typically ER-positive, and likewise carries a good prognosis with a low rate of nodal involvement and distant metastasis. Cribriform carcinoma (0.3

to 0.8%) shows a distinctive sieve-like or arch-like glandular growth pattern, is low-grade and ER-positive, and is frequently found admixed with tubular carcinoma, sharing its excellent prognosis. Invasive papillary carcinoma (0.5 to 1%) shows a papillary (finger-like projection) architecture, occurs predominantly in older women, is typically ER-positive, and also carries a favorable prognosis, generally with a central tumor location.

Medullary-like carcinoma (1 to 2%) is a distinctive and somewhat paradoxical entity: histologically, it shows pushing (rather than infiltrative) tumor borders and a dense lymphocytic infiltrate, and, despite typically being estrogen-receptor-negative, HER2-negative, and high-grade in appearance (features that would ordinarily predict an aggressive clinical course), medullary-like carcinoma paradoxically behaves BETTER than grade-matched ordinary invasive ductal carcinoma with the same immunophenotype -- the prevailing explanation is that the dense lymphocytic infiltrate reflects an effective host immune response against the tumor, an early example of the same tumor-immune-interaction biology that underlies the modern use of immunotherapy in triple-negative breast cancer (Section 13, Slide 52). Medullary-like carcinoma is also notably enriched among BRCA1-associated breast cancers, consistent with BRCA1's known association with basal-like, triple-negative tumor biology (Section 04, Slide 15).

At the opposite end of the prognostic spectrum, two special types carry a WORSE prognosis than ordinary invasive ductal carcinoma. Metaplastic carcinoma (0.2 to 1%) shows differentiation of the tumor along squamous, spindle-cell, chondroid (cartilaginous), or osseous (bony) lines -- a phenomenon reflecting aberrant differentiation of the malignant epithelial clone along mesenchymal-like lineages -- is almost always triple-negative, and behaves aggressively with notable resistance to standard cytotoxic chemotherapy, making it one of the more therapeutically challenging breast cancer subtypes despite its rarity. Invasive micropapillary carcinoma (1 to 2%) is characterized by small clusters of tumor cells suspended within clear stromal lacunar spaces (an artifactual appearance reflecting the tumor's distinctive growth pattern rather than true vascular invasion in every focus, though true lymphovascular invasion is also characteristically extensive in this subtype), is typically ER-positive, but carries a disproportionately high rate of lymphovascular invasion and lymph node metastasis relative to its often modest primary tumor size, translating into a comparatively poor prognosis that does not fully correlate with the tumor's apparent size or grade at initial assessment -- for this reason, micropapillary histology is itself often treated as an adverse prognostic factor warranting more aggressive locoregional and systemic management than the primary tumor size alone would suggest.

SLIDE 35 -- Special Histologic Types and Presentations

Slide content recap:

- IBC clinical diagnosis: rapid onset erythema $\geq 1/3$ breast +/- peau d'orange +/- nipple retraction
- IBC cause: dermal lymphatic invasion (NOT a skin infection -- no fever initially)
- IBC pathologic: may or may not see classic invasive ductal cells in biopsy specimen
- IBC ALWAYS cT4d stage; > Stage IIIB; no TNM "T" based on tumor size
- IBC management: NEOADJUVANT chemo first (NOT surgery first); then surgery then RT
- IBC prognosis: 5-year OS ~40%; TNBC and HER2+ subtypes common
- Phyllodes tumor: biphasic (epithelial + stromal); stromal overgrowth key
- Phyllodes classification: benign (50%), borderline (25%), malignant (25%)
- Malignant phyllodes: cellular stroma, high mitoses, permeative borders, stromal overgrowth
- Phyllodes metastasis: hematogenous (lung, bone) -- NOT lymphatic (no SLNB needed)
- Phyllodes treatment: wide excision ≥ 1 cm margin; mastectomy if large; NO chemotherapy (sarcoma)
- Phyllodes recurrence: all grades -> wide excision; malignant -> consider RT
- Paget's: large, pale cells (Paget cells) in nipple epidermis -- eczematous nipple rash
- Paget's always associated with underlying DCIS (100%) or invasive cancer (50%)
- Paget's location: central (retroareolar); unilateral, persistent, non-responsive to topical steroids

- Paget's diagnosis: biopsy of nipple epidermis or punch biopsy
- Paget's management: total mastectomy (if underlying invasive) or BCS + nipple excision + RT
- Paget's prognosis: depends on underlying lesion; without invasive Ca -> excellent
- Angiosarcoma primary (spontaneous): rare; young women; high-grade; no risk factors
- Angiosarcoma secondary (post-irradiation): most important form; 6-10 years after RT for BC
- Stewart-Treves syndrome: lymphedema-associated angiosarcoma (post-ALND)
- Angiosarcoma skin: blue-red discoloration, satellite nodules; rapidly enlarging
- Angiosarcoma treatment: mastectomy + wide excision; no role for sentinel node biopsy
- Angiosarcoma prognosis: poor; median OS 2-3 years; rapid metastasis to lungs

Detailed Explanation and Source Context

Inflammatory breast cancer (IBC) is emphasized repeatedly across this document (Section 07 Slide 26, Section 13 Slide 52, Section 14 Slide 56, Section 17 Slide 66) because it is a diagnosis made clinically, not histologically, and because failure to recognize it promptly and manage it in the correct sequence carries real consequences. The clinical diagnosis requires rapid-onset erythema involving at least one-third of the breast, often with accompanying peau d'orange and nipple retraction, developing over a course of days to weeks rather than the months typical of ordinary locally advanced breast cancer. The underlying pathophysiology is dermal lymphatic invasion by tumor emboli, producing lymphatic obstruction and the resulting skin edema and erythema -- critically, this is NOT an infectious process, and the absence of fever or the characteristic failure to respond to a trial of antibiotics is the key clinical clue that should prompt biopsy rather than continued treatment for presumed mastitis or abscess, a distinction already flagged in Section 07 (Slide 26). On biopsy, the underlying invasive ductal carcinoma cells may or may not be sampled, since the diagnosis of IBC is established by the clinical syndrome and the finding of dermal lymphatic tumor emboli on skin biopsy, not strictly by demonstrating invasive carcinoma in every sample obtained. Because the clinical presentation itself, rather than tumor size, defines the disease, IBC is always staged as cT4d regardless of the size of any underlying palpable mass, and this automatically places the disease at Stage IIIB or higher (Section 10, Slide 37) -- meaning a patient can present with IBC and a relatively small or even non-palpable underlying tumor and still carry a locally advanced stage designation. The single most important management principle, repeated at every point this diagnosis is revisited, is that surgery must NEVER be the first treatment step: neoadjuvant chemotherapy is mandatory first, followed by modified radical mastectomy (breast-conserving surgery is never appropriate for IBC) and then post-mastectomy radiation therapy to the chest wall and regional nodes -- operating first, before treating the diffuse dermal lymphatic disease systemically, would leave the extensive skin and lymphatic disease burden unaddressed and compromise both local control and the patient's ability to tolerate subsequent systemic therapy. IBC carries a guarded prognosis, with an approximate 40% five-year overall survival, and is disproportionately triple-negative or HER2-positive in molecular subtype, both of which are the more aggressive molecular categories discussed in Section 11 (Slide 42).

Phyllodes tumor is a distinctive biphasic neoplasm, meaning it contains both an epithelial and a stromal (connective tissue) component, with the key distinguishing pathologic feature relative to its benign look-alike, the fibroadenoma, being stromal overgrowth and cellularity. Phyllodes tumors are classified into benign (approximately 50% of cases), borderline (25%), and malignant (25%) categories based on stromal cellularity, mitotic rate, and the character of the tumor's border (well-circumscribed versus permeative/infiltrative into surrounding tissue) -- malignant phyllodes tumors specifically show markedly cellular stroma, a high mitotic rate, permeative (infiltrative) tumor borders, and pronounced stromal overgrowth relative to the epithelial component. A critical and frequently tested management point is that phyllodes tumors -- unlike epithelial breast carcinomas -- metastasize hematogenously (to the lungs and bone) rather than via the lymphatic system, meaning sentinel lymph node biopsy is not indicated regardless of tumor grade, since there is no meaningful risk of lymphatic spread to stage. Treatment for phyllodes tumors of any grade is wide local excision with a margin of at least 1

centimeter (wider than the "no ink on tumor" standard applied to epithelial invasive carcinoma, Section 12, Slide 44), reflecting phyllodes tumors' propensity for local recurrence if incompletely excised; mastectomy is reserved for very large tumors relative to breast size. A second critical management point is that phyllodes tumors -- being fundamentally a stromal/connective-tissue neoplasm, biologically closer to a sarcoma than to an epithelial carcinoma -- do NOT respond to standard breast cancer chemotherapy regimens and chemotherapy is not part of routine management. Local recurrence, when it occurs (across all grades), is managed with repeat wide excision, and radiation therapy may be considered as an adjunct specifically for malignant phyllodes tumors given their higher recurrence propensity.

Paget's disease of the nipple, already introduced anatomically in Section 02 (Slide 6) as arising from the central, retroareolar zone where the major lactiferous ducts converge, is caused by large, pale, atypical epithelial cells (Paget cells) infiltrating the nipple epidermis, producing a persistent, eczematous, often crusted or ulcerated nipple rash that is characteristically unilateral and, critically, fails to respond to topical steroid treatment -- this treatment-resistance is an important clinical clue distinguishing Paget's disease from true eczema or dermatitis, which would be expected to at least partially respond to topical steroids. A defining diagnostic fact is that Paget's disease is essentially always associated with an underlying malignancy: an underlying DCIS is found in essentially 100% of cases, and an underlying invasive carcinoma is found in approximately 50% of cases, reflecting the prevailing theory that Paget cells represent intraepidermal extension of an underlying intraductal (in situ or invasive) carcinoma migrating along the ductal system into the nipple epidermis, rather than an independently arising primary skin cancer. Diagnosis requires biopsy of the nipple epidermis itself (a punch biopsy is a common technique), and management is dictated entirely by the underlying lesion: total mastectomy is performed when an underlying invasive carcinoma is present, while breast-conserving surgery (central excision including the nipple-areolar complex) combined with whole-breast radiation is an option when the underlying lesion is DCIS without invasion -- this mirrors the DCIS management algorithm of Section 08 (Slide 32), simply centered on the nipple rather than a peripheral quadrant. Prognosis for Paget's disease is determined entirely by the biology and stage of its underlying lesion; when no invasive component is found, prognosis is excellent, essentially equivalent to isolated DCIS.

Angiosarcoma of the breast is a rare, high-grade malignant vascular neoplasm that occurs in two distinct clinical settings, and distinguishing which setting applies is itself diagnostically important. Primary (spontaneous) angiosarcoma arises without any identifiable predisposing risk factor, typically in young women, and is high-grade at presentation. Secondary angiosarcoma -- described here as the more clinically important of the two forms given its direct link to breast cancer treatment itself -- arises as a delayed complication of prior radiation therapy for breast cancer, typically developing 6 to 10 years after the original radiation exposure, meaning any patient who develops new, unexplained skin changes in a previously irradiated breast years after treatment for an earlier cancer should raise clinical suspicion for this specific, serious late complication (already flagged as a rare RT toxicity in Section 12, Slide 46). A specific, eponymous variant, Stewart-Treves syndrome, refers to angiosarcoma arising in the setting of chronic lymphedema following axillary lymph node dissection, illustrating that chronically lymphedematous tissue itself carries an elevated long-term malignant transformation risk independent of radiation exposure. Clinically, angiosarcoma presents with blue-red skin discoloration and satellite nodules that enlarge rapidly, a presentation that can be mistaken for a bruise or vascular malformation early in its course before the diagnosis becomes clinically evident. Treatment is surgical, with mastectomy and wide excision, and -- as with phyllodes tumor, reflecting angiosarcoma's fundamentally mesenchymal (sarcomatous) rather than epithelial biology -- there is no role for sentinel lymph node biopsy, since angiosarcoma, like other sarcomas, does not characteristically spread via the lymphatic system. Prognosis is poor, with a median overall survival of only 2 to 3 years and a strong propensity for rapid hematogenous metastasis to the lungs, making angiosarcoma one of the most aggressive entities discussed anywhere in this

document despite its rarity.

SECTION 10 -- AJCC Staging (TNM), 8th Edition

The T, N, and M category definitions, the resulting anatomic stage groupings and their associated survival, stage-appropriate staging workup, and the 8th edition innovation of incorporating biomarkers into a "prognostic stage" distinct from pure anatomic stage.

SLIDE 36 -- Section Divider -- AJCC Staging (TNM)

Detailed Explanation and Source Context

Staging is placed after the pathology sections because T and N categories depend directly on anatomic and histologic facts established there -- tumor size and invasion status (Section 08-09) determine T category, and the Berg nodal levels and internal mammary drainage patterns (Section 02, Slide 8) determine N category. This section is the hinge point of the entire document: everything before it describes what breast cancer IS, and everything after it (Sections 11 onward) describes what to DO about a given stage and biology.

SLIDE 37 -- AJCC 8th Edition -- T and N Categories

Slide content recap:

- Tis: in situ (DCIS, LCIS, Paget without invasive)
- T1mi: microinvasion ≤ 1 mm
- T1a: $>1\text{mm}-\leq 5\text{mm}$; T1b: $>5\text{mm}-\leq 10\text{mm}$; T1c: $>10\text{mm}-\leq 20\text{mm}$
- T2: $>20\text{mm}-\leq 50\text{mm}$; T3: $>50\text{mm}$
- T4a: chest wall invasion (ribs, intercostals, serratus -- NOT pectoralis)
- T4b: skin ulceration, satellite nodules, edema (peau d'orange)
- T4c: T4a + T4b; T4d: inflammatory breast cancer
- pN0: no nodal metastasis by H and E or IHC
- pN1mi: micrometastasis $>0.2\text{mm}$, $\leq 2\text{mm}$
- pN1a: 1-3 axillary nodes (macrometastasis)
- pN1b: internal mammary sentinel node
- pN1c: 1-3 axillary + internal mammary SLN
- pN2a: 4-9 axillary nodes; pN2b: internal mammary nodes (clinical)
- pN3a: ≥ 10 axillary nodes or infraclavicular (Level III)
- pN3b: IMN + ≥ 1 axillary node; or ≥ 4 axillary + IMN SLN
- pN3c: supraclavicular nodes
- KEY: chest wall = ribs/intercostals/serratus anterior. Pectoralis muscle involvement does NOT = T4a

Detailed Explanation and Source Context

The T category is a purely size-based, threshold-driven classification for invasive disease up through T3, and each threshold in the table should be understood as a strict cutoff rather than an approximation: T1 is subdivided into four progressively larger bands -- T1mi (microinvasion, 1 millimeter or less, the specific threshold already introduced in Section 08's DCIS discussion, Slide 32, as the boundary above which a DCIS lesion is upstaged out of pure Tis), T1a (greater than 1 mm up to 5 mm), T1b (greater than 5 mm up to 10 mm), and T1c (greater than 10 mm up to 20 mm) -- followed by T2 (greater than 20 mm up to 50 mm) and T3 (greater than 50 mm). T4, by contrast, is defined not by size at all but by specific patterns of local extension, and the four T4 subcategories each describe a distinct anatomic or clinical picture: T4a is chest wall invasion, and the single most commonly tested and most commonly misunderstood point on this entire slide is the explicit statement that "chest wall" for staging purposes means the ribs, the intercostal muscles, and the serratus anterior muscle -- invasion of the pectoralis major or minor muscle alone, without involvement of these deeper structures, does NOT qualify as

T4a, because the pectoral muscles are considered part of the breast's operative field (routinely encountered and, historically, deliberately resected in a Halsted radical mastectomy, Section 01) rather than "chest wall" in the T4a sense; T4b describes skin-level findings -- ulceration, ipsilateral satellite skin nodules, or skin edema (peau d'orange) -- caused by direct dermal or dermal lymphatic tumor involvement; T4c is simply the combination of T4a and T4b criteria both being present simultaneously; and T4d is reserved specifically for inflammatory breast cancer (Section 09, Slide 35), staged this way regardless of the size or even palpability of any underlying mass, since IBC is a clinical diagnosis based on the erythema/edema syndrome rather than a tumor-size-based determination.

The N category (specifically, the pathologic pN classification determined after surgical staging) is built directly on the Berg axillary level anatomy and internal mammary drainage pattern established in Section 02 (Slide 8), and its thresholds are defined by the number and location of involved nodes together with the size of the metastatic deposit within each node. pN0 indicates no nodal metastasis detectable by either standard hematoxylin-and-eosin staining or immunohistochemistry. pN1mi designates isolated micrometastases measuring more than 0.2 mm but no more than 2 mm within a node -- a category specifically carved out because deposits at this small size carry different prognostic weight than fully macrometastatic disease, even though they represent unambiguous nodal involvement rather than the isolated tumor cells (below the 0.2 mm threshold) that are not counted as nodal metastasis at all under the current system. pN1a designates one to three involved axillary nodes with macrometastatic disease (deposits larger than 2 mm), the most commonly encountered node-positive category and the one most directly relevant to the Z0011 trial's eligibility criteria (Section 12, Slide 45). pN1b designates involvement limited to an internal mammary sentinel node (detected, for instance, via internal mammary sentinel node biopsy technique, without axillary involvement), and pN1c designates the combination of one to three positive axillary nodes together with a positive internal mammary sentinel node. Moving to more extensive disease, pN2a designates four to nine involved axillary nodes, and pN2b designates clinically detected (rather than purely sentinel-node-detected) internal mammary node involvement without axillary involvement. At the most extensive end, pN3a designates ten or more involved axillary nodes, or involvement of the infraclavicular (Berg Level III) nodes specifically -- tying back to the "complete dissection" anatomic level discussed in Section 02, Slide 8; pN3b designates internal mammary node involvement occurring together with at least one positive axillary node, or, alternatively, four or more positive axillary nodes together with a positive internal mammary sentinel node -- this is the specific staging consequence, previewed in Section 02 (Slide 8), of the anatomic fact that internal mammary nodal involvement independently predicts a worse prognosis; and pN3c designates supraclavicular nodal involvement, the most distant nodal basin still classified as regional (rather than distant/M1) disease, reflecting the anatomic teaching from Section 02 that supraclavicular involvement indicates advanced regional, rather than systemic, spread.

This T and N framework is the direct anatomic and quantitative basis for the stage groupings and survival data presented on the next slide (Slide 38), and mastering the specific numeric thresholds here -- particularly the T4a chest-wall-versus-pectoralis distinction and the specific node-count boundaries separating N1 from N2 from N3 -- is a prerequisite for correctly assigning an anatomic stage to any given clinical scenario.

SLIDE 38 -- AJCC Staging Groups and Workup by Stage

Slide content recap:

- Stage 0 (Tis, N0, M0): ~100% 5-year OS
- Stage IA (T1, N0, M0): ~99%
- Stage IB (T0-1, N1mi, M0): ~98%
- Stage IIA (T0-1/T2, N1/N0, M0): ~91%

- Stage IIB (T2/T3, N1/N0, M0): ~80%
- Stage IIIA (T0-3/T3, N2/N1-2, M0): ~67%
- Stage IIIB (T4, N0-2, M0): ~52%
- Stage IIIC (any T, N3, M0): ~48%
- Stage IV (any T, any N, M1): ~27% (5-yr, improving)
- Stage I-IIA (asymptomatic): H and P + bilateral mammo + tumor biomarkers only; NO routine CT/bone scan
- Stage IIB-III: CT chest/abdomen/pelvis + bone scan OR PET-CT
- Stage IV: CT + bone scan or PET-CT; liver US; consider brain MRI if symptoms
- NCCN: routine CBC, LFTs, ALP; CXR; selective MRI breast for extent
- 8th ed introduced "Prognostic Stage Groups" incorporating biomarkers
- -> T, N, M + Grade + ER/PR/HER2 status + Oncotype DX RS (if used)
- Prognostic stage may differ from anatomic stage (e.g., T2N1 ER+/HER2-/Grade1 = Stage IIA anatomic -> Stage IB prognostic)
- Clinically significant: Stage I tumors may have very different outcomes by biology
- Important for treatment decisions, especially adjuvant systemic therapy

Detailed Explanation and Source Context

The stage groupings on this slide combine the T and N categories from Slide 37 with the M (metastasis) category into the familiar Stage 0 through Stage IV framework, and the accompanying five-year overall survival figures make explicit exactly what is at stake in each stage transition: survival falls from essentially 100% at Stage 0 (pure in situ disease) through the high-90s for Stage I, into the 80 to 91% range for Stage II, dropping more steeply through the 48 to 67% range across the three Stage III substages, and falling to a wide 27% (and improving, given the systemic therapy advances discussed throughout this document, particularly Section 15) at Stage IV. This same stage-survival relationship was already introduced from the natural-history perspective in Section 03 (Slide 12); here it is anchored to the precise T/N/M definitions.

Staging workup -- the set of tests ordered to determine whether metastatic (M1) disease is present -- is explicitly and deliberately NOT applied uniformly across all stages, and the principle governing this selectivity is one of the more commonly tested points in this section: for asymptomatic patients with Stage I or IIA disease, the pretest probability of occult distant metastasis is low enough that routine staging CT scans or bone scans are NOT recommended, since the yield of finding true metastatic disease is low while the rate of false-positive incidental findings requiring further (often invasive and anxiety-provoking) workup is comparatively high -- the appropriate workup at this early stage is limited to history and physical examination, bilateral mammography, and the tumor biomarker panel (ER, PR, HER2, discussed fully in Section 11). Once a patient reaches Stage IIB or Stage III, the pretest probability of occult metastatic disease rises sufficiently that CT of the chest, abdomen, and pelvis combined with either a bone scan or PET-CT becomes appropriate and recommended. For Stage IV disease -- where distant metastasis is already established by definition -- the same CT and bone scan (or PET-CT) combination is used, now for the purpose of fully characterizing the extent and specific sites of metastatic disease rather than for detecting its mere presence, supplemented by liver ultrasound and, when the patient has neurologic symptoms, brain MRI (reflecting the CNS metastasis risk pattern, particularly high in HER2-positive and triple-negative disease, already discussed in Section 03, Slide 12, and revisited in the ESMO 2026 update of Section 15). NCCN additionally recommends routine complete blood count, liver function tests, and alkaline phosphatase (as a general screen for hepatic or bony metastatic involvement) and a chest radiograph at any stage, together with selective use of breast MRI specifically to characterize the local extent of disease when that information would change surgical planning (Section 07, Slide 27), independent of the systemic staging workup decision.

The 8th edition of the AJCC staging manual introduced a genuinely significant conceptual innovation that distinguishes it from all prior editions: the "Prognostic Stage Group," which incorporates tumor grade and the ER, PR, and HER2 biomarker status (and, when performed, the Oncotype DX Recurrence Score, detailed in Section 11, Slide 41) directly into the stage assignment, alongside the traditional anatomic T, N, and M categories. This

means that two tumors with identical anatomic T and N categories can be assigned to different final "prognostic stages" if their underlying tumor biology differs -- the specific example given here is instructive and worth internalizing: a T2N1 tumor that is estrogen-receptor-positive, HER2-negative, and grade 1 would be classified as Stage IIA under the pure anatomic system, but is reclassified as the more favorable Stage IB under the prognostic system, reflecting the genuinely better expected outcome that low-grade, hormone-receptor-positive biology confers even at an anatomic stage that would otherwise suggest a less favorable prognosis. The clinical significance of this innovation is considerable: it formally acknowledges, within the staging system itself, that anatomic extent alone is an incomplete predictor of outcome, and that a nominal "Stage I" anatomic designation can, in fact, describe tumors with meaningfully different biological behavior and expected outcome depending on grade and receptor status -- this is precisely the same underlying principle, that biology modifies anatomy-based risk, that motivates the entire molecular subtyping and multigene assay framework developed in the next section (Section 11), and that directly informs which patients require adjuvant systemic therapy and which do not.

SECTION 11 -- Prognostic Factors and Molecular Subtypes

Traditional pathologic prognostic factors, predictive biomarkers and the concept of pathologic complete response, the validated multigene expression assays that refine chemotherapy decision-making beyond stage and receptor status alone, and the intrinsic molecular subtype classification -- including the newly clinically actionable "HER2-low" category -- that anchors nearly every systemic-therapy decision in Sections 13 through 15.

SLIDE 39 -- Section Divider -- Prognostic Factors and Molecular Subtypes

Detailed Explanation and Source Context

This section is the biological bridge between staging (Section 10, which describes anatomic extent) and treatment (Sections 12 through 15, which describe what to do about a given tumor). Every systemic therapy decision made later in this document traces back to a specific data point established here: ER/PR/HER2 status, grade, a genomic recurrence score, or an intrinsic molecular subtype label.

SLIDE 40 -- Traditional Prognostic and Predictive Factors

Slide content recap:

- Lymph node status: #1 predictor; each additional positive node decrements survival
- Tumor size: T1 < T2 < T3; continuous variable, not threshold-dependent
- Histologic grade (Elston-Ellis/Nottingham): tubule formation (1-3) + nuclear pleomorphism (1-3) + mitotic rate (1-3)
- Grade 1 (3-5 pts): low; Grade 2 (6-7): intermediate; Grade 3 (8-9): high
- Lymphovascular invasion (LVI): predicts nodal involvement + distant recurrence
- Surgical margins: adequate margin reduces IBTR risk (negative = "no ink on tumor")
- Age: younger patients (<35): increased recurrence risk; more aggressive biology; BRCA testing
- Race/ethnicity: Black women -> higher TNBC rate; biologic and systemic disparities
- Androgen receptor (AR): positive in 70-95% ER+ BC and ~50% TNBC (emerging target)
- TILs (tumor-infiltrating lymphocytes): high TILs -> improved pCR in TNBC/HER2+
- Residual cancer burden (RCB): RCB-0 (pCR) -> best OS; RCB-III -> worst
- ER (Estrogen Receptor): >=1% IHC positive -> endocrine therapy benefit (NCCN)
- ER-low (1-10%): technically positive; biologically heterogeneous -> full risk assessment
- PR (Progesterone Receptor): ER+/PR+ best response; ER+/PR- intermediate; correlates with HER2
- HER2 (ERBB2): IHC 3+ or FISH amplified -> trastuzumab + pertuzumab benefit
- HER2-low: IHC 1+ or 2+/FISH negative -> T-DXd eligibility (DESTINY-Breast04)
- Ki-67: surrogate for proliferative activity; >20-30% = high; threshold varies by lab
- pCR = no residual invasive cancer in breast + lymph nodes after neoadjuvant therapy
- Standard definition: ypT0 ypN0 (NSABP) or ypT0/Tis ypN0
- Surrogate endpoint for improved OS in TNBC and HER2+ (CREATE-X, KATHERINE)
- pCR rates: TNBC ~30-55% with platinum-containing NAC; HER2+ up to 60-70%
- Residual disease in TNBC -> capecitabine 6-8 cycles improves OS (CREATE-X)
- Residual disease in HER2+ -> T-DM1 (ado-trastuzumab emtansine) improves DFS (KATHERINE)

Detailed Explanation and Source Context

Lymph node status remains, even in an era of sophisticated genomic assays, the single most powerful pathologic prognostic factor in breast cancer -- this was already established quantitatively in Section 03 (Slide 12), where the specific ten-year disease-free survival figures by nodal burden (70% for node-negative, falling stepwise to 20% for ten or more positive nodes) were presented, and it is restated here as the anchor point

against which every other prognostic factor on this slide is a refinement, not a replacement. Tumor size behaves as a continuous prognostic variable rather than a simple threshold effect -- larger tumors carry progressively worse prognosis across the full T1-T2-T3 range, not merely at the specific T-category boundaries defined for staging purposes (Section 10, Slide 37).

Histologic grade is formally assessed using the Elston-Ellis modification of the Scarff-Bloom-Richardson system (commonly called the Nottingham grading system), which scores three independent histologic features, each on a 1-to-3 scale: the degree of tubule/gland formation, the degree of nuclear pleomorphism (variation in nuclear size and shape), and the mitotic rate (count of dividing cells per high-power field). The three component scores are summed to a total ranging from 3 to 9, which is then translated into the final grade: a total of 3 to 5 points designates Grade 1 (well-differentiated, low-grade), 6 to 7 points designates Grade 2 (moderately differentiated), and 8 to 9 points designates Grade 3 (poorly differentiated, high-grade) -- this same grading system was already referenced in the prognostic-staging discussion of Section 10 (Slide 38), where grade was shown to directly modify anatomic stage into the newer "prognostic stage" category. Lymphovascular invasion, the histologic finding of tumor cells within vascular or lymphatic channels (first introduced mechanistically in Section 03's metastatic cascade discussion, Slide 12), is an independent adverse prognostic factor predicting both occult nodal involvement (even in a sentinel-node-negative patient) and eventual distant recurrence. Surgical margin status is prognostically important specifically for local (rather than distant) recurrence risk, with the general standard for invasive carcinoma being "no ink on tumor" (any margin, however narrow, so long as tumor cells do not reach the inked resection surface) -- a standard explicitly distinguished from the wider, quantitative 2-millimeter margin required for DCIS (Section 08, Slide 32), reflecting DCIS's tendency toward underestimated true extent discussed there.

Age carries independent prognostic weight: patients diagnosed under age 35 face an elevated recurrence risk beyond what their stage and receptor status alone would predict, reflecting both an association with more aggressive underlying tumor biology at this young age and the higher pretest probability of an underlying hereditary predisposition (BRCA1/2 and other syndromes, Section 04) that should trigger genetic counseling referral regardless of family history in this age group specifically. Race and ethnicity carry prognostic implications primarily through their association with molecular subtype distribution rather than through any independent biological effect of race itself: Black women have a demonstrated higher frequency of triple-negative breast cancer (already quantified in Section 03, Slide 11, at 20 to 25% versus 10 to 15% in white women), and this biological disparity compounds with well-documented disparities in access to guideline-concordant treatment to produce the higher overall breast cancer mortality in Black women discussed in that same epidemiology slide. The androgen receptor (AR) is an emerging, actionable biomarker: it is expressed in the majority (70 to 95%) of ER-positive breast cancers, where its role is less independently prognostic, but is also expressed in roughly half of triple-negative breast cancers, where its presence identifies the specific "luminal androgen receptor" (LAR) molecular subclass discussed in Slide 42 and defines a population potentially responsive to anti-androgen therapy such as enzalutamide. Tumor-infiltrating lymphocytes (TILs), quantified on routine histology as the density of lymphocytic infiltration within and around the tumor, predict a higher likelihood of achieving pathologic complete response to neoadjuvant chemotherapy specifically in triple-negative and HER2-positive disease -- this immune-infiltration-predicts-response relationship is the same underlying biological principle already introduced with medullary-like carcinoma in Section 09 (Slide 35) and is directly relevant to understanding why immunotherapy (pembrolizumab) has been incorporated into TNBC treatment, discussed in Section 13 (Slide 52). Residual cancer burden (RCB), a formal, validated scoring system applied after neoadjuvant therapy that quantifies the amount of residual invasive tumor remaining in both the breast and the lymph nodes, stratifies patients into RCB-0 (equivalent to pathologic complete response, and associated with the best long-term outcome) through RCB-III (extensive residual disease, associated with the worst outcome) -- RCB is the more

granular, continuous version of the simpler pCR-versus-not-pCR binary distinction discussed later on this same slide.

The predictive biomarkers section of this slide addresses receptors that specifically predict RESPONSE to a targeted therapy, as distinct from the prognostic factors above that predict overall outcome regardless of treatment choice. Estrogen receptor positivity is defined, per current NCCN criteria, as 1% or greater nuclear staining by immunohistochemistry, and any ER-positive result at this threshold confers eligibility for endocrine therapy; a specific and increasingly recognized subcategory, "ER-low" (staining in the 1 to 10% range), is technically ER-positive by the formal cutoff but is biologically heterogeneous -- some ER-low tumors behave more like genuinely hormone-responsive luminal disease, while others behave more like triple-negative disease despite the technically positive ER result -- and this heterogeneity is the specific reason ER-low tumors warrant a fuller risk assessment (incorporating grade, proliferation markers, and potentially genomic assay results) rather than being managed on the assumption of robust endocrine responsiveness. Progesterone receptor status adds prognostic and predictive granularity beyond ER alone: ER-positive/PR-positive tumors have the best expected response to endocrine therapy, ER-positive/PR-negative tumors have an intermediate response, and PR loss in an ER-positive tumor also correlates with a higher likelihood of HER2 co-positivity, reflecting cross-talk between the HER2 and estrogen receptor signaling pathways. HER2 positivity -- defined as immunohistochemistry score 3+, or a 2+ (equivocal) result confirmed amplified by fluorescence in situ hybridization -- identifies the population that benefits from trastuzumab and pertuzumab, the anti-HER2 agents detailed in Section 13 (Slide 51); the newly clinically relevant "HER2-low" category (IHC score of 1+, or 2+ with a negative FISH result) was previously grouped together with genuinely HER2-negative disease as simply "not eligible for anti-HER2 therapy," but the DESTINY-Breast04 trial established that this HER2-low population does, in fact, benefit from the antibody-drug conjugate trastuzumab deruxtecan (T-DXd) -- this pivotal reclassification is developed in full in Slide 42 and in the ESMO 2026 update of Section 15. Ki-67, the proliferation marker already introduced mechanistically in Section 03 (Slide 12), is used clinically as a surrogate for tumor growth rate, with values above roughly 20 to 30% generally considered "high," though the precise threshold varies meaningfully by laboratory and staining protocol, a recognized limitation that has motivated some centers to prefer the more standardized multigene assays discussed in Slide 41 over Ki-67 alone for treatment decision-making.

Pathologic complete response (pCR) -- introduced conceptually in Section 07 (Slide 26) in the context of neoadjuvant monitoring -- is formally defined as the complete absence of residual invasive cancer in both the breast and the axillary lymph nodes following neoadjuvant systemic therapy, with the standard notation being ypT0 ypN0 (or the slightly broader ypT0/Tis ypN0, which also permits residual pure DCIS without invasive disease to still count as a pCR). pCR functions as a validated surrogate endpoint for improved long-term overall survival specifically in triple-negative and HER2-positive breast cancer -- the two subtypes with the fastest natural growth kinetics and the most aggressive early recurrence patterns (Section 03, Slide 12) -- and achieving it is directly actionable, since the CREATE-X and KATHERINE trials (both introduced earlier in this document and detailed fully in Section 13, Slide 52) demonstrated that patients who do NOT achieve pCR (that is, who have residual disease after neoadjuvant therapy) benefit from escalated adjuvant treatment: residual triple-negative disease is treated with six to eight cycles of adjuvant capecitabine, which improved overall survival in the CREATE-X trial, and residual HER2-positive disease is treated with adjuvant T-DM1 (ado-trastuzumab emtansine) in place of continued trastuzumab alone, which improved disease-free survival in the KATHERINE trial. Typical pCR rates with modern neoadjuvant regimens run from approximately 30 to 55% in triple-negative disease treated with platinum-containing chemotherapy, up to 60 to 70% in HER2-positive disease treated with dual anti-HER2 blockade -- these numbers, and the specific "escalate therapy if pCR is not achieved" principle, are the direct clinical payoff of everything this slide establishes about prognostic and predictive biomarkers, and they are revisited in full algorithmic detail in Section 13 (Slide 52) and Section 14

(Slide 55, 56).

SLIDE 41 -- Multigene Assays -- Oncotype DX and MammaPrint

Slide content recap:

- Recurrence Score (RS): 0-100 scale; validated in ER+/HER2-/N0 and N1 (1-3 nodes)
- RS <18 (low): endocrine therapy alone -- chemotherapy adds no benefit (TAILORx, RxPONDER)
- RS 18-30 (intermediate): chemo benefit uncertain; menopausal status critical
- RS >30 (high): chemotherapy adds significant benefit
- Premenopausal with RS 16-25: may benefit from chemo (partly via ovarian suppression)
- RS testing: pN0 or pN1 (1-3 nodes) ER+/HER2- invasive BC -- strongly recommended NCCN
- MammaPrint (70-gene) microarray: classifies BC as Low Risk (LR) vs High Risk (HR) for distant recurrence
- MINDACT trial: "Ultra-low risk" subset: clinical high / genomic low -> can omit chemo
- ~46% clinical-high/genomic-low: chemotherapy omission safe at 5 years
- MammaPrint best suited for: ER+/HER2- early BC when additional genomic context needed
- MammaPrint validated for both pre- and postmenopausal; pN0 and limited N1 disease
- RxPONDER Phase III: 5083 women, ER+/HER2-, N1(1-3 nodes), RS ≤25
- RxPONDER postmenopausal: no chemo benefit (adds only toxicity)
- RxPONDER premenopausal RS ≤25 N1: chemo improved iDFS (HR 0.64) -- possibly via ovarian suppression
- Clinical implication: postmenopausal N1 RS≤25 -> endocrine therapy alone safe
- Clinical implication: premenopausal N1 -> consider chemo + ovarian suppression
- PAM50/Prosigna (50 gene): reports intrinsic subtype + ROR (risk of recurrence) 0-100
- Endopredict (EP): 12-gene; binary low/high; used in postmenopausal N0/N1 ER+ HER2-
- Breast Cancer Index (BCI): predicts late recurrence (years 5-10); guides extended endocrine therapy
- All validated for chemotherapy decision in ER+/HER2- early BC -- not interchangeable
- NCCN prefers Oncotype DX but accepts validated alternatives for specific clinical scenarios

Detailed Explanation and Source Context

Multigene expression assays were developed to solve a specific clinical problem that traditional prognostic factors (Slide 40) could not fully resolve: among patients with hormone-receptor-positive, HER2-negative early breast cancer -- the single largest molecular subtype group (Slide 42) -- which specific patients truly need chemotherapy in addition to endocrine therapy, and which can safely be spared its toxicity because their tumor's underlying biology is genomically indolent despite an intermediate clinical risk profile? Oncotype DX, a 21-gene reverse-transcription PCR-based assay, is the most widely used and NCCN-preferred answer to this question, producing a Recurrence Score (RS) on a 0-to-100 scale that has been validated in both node-negative and limited node-positive (one to three positive nodes) ER-positive, HER2-negative disease. The pivotal TAILORx trial established that a low Recurrence Score (below 18) identifies patients for whom endocrine therapy alone is sufficient, with chemotherapy adding no measurable benefit -- allowing this substantial population to safely avoid chemotherapy entirely; a high Recurrence Score (above 30) identifies patients for whom chemotherapy does add significant benefit and should be given; and the intermediate range (18 to 30) is where the clinical decision becomes genuinely more nuanced, with menopausal status emerging as a critical modifying factor -- specifically, TAILORx's supplementary analysis found that premenopausal patients with a Recurrence Score in the 16-to-25 range may still derive benefit from chemotherapy, an effect now thought to be mediated substantially through chemotherapy-induced ovarian suppression (effectively adding an endocrine-pathway benefit) rather than through cytotoxic activity alone -- this is a subtle but important point, since it implies the chemotherapy "benefit" in this specific subgroup may be attributable to a hormonal, not purely cytotoxic, mechanism.

The RxPONDER trial extended this genomic decision-making framework specifically to the node-positive population (one to three positive nodes) with a Recurrence Score of 25 or below, enrolling over 5,000 women, and its findings sharpen the menopausal-status distinction already suggested by TAILORx into an even clearer,

directly actionable dichotomy: in postmenopausal women with one to three positive nodes and RS 25 or below, chemotherapy conferred NO benefit and added only toxicity, meaning endocrine therapy alone is the appropriate, safe treatment even in this node-positive population; in premenopausal women with the identical clinical and genomic profile, chemotherapy DID improve invasive disease-free survival (hazard ratio 0.64), an effect the trial investigators again attribute at least partly to chemotherapy-induced ovarian function suppression rather than purely cytotoxic mechanisms. The direct clinical translation of RxPONDER, restated explicitly on this slide, is one of the more consequential and precise treatment-selection rules in modern breast oncology: postmenopausal women with N1 disease and RS 25 or below can safely receive endocrine therapy alone, while premenopausal women with the same clinical and genomic profile should be considered for chemotherapy, generally combined with ovarian function suppression given the mechanistic hypothesis underlying the observed benefit -- this recommendation is directly incorporated into the stage-specific management algorithm of Section 14 (Slide 55).

MammaPrint, a 70-gene expression microarray assay, takes a different classification approach from Oncotype DX's continuous Recurrence Score, instead producing a binary Low Risk versus High Risk classification for distant recurrence. The pivotal MINDACT trial identified a specific and clinically valuable discordant subgroup -- patients with a clinically high-risk profile (based on traditional factors like tumor size and grade) but a genomically Low Risk MammaPrint result, sometimes termed "ultra-low risk" in this discordant context -- and found that in this specific subgroup, comprising approximately 46% of clinically high-risk patients, chemotherapy could be safely omitted without compromising five-year outcomes, demonstrating that genomic risk assessment can override a purely clinical high-risk designation in a meaningful proportion of patients. MammaPrint is best suited to exactly this scenario -- ER-positive, HER2-negative early breast cancer where the clinical risk assessment and the desire for additional genomic context to guide the chemotherapy decision are both present -- and, like Oncotype DX, is validated across both pre- and postmenopausal patients with node-negative or limited node-positive disease.

Three additional, less universally used but independently validated assays round out the genomic testing landscape, each with a somewhat different intended clinical application. PAM50 (marketed as the Prosigna assay), a 50-gene panel, reports both an intrinsic molecular subtype classification (directly corresponding to the Luminal A/B, HER2-enriched, and basal-like categories detailed in Slide 42) and a separate Risk of Recurrence (ROR) score on a 0-to-100 scale, making it unique among these assays in explicitly linking the genomic risk score to the intrinsic subtype framework. EndoPredict, a 12-gene assay, produces a simpler binary low-versus-high risk classification and is specifically validated in postmenopausal, node-negative or node-positive, ER-positive/HER2-negative disease. The Breast Cancer Index (BCI) serves a distinct and complementary clinical purpose from the other assays on this slide: rather than primarily informing the initial chemotherapy decision, BCI specifically predicts the risk of LATE recurrence, occurring between years five and ten after diagnosis -- directly relevant to the decision, discussed in Section 13 (Slide 49), of whether to extend adjuvant endocrine therapy beyond the traditional five-year duration, since BCI can identify which patients carry a genuinely elevated late-recurrence risk that would justify the additional years of endocrine therapy (and its associated toxicity) versus those whose late-recurrence risk is low enough that extended therapy would offer minimal additional benefit. The slide closes with an important practical caveat: while all of these assays are independently validated for chemotherapy decision-making in ER-positive, HER2-negative early breast cancer, they are NOT interchangeable with one another (each was validated in a somewhat different patient population and produces a differently scaled result), and while NCCN expresses a preference for Oncotype DX given its most extensive prospective randomized validation (TAILORx and RxPONDER both used it specifically), the guideline explicitly accepts the validated alternatives discussed here for specific clinical scenarios where they may be more appropriate or more readily available.

SLIDE 42 -- Molecular Subtypes (Intrinsic/Sorlie-Perou Classification)

Slide content recap:

- Luminal A: ER+/PR+, HER2-, low Ki-67, Grade 1-2; 40-50%; highly endocrine responsive; endocrine alone (+/- CDK4/6)
- Luminal B: ER+/PR+/-, HER2- high Ki-67 OR HER2+; 20-30%; more aggressive; endocrine + chemo +/- CDK4/6
- HER2-enriched: HER2+ (ER-, PR-); 15-20%; HER2 amplification dominant; anti-HER2 + chemo +/- endocrine
- Triple Negative (TNBC): ER-, PR-, HER2-; 10-15%; basal-like (mostly); BRCA1 assoc.; chemo +/- pembrolizumab +/- PARP inhibitor
- Normal breast-like: low expression all markers; rare; usually excluded from analysis
- BL1 (Basal-like 1): highest pCR; cell cycle genes; responds to platinum
- BL2 (Basal-like 2): growth factor signaling; lower pCR
- M (Mesenchymal): EMT-related; low pCR; responds to PI3K pathway inhibitors
- IM (Immunomodulatory): high TILs; responds to immunotherapy (pembrolizumab)
- LAR (Luminal AR): AR-positive; responds to anti-androgen therapy (enzalutamide)
- HER2-low definition: IHC 1+ or IHC 2+/ISH negative (previously called "HER2 negative")
- HER2-low frequency: ~55-60% of all HER2-negative BC are actually HER2-low
- T-DXd proven efficacy in HER2-low (DESTINY-Breast04)
- DESTINY-Breast04: HR 0.49 for PFS; HR 0.64 for OS in HR+ HER2-low MBC
- IMPORTANT: HER2-low is not a biologic subtype -- it is a treatment eligibility criterion

Detailed Explanation and Source Context

The intrinsic molecular subtype classification, originally derived from gene-expression microarray studies by Sorlie, Perou, and colleagues, groups breast cancer into biologically and clinically distinct categories that correlate closely with, but are not identical to, the routine immunohistochemical ER/PR/HER2 and Ki-67 panel -- this classification is the single most important organizing framework for systemic therapy selection in this entire document, and every treatment algorithm in Sections 13 through 15 is fundamentally structured around it. Luminal A, the most common subtype at 40 to 50% of cases, is defined by ER and PR positivity, HER2 negativity, low Ki-67 proliferation, and grade 1 to 2 histology; it is the most highly endocrine-responsive subtype and, reflecting its favorable underlying biology, is treated with endocrine therapy alone (with the addition of a CDK4/6 inhibitor in higher-risk cases) far more often than with chemotherapy -- this is the subtype for which the multigene assays of Slide 41 are most frequently used to confirm that chemotherapy can indeed be safely omitted. Luminal B (20 to 30% of cases) shares ER positivity with Luminal A but is distinguished by either a high Ki-67 proliferation index or HER2 co-positivity, and behaves more aggressively, generally requiring the addition of chemotherapy to endocrine therapy (again with CDK4/6 inhibition considered in higher-risk cases) -- the Luminal A versus Luminal B distinction is, in essence, a proliferation-based split within the broader ER-positive population, and is the molecular-biology explanation for why not all ER-positive tumors are treated identically. HER2-enriched tumors (15 to 20%) are characterized by HER2 gene amplification as the dominant molecular driver, with ER and PR typically negative, and are treated with anti-HER2 therapy combined with chemotherapy, with endocrine therapy added only in the minority of cases where hormone receptors are also expressed. Triple-negative breast cancer (10 to 15%) lacks ER, PR, and HER2 expression entirely, is predominantly (though not exclusively) of basal-like gene-expression profile, shows a strong association with BRCA1 mutation (Section 04, Slide 15), and lacking any of the three standard receptor targets, is treated with chemotherapy, with pembrolizumab immunotherapy and PARP inhibitors added in specific eligible subpopulations as detailed in Section 13 (Slide 52). A rare "normal breast-like" category, showing generally low expression across all markers, is now generally understood to reflect either specimen contamination with adjacent normal tissue or a technical artifact of the profiling method, and is typically excluded from formal subtype analysis rather than treated as a genuine distinct clinical entity.

Because triple-negative breast cancer, despite being unified by the absence of ER, PR, and HER2, is itself a biologically heterogeneous category, Lehmann and colleagues in 2011 proposed a further subclassification into distinct TNBC molecular subgroups, each with a characteristic gene-expression signature and a distinct pattern of therapeutic responsiveness. Basal-like 1 (BL1) is enriched for cell-cycle-related genes and shows the highest pathologic complete response rate among the TNBC subgroups, with particular responsiveness to platinum-based chemotherapy, consistent with the DNA-repair-pathway vulnerabilities characteristic of highly proliferative, genomically unstable basal-like tumors. Basal-like 2 (BL2) is characterized by active growth factor signaling pathways and shows a comparatively lower pCR rate than BL1. The Mesenchymal (M) subgroup shows a gene-expression signature dominated by epithelial-mesenchymal transition pathways (the same EMT process introduced mechanistically in Section 03's metastatic cascade discussion, Slide 12) and a correspondingly low pCR rate to standard chemotherapy, with emerging evidence suggesting potential responsiveness to PI3K pathway inhibition instead. The Immunomodulatory (IM) subgroup is characterized by high tumor-infiltrating lymphocyte density (the same TILs biomarker discussed prognostically in Slide 40) and shows particular responsiveness to immunotherapy, providing the biological rationale for pembrolizumab's efficacy specifically within this TNBC subpopulation. The Luminal Androgen Receptor (LAR) subgroup, despite being classified under the broader triple-negative umbrella by ER/PR/HER2 immunohistochemistry, is characterized by androgen receptor pathway activation and shows a distinct responsiveness to anti-androgen therapy such as enzalutamide, illustrating that "triple-negative" by the standard three-marker panel does not mean the tumor lacks ANY actionable receptor-driven biology -- it simply lacks the three specific receptors the panel tests for.

The slide closes with what may be the single most practice-changing molecular reclassification of the past several years: the recognition of "HER2-low" as a distinct, clinically actionable treatment-eligibility category, defined as an immunohistochemistry score of 1+, or a score of 2+ with a negative in situ hybridization (ISH/FISH) result -- both of these results were previously grouped together, without further distinction, simply as "HER2-negative," since neither met the threshold for benefit from trastuzumab or pertuzumab. It is now recognized that HER2-low disease is remarkably common, constituting an estimated 55 to 60% of all tumors that would previously have been categorized simply as HER2-negative -- meaning that a majority of what oncologists have historically called "HER2-negative" breast cancer in fact expresses a low but biologically and therapeutically meaningful level of HER2 protein. The DESTINY-Breast04 trial demonstrated that the antibody-drug conjugate trastuzumab deruxtecan (T-DXd) -- which delivers a cytotoxic payload via a HER2-targeting antibody, exploiting even this low level of HER2 expression to achieve targeted drug delivery -- produces substantial benefit in this HER2-low population, with a hazard ratio of 0.49 for progression-free survival and 0.64 for overall survival specifically in hormone-receptor-positive, HER2-low metastatic disease, a result significant enough that it fundamentally redrew the boundary of what "HER2-negative" means for treatment-eligibility purposes. It is important to hold two ideas simultaneously and not conflate them, exactly as this slide's final bullet insists: HER2-low is NOT a new intrinsic biological subtype standing alongside Luminal A/B, HER2-enriched, and TNBC in the Sorlie-Perou framework -- a HER2-low tumor can be, and usually is, otherwise classified as either luminal (if hormone-receptor-positive) or triple-negative (if hormone-receptor-negative) by the standard framework -- rather, "HER2-low" is purely a TREATMENT ELIGIBILITY CRITERION, identifying which patients, regardless of their underlying intrinsic subtype, can benefit from a HER2-targeted antibody-drug conjugate even though they do not have true HER2 gene amplification. This distinction, and the full metastatic-setting treatment algorithm built around it, is developed further in the ESMO 2026 update of Section 15 (Slide 60).

SECTION 12 -- Local-Regional Therapy

Breast-conserving surgery versus mastectomy, axillary surgical management and the Z0011 paradigm shift, radiation therapy indications and toxicity, and mastectomy technique and reconstruction options -- the direct surgical descendants of the Section 01 historical narrative and the Section 02 anatomy.

SLIDE 43 -- Section Divider -- Local-Regional Therapy

Detailed Explanation and Source Context

This section closes the loop opened in Section 01: the Halsted-to-Fisher historical arc and the Z0011/SLNB concept previewed there (Slide 4) and grounded anatomically in Section 02 (Slide 8) are given their full, current-practice treatment here as the local-regional therapy backbone that every stage-specific algorithm in Section 14 builds upon.

SLIDE 44 -- Breast-Conserving Surgery (BCS) versus Mastectomy

Slide content recap:

- BCS EQUIVALENT to mastectomy in OS (6 RCTs: NSABP B-06, Milan I, others)
- IDEAL candidates: single tumor with adequate breast-to-tumor ratio
- Must achieve negative margins (no ink on invasive tumor for IDC; ≥ 2 mm for DCIS)
- Post-BCS radiation: WBRT standard; partial breast RT in selected low-risk patients
- BCS PREFERRED when possible for quality of life; equivalent survival
- Oncoplastic techniques expand BCS eligibility (therapeutic mammoplasty)
- Total (simple) mastectomy: breast only; no axillary dissection; used for prophylaxis
- MRM (Modified Radical Mastectomy): breast + levels I-III axilla; preserve pec major
- Skin-sparing mastectomy: remove nipple-areolar complex; preserve breast skin envelope
- Nipple-sparing mastectomy (NSM): preserve NAC; oncologically acceptable if retroareolar margin negative
- Halsted radical (historical): also removes pec major/minor; rarely indicated today
- Absolute/relative contraindications to BCS: prior breast irradiation to the same breast
- Diffuse suspicious microcalcifications (multicentric extent)
- Two separate primary tumors in different quadrants (multicentric)
- Persistently positive margins despite re-excision
- Active systemic lupus or scleroderma (relative)
- Pregnancy (relative -- WBRT after delivery acceptable in 3rd trimester)
- Large tumor:breast ratio (cannot achieve negative margins with acceptable cosmesis)
- Strong patient preference (informed choice -- equivalent survival)
- Genetic predisposition (BRCA carrier): prophylactic contralateral mastectomy option
- Inflammatory BC: mastectomy AFTER neoadjuvant chemotherapy
- Multifocal/multicentric disease in different quadrants
- Failed prior BCS (persistent positive margins or IBTR)

Detailed Explanation and Source Context

The equivalence of breast-conserving surgery to mastectomy for overall survival is not a single-trial finding but a consistently replicated result across six independent randomized controlled trials, with NSABP B-06 (Fisher and colleagues) and Milan I (Veronesi and colleagues) being the two most historically prominent, both already introduced in Section 01's history (Slide 4) and Section 03's Halsted-Fisher discussion (Slide 12) -- the consistency of this finding across multiple independent trial populations is precisely what elevated breast-conserving surgery from an interesting hypothesis to the guideline-endorsed, preferred standard of care it is today whenever it is technically and oncologically feasible.

Ideal candidacy for breast-conserving surgery rests on achieving an adequate breast-to-tumor size ratio -- meaning the tumor can be excised with a cosmetically acceptable result while leaving enough surrounding breast tissue -- and, most critically, on the ability to achieve negative surgical margins: "no ink on tumor" for invasive ductal carcinoma (meaning tumor cells do not extend to the inked surgical resection surface, however narrow that margin may be), contrasted explicitly here with the wider, quantitative 2-millimeter margin required for DCIS, already established in Section 08 (Slide 32) and reflecting DCIS's characteristic underestimation of true extent on imaging. Whole-breast radiation therapy after breast-conserving surgery is the guideline-standard adjunct (detailed fully in Slide 46), with partial breast irradiation available as an alternative in carefully selected low-risk patients. Breast-conserving surgery is explicitly designated as the PREFERRED approach whenever it is oncologically appropriate, reflecting its equivalent survival combined with a meaningfully better quality-of-life and body-image outcome relative to mastectomy -- and oncoplastic surgical techniques (discussed further in Slide 47) have progressively expanded the population for whom breast-conserving surgery remains cosmetically feasible, by using volume-displacement and volume-replacement principles to permit larger-volume resections without compromising the final cosmetic outcome.

Mastectomy technique spans a spectrum of increasing tissue preservation that directly recapitulates the historical progression traced in Section 01: total (simple) mastectomy removes breast tissue alone, without any axillary dissection, and is used primarily for prophylactic indications (Section 05, Slide 19) or for palliation in Stage IV disease, where sentinel node biopsy can be added separately if axillary staging is still needed; modified radical mastectomy (MRM) removes the breast together with axillary contents at Berg levels I through III while preserving the pectoralis major muscle -- this is the direct lineal descendant of the Patey modification described in Section 01 (Slide 4) and remains the most commonly performed mastectomy for invasive breast cancer requiring mastectomy today; skin-sparing mastectomy removes the nipple-areolar complex and the underlying breast tissue and axillary contents as needed, while deliberately preserving the overlying breast skin envelope specifically to facilitate immediate reconstruction with excellent cosmetic results, and has been shown to achieve equivalent oncologic outcomes to standard MRM; nipple-sparing mastectomy (NSM) goes a step further, preserving the nipple-areolar complex itself in addition to the skin envelope, and is considered oncologically acceptable specifically when the retroareolar tissue margin (immediately beneath the nipple) is confirmed negative for tumor, since residual nipple-areolar tissue cannot be examined for margin status any more thoroughly than that; and the historical Halsted radical mastectomy, which additionally removed both the pectoralis major and pectoralis minor muscles, is now rarely indicated, reserved essentially only for the uncommon scenario of direct tumor invasion into the pectoralis muscle itself, since no survival benefit over MRM has ever been demonstrated to justify its added morbidity -- a direct echo of the NSABP B-06 and Milan I findings that motivated the entire shift toward less extensive surgery documented throughout Section 01.

The specific list of situations in which mastectomy is preferred over breast-conserving surgery functions as a practical checklist worth committing to memory in its entirety, since it represents the accumulated exceptions to the "BCS preferred when feasible" default rule. Prior breast irradiation to the same breast is an important contraindication because the skin and soft tissue's radiation tolerance has already been exhausted, making a second course of whole-breast radiation (which BCS would require) unsafe. Diffuse, multicentric suspicious microcalcifications, or two genuinely separate primary tumors located in different quadrants of the breast (true multicentricity, as opposed to a single larger or multifocal-but-localized process), generally cannot be adequately addressed by a single, cosmetically reasonable breast-conserving excision. Persistently positive margins despite one or more attempts at re-excision indicate that breast-conserving surgery has already been attempted and failed to achieve oncologic adequacy, at which point proceeding to mastectomy becomes the appropriate next step. Active systemic lupus erythematosus or scleroderma are relative (not universally absolute) contraindications, reflecting these connective tissue diseases' association with poor tolerance of radiation

therapy and increased risk of severe radiation-induced fibrosis or tissue injury. Pregnancy is a relative contraindication specifically because radiation therapy must be avoided during pregnancy (Section 17, Slide 65), though breast-conserving surgery performed during pregnancy with radiation deferred until after delivery (acceptable when diagnosis occurs in the third trimester, allowing a relatively short deferral interval) remains an option in appropriately selected cases. A tumor that is simply too large relative to the breast to permit clear margins with an acceptable cosmetic outcome is a straightforward anatomic contraindication, though neoadjuvant systemic therapy (Section 13, Slide 52) can sometimes downsize such a tumor sufficiently to convert a mastectomy-only candidate into a breast-conserving surgery candidate. Strong, well-informed patient preference for mastectomy is itself a legitimate and sufficient reason to proceed with mastectomy, given the demonstrated equivalence in survival -- this is a shared decision, not a purely algorithmic one. A confirmed BRCA (or other high-penetrance gene) mutation does not by itself mandate mastectomy for treatment of an existing cancer, but does make prophylactic contralateral mastectomy a reasonable option to discuss, given the substantially elevated risk of a second primary in this population (Section 05, Slide 19). Inflammatory breast cancer, as emphasized repeatedly throughout this document (Section 09, Slide 35; Section 14, Slide 56), requires mastectomy, but only AFTER neoadjuvant chemotherapy -- breast-conserving surgery is never appropriate for IBC given its diffuse dermal lymphatic involvement. True multifocal or multicentric disease spanning different quadrants, and a prior failed breast-conserving surgery attempt (whether from persistent positive margins or a subsequent ipsilateral breast tumor recurrence, discussed fully in Section 16, Slide 63), round out the list as scenarios where mastectomy becomes the appropriate, rather than merely an alternative, surgical approach.

SLIDE 45 -- Axillary Surgery -- SLNB and the Z0011 Trial

Slide content recap:

- Technique: dual tracer (Tc-99m sulfur colloid + isosulfan blue dye) -- sensitivity 97%
- False-negative rate: ~5-8% with dual tracer; acceptable for clinical practice
- cN0 disease: SLNB standard; if sentinel node negative -> ALND not needed
- Frozen section: controversial (defer final decision to permanent section in most centers)
- Learning curve: 20+ cases required for competency; must be >=95% identification rate
- Contraindications to SLNB: cN2-3, prior axillary surgery, inflammatory BC
- Initially cN1+: clip suspicious node at diagnosis biopsy
- After neoadjuvant: if pathologic N0 on imaging -> SLNB + remove clipped node (TAD)
- TAD reduces false-negative rate from 14% (SLNB alone) to ~2% (TAD) in post-neoadjuvant setting
- SENTINA/ACOSOG Z1071: validated TAD approach for ycN0 after chemo
- If 3+ SLN removed and targeted node negative -> may avoid ALND
- ACOSOG Z0011: cT1-2 cN0; BCS + WBRT; 1-2 positive SLNs (macro)
- ALND vs no ALND: no difference in OS or DFS at 10 years
- Z0011 criteria: cT1-2, clinically N0, BCS, receiving WBRT, 1-2 positive SLNs
- ALND NOT needed if Z0011 criteria met -- even with macrometastasis in SLNs
- Key: cannot apply Z0011 to mastectomy, neoadjuvant, or >2 positive SLNs
- ALND indications remaining: cN2-3 disease (upfront surgery patients)
- Post-neoadjuvant: residual nodal disease (ycN+) confirmed by TAD
- Failed SLNB mapping (no sentinel node identified)
- Inflammatory BC (after neoadjuvant therapy)
- >2 positive SLNs in patients NOT meeting Z0011 criteria
- ALND extent: levels I and II standard; level III if gross disease or frozen positive at level II

Detailed Explanation and Source Context

The sentinel lymph node biopsy technique, introduced conceptually in Section 02 (Slide 8), uses a dual-tracer localization method -- technetium-99m-labeled sulfur colloid combined with isosulfan blue dye, injected around

the tumor or subareolarly and taken up by the same lymphatic channels that would carry tumor cells -- achieving a combined sentinel node identification sensitivity of approximately 97%, with an accepted false-negative rate of 5 to 8% (meaning that in this proportion of node-positive patients, the sentinel node itself would be falsely reported negative, a number already introduced in Section 02 and repeated here as the benchmark against which the post-neoadjuvant targeted axillary dissection technique, discussed below, is measured and substantially improved upon). For clinically node-negative (cN0) disease, sentinel node biopsy is the standard of care, and a negative sentinel node result means completion axillary lymph node dissection is not needed. The use of intraoperative frozen section analysis of the sentinel node to guide an immediate decision about proceeding to completion axillary dissection in the same operative setting remains controversial and is not universally practiced; many centers now defer this decision to permanent section pathology, returning for a second axillary procedure only in the minority of cases where completion dissection is ultimately indicated, reflecting the increasing recognition (culminating in Z0011, discussed below) that many sentinel-node-positive patients do not require completion dissection at all. Sentinel node biopsy has a genuine and well-documented learning curve, requiring more than 20 supervised cases for a surgeon to achieve competency, defined by an identification rate of 95% or greater -- below this competency threshold, the false-negative rate rises unacceptably, which is why sentinel node biopsy credentialing typically requires demonstrated proficiency before independent practice. Sentinel node biopsy is not appropriate in every clinical scenario: it is contraindicated (or at least not validated as reliable) in clinically node-positive disease at the N2-N3 level (where the extent of disease is already established and sentinel mapping would not change management), in patients with prior axillary surgery (which disrupts the normal lymphatic drainage pathways the technique depends on), and in inflammatory breast cancer, where the diffuse dermal lymphatic involvement characteristic of the disease similarly disrupts reliable lymphatic mapping.

Targeted axillary dissection (TAD) is the specific technique developed to solve a problem that emerged once neoadjuvant chemotherapy became standard for many node-positive patients (Section 13, Slide 52): a patient who initially presents with a clinically positive axillary node (biopsy-confirmed and marked with a radiographic clip, as discussed in Section 07, Slide 27) may convert to clinically node-negative after neoadjuvant chemotherapy, but a standard sentinel node biopsy performed after this conversion carries an unacceptably high false-negative rate of approximately 14%, because the originally-involved node is not necessarily the same node that maps as the new sentinel node after treatment-induced changes in lymphatic drainage. Targeted axillary dissection solves this by combining standard sentinel node biopsy with deliberate identification and removal of the SPECIFIC node that was clipped at initial diagnosis, reducing the false-negative rate from 14% down to approximately 2% -- a dramatic improvement that was validated by the SENTINA and ACOSOG Z1071 trials specifically for this post-neoadjuvant, clinically-node-negative-after-treatment (ycN0) scenario. An additional refinement identified in this same body of evidence is that removing three or more sentinel nodes (rather than the one or two nodes sometimes retrieved in standard SLNB), combined with confirming the specifically targeted (clipped) node is negative, further improves the reliability of this approach and may allow completion axillary dissection to be avoided even in this more complex post-neoadjuvant setting.

The ACOSOG Z0011 trial, already introduced historically in Section 01 (Slide 4) and anatomically in Section 02 (Slide 8), is given its complete, precise eligibility criteria here, and getting these criteria exactly right is essential, because Z0011's findings apply ONLY within this specific population and are frequently misapplied if the criteria are not respected. The trial enrolled patients with clinical T1-2 tumors, clinically node-negative (cN0) disease at presentation, who were undergoing breast-conserving surgery with planned whole-breast radiation, and who were found to have one or two positive sentinel lymph nodes containing macrometastatic disease. Within this precisely defined population, the trial found no difference in either overall survival or disease-free survival at ten years of follow-up between patients who underwent completion axillary lymph node dissection and those who did

not -- establishing that, in this specific population, axillary lymph node dissection can be safely omitted even in the presence of one or two positive sentinel nodes, since the planned whole-breast radiation (which incidentally treats a portion of the low axilla) combined with adjuvant systemic therapy appears to adequately control any residual disease left behind in the unresected axillary nodes. The critical caveat, restated explicitly and emphatically on this slide precisely because it is so frequently tested and so consequential if misapplied, is that Z0011's findings CANNOT be extrapolated to patients undergoing mastectomy (who do not receive the incidental low-axillary radiation dose that whole-breast radiation provides), to patients who received neoadjuvant therapy (a fundamentally different biological and staging scenario, addressed instead by the targeted axillary dissection approach above), or to patients with more than two positive sentinel nodes -- in any of these three scenarios, Z0011 simply does not apply, and standard axillary staging principles (including, where indicated, completion axillary dissection) should instead be followed.

The remaining, current indications for completion axillary lymph node dissection are therefore precisely the scenarios that fall outside both the sentinel-node-negative default and the Z0011 exception: clinically N2 or N3 disease at initial presentation in patients proceeding directly to upfront surgery (where the extent of nodal disease is already established as substantial); residual nodal disease confirmed by targeted axillary dissection after neoadjuvant therapy (ycN-positive); failed sentinel node mapping, where no sentinel node can be reliably identified intraoperatively despite adequate technique; inflammatory breast cancer, even after neoadjuvant therapy, given the previously discussed unreliability of lymphatic mapping in this disease; and more than two positive sentinel nodes found in a patient who does not otherwise meet the full Z0011 eligibility criteria (for instance, a patient undergoing mastectomy with three positive sentinel nodes would require completion dissection, since neither the whole-breast-radiation-dependent Z0011 exception nor the two-node ceiling would be satisfied). When axillary lymph node dissection is performed, its standard extent is Berg levels I and II (recall Section 02, Slide 8), with extension to level III reserved for cases where grossly evident disease is found at that level intraoperatively, or where frozen section analysis reveals positive disease at level II that raises concern for skip involvement or extension to level III.

SLIDE 46 -- Radiation Therapy in Breast Cancer

Slide content recap:

- WBRT standard after BCS: reduces IBTR by ~50-70% (EBCTCG meta-analysis)
- Dose: 40-50 Gy in 15-25 fractions; boost 10-16 Gy to tumor bed (high-risk)
- Hypofractionation (40 Gy in 15 fractions): equivalent to conventional; preferred
- RT omission: age ≥ 65 -70, T1, ER+/HER2-, negative nodes, on endocrine therapy (PRIME II)
- DCIS after BCS: RT standard; margin ≥ 2 mm critical
- Partial Breast Irradiation (APBI): selected low-risk patients; ASTRO criteria
- Regional Nodal Irradiation (RNI): include IMN and supraclavicular nodes when regional disease risk is high
- MA.20 trial: WBRT + RNI vs WBRT alone: increased DFS but not OS; increased toxicity
- DBCG-IMN: IMN RT improved OS in right-sided tumors with positive nodes
- RNI indications: ≥ 4 positive nodes; 1-3 nodes + medial/central tumor; high-risk N1
- PMRT indicated for: T3-4 tumors; ≥ 4 positive nodes; 1-3 positive nodes in high-risk (NCCN)
- 1-3 positive nodes: PMRT reduced locoregional recurrence (EBCTCG 2014 meta-analysis)
- PMRT dose: 46-50 Gy to chest wall +/- regional nodes
- PMRT target volumes: chest wall + IMN + supraclavicular nodes (varies by center)
- Reconstructed patients: implant-based reconstruction + PMRT -> higher complication rate
- PMRT timing: after completion of chemotherapy; before hormonal therapy in some protocols
- RT toxicity skin: early acute radiodermatitis; late fibrosis, telangiectasia
- Lymphedema: WBRT + ALND: 20-30%; WBRT + SLNB: 5-7%
- Cardiac: left-sided RT historical -> coronary artery disease, pericarditis (now reduced)
- Pulmonary: radiation pneumonitis (1-3%); late fibrosis

- Brachial plexus neuropathy: <1% with modern techniques
- Secondary malignancy: angiosarcoma (rare but serious; 6-10 yrs post-RT)

Detailed Explanation and Source Context

Whole-breast radiation therapy after breast-conserving surgery reduces ipsilateral breast tumor recurrence by 50 to 70%, a figure derived from the Early Breast Cancer Trialists' Collaborative Group (EBCTCG) meta-analyses that pool data across dozens of randomized trials and represent the single most authoritative evidence base in breast radiation oncology. Standard dosing delivers 40 to 50 Gy across 15 to 25 fractions, with an additional boost of 10 to 16 Gy directed specifically to the tumor bed in patients at higher risk of local recurrence (younger age, close margins, higher grade). Hypofractionated regimens -- delivering a similar biologically effective dose in fewer, larger fractions, such as 40 Gy in 15 fractions rather than the more protracted conventional schedule -- have been shown to be equivalent in efficacy and toxicity to conventional fractionation and are now generally the preferred approach, offering the practical advantage of a shorter overall treatment course. Radiation can be safely omitted in a specific, well-defined low-risk population, per the PRIME II trial: patients aged 65 to 70 or older, with T1 tumors, estrogen-receptor-positive and HER2-negative disease, node-negative status, who will receive adjuvant endocrine therapy -- in this specific combination of favorable factors, the absolute benefit of radiation is small enough that its omission is a reasonable, guideline-supported option, sparing this population the time, cost, and toxicity burden of a full course of radiation. For DCIS treated with breast-conserving surgery, radiation remains the standard adjunct (as already established in Section 08, Slide 32), with the 2-millimeter margin standard being a critical prerequisite for radiation's benefit to be fully realized.

Partial breast irradiation (also called accelerated partial breast irradiation, APBI) delivers radiation to only the tumor bed and its immediate surrounding tissue, rather than the entire breast, over a substantially shorter treatment course; it is reserved for carefully selected low-risk patients meeting specific criteria established by professional societies such as ASTRO, reflecting an accumulating evidence base that a subset of low-risk patients can achieve equivalent local control with this more limited, shorter treatment approach.

Regional nodal irradiation (RNI) extends the radiation field beyond the breast or chest wall itself to include the regional nodal basins -- specifically the internal mammary chain and the supraclavicular nodes -- when the risk of occult regional nodal disease is judged sufficiently high to warrant treatment. The MA.20 trial, a landmark study in this area, found that adding regional nodal irradiation to whole-breast radiation improved disease-free survival but did NOT improve overall survival, while increasing treatment-related toxicity -- a nuanced result that has shaped a more selective approach to RNI, reserving it for patients where the disease-free-survival benefit is judged to outweigh the added toxicity, rather than applying it universally. The DBCG-IMN trial, a Danish cohort study, found that specifically internal mammary node irradiation improved overall survival in patients with right-sided tumors and positive axillary nodes -- the right-sided specification in this trial's design and results reflects the historical concern about excess cardiac radiation dose with left-sided internal mammary irradiation, an asymmetry in the evidence base that persists in how RNI is applied in practice. Standard indications for adding regional nodal irradiation are four or more positive axillary nodes, one to three positive nodes when the tumor is medially or centrally located (reflecting the higher probability of internal mammary drainage from these locations, as established anatomically in Section 02, Slide 8), or other high-risk features in the setting of N1 disease.

Post-mastectomy radiation therapy (PMRT) is indicated for T3 or T4 tumors, four or more positive axillary nodes, and, per current NCCN guidance, selected patients with one to three positive nodes who carry additional high-risk features -- this last, more nuanced indication reflects the EBCTCG 2014 meta-analysis finding that PMRT reduces locoregional recurrence even in this more limited one-to-three-node population, a finding that has progressively broadened PMRT's use beyond the more clearly established four-or-more-node indication.

Standard PMRT dosing is 46 to 50 Gy delivered to the chest wall, with or without the regional nodal basins, and target volumes typically include the chest wall together with the internal mammary and supraclavicular nodal regions, though the precise target volume design varies somewhat by treating center and by the specific pattern of nodal involvement. In patients undergoing implant-based breast reconstruction, the combination of PMRT with an implant is associated with a meaningfully higher rate of reconstructive complications (capsular contracture, implant malposition, and higher failure rates) than reconstruction without radiation exposure -- this interaction is an important factor in reconstructive planning and patient counseling, discussed further in Slide 47. PMRT timing is generally sequenced after the completion of chemotherapy, and, in some institutional protocols, before the initiation of hormonal (endocrine) therapy, though the precise sequencing relative to endocrine therapy varies somewhat across practice settings without a single universally mandated order.

Radiation-related toxicity spans a spectrum from common, generally self-limited acute effects to rare but serious late complications. Skin toxicity progresses from acute radiodermatitis (erythema and desquamation) during and immediately after treatment to late fibrosis and telangiectasia developing over subsequent months to years. Lymphedema risk is strongly modified by the extent of concurrent axillary surgery: the combination of whole-breast radiation with a full axillary lymph node dissection carries a 20 to 30% lymphedema risk, substantially higher than the 5 to 7% risk seen when whole-breast radiation is combined with sentinel node biopsy alone -- this striking difference is itself one of the strongest practical arguments for the axillary de-escalation achieved by Z0011 and sentinel node biopsy more broadly (Slide 45), since avoiding unnecessary completion axillary dissection meaningfully reduces this specific, often permanent and quality-of-life-limiting complication. Cardiac toxicity was a significant historical concern with left-sided breast radiation, associated with increased rates of coronary artery disease and pericarditis from incidental cardiac radiation exposure, though modern radiation planning techniques (including breath-hold techniques that physically displace the heart away from the treatment field during left-sided treatment) have substantially reduced this risk in contemporary practice. Pulmonary toxicity, in the form of radiation pneumonitis, occurs in 1 to 3% of patients acutely, with late pulmonary fibrosis possible in the treated lung volume. Brachial plexus neuropathy is now rare, occurring in less than 1% of patients treated with modern radiation techniques, reflecting improved dose planning and constraint-setting around this critical structure relative to older radiation techniques. Finally, secondary malignancy -- specifically, radiation-induced angiosarcoma, already discussed in Section 09 (Slide 35) -- is a rare but serious late complication, typically arising 6 to 10 years after radiation treatment, and any new, unexplained skin finding in a previously irradiated breast at this time interval after treatment should prompt specific clinical suspicion for this diagnosis.

SLIDE 47 -- Mastectomy Types and Reconstruction

Slide content recap:

- Total (Simple): breast tissue only; no ALND; prophylaxis, DCIS, stage IV palliation
- MRM: breast + ALND (levels I-III); preserve pec major; standard for invasive BC requiring mastectomy
- Skin-sparing: MRM + preserve breast skin (remove NAC); immediate reconstruction; excellent cosmesis
- Nipple-sparing (NSM): preserve NAC; remove all breast tissue; prophylaxis; small central-quadrant tumors; BRCA
- Halsted Radical (historical): breast + pec major + pec minor + ALND; pec muscle involvement by tumor (rare)
- Implant-based reconstruction: tissue expander -> definitive implant (2 stages); most common
- Direct-to-implant: single-stage; requires adequate soft tissue coverage
- Latissimus dorsi (LD) flap: pedicled; provides additional soft tissue for coverage
- TRAM/DIEP flap: autologous; gold standard for natural feel; higher morbidity
- Volume displacement: tissue rearrangement within the breast (therapeutic mammoplasty)
- Volume replacement: local/regional flap to fill defect (latissimus mini-flap)
- Oncoplastic BCS allows larger resections with acceptable cosmesis; reduces positive margin rates

- Requires oncoplastic training; contralateral symmetrization often needed

Detailed Explanation and Source Context

This slide consolidates the mastectomy technique spectrum already introduced in Slide 44, adding the specific indication-matching logic and the corresponding reconstruction options for each. Total (simple) mastectomy, removing breast tissue alone without axillary dissection, is matched to prophylactic risk-reducing surgery (Section 05, Slide 19), to pure DCIS management when mastectomy is chosen (Section 08, Slide 32), and to palliative surgery in Stage IV disease where local control of a symptomatic primary tumor is the goal rather than curative nodal staging. Modified radical mastectomy, combining breast removal with a full Berg level I-III axillary dissection while preserving the pectoralis major, remains the standard operation for invasive breast cancer in patients who are not candidates for (or who choose not to pursue) breast-conserving surgery, and, reflecting its practical predominance across the full range of invasive-disease mastectomy indications, is the most commonly performed mastectomy variant in general practice. Skin-sparing mastectomy, which removes the nipple-areolar complex and underlying glandular tissue while preserving the breast's overlying skin envelope, is specifically matched to patients planning immediate reconstruction, since the preserved skin envelope provides the reconstructive surgeon with a natural, well-vascularized "pocket" that yields excellent cosmetic results, and this technique achieves oncologic outcomes equivalent to standard MRM. Nipple-sparing mastectomy extends this same principle to preserve the nipple-areolar complex itself, matched specifically to prophylactic risk-reducing surgery and to small tumors located away from the central/subareolar region (since a central or retroareolar tumor would compromise the ability to achieve an oncologically clear retroareolar margin, as already discussed in Slide 44), and is an increasingly utilized option in appropriately selected BRCA carriers undergoing risk-reducing surgery. The historical Halsted radical mastectomy, additionally removing both pectoral muscles, is reserved for the rare scenario of direct tumor invasion into the pectoralis muscle itself, since -- as established repeatedly throughout Section 01 and this section -- no survival benefit over MRM has ever been demonstrated for this more extensive, higher-morbidity operation.

Reconstruction options are chosen based on the mastectomy defect, prior or planned radiation exposure, patient body habitus, and patient preference, spanning implant-based and autologous (the patient's own tissue) approaches. Implant-based reconstruction is most commonly performed as a two-stage procedure: a temporary tissue expander is placed at the time of mastectomy (or shortly after) and gradually inflated over subsequent weeks to stretch the overlying skin and soft tissue envelope, followed by a second surgical stage to exchange the expander for a definitive, permanent implant -- this staged approach remains the most frequently chosen reconstruction pathway overall, given its comparatively shorter individual operative times and the wide availability of the technique. Direct-to-implant reconstruction places the definitive implant in a single stage at the time of mastectomy itself, but requires adequate native soft tissue coverage over the implant to be safely performed, making careful patient selection (particularly regarding skin flap thickness and vascularity) essential. The latissimus dorsi flap is a pedicled (meaning its blood supply, via the thoracodorsal artery and vein already introduced anatomically in Section 02, Slide 7, remains attached and is rotated into position rather than being surgically detached and reattached) myocutaneous flap that provides additional vascularized soft tissue coverage, frequently used to supplement an implant-based reconstruction in patients with thin or radiated tissue where implant coverage alone would be inadequate -- and, as already flagged in Section 02 (Slide 7), preserving the thoracodorsal nerve and vessels during any prior axillary dissection is a specific prerequisite for this flap to remain a viable reconstructive option later. The TRAM (transverse rectus abdominis myocutaneous) flap and its more refined, muscle-sparing successor the DIEP (deep inferior epigastric perforator) flap are autologous reconstruction options that transfer the patient's own abdominal tissue (with or without the underlying rectus muscle, depending on the specific flap variant) to reconstruct the breast, generally regarded as the gold standard for achieving a natural look and feel that most closely approximates native breast tissue over the long

term, but at the cost of a more extensive, technically demanding operation with correspondingly higher perioperative morbidity and a donor-site (abdominal wall) recovery to manage in addition to the breast reconstruction itself.

Oncoplastic breast-conserving surgery techniques deserve specific mention as the mechanism by which the eligibility for breast-conserving surgery itself (Slide 44) has been meaningfully expanded in recent years, using two complementary strategic principles. Volume displacement techniques rearrange the patient's own remaining breast tissue within the breast itself (a strategy formally termed therapeutic mammoplasty) to fill the defect left by tumor excision, essentially borrowing reconstructive-surgery closure principles to permit larger-volume breast-conserving resections while still achieving an acceptable cosmetic result -- this approach often requires a matching symmetrization procedure on the contralateral breast to maintain overall symmetry. Volume replacement techniques instead bring in tissue from outside the immediate resection site, typically a local or regional flap such as a latissimus dorsi mini-flap, to fill a defect that is too large to be closed by simple tissue rearrangement alone. Together, these oncoplastic techniques allow surgeons to perform larger, more generous resections than would otherwise be attempted with a standard lumpectomy technique, which in turn has been shown to reduce the rate of positive margins requiring re-excision -- but they require dedicated oncoplastic surgical training beyond standard general or breast surgical oncology training, and require careful pre-operative planning, including the frequent need to plan a contralateral symmetrization procedure as part of the overall reconstructive strategy.

SECTION 13 -- Systemic Therapy (Adjuvant)

Adjuvant endocrine therapy and its extended-duration and CDK4/6-inhibitor variants, standard adjuvant chemotherapy regimens, the anti-HER2 targeted therapy armamentarium, and the principles governing neoadjuvant (preoperative) systemic therapy including the practice-changing addition of pembrolizumab to early triple-negative disease.

SLIDE 48 -- Section Divider -- Systemic Therapy (Adjuvant)

Detailed Explanation and Source Context

This section is where the biomarker and molecular-subtype framework of Section 11 becomes concrete treatment: every drug and regimen here is selected based on a specific ER/PR/HER2/genomic-assay result established earlier in this document, and the section is organized in the same conceptual order as a real treatment-planning conversation -- endocrine therapy for hormone-receptor-positive disease, chemotherapy regimen selection, anti-HER2 targeted therapy, and finally the neoadjuvant-versus-adjuvant sequencing decision that determines when these therapies are given relative to surgery.

SLIDE 49 -- Adjuvant Endocrine Therapy

Slide content recap:

- Tamoxifen 20mg (SERM): Pre/Postmenopausal ER+; 5-10 yrs; ATLAS: 10 yrs -> OS benefit
- Anastrozole 1mg (NSAI): Postmenopausal ER+; 5 yrs (+up to 5 more); ATAC: superior DFS vs Tam
- Letrozole 2.5mg (NSAI): Postmenopausal ER+; 5 yrs (+up to 5 more); BIG 1-98: superior to Tam alone
- Exemestane 25mg (SAI): Postmenopausal ER+; 5 yrs; after 2-3 yrs Tam -> IES trial
- Tamoxifen + OFS: Premenopausal high-risk ER+; 5 yrs OFS; SOFT/TEXT: OFS+Tam improves DFS vs Tam alone
- Exemestane + OFS: Premenopausal high-risk ER+; 5 yrs; SOFT/TEXT: OFS+Exe superior (younger/higher-risk)
- Abemaciclib + ET (CDK4/6i): HR+/HER2- high-risk early BC; 2 yrs abema + 5+ yrs ET; monarchE: iDFS benefit in N+ high Ki-67
- MA.17R: letrozole extended to 10 yrs: decreased contralateral BC; increased bone loss
- NCCN: consider 7.5-10 yrs AI in high-risk N+ postmenopausal patients
- Abemaciclib: N2+ or N1+ with G3 or Ki-67 $\geq 20\%$ + T ≥ 5 cm criteria
- 2 yrs abemaciclib -> iDFS HR 0.664; effect maintained beyond treatment

Detailed Explanation and Source Context

Adjuvant endocrine therapy is the systemic-therapy backbone for essentially all hormone-receptor-positive breast cancer, and the drug and duration selected depend on menopausal status, baseline recurrence risk, and, increasingly, whether CDK4/6 inhibition is added for the highest-risk patients. Tamoxifen, a selective estrogen receptor modulator (SERM) that competitively blocks estrogen binding at the receptor in breast tissue, is unique among the agents on this slide in being effective in BOTH pre- and postmenopausal patients, since its mechanism (direct receptor antagonism) does not depend on the postmenopausal hormonal milieu that the aromatase inhibitors require. The ATLAS trial established that extending tamoxifen duration from the traditional five years to ten years produces a genuine overall survival benefit, though at the cost of an extended exposure to tamoxifen's known risks (venous thromboembolism, endometrial cancer) -- this extended-duration option is revisited below alongside its aromatase-inhibitor counterpart.

The non-steroidal aromatase inhibitors, anastrozole and letrozole, work by an entirely different mechanism -- blocking the peripheral (extra-ovarian) conversion of androgens to estrogen via aromatase inhibition, a mechanism that only meaningfully suppresses circulating estrogen in a postmenopausal patient, since a

premenopausal patient's ovaries would otherwise continue producing estrogen independent of peripheral aromatization, actually causing a compensatory rise in gonadotropins and paradoxically increased ovarian estrogen output if given without concurrent ovarian suppression -- this is precisely why aromatase inhibitors are restricted to postmenopausal use unless combined with ovarian function suppression, discussed below. The ATAC trial established anastrozole's superiority over tamoxifen in disease-free survival for postmenopausal patients, and the BIG 1-98 trial established the same superiority for letrozole -- these two trials, together, are the evidentiary basis for aromatase inhibitors having generally supplanted tamoxifen as the preferred initial adjuvant endocrine agent in postmenopausal, hormone-receptor-positive disease. Exemestane, a steroidal aromatase inactivator (mechanistically distinct from the non-steroidal anastrozole and letrozole, though achieving a similar net estrogen-suppression effect), was specifically validated in the IES trial in a sequential regimen -- switching to exemestane after an initial two to three years of tamoxifen -- demonstrating that this sequential switching strategy improves outcomes relative to five years of tamoxifen alone.

For premenopausal patients with higher-risk disease, the SOFT and TEXT trials (already introduced in Section 01's historical timeline, Slide 4) established that adding ovarian function suppression (OFS, typically via a GnRH agonist) to endocrine therapy improves disease-free survival relative to tamoxifen alone -- and, further, that within the higher-risk, generally younger subgroup, ovarian suppression combined with exemestane achieves superior outcomes to ovarian suppression combined with tamoxifen, establishing exemestane plus OFS as the preferred combination specifically for the higher-risk premenopausal population, while tamoxifen plus OFS (or tamoxifen alone, for lower-risk premenopausal patients) remains appropriate for less aggressive disease.

Two extended-duration and treatment-intensification strategies are detailed for the highest-risk patients. The MA.17R trial demonstrated that extending letrozole therapy out to ten years total duration (rather than stopping at five years) reduces the rate of contralateral breast cancer, though at the cost of accelerated bone density loss -- a tradeoff that NCCN addresses by recommending consideration of extended aromatase inhibitor therapy, out to 7.5 to 10 years total, specifically in postmenopausal patients with node-positive, higher-risk disease, where the added recurrence-reduction benefit is judged most likely to outweigh the bone-health cost. The monarchE trial established a genuinely new treatment paradigm: adding the CDK4/6 inhibitor abemaciclib to standard adjuvant endocrine therapy, for a defined two-year duration, in a specifically defined high-risk population -- patients with N2 (four or more positive nodes) disease, or N1 (one to three positive nodes) disease combined with either grade 3 histology, a Ki-67 proliferation index of 20% or greater, or a primary tumor size of 5 centimeters or greater -- produced a clinically meaningful improvement in invasive disease-free survival, with a hazard ratio of 0.664, and notably this benefit has been shown to persist and even grow after the two-year abemaciclib treatment course itself ends, a "carry-over" effect suggesting abemaciclib may be eradicating micrometastatic disease during the treatment window rather than merely suppressing it for as long as the drug is administered. This monarchE-defined high-risk population and its specific eligibility criteria (N2, or N1 with grade 3/high Ki-67/large tumor size) are the exact numeric thresholds used to identify candidates for adjuvant CDK4/6 inhibition in the stage-specific management algorithm of Section 14 (Slide 55).

SLIDE 50 -- Adjuvant Chemotherapy Regimens

Slide content recap:

- ddAC->ddT (dose-dense): ddAC x 4 -> ddPaclitaxel x 4 (q14d); all subtypes standard; CALGB 9741: decreased recurrence vs q21d; G-CSF required
- TC x 4-6 (taxane+cyclophosphamide): docetaxel+cyclophosphamide q21d x6; lower-risk ER+ HER2-; avoid anthracyclines; USO 9735: TC x6 > AC x4 (DFS)

- AC x 4 (anthracycline only): doxorubicin 60 + cyclophosphamide 600 q21d; lower-risk patients; older regimen, often replaced by TC
- Carboplatin addition (TNBC): carboplatin + standard chemo; TNBC especially BRCA1/2; BrighTNess: pCR 57% vs 31% without
- Capecitabine (residual TNBC): capecitabine 1000-1250 mg/m² PO bid x 6-8 cycles; CREATE-X: 74% -> 78% 5-yr DFS; OS benefit (HR 0.59)
- CMF (historical): cyclophosphamide + methotrexate + 5FU; rarely used now; Cooper regimen era (1970s)
- Toxicity monitoring: LVEF (anthracyclines), neuropathy (taxanes), neutropenia
- G-CSF required with dose-dense regimens
- Cardiomyopathy risk: lifetime doxorubicin dose <550 mg/m²

Detailed Explanation and Source Context

Dose-dense AC followed by dose-dense paclitaxel (ddAC to ddT) -- four cycles of doxorubicin plus cyclophosphamide followed by four cycles of paclitaxel, each given every 14 days rather than the traditional 21-day interval -- represents the current standard-intensity chemotherapy backbone applicable across all molecular subtypes when a full anthracycline-and-taxane regimen is indicated. The CALGB 9741 trial established that compressing the treatment interval from the traditional 21 days to 14 days (hence "dose-dense") reduces the rate of recurrence compared with the same total drug doses given on the traditional, more spread-out schedule -- the biological rationale is that shortening the interval between cycles allows less time for surviving tumor cells to repopulate between treatments, consistent with the Norton-Simon mathematical model of tumor growth kinetics. This compressed schedule is only tolerable, however, with routine use of prophylactic granulocyte colony-stimulating factor (G-CSF) support, since the shortened interval between cycles would otherwise not allow adequate neutrophil recovery, making G-CSF administration a mandatory, not optional, component of any dose-dense regimen.

For lower-risk patients, particularly those with hormone-receptor-positive, HER2-negative disease where the absolute benefit of chemotherapy is more modest, a non-anthracycline regimen combining a taxane with cyclophosphamide (TC), typically docetaxel plus cyclophosphamide given every 21 days for four to six cycles, is frequently preferred specifically to avoid anthracycline-associated cardiotoxicity (discussed below) in a population where the added efficacy of anthracycline exposure may not be worth this specific risk. The US Oncology 9735 trial established that six cycles of TC achieved superior disease-free survival compared with four cycles of the older anthracycline-only AC regimen (doxorubicin plus cyclophosphamide), a finding that has progressively shifted lower-risk patients away from anthracycline-containing regimens toward TC specifically, with AC alone now generally considered a legacy regimen retained mainly in settings where TC is not appropriate or available.

For triple-negative breast cancer, particularly in patients with BRCA1/2 mutations (reflecting the platinum-sensitivity mechanism already discussed in Section 05, Slide 19), the addition of carboplatin to a standard anthracycline-taxane backbone increases pathologic complete response rates substantially -- the BrighTNess trial demonstrated a pCR rate of 57% with carboplatin added, compared with 31% without it, establishing platinum addition as now a standard consideration in neoadjuvant TNBC regimens, particularly (though not exclusively) in BRCA-mutated disease. For patients found to have residual invasive disease in the breast or nodes after neoadjuvant chemotherapy specifically in the triple-negative subtype, the CREATE-X trial (already introduced in Section 05, Slide 19, and Section 11, Slide 40) established that adding six to eight cycles of adjuvant capecitabine, an oral fluoropyrimidine, improved five-year disease-free survival from 74% to 78% and produced a statistically significant overall survival benefit (hazard ratio 0.59) -- this is the direct evidentiary basis for the "escalate therapy for residual disease" principle already introduced conceptually in Section 11 (Slide 40) and applied throughout the stage-specific algorithms of Section 14.

CMF (cyclophosphamide, methotrexate, and 5-fluorouracil), the so-called "Cooper regimen" dating from the 1970s, is included here principally for historical context as one of the earliest validated adjuvant chemotherapy regimens in breast cancer, and is now rarely used in contemporary practice, having been supplanted by the more effective anthracycline- and taxane-containing regimens described above.

Toxicity monitoring across these regimens follows a predictable, drug-class-specific pattern that should guide surveillance during treatment: anthracyclines (doxorubicin) require monitoring of left ventricular ejection fraction, given their well-established dose-dependent cardiotoxicity risk, with the cumulative lifetime doxorubicin dose generally kept below 550 mg/m² specifically to minimize the risk of doxorubicin-induced cardiomyopathy, a risk that rises sharply above this cumulative threshold; taxanes (paclitaxel, docetaxel) require monitoring for peripheral neuropathy, a common, sometimes dose-limiting and occasionally persistent toxicity of this drug class; and neutropenia requires routine monitoring across essentially all cytotoxic regimens, with G-CSF support specifically mandated for dose-dense schedules as already discussed, and used more selectively (based on individual neutropenia risk assessment) with standard-interval regimens.

SLIDE 51 -- HER2-Targeted Therapy

Slide content recap:

- Trastuzumab (Herceptin): anti-HER2 mAb; ADCC + HER2 signaling block; HER2+ adjuvant (12 months) + metastatic; HERA: 50% decreased recurrence
- Pertuzumab (Perjeta): anti-HER2 domain 2 mAb; blocks HER2-HER3 dimerization; HER2+ stage III or high-risk II; APHINITY: increased iDFS in N+
- T-DM1/ado-trastuzumab: ADC (trastuzumab+DM1); residual HER2+ after neoadjuvant; KATHERINE: iDFS HR 0.50 vs trastuzumab alone; standard of care
- T-DXd (trastuzumab deruxtecan): ADC (trastuzumab+deruxtecan, topo I inhibitor); 2nd line+ HER2+ MBC; HER2-low MBC; DESTINY trials; ILD risk
- Lapatinib (Tykerb): dual EGFR/HER2 TKI (reversible); HER2+ MBC (now rarely first-line)
- Neratinib (Nerlynx): irreversible pan-HER TKI; extended adjuvant after 1 yr trastuzumab; ExteNET: increased iDFS especially HR+ HER2+
- Tucatinib (Tukysa): HER2-selective TKI; HER2+ MBC (+trastuzumab+capecitabine); HER2CLIMB: OS benefit incl. brain mets
- Margetuximab: Fc-engineered anti-HER2 mAb; HER2+ MBC (3rd line+); SOPHIA: modest PFS benefit
- [UPDATE] ESMO 2026: T-DXd preferred 2nd-line HER2+ MBC over other anti-HER2 options
- [UPDATE] Tucatinib+Tras+Cape: preferred for brain metastases

Detailed Explanation and Source Context

This slide catalogues the anti-HER2 targeted-therapy armamentarium, and understanding it requires distinguishing three distinct mechanistic classes: monoclonal antibodies that bind the HER2 receptor's extracellular domain, antibody-drug conjugates that use a HER2-targeting antibody to deliver a cytotoxic payload, and small-molecule tyrosine kinase inhibitors that block HER2's intracellular signaling directly.

Trastuzumab, the original and still foundational anti-HER2 monoclonal antibody, binds the HER2 extracellular domain, producing both direct inhibition of HER2 downstream signaling and antibody-dependent cellular cytotoxicity (ADCC, recruiting the immune system's natural killer cells to attack HER2-expressing tumor cells). As already established in Section 01 (Slide 4) and Section 11 (Slide 40), the HERA trial demonstrated an approximate 50% reduction in recurrence risk with one year of adjuvant trastuzumab, a result so decisive that it established HER2 testing and trastuzumab-based treatment as mandatory standard practice for HER2-positive disease essentially overnight, and trastuzumab is used both in the adjuvant setting (standard duration: 12 months) and throughout the metastatic HER2-positive treatment algorithm (Section 14, Slide 57). Pertuzumab binds a different site on the HER2 extracellular domain (domain II) than trastuzumab, specifically blocking HER2's ability to dimerize with HER3 -- a distinct and complementary mechanism to trastuzumab's own action,

which is why the two antibodies are frequently used together ("dual HER2 blockade") rather than as alternatives to one another. The APHINITY trial demonstrated that adding pertuzumab to standard adjuvant trastuzumab and chemotherapy improves invasive disease-free survival specifically in node-positive HER2-positive disease (hazard ratio 0.76), though the overall benefit across the full trial population was more modest, which is why pertuzumab's adjuvant use is generally reserved for Stage III disease or higher-risk Stage II disease, both in the neoadjuvant and adjuvant settings, rather than applied universally to all HER2-positive early breast cancer.

T-DM1 (ado-trastuzumab emtansine), an antibody-drug conjugate linking trastuzumab to the cytotoxic microtubule-disrupting agent DM1 (a maytansinoid), delivers this cytotoxic payload specifically to HER2-expressing cells via trastuzumab-mediated internalization. As already established in Section 11 (Slide 40), the KATHERINE trial demonstrated that switching to adjuvant T-DM1, in place of continued trastuzumab, for patients found to have residual invasive HER2-positive disease after neoadjuvant therapy improves invasive disease-free survival with a hazard ratio of 0.50 relative to continued trastuzumab alone -- this dramatic benefit has established T-DM1 as the standard of care specifically for this residual-disease population, directly paralleling the analogous CREATE-X/capecitabine strategy for residual triple-negative disease (Slide 50). T-DXd (trastuzumab deruxtecan), a newer-generation antibody-drug conjugate linking trastuzumab to a topoisomerase I inhibitor payload (deruxtecan), has demonstrated substantial efficacy across the DESTINY-Breast trial series, both in later-line HER2-positive metastatic disease and, as established in Section 11 (Slide 42), in the newly recognized HER2-low population -- T-DXd carries a specific and clinically important toxicity requiring active monitoring, interstitial lung disease (ILD), occurring in an estimated 2 to 13% of treated patients and requiring vigilant surveillance (typically with periodic CT imaging) and prompt dose modification or discontinuation if ILD develops, given its potential severity.

Among the small-molecule tyrosine kinase inhibitors, lapatinib, a reversible dual inhibitor of both EGFR and HER2, was an earlier-generation agent for HER2-positive metastatic disease that is now used only rarely as first-line therapy, having been largely supplanted by the newer, more effective and better-tolerated agents on this slide. Neratinib, an irreversible pan-HER family inhibitor, is used specifically as extended adjuvant therapy following completion of a standard one-year trastuzumab course; the ExteNET trial demonstrated an improvement in invasive disease-free survival with this extended-adjuvant approach, with the benefit particularly pronounced in the hormone-receptor-positive, HER2-positive subgroup, though neratinib's use is tempered in practice by its associated high rate of diarrhea, a common and sometimes treatment-limiting toxicity. Tucatinib, a HER2-selective tyrosine kinase inhibitor (notably more HER2-selective, and consequently better tolerated, than the older dual-target lapatinib), is used in combination with trastuzumab and capecitabine for HER2-positive metastatic disease; the HER2CLIMB trial demonstrated an overall survival benefit with this combination that notably extended to patients with brain metastases -- tucatinib's demonstrated central nervous system penetration and activity is the specific reason it is preferred, ahead of other anti-HER2 options, in patients with brain metastases, a subtype-selective CNS-metastasis risk pattern already established in Section 03 (Slide 12). Margetuximab, an Fc-engineered anti-HER2 monoclonal antibody (engineered specifically to enhance its ADCC activity relative to standard trastuzumab), showed a modest progression-free survival benefit in the SOPHIA trial in third-line-or-later HER2-positive metastatic disease, with a specific Fc-gamma receptor polymorphism identified as predicting greater benefit in some patients -- reflecting an early example of pharmacogenomic patient selection within the anti-HER2 treatment landscape.

[UPDATE] The ESMO 2026 guideline update, developed in full in Section 15 (Slide 60), establishes two specific sequencing preferences among this now-crowded anti-HER2 therapeutic landscape that represent genuine, current-cycle guidance beyond what the base DeVita and Schwartz chapters would specify: T-DXd is now preferred specifically as second-line therapy for HER2-positive metastatic breast cancer, ahead of other

anti-HER2 options at that treatment line, reflecting its superior efficacy demonstrated in head-to-head comparison; and the tucatinib-trastuzumab-capecitabine combination is preferred specifically in patients with brain metastases, given its demonstrated CNS activity and overall survival benefit in this specific, historically difficult-to-treat population.

SLIDE 52 -- Neoadjuvant (Preoperative) Systemic Therapy

Slide content recap:

- Standard for: IBC (inflammatory BC -- always neoadjuvant first)
- Stage III (LABC): downstage tumor + nodal disease
- Stage II HER2+ or TNBC: improve pCR rate; guide post-neoadjuvant therapy
- Goal BCS: large tumor -> relative contraindication -> neoadjuvant to achieve BCS eligibility
- Research/prognostic: pCR correlates with improved DFS/OS in TNBC and HER2+
- Response-adaptive treatment: residual disease -> escalate (CREATE-X, KATHERINE)
- [UPDATE] KEYNOTE-522: Stage II-III TNBC; pembrolizumab x 8 cycles neoadjuvant + 9 cycles adjuvant
- KEYNOTE-522 pCR: 64.8% (pembro) vs 51.2% (placebo); EFS HR 0.63
- FDA approved 2021: pembrolizumab + chemo -> surgery -> pembrolizumab for 9 cycles
- irAE monitoring: immune-related toxicities (endocrinopathies, pneumonitis)
- Note: PD-L1 testing NOT required for neoadjuvant eligibility in TNBC
- Olaparib adjuvant: for BRCA1/2 germline, HER2-neg, residual disease or high-risk
- TNBC neoadjuvant: ddAC + paclitaxel +/- pembrolizumab (KEYNOTE-522)
- TNBC BRCA1/2 neoadjuvant: add carboplatin; pCR 55-60%
- HER2+ neoadjuvant: docetaxel + carbo + trastuzumab + pertuzumab (TCbHP -- NEOSPHERE)
- ER+/HER2- neoadjuvant: chemo before surgery; endocrine for postmenopausal elderly if non-responders
- Residual TNBC after NAC: capecitabine x 6-8 cycles (CREATE-X)
- Residual HER2+ after NAC: T-DM1 x 14 cycles (KATHERINE)
- Monitoring: clinical tumor size (calipers/US) every 2 cycles; reassess at mid-point
- Imaging: MRI best for residual disease (85% correlation with pathology)
- Axillary: US +/- FNA for suspicious nodes; place clip for TAD
- Insufficient response: switch regimen vs proceed to surgery (>=4-6 cycles typical)
- pCR assessment at surgery: ypT0/Tis ypN0 = pCR (NSABP/CTRC-AACR definition)

Detailed Explanation and Source Context

Neoadjuvant (preoperative) systemic therapy has four distinct, non-mutually-exclusive rationales that together explain its expanding role in modern breast cancer management. First, and most absolutely, it is mandatory -- not merely preferred -- for inflammatory breast cancer, where, as established repeatedly throughout this document (Section 09, Slide 35; Section 12, Slide 44; Section 14, Slide 56), operating first before addressing the diffuse dermal lymphatic disease burden systemically would be an incorrect sequence of care. Second, for Stage III (locally advanced) disease, neoadjuvant therapy serves to downstage both the primary tumor and any bulky nodal disease, potentially converting an initially unresectable or marginally resectable presentation into one that can be more safely and completely resected. Third, for Stage II HER2-positive or triple-negative disease specifically, neoadjuvant therapy serves a response-adaptive purpose: achieving pathologic complete response (or, short of that, quantifying the residual cancer burden) directly informs the subsequent adjuvant treatment escalation decisions detailed throughout this slide, an option that is only available if systemic therapy is given before, rather than after, surgery. Fourth, when a tumor is too large relative to the breast to permit breast-conserving surgery with an acceptable cosmetic result (a relative contraindication to BCS, per Section 12 Slide 44), neoadjuvant therapy can shrink the tumor sufficiently to convert an initial mastectomy-only candidate into a breast-conserving surgery candidate, expanding surgical options without requiring any change to the eventual systemic treatment plan. Beyond these direct clinical indications, pathologic complete response itself carries research and prognostic value, as already established in Section 11 (Slide 40): pCR correlates with

improved disease-free and overall survival specifically in triple-negative and HER2-positive disease, and the resulting response-adaptive treatment paradigm -- escalating adjuvant therapy specifically for patients who do NOT achieve pCR, using the CREATE-X (capecitabine, TNBC) and KATHERINE (T-DM1, HER2-positive) strategies already introduced in Slide 50 and Slide 51 -- is only possible because neoadjuvant sequencing provides this direct pathologic readout of treatment response before the adjuvant phase of care begins.

[UPDATE] The KEYNOTE-522 trial represents one of the most consequential recent additions to neoadjuvant treatment, extending immunotherapy into early-stage (rather than only metastatic) triple-negative breast cancer for the first time. The trial design added pembrolizumab, an anti-PD-1 checkpoint inhibitor, to standard neoadjuvant chemotherapy for eight cycles, continuing pembrolizumab as monotherapy for an additional nine cycles after surgery, in patients with Stage II to III triple-negative breast cancer. The result was a substantial and clinically meaningful improvement in pathologic complete response, from 51.2% with chemotherapy alone to 64.8% with the addition of pembrolizumab, together with a statistically significant improvement in event-free survival (hazard ratio 0.63) -- this combination of results led to FDA approval in 2021 of the full KEYNOTE-522 regimen: pembrolizumab plus chemotherapy neoadjuvantly, followed by surgery, followed by nine additional cycles of adjuvant pembrolizumab monotherapy. Because pembrolizumab is an immune checkpoint inhibitor rather than a conventional cytotoxic agent, its use requires a distinct toxicity monitoring paradigm centered on immune-related adverse events (irAEs) -- endocrinopathies (thyroiditis, hypophysitis, adrenal insufficiency), pneumonitis, colitis, and hepatitis being among the most clinically significant categories requiring active surveillance throughout treatment. A specific and important point, worth committing to memory precisely because it differs from immunotherapy's use in some other tumor types, is that PD-L1 testing is explicitly NOT required to establish eligibility for neoadjuvant pembrolizumab in triple-negative breast cancer -- unlike some other checkpoint-inhibitor indications where a minimum PD-L1 expression level gates eligibility, the KEYNOTE-522 trial enrolled patients regardless of PD-L1 status, and the resulting approval likewise does not require PD-L1 testing as a prerequisite. Finally, for patients with confirmed germline BRCA1/2 mutations and HER2-negative disease who have either high-risk features or residual disease after neoadjuvant chemotherapy, adjuvant olaparib is added per the OlympiA trial design, extending the PARP-inhibitor synthetic-lethality principle already introduced in Section 05 (Slide 19) into this specific early-stage, post-neoadjuvant treatment scenario.

Neoadjuvant regimen selection follows the same subtype-driven logic established throughout Section 11 and this section: triple-negative disease is treated with dose-dense AC followed by paclitaxel, with pembrolizumab added per KEYNOTE-522 as just described; BRCA1/2-mutated triple-negative disease additionally receives carboplatin, achieving pathologic complete response rates of 55 to 60% reflecting the platinum-sensitivity mechanism discussed throughout Section 05; HER2-positive disease is treated with a docetaxel-carboplatin-trastuzumab-pertuzumab combination (commonly abbreviated TCbHP), validated by the NEOSPHERE trial as an effective dual-anti-HER2-blockade neoadjuvant backbone; and hormone-receptor-positive, HER2-negative disease is generally treated with standard chemotherapy before surgery, with neoadjuvant endocrine therapy reserved as an alternative specifically for postmenopausal, elderly, or otherwise chemotherapy-intolerant patients whose tumors are not expected to respond robustly to chemotherapy given their favorable underlying biology. As already established in Slides 50 and 51, residual triple-negative disease after neoadjuvant chemotherapy is treated with six to eight cycles of adjuvant capecitabine per CREATE-X, and residual HER2-positive disease is treated with fourteen cycles of adjuvant T-DM1 per KATHERINE.

Monitoring response during the neoadjuvant treatment course itself follows a structured, multi-modal approach: clinical assessment of tumor size, by caliper measurement or ultrasound, is performed approximately every two treatment cycles, with a more formal response reassessment typically conducted at the treatment midpoint to

confirm the tumor is responding adequately before committing to the full planned course. Among imaging modalities, MRI provides the best correlation with actual pathologic residual disease at the time of eventual surgery, achieving approximately 85% concordance, making it the preferred imaging modality specifically for assessing extent of residual disease before surgical planning (as already established in Section 07, Slide 27). Axillary assessment during and after neoadjuvant therapy uses ultrasound, with fine-needle aspiration of any persistently suspicious node, and, as detailed in Section 12 (Slide 45), any node found positive before neoadjuvant therapy should be marked with a radiographic clip to enable subsequent targeted axillary dissection. If a tumor shows an insufficient response after an adequate treatment trial (typically four to six cycles), the treating team must decide between switching to an alternative, non-cross-resistant chemotherapy regimen or proceeding directly to surgery rather than persisting with a regimen that is demonstrably not achieving an adequate response -- this decision requires individualized clinical judgment weighing the specific degree of response observed, the tumor's molecular subtype, and the patient's tolerance of treatment to date. Finally, pathologic complete response is formally assessed at the time of definitive surgery using the ypT0/Tis ypN0 criterion (the "yp" prefix denoting post-neoadjuvant-therapy pathologic staging, as already introduced in Section 11, Slide 40), which then directly determines whether the response-adaptive adjuvant escalation strategies discussed throughout this slide -- capecitabine for residual TNBC, T-DM1 for residual HER2-positive disease -- should be applied.

SECTION 14 -- Management by Stage

Stage-specific treatment algorithms from pure in situ disease through metastatic breast cancer, synthesizing the surgical (Section 12), systemic (Section 13), and biomarker (Section 11) building blocks established earlier into complete, actionable pathways.

SLIDE 53 -- Section Divider -- Management by Stage

Detailed Explanation and Source Context

This section is the synthesis point of the entire document: every building block established so far -- staging (Section 10), molecular subtype and genomic assays (Section 11), surgical technique (Section 12), and systemic therapy (Section 13) -- is now assembled into a single, stage-organized treatment algorithm proceeding from Stage 0 through Stage IV, culminating in Stage IV's handoff to the ESMO 2026 metastatic updates of Section 15.

SLIDE 54 -- Stage 0 -- In Situ Disease Management

Slide content recap:

- DCIS: BCS + WBRT standard (Cat 1); reduces IBTR ~50-70%
- DCIS: BCS alone (selected low-risk): VNPI 3-4; low-grade; <2.5 cm; margin ≥10 mm
- DCIS: mastectomy if multicentric; large size; cosmesis not achievable; patient preference
- DCIS margin: ≥2 mm required with WBRT; ≥3 mm for BCS alone (B-17 criteria)
- DCIS SLNB: not routine for pure DCIS; YES with mastectomy (TAD becomes difficult later)
- DCIS endocrine therapy: ER+ DCIS → tamoxifen 5 yrs (decreases recurrence + contralateral risk)
- Paget's pure (no underlying invasive): central excision + WBRT
- Paget's + underlying DCIS: same management as DCIS +/- Paget's-specific approach
- Paget's mastectomy: Paget's + multifocal DCIS or if BCS not achievable
- Paget's SLNB: as for DCIS/invasive component (perform with mastectomy)
- LCIS classic: observation + chemoprevention + annual mammography
- LCIS: no surgical re-excision required for classic LCIS at core biopsy margin
- LCIS chemoprevention: tamoxifen or raloxifene (NSABP P-2); reduces bilateral risk ~50%
- LCIS pleomorphic: excision to clear margins (similar to high-grade DCIS)
- LCIS bilateral risk-reducing mastectomy: option for highest-risk patients (strong FH + BRCA)
- LCIS annual MRI: consider for LCIS patients with additional risk factors
- Stage 0 endocrine therapy: tamoxifen 20mg/day x 5 yrs; AI (postmenopausal): anastrozole 1mg x 5 yrs (IBIS-II DCIS)
- No role for chemotherapy in pure DCIS/LCIS
- Follow-up: annual mammo; H and P every 6 months x 5 yrs then annually

Detailed Explanation and Source Context

Stage 0 management brings together, in algorithmic form, everything already established individually about DCIS (Section 08, Slide 32), LCIS (Section 08, Slide 31), and Paget's disease (Section 09, Slide 35), organizing each into a concrete treatment pathway. For DCIS, breast-conserving surgery combined with whole-breast radiation is the Category 1, guideline-preferred standard, reducing ipsilateral breast tumor recurrence by 50 to 70% as already quantified in Section 08; a carefully selected subset of low-risk patients -- specifically those with a Van Nuys Prognostic Index of 3 to 4, low-grade histology, a lesion under 2.5 cm, and a margin of 10 mm or more (a considerably wider margin standard than the 2 mm minimum used when radiation will be given) -- may be managed with breast-conserving surgery alone, omitting radiation, per the criteria established in the NSABP B-17 trial's follow-up analyses. Mastectomy remains the appropriate choice for DCIS that is multicentric, too extensive relative to breast size to achieve adequate margins with acceptable cosmesis, or for patients who

prefer this approach; the 2-millimeter margin standard applies when radiation follows breast-conserving surgery, versus the wider 3-millimeter (or greater) standard referenced for the B-17 criteria governing surgery-alone management. Sentinel lymph node biopsy is not routinely performed for pure DCIS, precisely because the diagnosis by definition excludes invasive disease and therefore excludes any risk of nodal metastasis -- but sentinel node biopsy IS performed at the time of mastectomy for DCIS specifically because, if an occult invasive component is subsequently found on the mastectomy specimen (recall the approximately 20% DCIS-to-invasive upgrade rate from Section 08, Slide 32), it would be technically far more difficult to return later and perform a sentinel node procedure once the breast tissue and its lymphatic drainage pathways have already been removed -- this pragmatic, forward-looking logic is exactly why the sentinel-node decision differs between breast-conserving surgery and mastectomy for the identical DCIS diagnosis. For estrogen-receptor-positive DCIS, five years of adjuvant tamoxifen reduces both ipsilateral recurrence and contralateral new primary risk, mirroring its role in invasive disease (Section 13, Slide 49) though generally understood to confer a more modest absolute benefit given DCIS's inherently lower baseline recurrence risk than invasive disease.

Paget's disease management, as already established in Section 09 (Slide 35), is entirely dictated by the underlying lesion rather than by the nipple finding itself: pure Paget's disease without any underlying invasive component is managed with central excision (including the nipple-areolar complex) combined with whole-breast radiation, directly analogous to standard breast-conserving DCIS management simply centered on the nipple; Paget's disease found together with an underlying DCIS component follows the same management algorithm as DCIS generally, adapted as needed for the specific central location; mastectomy is reserved for Paget's disease combined with multifocal DCIS elsewhere in the breast, or when breast-conserving surgery cannot otherwise achieve an adequate result; and sentinel lymph node biopsy follows the same DCIS-versus-invasive-component logic already discussed, performed specifically when mastectomy is chosen or when an invasive component is identified.

LCIS management, as already established in full in Section 08 (Slide 31), centers on observation, risk stratification, and chemoprevention for the classic variant rather than surgical eradication: annual mammography and clinical examination constitute routine surveillance, and no surgical re-excision is required when classic LCIS is present at a biopsy margin. Chemoprevention with tamoxifen, or with raloxifene as validated in the NSABP P-2 (STAR) trial, reduces subsequent bilateral breast cancer risk by approximately 50%, reflecting LCIS's role as a bilateral risk marker rather than a unilateral anatomic precursor. Pleomorphic LCIS, by contrast, is managed with surgical excision to achieve clear margins, mirroring high-grade DCIS management given its more aggressive histologic appearance, exactly as already detailed in Section 08. For the highest-risk patients -- classic LCIS combined with a strong family history or confirmed BRCA mutation -- bilateral risk-reducing mastectomy remains an option, as does consideration of annual MRI surveillance for LCIS patients who carry additional compounding risk factors, extending the general high-risk surveillance principles of Section 05 (Slide 19) to this specific population.

The endocrine therapy options available at Stage 0 -- tamoxifen 20 mg daily for five years across both pre- and postmenopausal patients, or, in postmenopausal patients specifically, anastrozole 1 mg daily for five years as validated in the IBIS-II DCIS trial -- mirror the invasive-disease endocrine therapy options of Section 13 (Slide 49), simply applied to the lower-risk, purely in situ disease setting. There is no role for cytotoxic chemotherapy in the management of pure DCIS or LCIS at any point in this algorithm, since neither entity carries any risk of distant (systemic) spread by definition, and chemotherapy's entire rationale throughout this document is addressing occult or established systemic micrometastatic disease -- a risk that simply does not exist for disease that has not yet, or (in the case of LCIS) may never, breach the basement membrane. Follow-up for Stage 0 disease consists of annual mammography combined with history and physical examination every six months for

the first five years, transitioning to annual physical examination thereafter -- this follow-up schedule is the Stage 0-specific instance of the general follow-up principles developed comprehensively in Section 19 (Slide 72).

SLIDE 55 -- Stage I-IIB -- Curative Intent Management

Slide content recap:

- Work-up: triple assessment, staging, biomarkers, genetics
- Neoadjuvant preferred: HER2+ Stage II+, TNBC Stage II+, LABC, goal BCS
- Adjuvant preferred: pT1 pN0 low-risk ER+/HER2-, older/frail patients
- Stage IIA ER+/HER2- N0: Oncotype DX RS guides chemo decision (TAILORx)
- Stage IIB-N1 ER+/HER2-: consider neoadjuvant or direct surgery then adjuvant
- T1N0 ER+ HER2-: excellent prognosis; chemo often not needed if RS≤25
- T2N1 HER2+: neoadjuvant HP-based; residual → T-DM1
- T2N1 TNBC: neoadjuvant pembro-chemo; residual → capecitabine
- T1-2 N0 TNBC RS<50: adjuvant chemo; pembro only with neoadjuvant path
- Fertility preservation: discuss before chemotherapy in premenopausal patients
- GnRH agonist during chemo: reduces premature ovarian failure (POEMS trial)
- BCS eligible → BCS + SLNB; not eligible → mastectomy + SLNB (+/- ALND)
- Adjuvant RT: WBRT after BCS; PMRT if T3/T4 or ≥4 nodes
- ER+/HER2- systemic: endocrine +/- chemo (by Oncotype DX RS); CDK4/6 (abemaciclib) if high-risk N+
- HER2+ systemic: chemo + trastuzumab +/- pertuzumab (N+ or high risk T2+)
- TNBC Stage II systemic: chemo +/- pembrolizumab neoadjuvant (preferred); capecitabine if residual
- BRCA1/2 systemic: olaparib adjuvant 1 yr if residual disease or pN+ after NAC
- Duration endocrine: 5-10 yrs depending on risk; OFS for premenopausal high-risk

Detailed Explanation and Source Context

Every Stage I to IIB patient begins with the identical workup foundation established in Sections 07, 10, and 11: triple assessment to confirm diagnosis, full anatomic staging, complete biomarker panel (ER/PR/HER2/Ki-67), and genetic counseling referral when criteria are met (Section 04, Slide 15; Section 05, Slide 20) -- this workup is what generates the specific data points that determine every subsequent branch point in the algorithm below.

The neoadjuvant-versus-adjuvant sequencing decision, developed conceptually in Section 13 (Slide 52), is applied concretely here: neoadjuvant therapy is preferred for HER2-positive disease at Stage II or higher, triple-negative disease at Stage II or higher, locally advanced disease, and whenever achieving breast-conserving surgery eligibility is a specific goal that upfront tumor size would otherwise preclude -- in each of these scenarios, the response-adaptive information neoadjuvant sequencing provides (pathologic complete response versus residual disease, directly informing the escalation strategies of Section 13) offers genuine clinical value beyond simply reordering the same treatments. Adjuvant-first sequencing (surgery before systemic therapy) remains preferred for lower-risk disease -- specifically small (T1), node-negative, hormone-receptor-positive, HER2-negative tumors, where neoadjuvant sequencing's chief benefit (response-adaptive escalation) is less relevant given the already-favorable expected biology -- and for older or frailer patients for whom the practical benefits of establishing a definitive pathologic stage promptly, without the extended treatment timeline neoadjuvant therapy entails, may be prioritized.

For Stage IIA, node-negative, hormone-receptor-positive, HER2-negative disease specifically, the Oncotype DX Recurrence Score (Section 11, Slide 41) is the central decision-making tool, directly implementing the TAILORx trial's findings: a low score supports endocrine therapy alone, a high score supports adding chemotherapy, and the intermediate range requires incorporating menopausal status into the decision, exactly as detailed in Section 11. For Stage IIB disease with N1 nodal involvement and the same hormone-receptor-positive, HER2-negative biology, either neoadjuvant sequencing or direct surgery followed by adjuvant therapy remains a reasonable

approach, with the specific choice individualized based on additional clinical factors including the RxPONDER-derived menopausal-status-specific recommendations already detailed in Section 11 (Slide 41).

The specific numeric decision points summarized on this slide function as worked clinical examples applying the principles established throughout Sections 11 and 13: a T1N0, estrogen-receptor-positive, HER2-negative tumor carries an excellent prognosis, and chemotherapy is frequently unnecessary when the Recurrence Score is 25 or below, following the TAILORx-established low-and-intermediate-risk pathway; a T2N1, HER2-positive tumor is treated with a neoadjuvant trastuzumab-and-pertuzumab-based (HP-based) regimen, with any residual disease at surgery escalated to adjuvant T-DM1 per the KATHERINE trial (Section 13, Slide 51); a T2N1 triple-negative tumor is treated with neoadjuvant pembrolizumab plus chemotherapy per the KEYNOTE-522 regimen, with residual disease escalated to adjuvant capecitabine per CREATE-X; and T1-2, node-negative triple-negative disease with a low genomic risk profile is generally treated with adjuvant chemotherapy alone (via the adjuvant-first pathway), reserving pembrolizumab specifically for patients who proceed via the neoadjuvant pathway, consistent with pembrolizumab's FDA-approved indication being tied specifically to the full KEYNOTE-522 neoadjuvant-plus-adjuvant sequence rather than to an adjuvant-only administration.

Because this stage range predominantly affects women of reproductive age, fertility preservation counseling should be offered and discussed BEFORE initiating chemotherapy in any premenopausal patient, given chemotherapy's well-established risk of accelerating ovarian failure and reducing future fertility -- and, as an adjunctive (though not fully replacement) strategy, administering a GnRH agonist concurrently during chemotherapy has been shown, per the POEMS trial, to reduce the rate of premature ovarian failure, providing at least a partial protective option for patients who are unable to pursue formal fertility preservation (egg or embryo cryopreservation) before treatment begins.

The surgical pathway itself follows directly from Section 12's breast-conserving-surgery-versus-mastectomy decision framework (Slide 44): patients who are breast-conserving-surgery-eligible proceed to lumpectomy combined with sentinel lymph node biopsy, while patients who are not eligible (per the specific contraindication list detailed in Section 12) proceed to mastectomy with sentinel node biopsy, adding completion axillary dissection as indicated per the Z0011-informed criteria of Section 12 (Slide 45). Adjuvant radiation follows the same principles established in Section 12 (Slide 46): whole-breast radiation is standard after breast-conserving surgery, and post-mastectomy radiation is added specifically for T3/T4 tumors or four or more positive nodes, per the criteria detailed there.

Systemic therapy selection by subtype at this stage range is, in essence, the direct application of everything established across Section 13: hormone-receptor-positive, HER2-negative disease receives endocrine therapy with or without chemotherapy (guided by the Oncotype DX Recurrence Score), with the addition of adjuvant abemaciclib specifically for the high-risk, node-positive population defined by the monarchE trial criteria (Section 13, Slide 49); HER2-positive disease receives chemotherapy plus trastuzumab, with pertuzumab added for node-positive or otherwise high-risk T2-or-larger disease per the APHINITY-derived criteria (Section 13, Slide 51); Stage II triple-negative disease receives chemotherapy, with neoadjuvant pembrolizumab preferred where feasible and adjuvant capecitabine added for residual disease; and confirmed germline BRCA1/2 carriers receive one year of adjuvant olaparib specifically when residual disease remains after neoadjuvant chemotherapy or when node-positive disease is found, per the OlympiA trial design already introduced in Section 13 (Slide 52). Endocrine therapy duration, as established in Section 13 (Slide 49), ranges from five to ten years depending on the individually assessed recurrence risk, with ovarian function suppression added specifically for higher-risk premenopausal patients per the SOFT/TEXT trial criteria.

SLIDE 56 -- Stage III -- Locally Advanced Breast Cancer (LABC)

Slide content recap:

- Stage IIIA: T3N1; T0-3N2 (fixed/matted axillary or IMN nodes)
- Stage IIIB: T4 (any) including IBC (T4d); T4N0-2
- Stage IIIC: any T, N3 (infraclavicular, supraclavicular, IMN + axillary)
- Inflammatory BC (T4d): rapid onset; clinical diagnosis; always Stage $\geq$ IIIB
- Workup: CT C/A/P + bone scan or PET-CT; brain MRI if symptomatic
- NEVER surgery first in IBC -- neoadjuvant chemotherapy MANDATORY
- Anthracycline + taxane backbone +/- platinum +/- anti-HER2 (if HER2+)
- Surgery: MRM after $\geq$ 4-6 cycles (even if "complete clinical response")
- PMRT: ALL IBC patients (even if pCR); target chest wall + regional nodes
- Adjuvant: capecitabine (TNBC residual); T-DM1 (HER2+ residual); endocrine
- Prognosis: 5-yr OS ~40%; TNBC-IBC worst; HER2+ IBC improved with targeted
- Step 1: biopsy + biomarkers (ER/PR/HER2/Ki-67) + full staging
- Step 2: neoadjuvant systemic therapy (NAC +/- targeted) -- ALL subtypes
- Step 3: MRM after neoadjuvant (BCS if excellent response; IBC -> MRM mandatory)
- Step 4: axillary dissection (ALND level I-II) for all N+ cases
- Step 5: PMRT (chest wall + regional nodes) -- ALL Stage III patients
- Step 6: adjuvant systemic (response-adapted): T-DM1 if HER2+ residual; capecitabine if TNBC residual; endocrine for ER+
- PMRT: all Stage III patients -- both after BCS and mastectomy
- PMRT dose: 46-50 Gy to chest wall + 10 Gy boost for close margins
- Target volumes: chest wall/residual breast + axilla + SCF + IMN
- PMRT timing: after completion of chemotherapy; before or concurrent with endocrine
- Reconstruction + PMRT: expander should be partially deflated during RT; complications
- Hypofractionation: FAST-Forward shows 26 Gy x 5 non-inferior for WBRT (selected)

Detailed Explanation and Source Context

Stage III, or locally advanced breast cancer (LABC), is defined by three anatomic substages that directly apply the T and N category definitions of Section 10 (Slide 37): Stage IIIA encompasses T3 tumors with N1 nodal involvement, or any T0-3 tumor with N2 involvement (four to nine axillary nodes, or clinically detected internal mammary nodes, reflecting fixed or matted axillary disease or internal mammary involvement); Stage IIIB encompasses any T4 tumor (chest wall invasion, skin involvement, or inflammatory breast cancer) combined with N0 to N2 nodal status; and Stage IIIC encompasses any T category combined with N3 nodal involvement (ten or more axillary nodes, infraclavicular involvement, internal mammary involvement combined with axillary disease, or supraclavicular involvement). Inflammatory breast cancer, staged as T4d regardless of underlying tumor size as already established in Section 09 (Slide 35), is therefore always at minimum Stage IIIB, and can be staged higher still if accompanying nodal involvement meets N2 or N3 criteria.

Staging workup for Stage III disease, following the stage-adapted principles established in Section 10 (Slide 38), includes CT of the chest, abdomen, and pelvis combined with either a bone scan or PET-CT, given the meaningfully elevated pretest probability of occult distant metastasis at this more advanced anatomic stage, with brain MRI reserved specifically for patients with neurologic symptoms rather than performed routinely in asymptomatic patients.

Inflammatory breast cancer receives dedicated, specific management attention on this slide because its treatment sequence departs so fundamentally from standard Stage III management, restating and reinforcing the principle already emphasized repeatedly in this document (Section 09, Slide 35; Section 12, Slide 44; Section 13, Slide 52): surgery must NEVER be performed first in IBC, and neoadjuvant chemotherapy -- built on an anthracycline-and-taxane backbone, with platinum added for triple-negative disease and anti-HER2 therapy added for HER2-positive disease, following the same subtype-specific regimen selection principles detailed in Section 13 -- is mandatory as the first treatment step. Modified radical mastectomy follows after at least four to

six cycles of neoadjuvant therapy, and, critically, mastectomy is performed EVEN IF the patient appears to have achieved a complete clinical response to neoadjuvant therapy, since IBC's diffuse dermal lymphatic involvement means that surgical removal of the entire breast and its overlying skin remains necessary for adequate local control regardless of how favorably the disease has responded clinically -- breast-conserving surgery is never an option for IBC, a point already established but worth restating in this stage-specific context. Post-mastectomy radiation therapy is likewise mandatory for ALL IBC patients, again regardless of the degree of pathologic response achieved (including even in patients who achieve a pathologic complete response), targeting the chest wall and regional nodal basins, reflecting the recognized risk of local-regional recurrence that persists in this disease even after an apparently excellent systemic treatment response. Adjuvant systemic therapy is then response-adapted using the same principles established throughout Section 13: capecitabine for residual triple-negative disease, T-DM1 for residual HER2-positive disease, and endocrine therapy for hormone-receptor-positive disease. IBC's overall prognosis remains guarded even with this comprehensive, correctly sequenced multimodal approach, with an approximate 40% five-year overall survival; triple-negative IBC carries the worst prognosis among IBC subtypes, while HER2-positive IBC, benefiting from the dramatic efficacy of modern anti-HER2 targeted therapy (Section 13, Slide 51), has shown the most meaningful improvement in outcomes over the past two decades.

The general Stage III treatment algorithm, applicable across all subtypes beyond the IBC-specific pathway just described, proceeds through six sequential steps that synthesize the entirety of Sections 11 through 13 into a single ordered pathway: first, biopsy confirmation together with the complete biomarker panel (ER/PR/HER2/Ki-67, Section 11) and full staging workup as described above; second, neoadjuvant systemic therapy, selected by subtype and incorporating targeted therapy when appropriate, applied to essentially ALL Stage III patients regardless of subtype, reflecting the near-universal benefit of downstaging at this locally advanced anatomic extent; third, modified radical mastectomy following neoadjuvant therapy, with breast-conserving surgery reserved for patients who achieve an excellent response and meet the standard BCS eligibility criteria of Section 12, while IBC mandates mastectomy regardless of response as just discussed; fourth, axillary lymph node dissection at Berg levels I and II for all node-positive cases, reflecting the reality that essentially all Stage III patients present with, or are treated presuming, significant nodal involvement that falls outside the Z0011 and targeted-axillary-dissection exceptions detailed in Section 12 (Slide 45); fifth, post-mastectomy radiation therapy targeting the chest wall and regional nodal basins for ALL Stage III patients, without the more selective, risk-stratified approach applied at earlier stages (Section 12, Slide 46); and sixth, response-adapted adjuvant systemic therapy -- T-DM1 for residual HER2-positive disease, capecitabine for residual triple-negative disease, and endocrine therapy for hormone-receptor-positive disease -- directly applying the escalation principles established throughout Section 13.

The near-universal application of post-mastectomy radiation at Stage III, restated explicitly here, applies both after mastectomy and, in the minority of patients who achieve breast-conserving surgery eligibility after an excellent neoadjuvant response, after breast-conserving surgery as well -- Stage III disease's baseline locoregional recurrence risk is simply high enough that radiation is essentially universal at this stage, in contrast to the more selective, risk-stratified radiation decisions applied at Stage I-II B (Section 12, Slide 46). Standard dosing is 46 to 50 Gy to the chest wall, with an additional 10 Gy boost specifically for patients with close surgical margins, and target volumes span the chest wall (or residual breast tissue, in the minority treated with BCS), the axilla, the supraclavicular fossa, and the internal mammary nodal chain. Radiation timing follows chemotherapy completion, with sequencing relative to endocrine therapy either following it or being given concurrently depending on institutional protocol, consistent with the general PMRT timing principles established in Section 12 (Slide 46). In patients with implant-based reconstruction who require post-mastectomy radiation, the tissue expander is typically partially deflated during the radiation course to reduce the radiation dose delivered through

the expanded tissue and to reduce the risk of radiation-related reconstructive complications, though, as already noted in Section 12 (Slide 47), the combination of implant-based reconstruction with radiation therapy carries an inherently higher complication rate regardless of this mitigation strategy. Finally, the FAST-Forward trial has demonstrated that an even more accelerated hypofractionation schedule -- 26 Gy delivered across just 5 fractions -- is non-inferior to conventional whole-breast radiation fractionation in selected patients, extending the hypofractionation principle already established in Section 12 (Slide 46) toward an even more compressed, patient-convenient treatment schedule.

SLIDE 57 -- Stage IV -- Metastatic Breast Cancer (MBC)

Slide content recap:

- Metastatic BC: incurable but treatable; goal = extend survival + maintain QoL
- Complete reassessment: rebiopsy metastatic site for ER/PR/HER2 (15-25% discordance)
- Test for actionable mutations: ESR1 (cfDNA), PIK3CA/PTEN/AKT1, BRCA1/2 (germline+somatic)
- Preferred sites for rebiopsy: bone (decalsified limited), liver, skin/lymph node
- ECOG performance status: critical for treatment selection
- Bone-only MBC: best prognosis; bone-modifying agents (zoledronic acid/denosumab)
- HR+/HER2- postmeno 1st line: CDK4/6i (ribociclib/abema/palbociclib) + AI; 2nd line: fulvestrant +/- inavolisib; elacestrant/implunestrant if ESR1 mut
- HR+/HER2- premeno 1st line: CDK4/6i + AI + OFS; 2nd line: capivasertib + fulvestrant (PIK3CA/AKT1/PTEN mut); ET options
- HER2+ 1st line: pertuzumab + trastuzumab + docetaxel (CLEOPATRA); 2nd line: T-DXd (preferred, DESTINY-Breast03)
- HER2+ brain mets 1st line: HP + docetaxel + trastuzumab; 2nd line: tucatinib + tras + capecitabine (HER2CLIMB; preferred CNS)
- TNBC PD-L1+ (CPS $\geq$ 10) 1st line: pembrolizumab + chemo (KEYNOTE-522); 2nd line: sacituzumab govitecan (SG); dato-DXd
- TNBC PD-L1- 1st line: chemo (gem/carbo; capecitabine); 2nd line: SG; T-DXd (HER2-low); dato-DXd
- BRCA1/2 mutated 1st line: olaparib or talazoparib (HER2-neg); 2nd line: SG; chemo
- [UPDATE] See ESMO 2026 slides for full updated algorithms with new agents

Detailed Explanation and Source Context

Stage IV, metastatic breast cancer, marks a fundamental shift in treatment philosophy from every preceding stage in this section: it is explicitly, and importantly for patient counseling, framed as incurable but treatable, with the goal of care shifting from eradication of disease to extending survival while maintaining quality of life -- this reframing has direct implications for how aggressively local therapies are pursued (generally reserved for symptom palliation rather than curative intent) and how treatment toxicity is weighed against expected benefit throughout the remainder of the patient's care.

A critical, easily overlooked but consequential first step upon a new metastatic diagnosis (whether at initial presentation or as a recurrence after earlier-stage treatment) is complete biomarker reassessment: rebiopsy of a metastatic site to reconfirm ER, PR, and HER2 status is recommended because receptor status discordance between the primary tumor and a metastatic site occurs in a clinically significant 15 to 25% of cases -- a tumor that was originally ER-positive can lose ER expression at a metastatic site (or, less commonly, the reverse can occur), and treating based on outdated, primary-tumor biomarker results in the face of such discordance would risk selecting an ineffective treatment strategy. Beyond standard receptor testing, metastatic tissue (or circulating cell-free DNA, "liquid biopsy") should be tested for additional actionable molecular alterations that specifically open metastatic-setting treatment options detailed later on this slide and in Section 15: ESR1 mutations (typically tested via cell-free DNA given their tendency to arise as an acquired resistance mechanism, meaning they may not be present in archival primary tumor tissue), PIK3CA, PTEN, and AKT1 alterations (governing eligibility for the PI3K/AKT pathway-targeted agents detailed in Section 15, Slide 59), and BRCA1/2 status, now tested for both germline and tumor-acquired (somatic) mutations, since either can confer PARP

inhibitor eligibility. Preferred anatomic sites for metastatic rebiopsy, when a choice is available among multiple metastatic sites, are bone (though decalcification procedures required for processing bone specimens can sometimes limit the quality of biomarker testing achievable), liver, and skin or lymph node metastases, the latter two generally providing the most straightforward and highest-quality tissue for comprehensive testing.

Baseline ECOG performance status -- a standardized functional assessment scale used broadly across oncology -- is a critical input into metastatic treatment selection, since it directly predicts a patient's ability to tolerate more intensive combination regimens versus requiring a better-tolerated, less aggressive approach, and is incorporated into essentially every major clinical trial's eligibility criteria underlying the treatment algorithms on this slide. Among the different patterns of metastatic spread, bone-only metastatic disease, as already established in Section 03 (Slide 12), carries the most favorable prognosis of any metastatic presentation, and its management is supplemented by bone-modifying agents -- zoledronic acid (a bisphosphonate) or denosumab (a RANK-ligand inhibitor) -- which reduce the rate of skeletal-related events (pathologic fracture, spinal cord compression, need for radiation or surgery to bone) even though they do not treat the underlying malignancy itself.

The subtype-and-line-specific treatment table on this slide is organized by the same molecular framework established throughout Section 11, now extended into the metastatic setting, and functions as a master summary that Section 15's dedicated ESMO 2026 slides will develop in substantially greater mechanistic and trial-data detail. For hormone-receptor-positive, HER2-negative disease in postmenopausal patients, first-line therapy is a CDK4/6 inhibitor (ribociclib, abemaciclib, or palbociclib) combined with an aromatase inhibitor, with second-line options including fulvestrant (a selective estrogen receptor degrader) with or without inavolisib, or the newer oral SERDs elacestrant or imlunestrant specifically when an ESR1 mutation is identified -- these newer agents are introduced here and given their full ESCAT/MCBS-scored evidentiary treatment in Section 15 (Slide 59). In premenopausal patients with the same hormone-receptor-positive, HER2-negative biology, first-line therapy adds ovarian function suppression to the CDK4/6-inhibitor-plus-aromatase-inhibitor backbone, with second-line options including capivasertib combined with fulvestrant specifically for tumors harboring PIK3CA, AKT1, or PTEN alterations, alongside other endocrine therapy options. For HER2-positive metastatic disease, first-line therapy is the CLEOPATRA regimen -- pertuzumab, trastuzumab, and docetaxel -- and second-line therapy is T-DXd, now preferred based on the DESTINY-Breast03 trial's head-to-head superiority over the previously standard T-DM1, exactly as already established in Section 13 (Slide 51). For HER2-positive disease specifically complicated by brain metastases, first-line therapy remains the same HP-plus-docetaxel backbone, but second-line therapy shifts specifically to the tucatinib-trastuzumab-capecitabine combination, preferred in this population for its demonstrated CNS activity per the HER2CLIMB trial, again as already established in Section 13 (Slide 51). For triple-negative disease with high PD-L1 expression (combined positive score of 10 or greater), first-line therapy incorporates pembrolizumab with chemotherapy, with second-line options including the TROP2-targeted antibody-drug conjugates sacituzumab govitecan and datopotamab deruxtecan; for PD-L1-negative triple-negative disease, first-line therapy is chemotherapy alone (gemcitabine-carboplatin or capecitabine-based regimens), with the same TROP2-targeted agents available second-line, alongside T-DXd specifically for HER2-low triple-negative disease per the DESTINY-Breast04 findings already discussed in Section 11 (Slide 42). Finally, for confirmed BRCA1/2-mutated, HER2-negative metastatic disease, olaparib or talazoparib monotherapy is preferred first-line over standard chemotherapy, per the OlympiAD and EMBRACA trials already introduced in Section 05 (Slide 19), with sacituzumab govitecan or standard chemotherapy available as second-line options upon progression.

[UPDATE] This slide's table represents the master metastatic treatment framework as of the current (2026) guideline cycle, but it deliberately defers the full trial-data detail, ESCAT actionability scoring, and Magnitude of

Clinical Benefit Scale (MCBS) evaluation of each of these newer agents to the dedicated ESMO 2026 update slides that immediately follow in Section 15 -- the reader should treat this slide as the organizing map of the metastatic treatment landscape, and Section 15 as the detailed legend explaining exactly how strong the evidence is behind each specific new-agent recommendation.

SECTION 15 -- ESMO 2026 -- Metastatic Breast Cancer Updates

The full evidentiary detail behind the newest metastatic-setting agents introduced in Section 14s master treatment table -- the ESR1/PIK3CA-AKT pathway agents for hormone-receptor-positive disease, the antibody-drug conjugate era for HER2-positive and triple-negative disease, the ESCAT and MCBS v2.0 evidence-grading frameworks that structure ESMO's 2026 recommendations, and the explicit [ETHIOPIA] gap this creates for a resource-limited health system.

SLIDE 58 -- Section Divider -- ESMO 2026 MBC Updates

Detailed Explanation and Source Context

[UPDATE] This section, together with Section 05's genetic-risk update (Slide 20) and Section 18's Ethiopian guideline integration, represents the content in this document that most directly reflects material postdating the base Schwartz and DeVita chapters, drawn from the ESMO 2026 Metastatic Breast Cancer guideline supplied specifically for this revision. Its three slides map directly onto the three molecularly distinct metastatic populations introduced in Section 14 (Slide 57): hormone-receptor-positive/HER2-negative disease (Slide 59), HER2-positive and triple-negative disease (Slide 60), and the newly recognized HER2-low population that cuts across both of the latter two categories (also Slide 60).

SLIDE 59 -- ESMO 2026 -- HR+/HER2- MBC: ESR1 and PI3K/AKT Updates

Slide content recap:

- ESR1 mutations: acquired endocrine resistance mechanism; present in ~40% post-CDK4/6
- Test: liquid biopsy (ctDNA) for ESR1 Y537S/D538G mutations before 2nd-line ET
- Elacestrant (EMERALD trial): oral SERD; ESR1-mutated: PFS HR 0.55 vs standard ET
- Imlunestrant (EMBER-3 trial): next-gen oral SERD; ESR1-mutated: increased PFS vs ET
- ESCAT level: IA for ESR1-mutated (elacestrant); oral SERDs preferred over injections
- Key: cfDNA testing (cell-free DNA liquid biopsy) before selecting 2nd-line ET
- ESCAT = ESMO Scale for Clinical Actionability of Molecular Targets
- Tier I: ready for routine use in clinical practice (level IA/IB)
- Tier II: promising but lacking definitive evidence
- Tier III: uncertain clinical utility -- investigational
- Tier IV: evidence for lack of utility -- do not use
- Purpose: triage molecular test findings for clinical decision-making
- Used to prioritize which actionable alterations drive treatment selection
- PIK3CA mutation: ~35-40% of ER+/HER2- BC; activates PI3K/AKT/mTOR pathway
- Inavolisib + palbociclib + fulvestrant: for PIK3CA-mutated, CDK4/6-pretreated
- INAVO120: PFS median 15.0 vs 7.3 months (HR 0.43) -- practice-changing
- Capivasertib + fulvestrant: AKT1/PIK3CA/PTEN-altered; CAPItello-291 trial (PFS HR 0.60)
- Everolimus + exemestane: mTOR inhibitor; BOLERO-2 (older; still used)
- MCBS v2.0: scores benefit of new agents on 1-5 (palliative) or A-C (curative)
- MCBS Score 5 (highest) = meets ESMO threshold for substantial benefit: PFS HR <0.65 or ≥6 mo improvement
- T-DXd HER2+: Score 5; Tucatinib+T+C: Score 4; Capivasertib: Score 3-4
- Guides formulary decisions in resource-limited settings (e.g., Ethiopia)
- Elacestrant ESR1-mutated: MCBS Score 3 -- meaningful but limited benefit magnitude

Detailed Explanation and Source Context

[UPDATE] ESR1 mutations represent the single most important acquired mechanism of endocrine resistance in hormone-receptor-positive metastatic breast cancer, arising in an estimated 40% of tumors following progression on a CDK4/6 inhibitor -- these mutations, most commonly the specific hotspot variants Y537S and D538G, alter the estrogen receptor's ligand-binding domain in a way that renders it constitutively active even in the near-total absence of circulating estrogen that an aromatase inhibitor would otherwise achieve, explaining why aromatase inhibitor therapy specifically (rather than endocrine therapy in general) loses effectiveness once an ESR1 mutation emerges. Because ESR1 mutations characteristically arise as an ACQUIRED resistance mechanism during treatment (rather than being present in the original, treatment-naïve tumor), testing for them is performed via liquid biopsy -- circulating tumor DNA (ctDNA) obtained from a blood draw -- specifically at the point of considering second-line endocrine therapy after CDK4/6 inhibitor progression, since testing archival primary-tumor tissue obtained before treatment would systematically miss mutations that only develop under the selective pressure of aromatase inhibitor exposure.

Two new oral selective estrogen receptor degraders (SERDs) have been specifically developed to remain effective in the presence of ESR1 mutations, unlike aromatase inhibitors, because SERDs work by directly binding and degrading the estrogen receptor protein itself rather than by depleting circulating estrogen, a mechanism that remains effective even against the constitutively-active mutant receptor. Elacestrant, validated in the EMERALD trial, achieved a progression-free survival hazard ratio of 0.55 compared with standard endocrine therapy specifically in the ESR1-mutated population. Imlunestrant, a next-generation oral SERD validated in the EMBER-3 trial, likewise demonstrated a progression-free survival improvement over standard endocrine therapy in ESR1-mutated disease. Both agents have achieved the highest ESCAT actionability tier (IA) specifically for the ESR1-mutated population, and oral SERDs are now generally preferred over the older injectable SERD (fulvestrant) in this specific molecularly-defined scenario -- though fulvestrant retains an important role in other treatment combinations discussed elsewhere in this document (Section 14, Slide 57). The overarching clinical principle this evidence establishes is that cell-free DNA liquid biopsy testing should be performed specifically before selecting second-line endocrine therapy after CDK4/6 inhibitor progression, since the presence or absence of an ESR1 mutation directly determines whether an oral SERD or an alternative approach represents the optimal next treatment choice.

The ESCAT framework (ESMO Scale for Clinical Actionability of Molecular Targets) is introduced here specifically because it is the standardized evidence-grading system ESMO uses throughout its 2026 guideline update to communicate exactly how strong the evidence is behind any given molecularly-targeted treatment recommendation, and understanding its four tiers is essential to correctly interpreting every subsequent agent discussed in this section. Tier I designates alterations and their matched therapies as ready for routine use in clinical practice, based on the strongest available evidence (levels IA and IB, reflecting randomized trial data of the highest quality) -- ESR1-mutated disease treated with elacestrant, just discussed, exemplifies this tier. Tier II designates promising but not yet definitively established associations, where the biological rationale is sound but the clinical evidence base remains less mature. Tier III designates alterations of genuinely uncertain clinical utility, appropriately considered investigational rather than actionable in routine practice. Tier IV designates alterations for which evidence specifically demonstrates a LACK of clinical utility -- a genuinely useful negative category, since it actively steers clinicians away from pursuing a treatment strategy that has been specifically shown not to work, rather than simply reflecting an absence of evidence either way. The overarching purpose of ESCAT is to triage the growing volume of molecular test findings that a genomic or liquid biopsy panel can generate into a clear, standardized hierarchy of clinical actionability, helping clinicians prioritize which of potentially several simultaneously-identified alterations should actually drive the treatment selection decision.

The PIK3CA/AKT1/PTEN pathway represents the second major genomically-targetable axis in hormone-receptor-positive, HER2-negative metastatic disease. PIK3CA mutations, activating the PI3K/AKT/mTOR intracellular signaling pathway, are found in an estimated 35 to 40% of ER-positive, HER2-negative breast cancers, making this one of the most common actionable alterations in this disease subtype. The triplet combination of inavolisib, palbociclib, and fulvestrant is specifically indicated for PIK3CA-mutated disease that has already been treated with (and progressed on) a CDK4/6 inhibitor, and the INAVO120 trial demonstrated a dramatic improvement in this specific population, with median progression-free survival extending from 7.3 months with the comparator regimen to 15.0 months with the inavolisib-containing triplet (hazard ratio 0.43) -- a result substantial enough to be characterized as practice-changing for this specific molecularly-defined population. Capiivasertib, combined with fulvestrant, offers a related but distinct option, targeting the pathway at the level of AKT1 specifically (as well as being effective in PIK3CA- or PTEN-altered disease more broadly), validated in the CAPitello-291 trial with a progression-free survival hazard ratio of 0.60. Everolimus, an mTOR inhibitor acting further downstream in this same signaling pathway, combined with exemestane, represents an older but still clinically utilized option, validated originally in the BOLERO-2 trial that predates the newer, more precisely biomarker-selected agents just discussed -- everolimus is generally considered a less molecularly-targeted, broader-spectrum option within this same overall pathway compared with the newer PIK3CA/AKT1-specific agents.

The Magnitude of Clinical Benefit Scale, version 2.0 (MCBS v2.0), is ESMO's complementary framework to ESCAT, addressing a distinct question: not whether a molecular alteration is actionable (ESCAT's purpose), but how LARGE the clinical benefit of a given new agent actually is, scored on a 1-to-5 scale for the palliative (non-curative, metastatic) treatment setting or an A-to-C scale for the curative treatment setting. A score of 5, the highest palliative-setting score, requires meeting a specific, explicit quantitative threshold: a progression-free survival hazard ratio below 0.65, or an improvement in median survival of six months or more -- this is a genuinely high bar, deliberately set to distinguish agents offering substantial, clinically meaningful benefit from those offering only marginal or statistically-significant-but-clinically-modest improvement. Applying this scale to the agents discussed throughout Section 13 through 15: T-DXd in HER2-positive disease achieves the maximum score of 5, reflecting its dramatic efficacy already discussed in Section 13 (Slide 51); the tucatinib-trastuzumab-capecitabine combination achieves a score of 4; and capivasertib achieves a score of 3 to 4, depending on the specific population and endpoint considered. The explicit, practical purpose of MCBS scoring, stated directly on this slide, is to guide formulary and resource-allocation decisions specifically in resource-limited settings such as Ethiopia -- when a health system cannot afford to make every new agent universally available, MCBS scoring provides an evidence-based framework for prioritizing which agents offer the largest genuine clinical benefit per unit of resource invested, a principle directly relevant to the Ethiopian guideline's own resource-prioritization challenges discussed in Section 18. Elacestrant in ESR1-mutated disease, despite achieving the highest ESCAT actionability tier (IA) as already discussed, receives a more modest MCBS score of 3 -- illustrating the important point that ESCAT (actionability, i.e., is there good evidence this specific biomarker-drug pairing works) and MCBS (magnitude, i.e., how large is the benefit once it is used) are two independent, complementary axes of evaluation, and a high score on one scale does not automatically imply an equally high score on the other.

SLIDE 60 -- ESMO 2026 -- HER2+ and TNBC MBC: New Agent Algorithms

Slide content recap:

- 1st line HER2+: pertuzumab + trastuzumab + taxane (CLEOPATRA -- OS benefit maintained)
- 2nd line HER2+: T-DXd (trastuzumab deruxtecan) -- DESTINY-Breast03: PFS HR 0.28 vs T-DM1

- -> T-DXd: PREFERRED 2nd line HER2+ by both ESMO and NCCN 2026
- 3rd line HER2+: tucatinib + trastuzumab + capecitabine (HER2CLIMB)
- Brain metastases present: tucatinib preferred (CNS penetration; intracranial ORR 47%)
- ILD: T-DXd-associated interstitial lung disease (2-13%); monitor with CT; dose modify
- TNBC PD-L1+ (CPS ≥ 10): pembrolizumab + chemotherapy (KEYNOTE-119/522)
- TNBC PD-L1-: chemotherapy (gem/carbo; capecitabine/vinorelbine); TROP2-targeted ADC
- Sacituzumab govitecan (SG): anti-TROP2 ADC; ASCENT: OS HR 0.48 vs single chemo
- Dato-DXd (datopotamab deruxtecan): anti-TROP2 ADC; TROPION-Breast01: PFS HR 0.63
- Pembrolizumab + SG: KEYNOTE-890 ongoing -- promising combination
- BRCA1/2 mut TNBC: olaparib or talazoparib -> PARP inhibitor first in BRCA carriers
- HER2-low = IHC 1+ or IHC 2+/FISH-negative
- DESTINY-Breast04: T-DXd vs physician's choice chemo in HER2-low MBC (previously treated)
- HR+ HER2-low: PFS HR 0.51; OS HR 0.64
- TNBC HER2-low: PFS HR 0.46 -- also significant benefit
- Now standard 2nd/3rd line for HER2-low MBC (HR+ and TNBC) -- ESMO Grade A
- Must test HER2 by IHC to identify HER2-low -- IHC 0 is NOT eligible for T-DXd
- [ETHIOPIA] Most ESMO 2026 agents (T-DXd, inavolisib, elacestrant, SG) NOT yet available in Ethiopia
- Ethiopian approach: established regimens (AC->T, CMF, tamoxifen, trastuzumab where available)
- Trastuzumab access: expanding at tertiary centers; not universally available
- CDK4/6 inhibitors: not yet routinely available in Ethiopian public health system
- MCBS scoring: guides priority for import/affordable access negotiations (WHO EML)
- [ETHIOPIA] Gap: Ethiopia is approx 2010-era treatment availability vs 2026 international standards

Detailed Explanation and Source Context

[UPDATE] The HER2-positive metastatic treatment sequence detailed here restates and extends the summary already presented in Section 13 (Slide 51) and Section 14 (Slide 57), now with the full trial-data detail behind each sequential line of therapy. First-line therapy remains the CLEOPATRA regimen -- pertuzumab, trastuzumab, and a taxane -- and the overall survival benefit first demonstrated when this trial was originally reported has been shown to be durable and maintained with longer-term follow-up, cementing this combination's continued first-line status even as the agents available in later lines have evolved substantially. Second-line therapy has shifted decisively to T-DXd, based on the DESTINY-Breast03 trial's finding of a progression-free survival hazard ratio of just 0.28 compared with the previously standard second-line agent, T-DM1 -- a magnitude of benefit large enough that T-DXd is now explicitly preferred as second-line therapy by both ESMO and NCCN in their 2026 guideline cycles, effectively displacing T-DM1 from the second-line position it held for most of the preceding decade (T-DM1 remains important, however, in the adjuvant, residual-disease setting per KATHERINE, Section 13 Slide 51, which is a distinct clinical scenario from this metastatic-setting sequencing question). Third-line therapy is tucatinib combined with trastuzumab and capecitabine, per the HER2CLIMB trial already introduced in Section 13 (Slide 51), and this combination is specifically preferred, ahead of its position in the general sequence, whenever brain metastases are present, reflecting its demonstrated intracranial objective response rate of 47% -- a genuinely high figure for a population that has historically had very limited systemic treatment options once brain metastases develop. T-DXd's specific, actively-monitored toxicity, interstitial lung disease, occurs in an estimated 2 to 13% of treated patients and requires structured surveillance with periodic CT imaging together with prompt dose modification or treatment discontinuation should ILD develop, given its potential for serious, even fatal, outcomes if not identified and managed promptly -- this toxicity profile, already flagged in Section 13 (Slide 51), is restated here because it becomes increasingly relevant as T-DXd's use expands into progressively earlier treatment lines and larger patient populations.

For triple-negative metastatic disease, treatment selection bifurcates based on PD-L1 expression status: PD-L1-positive disease (defined by a combined positive score of 10 or greater) is treated first-line with pembrolizumab combined with chemotherapy, per the KEYNOTE-119 and KEYNOTE-522 (this latter trial's design already detailed in the neoadjuvant, early-stage context in Section 13, Slide 52) evidence base, while

PD-L1-negative disease is treated with chemotherapy alone -- typically gemcitabine-carboplatin or capecitabine/vinorelbine-based regimens -- reflecting the absence of a clear immunotherapy benefit signal in the PD-L1-negative population. Beyond first-line therapy, TROP2-targeted antibody-drug conjugates have emerged as an important new therapeutic class applicable regardless of PD-L1 status: sacituzumab govitecan, delivering the cytotoxic payload SN-38 via a TROP2-targeting antibody, demonstrated an overall survival hazard ratio of 0.48 compared with single-agent chemotherapy in the ASCENT trial, and datopotamab deruxtecan, a related but structurally distinct TROP2-targeted antibody-drug conjugate (using deruxtecan, the same topoisomerase I inhibitor payload employed by T-DXd, rather than SN-38), demonstrated a progression-free survival hazard ratio of 0.63 in the TROPION-Breast01 trial. The combination of pembrolizumab with sacituzumab govitecan is under active investigation in the ongoing KEYNOTE-890 trial, representing a promising but not yet fully established combination strategy pairing immunotherapy with a TROP2-targeted antibody-drug conjugate. For confirmed BRCA1/2-mutated triple-negative disease, the PARP inhibitors olaparib and talazoparib remain the preferred first-line approach ahead of standard chemotherapy or the newer antibody-drug conjugates just discussed, exactly as already established in Section 05 (Slide 19) and Section 14 (Slide 57).

HER2-low disease, defined as an immunohistochemistry score of 1+, or a score of 2+ with a negative FISH result (the identical definition already established in Section 11, Slide 42), cuts across both the hormone-receptor-positive and triple-negative categories just discussed, since HER2-low status is independent of, and can co-occur with, either hormone-receptor status. The pivotal DESTINY-Breast04 trial, comparing T-DXd against physician's choice of standard chemotherapy specifically in previously-treated HER2-low metastatic disease, demonstrated a progression-free survival hazard ratio of 0.51 and an overall survival hazard ratio of 0.64 in the hormone-receptor-positive, HER2-low subgroup, with a similarly significant progression-free survival hazard ratio of 0.46 observed in the triple-negative, HER2-low subgroup -- establishing that the HER2-low treatment benefit is not restricted to hormone-receptor-positive disease alone. T-DXd is consequently now established as standard second- or third-line therapy for HER2-low metastatic disease across BOTH the hormone-receptor-positive and triple-negative populations, carrying ESMO's highest evidence grade (Grade A) for this specific indication. A critical practical point, worth restating with emphasis exactly as this slide does, is that identifying HER2-low status absolutely requires HER2 immunohistochemistry testing on every metastatic (or, at minimum, every newly biopsied) specimen -- a tumor scored as IHC 0 (no detectable HER2 expression at all) is explicitly NOT eligible for T-DXd under this indication, since "HER2-low" specifically requires SOME detectable HER2 expression (IHC 1+, or 2+ with negative amplification), distinguishing it from the smaller subset of truly HER2-null tumors for which T-DXd's targeted delivery mechanism would have no molecular target to engage.

[ETHIOPIA] This slide closes with an explicit, unflinching acknowledgment of the resource gap this entire wave of new-agent development creates for a health system such as Ethiopia's: most of the agents discussed throughout this ESMO 2026 update -- T-DXd, inavolisib, elacestrant, and sacituzumab govitecan chief among them -- are not yet available in Ethiopia, meaning the metastatic treatment algorithms just described represent an aspirational international standard rather than the treatment reality currently accessible to Ethiopian patients. The current Ethiopian approach to metastatic breast cancer instead relies on the more established, older-generation regimens already discussed throughout this document: anthracycline-taxane sequential chemotherapy (AC followed by a taxane), the older CMF regimen where still used, tamoxifen for hormone-receptor-positive disease, and trastuzumab specifically where it is available. Trastuzumab access itself, while expanding at Ethiopian tertiary centers, remains inconsistent and is not yet universally available across the full national health system -- and, further down the treatment-sophistication spectrum, CDK4/6 inhibitors, now considered the first-line standard for hormone-receptor-positive metastatic disease

internationally (Section 14, Slide 57), are not yet routinely available within the Ethiopian public health system at all. The MCBS scoring framework introduced in Slide 59 is presented here specifically as a practical tool for navigating this resource gap: by quantifying which new agents offer the largest genuine clinical benefit, MCBS scores can guide prioritization for future import and affordable-access negotiations, for instance through the WHO Essential Medicines List (EML) process, helping a resource-constrained health system decide which of the many new agents described in this section would be worth prioritizing first if and when expanded access becomes feasible. The slide's closing assessment is stated with a directness that avoids euphemism specifically because the guideline authors judge this gap important enough to name explicitly rather than leave implicit: Ethiopia's currently accessible metastatic breast cancer treatment landscape approximates the international standard of care as it existed around 2010, rather than the 2026 standard detailed throughout this section -- a sixteen-year treatment-access gap that frames the urgency behind the health-system strengthening priorities detailed comprehensively in Section 18.

SECTION 16 -- Recurrence -- Types and Management

A dedicated classification of recurrence by anatomic type -- local, regional, and distant -- with precise definitions distinguishing true ipsilateral recurrence from a new primary, drawn from standard surgical oncology teaching, followed by site-specific management algorithms for each recurrence pattern.

SLIDE 61 -- Section Divider -- Recurrence, Types and Management

Detailed Explanation and Source Context

This section was added to this revision specifically to consolidate recurrence terminology and management that, in the base Schwartz and DeVita chapters, is often discussed piecemeal alongside the original stage-specific treatment sections rather than presented as a unified topic. The definitions established here -- particularly the specific criteria distinguishing a true ipsilateral breast tumor recurrence from an entirely new ipsilateral primary -- are drawn from standard surgical and radiation oncology teaching and are essential for correctly interpreting a patient's disease course and selecting appropriate salvage therapy.

SLIDE 62 -- Recurrence -- Classification and Definitions

Slide content recap:

- IBTR (true local): ipsilateral breast recurrence after BCS -- same site, same histology; salvageable; 60-70% 10-yr OS
- New ipsilateral primary: different quadrant or ≥ 5 yrs or different biomarkers; treat as new primary -- better prognosis
- Chest wall (post-mastectomy): skin/soft tissue at mastectomy site; 25-40% distant mets within 2 yrs
- Regional (nodal): ipsilateral axillary, SCF, IMN recurrence; 20-30% eventually develop distant mets
- Distant metastatic: any site beyond locoregional; median OS 2-5 yrs (improving)
- Local recurrence: at the ipsilateral breast (after BCS) or chest wall (after mastectomy)
- Regional recurrence: ipsilateral axillary, supraclavicular, infraclavicular, or IMN nodes
- LRR incidence: ~5-10% at 10 years with modern therapy
- IBTR: at original tumor site vs new primary? True recurrence: within 2-3 cm of original site, same histology, < 5 years
- New primary: different quadrant, OR ≥ 5 years, OR different histology/biomarkers
- Distant metastasis: any organ outside the breast/regional nodes
- Bone: most common (50-70% of distant recurrence); may be first site
- Visceral: liver (20-30%), lung (10-15%), brain (especially HER2+ and TNBC)
- Leptomeningeal: rare; HER2+ and TNBC; poor prognosis
- Time to recurrence: ER+ late (years 5-20+); TNBC/HER2+ early (< 3 yrs)
- Annual hazard rate: peaks at ~2 yrs for TNBC, ~3-4 yrs for HER2+, bimodal ER+

Detailed Explanation and Source Context

This slide establishes the precise vocabulary and diagnostic criteria that distinguish the several patterns of breast cancer recurrence, and getting the terminology exactly right matters clinically because each category carries a genuinely different prognosis and management pathway, developed fully in the algorithm slide that follows (Slide 63). Local recurrence refers to disease recurring at the ipsilateral breast itself following breast-conserving surgery, or at the chest wall following mastectomy; regional recurrence refers to disease recurring in the ipsilateral axillary, supraclavicular, infraclavicular, or internal mammary nodal basins -- the same anatomic basins established in Section 02's lymphatic drainage discussion (Slide 8) and formalized into the N-category system in Section 10 (Slide 37). Taken together, local and regional recurrence (jointly termed locoregional recurrence, LRR) occurs in approximately 5 to 10% of patients at ten years with modern,

guideline-concordant therapy -- a meaningfully lower rate than in the pre-modern-radiation, pre-modern-systemic-therapy era, reflecting the cumulative benefit of the local and systemic therapy advances documented throughout Sections 12 and 13.

The single most important, and most frequently misunderstood, distinction on this slide is between a true ipsilateral breast tumor recurrence (IBTR) and an entirely new ipsilateral primary tumor arising independently in the same, previously treated breast -- these are not the same clinical entity, despite both technically representing "cancer found again in the same breast," and distinguishing them requires evaluating three specific criteria together: anatomic location, time interval, and tumor biology. A true recurrence is characterized by a new lesion arising within approximately 2 to 3 centimeters of the original tumor bed, sharing the same histologic type and biomarker profile as the original tumor, and occurring within less than five years of the original diagnosis -- these features together indicate that the "new" tumor almost certainly represents residual or re-emergent disease from the original malignant clone rather than a genuinely independent carcinogenic event. A new ipsilateral primary, by contrast, is indicated by any one of three alternative features: a different quadrant location from the original tumor (implying an anatomically distinct site of origin), an interval of five years or more since the original diagnosis (long enough that residual micrometastatic disease from the original tumor becoming clinically evident this late is less likely than an independent new carcinogenic event, though this is a probabilistic rather than absolute distinction), or different histology or biomarker profile from the original tumor (direct evidence, via pathology, that this is biologically a distinct tumor). This distinction carries direct prognostic and management significance, restated in the summary table at the top of this slide: a true IBTR, while requiring salvage treatment, still carries a relatively favorable prognosis, with 60 to 70% ten-year overall survival achievable with appropriate salvage therapy (detailed in Slide 63); a new ipsilateral primary is managed, and generally carries a prognosis, akin to any newly diagnosed primary breast cancer -- meaning it should be worked up, staged, and treated according to its own newly determined stage and biology, rather than being conflated with, or treated as an extension of, the original cancer.

Chest wall recurrence, occurring specifically after mastectomy (since there is no remaining breast tissue for a true IBTR-pattern recurrence to occur within), presents as skin or soft tissue disease at or near the mastectomy scar, and carries a meaningfully guarded prognosis: 25 to 40% of patients with an isolated chest wall recurrence will develop distant metastatic disease within two years, reflecting that a chest wall recurrence, while itself locoregional, often signals a more widespread underlying disease process that has not yet become clinically detectable at distant sites. Regional (nodal) recurrence -- affecting the ipsilateral axillary, supraclavicular, or internal mammary basins -- likewise carries meaningful systemic implications, with 20 to 30% of patients eventually developing distant metastatic disease, again reflecting that isolated regional nodal recurrence, much like chest wall recurrence, often heralds broader systemic disease.

Distant metastatic recurrence -- disease at any site beyond the locoregional basins already discussed -- carries the most guarded prognosis of the recurrence categories on this slide, with median overall survival currently in the range of 2 to 5 years, a figure that, importantly, continues to improve as the systemic therapy advances detailed throughout Section 13 through 15 progressively extend survival even in this incurable disease setting. The specific pattern of distant spread mirrors exactly what was established in Section 03's metastatic cascade discussion (Slide 12): bone is the single most common site of distant recurrence, accounting for 50 to 70% of cases and frequently representing the first site of distant disease detected; visceral sites follow in a predictable order of relative frequency, with the liver involved in 20 to 30% of cases and the lung in 10 to 15%, while brain metastasis, though less common in raw frequency terms, shows the same striking subtype-selectivity already established in Section 03 -- occurring disproportionately in HER2-positive and triple-negative disease. Leptomeningeal disease, while rare in absolute terms, likewise shows this same HER2-positive and

triple-negative subtype predilection and carries a particularly poor prognosis even relative to other patterns of central nervous system involvement.

Finally, this slide restates and reinforces the subtype-specific temporal recurrence pattern already established in Section 03 (Slide 12), because it directly informs the follow-up intensity and duration discussed in Section 19 (Slide 72): estrogen-receptor-positive tumors show a distinctive capacity for late recurrence, occurring anywhere from five to twenty or more years after initial diagnosis, with the annual hazard of recurrence following a bimodal pattern across this extended timeframe -- this is the direct biological rationale for extended-duration adjuvant endocrine therapy (Section 13, Slide 49) and for the recognition that an ER-positive patient's disease-free interval does not "reset" her risk to zero even many years after diagnosis. Triple-negative and HER2-positive tumors, by contrast, show a front-loaded recurrence pattern, with the annual hazard peaking early -- around two years post-diagnosis for triple-negative disease and three to four years for HER2-positive disease -- before falling off sharply thereafter, meaning a patient who remains disease-free at five years with either of these more aggressive subtypes has a substantially more reassuring long-term prognosis than an ER-positive patient reaching the same five-year milestone, whose late-recurrence risk window remains open for a much longer subsequent period.

SLIDE 63 -- Locoregional Recurrence -- Management Algorithm

Slide content recap:

- High grade + large tumor size (T3-4)
- Positive or close surgical margins after prior BCS
- No prior radiation therapy to the breast/chest wall
- Young age (<40): higher absolute LRR risk independent of other factors
- Lymphovascular invasion (LVI): strong predictor of nodal/locoregional recurrence
- Triple negative and HER2-enriched subtypes: early LRR within 2-3 years
- ER-negative tumors: front-loaded recurrence pattern; no late recurrences
- Full staging workup before local treatment
- Isolated chest wall: excision (wide local) + RT (if not previously irradiated)
- Previously irradiated field: re-irradiation possible at expert centers (IMRT/SBRT)
- Systemic therapy: by current tumor biomarkers; resistance patterns
- Prognosis: isolated chest wall LRR -> 25-35% 5-yr OS
- Concurrent distant mets found: systemic therapy is primary; local control adjunctive
- re-staging: CT C/A/P + bone scan (or PET) -> exclude distant disease
- Surgical salvage: mastectomy (standard for IBTR after BCS + prior RT)
- Second BCS: possible if NO prior RT and lesion is amenable to resection with clear margins
- Post-mastectomy RT: required if chest wall not previously irradiated
- Systemic therapy: treat by new biomarker profile (rebiopsied sample)
- Prognosis: 5-yr OS ~60-70% for isolated IBTR (better if ER+, long disease-free interval)
- Ipsilateral axillary recurrence after prior SLNB: full ALND (levels I-III)
- Axillary recurrence after prior ALND: surgical excision +/- radiotherapy
- Supraclavicular recurrence: radiotherapy +/- systemic; surgical rarely feasible
- All regional recurrences: full staging to exclude distant mets
- Systemic therapy: based on new biomarker testing + prior treatment history
- Isolation perfusion: selective for in-transit (rare in BC); HILP (hyperthermic limb)

Detailed Explanation and Source Context

The risk factors for locoregional recurrence catalogued on this slide draw together adverse prognostic features already established individually across earlier sections, now specifically framed around their predictive value for recurrence rather than for initial-treatment decision-making: high tumor grade and large tumor size (T3-4, Section 10, Slide 37), positive or close surgical margins remaining after breast-conserving surgery (directly

related to the margin adequacy standards of Section 12, Slide 44), the absence of prior radiation therapy to the breast or chest wall (reflecting radiation's demonstrated recurrence-reduction benefit, Section 12, Slide 46), young age under 40 (which independently elevates absolute locoregional recurrence risk beyond what is explained by other correlated factors alone), lymphovascular invasion (already established as a strong predictor of both nodal involvement and distant recurrence in Section 11, Slide 40, and shown here to predict locoregional recurrence specifically as well), and the triple-negative and HER2-enriched molecular subtypes, which -- consistent with the temporal recurrence pattern established in Slide 62 -- show their locoregional recurrences concentrated early, typically within two to three years of initial treatment, with estrogen-receptor-negative tumors broadly following this same front-loaded pattern without the late-recurrence tail characteristic of ER-positive disease.

Before initiating any local treatment for a suspected locoregional recurrence, full re-staging is mandatory precisely because, as already established in Slide 62, isolated locoregional recurrence often heralds occult distant disease -- proceeding directly to local salvage surgery without first excluding distant metastasis would risk subjecting a patient with unrecognized Stage IV disease to a potentially unnecessary or inappropriately extensive local procedure, when systemic therapy should instead be the primary treatment focus in that scenario. For an isolated chest wall recurrence following mastectomy, management consists of wide local excision combined with radiation therapy, provided the chest wall has not been previously irradiated; if the field was previously irradiated (for instance, if post-mastectomy radiation was already given as part of initial treatment), re-irradiation remains possible at centers with expertise in highly conformal techniques such as intensity-modulated radiation therapy (IMRT) or stereotactic body radiation therapy (SBRT), which can deliver a therapeutic dose to the recurrence while limiting cumulative dose to previously irradiated normal tissue. Systemic therapy for chest wall recurrence should be selected based on the CURRENT tumor's biomarker profile (obtained by rebiopsy, following the same principle already established for metastatic disease in Section 14, Slide 57) rather than assuming the original tumor's biomarker profile remains unchanged, since resistance mechanisms and biomarker discordance can develop over the intervening treatment course. Isolated chest wall locoregional recurrence, when treated with this combined local-and-systemic approach, achieves a 25 to 35% five-year overall survival -- a guarded but not uniformly hopeless prognosis, distinctly better than that of frankly distant metastatic disease though worse than that of a true isolated IBTR discussed next. When re-staging reveals concurrent distant metastatic disease at the time a locoregional recurrence is identified, the treatment paradigm shifts: systemic therapy becomes the primary treatment modality, following the metastatic-disease algorithms of Section 14 (Slide 57) and Section 15, with local treatment of the locoregional site becoming an adjunctive measure aimed at symptom control and local disease management rather than curative intent.

Ipsilateral breast tumor recurrence after breast-conserving surgery follows its own specific management pathway, beginning, as with chest wall recurrence, with full re-staging (CT of the chest, abdomen, and pelvis, combined with a bone scan or PET-CT) to exclude distant disease before committing to local salvage therapy. Surgical salvage for IBTR occurring after prior whole-breast radiation is standardly mastectomy, since the breast tissue's radiation tolerance has already been used and a second course of whole-breast radiation (which a second breast-conserving procedure would require) is not safely deliverable -- this mirrors exactly the "prior breast irradiation" contraindication to breast-conserving surgery already established in Section 12 (Slide 44), simply applied now in the specific context of a recurrence rather than an initial diagnosis. A second breast-conserving procedure remains a possible option specifically in the less common scenario where NO prior radiation was given (for instance, in an older patient who was managed with breast-conserving surgery alone, without radiation, per the selective radiation-omission criteria of Section 12, Slide 46) and where the recurrent lesion is amenable to complete resection with clear margins. When mastectomy is chosen as salvage therapy, post-mastectomy radiation is required specifically if the chest wall has not been previously irradiated, following

the same general PMRT indication principles established in Section 12 (Slide 46). As with chest wall recurrence, systemic therapy selection should be guided by the biomarker profile of the newly rebiopsied recurrent tumor rather than assumed from the original diagnosis. Prognosis for an isolated, true IBTR treated with appropriate salvage therapy is meaningfully better than for chest wall recurrence, at approximately 60 to 70% five-year overall survival, with outcomes further improved specifically among patients with estrogen-receptor-positive disease and a longer original disease-free interval before the recurrence was detected -- both findings consistent with the general principle, established throughout this document, that favorable underlying tumor biology and a longer initial disease-free interval both predict a more favorable subsequent clinical course even after a recurrence has occurred.

Regional (nodal) recurrence management depends on the specific nodal basin involved and the extent of any prior axillary surgery. Ipsilateral axillary recurrence occurring after an initial sentinel lymph node biopsy alone (without completion dissection, as would occur, for instance, in a Z0011-eligible patient per Section 12, Slide 45, who subsequently develops an axillary recurrence) is managed with full axillary lymph node dissection at levels I through III, essentially "completing" the axillary staging and treatment that was appropriately deferred at initial diagnosis but is now indicated given the recurrence. Axillary recurrence occurring after a prior COMPLETE axillary dissection (a more challenging scenario, since the standard surgical treatment has already been performed once) is managed with surgical excision of the recurrent disease, with or without adjuvant radiotherapy, individualized based on the extent of recurrent disease and whether radiation has previously been delivered to that field. Supraclavicular nodal recurrence is managed primarily with radiotherapy, with or without systemic therapy, since surgical resection of supraclavicular disease is only rarely anatomically and oncologically feasible given the region's complex neurovascular anatomy and its typical association with more extensive underlying disease. As with the other recurrence categories on this slide, full staging to exclude concurrent distant metastatic disease is mandatory before committing to local treatment of any regional nodal recurrence, and systemic therapy selection should again be based on newly obtained biomarker testing together with careful consideration of the patient's prior treatment history and likely resistance patterns. Finally, isolated limb perfusion techniques (specifically hyperthermic isolated limb perfusion, HILP) are mentioned here only for completeness, as a technique more classically associated with managing in-transit melanoma metastases; their application in breast cancer is highly selective and rare, relevant only to unusual, isolated extremity recurrence patterns that fall well outside the standard locoregional recurrence scenarios this slide otherwise addresses.

SECTION 17 -- Special Situations

Pregnancy-associated breast cancer, male breast cancer treatment specifics, the diagnostic approach to nipple discharge, inflammatory breast cancer summarized as a cross-reference, occult primary breast cancer, bilateral breast cancer, and a catalogue of genuinely rare breast presentations that nonetheless appear with some regularity on examinations and in clinical practice.

SLIDE 64 -- Section Divider -- Special Situations

Detailed Explanation and Source Context

This section gathers the clinical scenarios that do not fit neatly into the stage-based algorithm of Section 14 because they require modification of standard management for a specific reason -- physiologic (pregnancy), demographic (male sex), diagnostic (occult primary, nipple discharge), or presentational (bilateral disease, rare histologies) -- and each entry cross-references back to the relevant foundational discussion earlier in this document rather than repeating it.

SLIDE 65 -- Special Situations: Pregnancy, Male BC, and Nipple Discharge

Slide content recap:

- PABC incidence: 1 in 3,000 pregnancies; most common cancer in pregnancy
- PABC diagnosis: US (safe in pregnancy); MRI without gadolinium acceptable
- PABC mammography: low radiation dose to fetus (shields); if clinically warranted -- OK
- PABC surgery: all trimesters safe; avoid sentinel node mapping with blue dye (Tc-99m acceptable)
- PABC chemotherapy: AVOID in 1st trimester (organogenesis); AC-T safe after 14 weeks
- PABC termination: NOT recommended for cancer treatment; does not improve prognosis
- 1% of all BC is male; higher in Sub-Saharan Africa (~6%)
- Male BC risk factors: BRCA2 (most important gene), Klinefelter (XXY), exogenous estrogen, radiation
- Male BC biology: ER+ in ~90%; HER2+ uncommon; TNBC rare
- Male BC presentation: central retroareolar mass, often later than female BC
- Male BC treatment: MRM (standard); BCS rare (limited breast tissue); tamoxifen for ER+
- Aromatase inhibitors less effective in men (androgens compete); tamoxifen preferred
- NCCN 2026: BRCA2 male carriers -> annual PSA from age 40-45 (prostate Ca risk)
- PABC taxanes: use with caution in 2nd/3rd trimester; limited data
- PABC trastuzumab: CONTRAINDICATED in pregnancy (oligohydramnios, fetal renal failure)
- PABC radiation: AVOID during pregnancy (fetal dose exposure); defer to postpartum
- PABC tamoxifen: TERATOGENIC -- contraindicated; start after delivery/breastfeeding cessation
- PABC breastfeeding: not recommended during chemotherapy
- PABC prognosis: stage-for-stage equivalent to non-pregnant; delayed diagnosis -> worse outcomes
- Nipple discharge benign: bilateral, milky, multiple ducts, non-spontaneous -> hormonal/galactorrhea
- Nipple discharge concerning: unilateral, bloody/serosanguineous, single duct, spontaneous
- Bloody discharge: #1 cause = intraductal papilloma; #2 = DCIS (must exclude)
- Workup: bilateral mammogram + US +/- ductography +/- MRI +/- ductoscopy
- Subareolar excision: for persistent single-duct discharge after imaging
- Pathology: exclude DCIS; papilloma excision -> 10% upgrade to DCIS or invasive

Detailed Explanation and Source Context

Pregnancy-associated breast cancer (PABC), defined as breast cancer diagnosed during pregnancy or within the first year postpartum, occurs in approximately 1 in 3,000 pregnancies and is, notably, the most common malignancy diagnosed during pregnancy overall. Diagnostic imaging must be adapted to pregnancy's specific safety constraints: ultrasound, using no ionizing radiation, is entirely safe and is the preferred initial imaging

modality; MRI can be performed but should avoid gadolinium contrast, which crosses the placenta and has uncertain fetal safety; and mammography, despite involving ionizing radiation, delivers a fetal radiation dose low enough (particularly with appropriate abdominal shielding) to be considered acceptable when clinically indicated, reflecting a careful risk-benefit calculation that favors obtaining necessary diagnostic information over a largely theoretical fetal radiation concern at these low doses. Surgery is considered safe throughout all three trimesters of pregnancy, with one specific technical modification: sentinel lymph node mapping using isosulfan blue dye should be avoided, given case reports of maternal anaphylactoid reactions to the dye and theoretical concerns about fetal exposure, while the radioactive tracer technetium-99m sulfur colloid is considered acceptable, delivering a fetal radiation dose low enough to be safely used for this specific, targeted purpose. Chemotherapy timing follows the principle of avoiding first-trimester exposure specifically because this is the period of fetal organogenesis, when teratogenic risk is highest; the standard AC-then-taxane regimen (Section 13, Slide 50) is considered safe when initiated after 14 weeks of gestation, once organogenesis is complete, reflecting a substantial and reassuring body of accumulated experience with this specific regimen in pregnancy. Pregnancy termination is explicitly NOT recommended as a component of cancer treatment strategy, since terminating the pregnancy does not improve the mother's oncologic prognosis -- treatment can generally be appropriately sequenced and modified to address the cancer while continuing the pregnancy, as this slide's remaining content details.

Specific drugs and radiation therapy require particular caution or are outright contraindicated during pregnancy. Taxanes should be used with caution, and only in the second or third trimester, given the comparatively more limited safety data available for this drug class in pregnancy relative to the more extensively studied anthracycline-cyclophosphamide backbone. Trastuzumab is explicitly CONTRAINDICATED throughout pregnancy, given a well-documented association with oligohydramnios (reduced amniotic fluid) and fetal renal failure, reflecting HER2's important physiologic role in fetal renal development that trastuzumab's HER2-blocking mechanism disrupts. Radiation therapy should be avoided entirely during pregnancy given fetal radiation dose exposure concerns, with any radiation indicated by the patient's stage and treatment plan deferred until after delivery. Tamoxifen is teratogenic and therefore contraindicated throughout pregnancy, appropriately initiated only after delivery and after the cessation of any breastfeeding. Breastfeeding itself is not recommended during active chemotherapy administration, given the secretion of cytotoxic agents into breast milk. Despite these numerous necessary treatment modifications, pregnancy-associated breast cancer's prognosis, stage-for-stage, is equivalent to that of non-pregnant breast cancer -- the guarded outcomes sometimes observed in this population are attributable specifically to diagnostic delay (symptoms attributed to pregnancy-related breast changes, or diagnostic imaging deferred out of excessive caution) rather than to any inherently more aggressive tumor biology associated with pregnancy itself, underscoring the importance of prompt, appropriately-adapted (rather than avoided or excessively delayed) workup of any concerning breast finding during pregnancy.

Male breast cancer, already introduced physiologically and epidemiologically in Section 02 (Slide 9) and Section 03 (Slide 11), is restated here with its specific treatment considerations consolidated. Accounting for approximately 1% of all breast cancer cases globally, with a markedly higher relative frequency (approximately 6%) in Sub-Saharan Africa, male breast cancer's dominant risk factors are germline BRCA2 mutation (the single most important known genetic risk factor, as already established in Section 04, Slide 15), Klinefelter syndrome, exogenous estrogen exposure, and radiation exposure. Biologically, male breast cancer is overwhelmingly estrogen-receptor-positive (approximately 90% of cases), with HER2 positivity comparatively uncommon and triple-negative disease rare -- a biological profile that, together with the typically older age at diagnosis and the tendency toward central, retroareolar presentation (reflecting the smaller overall volume of male breast tissue, as already discussed in Section 02, Slide 9), shapes a treatment approach centered heavily on endocrine therapy. Given this limited breast tissue volume, modified radical mastectomy remains the standard surgical approach,

with breast-conserving surgery only rarely a feasible option; tamoxifen, rather than an aromatase inhibitor, is the preferred adjuvant endocrine agent specifically because aromatase inhibitors are less effective in men, since ongoing testicular and adrenal androgen production continues to provide substrate for peripheral aromatization to estrogen even when peripheral aromatase activity itself is pharmacologically blocked -- a mechanistic distinction from postmenopausal women, in whom ovarian estrogen production has already ceased, making aromatase inhibition far more completely effective at suppressing the total estrogen pool. The NCCN 2026 recommendation, already introduced in Section 02 (Slide 9), Section 04 (Slide 15), and Section 05 (Slide 20), that male BRCA2 carriers undergo annual PSA screening beginning at age 40 to 45 is restated here as a component of comprehensive male breast cancer survivorship care, reflecting the dual breast-and-prostate cancer risk this specific genetic finding confers.

Nipple discharge evaluation, introduced physiologically in Section 02 (Slide 9) and diagnostically in Section 07 (Slide 26), is given its complete, consolidated workup algorithm here. The clinical features distinguishing benign from concerning discharge are worth committing to memory as a paired contrast: benign discharge is typically bilateral, milky in character, expressed from multiple ducts, and elicited only with manual expression (non-spontaneous) -- a pattern most consistent with a hormonal or galactorrhea-type mechanism, as detailed physiologically in Section 02; concerning discharge, by contrast, is typically unilateral, bloody or serosanguineous, arising from a single duct, and spontaneous (occurring without manual expression). When bloody discharge specifically is present, the single most common underlying cause is a benign intraductal papilloma, but ductal carcinoma in situ is the second most common cause and must always be actively excluded before a benign etiology is accepted -- this is the identical principle already emphasized in Section 02 and Section 07. Workup proceeds with bilateral mammography and ultrasound as first-line imaging, with galactography (contrast injected directly into the discharging duct to outline any intraductal filling defect), MRI, or ductoscopy (direct endoscopic visualization of the ductal lumen) reserved as second-line studies when standard imaging is inconclusive. When discharge persists from a single, identifiable duct despite unrevealing imaging, subareolar excision of the involved duct (a microdochectomy) provides both definitive diagnosis and, if the underlying lesion is benign, definitive treatment. Pathologic evaluation of any excised papillary lesion must specifically exclude an associated DCIS or invasive component, since excision of a papilloma is found to reveal an upgrade to DCIS or invasive carcinoma in approximately 10% of cases -- directly paralleling the analogous upgrade-risk principle already established for LCIS in Section 08 (Slide 31) and for DCIS itself in Section 08 (Slide 32), reinforcing the broader theme throughout this document that percutaneous sampling of any breast lesion carries an inherent risk of under-representing more advanced disease present elsewhere within that same lesion.

SLIDE 66 -- Inflammatory BC, Local-Regional Recurrence, and Rare Cancers

Slide content recap:

- IBC clinical diagnosis: rapid onset; $\geq 1/3$ breast erythema + peau d'orange +/- nipple retraction
- IBC often misdiagnosed as mastitis/abscess -- key: NO fever response to antibiotics
- IBC always T4d; Stage $\geq$ IIIB at minimum
- IBC mandatory sequence: biopsy -> biomarkers -> neoadjuvant chemo -> MRM -> PMRT
- IBC NEVER primary surgery; surgery without prior chemo = wrong approach
- HER2+ IBC: add trastuzumab + pertuzumab (excellent neoadjuvant response ~60-70% pCR)
- Occult primary BC (OPC) definition: axillary nodal metastasis with no identifiable breast primary on standard imaging
- OPC incidence: ~1% of breast cancer presentations
- OPC workup: mammogram + US -> MRI breast (identifies primary in 70% of OPC)
- If MRI identifies primary: treat as standard invasive BC
- If MRI negative: still likely breast origin (ER/PR/GCDFP-15 IHC)

- OPC management: ALND + whole-breast RT (vs MRM); systemic therapy as per biology
- Synchronous bilateral BC: diagnosed simultaneously (within 3 months)
- Metachronous bilateral BC: second primary BC ≥ 3 months after first
- Lifetime risk of contralateral primary BC: 0.5-1% per year after first BC
- Higher contralateral risk in: BRCA carriers, younger patients, lobular histology
- Bilateral BC staging: both primaries staged independently; combined stage uses worst primary
- Bilateral BC treatment: both breast surgeries staged (immediate or sequential); systemic therapy by higher-risk tumor
- Bilateral synchronous IBC: extremely rare; treat each breast with neoadjuvant
- Dermal metastasis (breast carcinoma en cuirasse): skin involvement by lymphatic spread
- Lymphoma of breast: rare primary; biopsy essential; not amenable to surgery (treat systemically)
- Sarcoma of breast: excision with wide margins; no sentinel node; no routine chemotherapy
- Granular cell tumor: benign; subareolar; mimics BC clinically -- excision curative
- Hamartoma: mammographically distinct "breast-within-breast"; excision for diagnosis

Detailed Explanation and Source Context

This slide's summary of inflammatory breast cancer restates, in condensed cross-reference form, the complete IBC discussion already developed in full across Section 09 (Slide 35), Section 12 (Slide 44), Section 13 (Slide 52), and Section 14 (Slide 56) -- its repeated appearance across so many sections of this document is itself instructive, reflecting how many distinct aspects of breast cancer management (pathology, surgical technique, systemic therapy sequencing, and stage-specific algorithms) IBC's unique clinical behavior touches and modifies. The core diagnostic features -- rapid-onset erythema involving at least one-third of the breast, accompanied by peau d'orange and often nipple retraction -- remain the clinical hallmark, and the recurring emphasis that IBC is frequently misdiagnosed as mastitis or a breast abscess, distinguished by the critical clue of no fever response to a trial of antibiotics, is repeated here because this diagnostic pitfall, if not actively guarded against, is precisely what delays the correct diagnosis and the correspondingly urgent, correctly-sequenced treatment this disease requires. IBC's automatic T4d, Stage IIIB-or-higher staging regardless of underlying tumor size, and its absolute, non-negotiable mandatory treatment sequence -- biopsy and full biomarker confirmation, then neoadjuvant chemotherapy, then modified radical mastectomy, then post-mastectomy radiation -- are restated with the same emphasis given throughout this document precisely because operating first, without preceding systemic therapy, represents a genuinely incorrect and consequential sequencing error rather than a minor variation in acceptable practice. For HER2-positive IBC specifically, adding trastuzumab and pertuzumab to the neoadjuvant chemotherapy backbone achieves a notably favorable pathologic complete response rate of approximately 60 to 70%, representing one of the more encouraging outcomes achievable within this generally high-risk disease category, and reflecting the broader theme, established throughout Section 13, that HER2-targeted therapy has meaningfully improved outcomes specifically within the HER2-positive subset of an otherwise more guarded-prognosis disease presentation.

Occult primary breast cancer is a distinct and genuinely important diagnostic scenario, defined as the presentation of axillary nodal metastasis with no identifiable primary tumor detectable on standard breast imaging (mammography and ultrasound) -- this scenario accounts for approximately 1% of breast cancer presentations and represents exactly the clinical situation already flagged in Section 07 (Slide 26) as warranting breast MRI before proceeding to biopsy of the axillary node. The recommended workup sequence proceeds from standard mammography and ultrasound (which, by the definition of this clinical scenario, fail to identify a primary lesion) to breast MRI, which -- given its superior sensitivity, already established in Section 07 (Slide 27) -- successfully identifies the occult primary tumor in approximately 70% of these cases. When MRI does identify a primary lesion, the patient is treated according to the standard invasive breast cancer algorithms developed throughout Sections 12 through 14, now directed at a specifically identified, localized target rather than proceeding blind. When MRI fails to identify a primary lesion (the remaining approximately 30% of cases), the axillary metastasis is nonetheless still considered most likely to represent an occult breast primary, based on

immunohistochemical staining of the metastatic tissue for markers such as estrogen receptor, progesterone receptor, and GCDFP-15 (a marker relatively specific for breast, and specifically apocrine, origin) that support a breast-origin diagnosis even in the absence of a radiographically identifiable primary tumor. Management of true occult primary breast cancer, when no primary is ever identified despite thorough workup, consists of axillary lymph node dissection combined with whole-breast radiation therapy (as an alternative to mastectomy, since there is no localized mass to excise via mastectomy in the first place), together with systemic therapy selected according to the biology (receptor status) of the metastatic tissue that has been sampled.

Bilateral breast cancer is classified by timing relative to the first diagnosis: synchronous bilateral disease is diagnosed within three months of the index diagnosis (essentially, discovered together, or very shortly after one another), while metachronous disease represents a second, contralateral primary arising three months or more after the first diagnosis -- this same synchronous-versus-metachronous distinction and its approximate 1 to 3% prevalence for the synchronous form was already introduced in the clinical examination discussion of Section 07 (Slide 26). The ongoing lifetime risk of developing a metachronous contralateral primary after an initial breast cancer diagnosis is approximately 0.5 to 1% per year, persisting indefinitely rather than declining to zero after some fixed interval -- directly consistent with the contralateral-breast-cancer risk figure already established as a risk factor in its own right in Section 04 (Slide 14). This contralateral risk is further elevated specifically in BRCA carriers (reflecting the bilateral gene-driven risk already discussed in Section 05), in younger patients, and in patients with lobular histology (reflecting invasive lobular carcinoma's demonstrated propensity toward bilateral and multifocal disease, already established in Section 08, Slide 31). When bilateral breast cancer is identified, each primary tumor is staged entirely independently using its own T, N, and M findings, though the overall, combined patient-level stage assigned for prognostic and broad treatment-planning purposes uses whichever of the two primaries carries the WORSE (higher) individual stage. Treatment likewise addresses each breast's surgical management as its own staged procedure (which may be performed simultaneously or sequentially, depending on surgical planning and patient preference), while systemic therapy selection is generally guided by the biology of the higher-risk of the two tumors, ensuring the systemic treatment plan adequately addresses the more aggressive of the two disease processes. The specific, extremely rare scenario of bilateral synchronous inflammatory breast cancer, while exceptionally uncommon, is managed by applying the standard, mandatory IBC neoadjuvant-first treatment sequence independently to each affected breast.

The slide closes with a brief catalogue of genuinely rare breast presentations, included specifically because their rarity does not eliminate their clinical or examination relevance. Dermal metastasis, sometimes termed breast carcinoma en cuirasse (reflecting its classic appearance resembling a tight, armor-like band of thickened, indurated skin), results from extensive lymphatic spread of tumor within the dermis, representing a particularly advanced and difficult-to-manage local disease pattern. Primary lymphoma of the breast is a rare entity requiring biopsy for definitive diagnosis (since it can mimic a primary breast carcinoma both clinically and radiographically) and, once correctly diagnosed as lymphoma rather than carcinoma, is treated with systemic lymphoma-directed therapy rather than with the surgical and breast-cancer-specific systemic approaches detailed throughout this document -- surgery has no primary therapeutic role in breast lymphoma. Primary sarcoma of the breast, similarly rare, is managed surgically with wide excision margins, without sentinel lymph node biopsy (reflecting sarcomas' characteristic hematogenous, rather than lymphatic, metastatic pattern, exactly as already established for phyllodes tumor and angiosarcoma in Section 09, Slide 35) and without routine chemotherapy, following general soft-tissue sarcoma management principles rather than breast carcinoma treatment principles. Granular cell tumor is a benign, typically subareolar lesion that can clinically mimic breast cancer (presenting as a firm, sometimes fixed-feeling mass) but is cured by simple excision once its benign nature is confirmed pathologically. Hamartoma of the breast presents with a mammographically distinctive "breast-within-a-breast" appearance (reflecting a well-circumscribed lesion containing a mixture of normal glandular, fibrous, and fatty

tissue in an abnormal arrangement, encapsulated within otherwise normal surrounding breast parenchyma) and, like granular cell tumor, is managed with excision primarily to confirm the diagnosis, after which no further specific treatment is required.

SECTION 18 -- Ethiopian Guideline 2024 Integration

The full structural architecture of Ethiopia's first-ever national breast cancer guideline -- its three pillars, its scope-of-practice mapping across the health system's three tiers, its multidisciplinary team and patient navigation approach, and a direct, unambiguous comparison against the international standard of care naming each specific gap and its priority action.

SLIDE 67 -- Section Divider -- Ethiopian Guideline 2024

Detailed Explanation and Source Context

[ETHIOPIA] While Ethiopian guideline notes have appeared at specific points throughout this document (Sections 02, 03, 06, 07, and 15), this section consolidates the guideline's own overall architecture as a unified whole, rather than presenting it only piecemeal alongside the international-standard topic it most directly modifies. This structural, whole-guideline view is essential for understanding the FMOH 2024 guideline as a coherent national health-system strategy, not merely a list of scattered, topic-specific adaptations.

SLIDE 68 -- Ethiopian BC Guideline 2024 -- Overview and Three Pillars

Slide content recap:

- [ETHIOPIA] First-ever national BC guideline for Ethiopia (2024-2028)
- 16,133 new BC cases/year; ASI 41.5/100,000 -- #1 cancer in Ethiopian women
- Overall Goal: reduce BC mortality aligned with WHO Global Breast Cancer Initiative (GBCI)
- Three-tier health system: primary (PHCU) -> secondary (general hosp.) -> tertiary (specialized)
- 3 Targets (60:60:80 aligned with WHO GBCI): 60% of invasive BC diagnosed at Stage I or II
- 60-day turnaround from referral to full diagnosis + start treatment
- 80% of patients completing multidisciplinary treatment
- Pillar 1 -- Health Promotion and Early Detection: Goal >60% breast cancer detected at Stage I and II
- CBE from age >=30 annually (no population mammography)
- BSE (breast self-exam) education: women >=20 years monthly
- Integration with MCH services: ANC, postnatal, family planning
- Health Extension Workers (HEWs): trained for CBE and referral
- School education on BC risk factors for children/adolescents
- Community mobilization: faith-based, community-based orgs
- Pillar 2 -- Timely Diagnosis: Goal = full diagnosis (evaluation + imaging + pathology) within 60 days
- Triple assessment: CBE + US +/- mammography + FNAC/CNB
- FNAC first approach (resource-pragmatic) -> CNB if needed
- Diagnostic algorithm: women >=30 with lump/skin change/nipple change -> US + FNAC -> confirm/refer
- Secondary hospitals: ideal location for diagnostic services (geographic access)
- Pathology: histology + ER/PR; HER2 testing expanding at tertiary
- Pillar 3 -- Comprehensive BC Management: Goal 80% completing multidisciplinary treatment
- MDT approach: surgeon + oncologist + pathologist + radiologist + social worker + nurse
- MDT currently only at tertiary centers (Tikur Anbessa, JUSH, ALERT)
- Treatment available: surgery (MRM standard), chemotherapy (24 public hospitals)
- Radiotherapy: available at TASH + 2 regional hospitals; 4 more in development
- Hormonal therapy: tamoxifen routinely available
- HER2-targeted: expanding but not universally available

Detailed Explanation and Source Context

[ETHIOPIA] The Ethiopian Federal Ministry of Health's 2024 to 2028 National Breast Cancer Guideline represents the country's first-ever dedicated, national-level breast cancer clinical guideline, developed in direct

response to the epidemiologic reality already established in Section 03 (Slide 11): 16,133 new cases annually, an age-standardized incidence rate of 41.5 per 100,000, and breast cancer's position as the single most common cancer diagnosed among Ethiopian women. The guideline's overall goal -- reducing breast cancer mortality -- is explicitly aligned with, and structured around, the World Health Organization's Global Breast Cancer Initiative (GBCI) framework, and is deployed across Ethiopia's existing three-tier health system architecture, moving from primary health care units, through general (secondary) hospitals, to specialized (tertiary) hospitals -- this tiered structure, and the specific scope of practice assigned to each tier, is developed in complete, granular detail in the very next slide (Slide 69).

The guideline's central, quantified strategy is the WHO GBCI-aligned "60:60:80" target framework, already introduced in Section 03 (Slide 11) and restated here as the organizing structure around which the guideline's three pillars are each built: at least 60% of invasive breast cancers should be diagnosed at Stage I or II (an early-detection target, addressed by Pillar 1), a maximum of 60 days should elapse from initial referral to completion of full diagnostic workup and treatment initiation (a diagnostic-efficiency target, addressed by Pillar 2), and at least 80% of diagnosed patients should complete their full, multidisciplinary treatment course (a treatment-completion target, addressed by Pillar 3) -- these three numeric targets are not independent goals but a deliberately sequenced continuum, since a patient must first be found early (Pillar 1), then diagnosed promptly (Pillar 2), before treatment completion (Pillar 3) becomes a meaningful, achievable goal at all.

Pillar 1, Health Promotion and Early Detection, targets a shift toward greater than 60% of breast cancer being detected at Stage I or II, and is built on the CBE-centered screening strategy already detailed in full in Section 06 (Slide 24): clinical breast examination from age 30 upward performed annually, breast self-examination education for women from age 20 upward performed monthly, deliberate integration of breast health messaging and CBE delivery into existing maternal and child health service touchpoints (antenatal care, postnatal care, and family planning clinics), and a workforce strategy that specifically trains Health Extension Workers -- Ethiopia's community-level health workforce cadre -- in CBE technique and appropriate referral pathways. Pillar 1 additionally extends beyond the clinical encounter into broader public health messaging: school-based education on breast cancer risk factors for children and adolescents, aimed at building long-term population health literacy, and community mobilization efforts explicitly engaging faith-based and community-based organizations, recognizing that trusted community and religious institutions can meaningfully extend the reach of health messaging beyond what the formal health system alone can achieve, particularly in a setting where trust in and access to formal health services may be variable.

Pillar 2, Timely Diagnosis, targets completion of the full diagnostic evaluation -- clinical assessment, imaging, and pathology -- within 60 days, and restates the diagnostic algorithm already detailed in full in Section 07 (Slide 29): triple assessment combining clinical breast examination, ultrasound (with mammography where available) as the imaging component, and fine-needle aspiration cytology as the initial pathology modality (escalating to core needle biopsy when needed), applied to any woman aged 30 or older presenting with a breast lump, skin change, or nipple change, following the same referral algorithm already established in Section 07. Secondary (general) hospitals are specifically identified as the ideal location for delivering this diagnostic service tier, reflecting a deliberate health-system-design choice to position diagnostic capacity at a level of the system that is geographically more accessible to the population than tertiary centers alone would be, while still commanding more diagnostic resources and capability than the most peripheral primary-care level. Pathology capacity at this stage of the pathway currently includes routine histology together with estrogen and progesterone receptor testing, with HER2 testing capacity described as still expanding, currently concentrated at tertiary centers -- this specific, partial HER2 testing gap was already flagged explicitly as one of the named comparison points in Section 07 (Slide 29) and is revisited again in the direct gap-comparison table of Slide 70.

Pillar 3, Comprehensive Breast Cancer Management, targets 80% of diagnosed patients successfully completing their full multidisciplinary treatment course, and is built around a multidisciplinary team model bringing together a breast surgeon, medical oncologist, pathologist, radiologist, social worker, and specialized nursing support -- a composition directly analogous to the multidisciplinary team model expected at any well-resourced international cancer center, but, as this slide states directly, currently available in Ethiopia only at a small number of tertiary institutions, specifically named here as Tikur Anbessa Specialized Hospital, Jimma University Specialized Hospital, and ALERT Hospital. Treatment capacity across the broader health system is described with specific figures that make the scope of current capability, and its limitations, concrete: surgery, with modified radical mastectomy as the standard operation, is described as broadly available; chemotherapy is available at 24 public hospitals nationally -- a specific, named number that gives a tangible sense of current treatment infrastructure scale relative to the country's population and disease burden; radiotherapy, by contrast, remains far more geographically concentrated, currently available only at the Tikur Anbessa Specialized Hospital together with two additional regional hospitals, with four further radiotherapy centers described as currently in development, reflecting an active, ongoing capacity-expansion effort rather than a static, unchanging gap; hormonal therapy, specifically tamoxifen, is described as routinely available, reflecting both its low cost and its long-established position in the essential medicines supply chain; and HER2-targeted therapy is described as an expanding but not yet universally available service, directly consistent with the trastuzumab-access gap already discussed in Section 15 (Slide 60).

SLIDE 69 -- Ethiopia -- Treatment Scope by Healthcare Tier

Slide content recap:

- Health Post/Health Center (3,000-40,000 population): CBE training, BHA education, identify suspicious findings, first contact -> refer to Primary Hospital or higher
- Primary Hospital/PHCU (60,000-100,000 population): CBE, risk factor counseling, basic breast US (if available), FNAC (selected), initiate referral -> General Hospital for imaging/biopsy/staging
- General/Secondary Hospital (500,000-1.5 million population): full diagnostic imaging (US, mammo), CNB, staging, basic chemotherapy, MRM/BCS surgery, follow-up -> Specialized Hospital for complex surgery/RT/HER2/MDT
- Specialized/Tertiary Hospital (3.5-5 million population): full MDT, complex surgery (reconstruction), radiotherapy, all systemic therapies, advanced diagnostics, research; hub for outreach to secondary level
- Multidisciplinary Team: required for all BC management decisions at tertiary
- MDT members: breast surgeon, medical oncologist, radiation oncologist, radiologist, pathologist, clinical nurse specialist, social worker
- Second consultation required when discussing cancer management plans
- Patient navigation program: standardized follow-up contact to reduce loss to follow-up
- Navigator: reduces loss to follow-up between referral levels
- Clear referral pathway written: from primary -> tertiary care
- Appointment calendar: standardized scheduling tools (Appendix D)
- Psychosocial support: integrate social worker into MDT
- Patient support groups: connect survivors with newly diagnosed patients
- Financial navigation: high cost is key barrier to treatment completion

Detailed Explanation and Source Context

[ETHIOPIA] This slide provides the granular, tier-by-tier scope-of-practice mapping that operationalizes the three-tier health system architecture introduced in Slide 68, and its inclusion of specific population-catchment figures for each tier is a deliberate design choice that reveals the underlying health-system planning logic: capability and resource intensity increase precisely as population catchment (and, implicitly, the case volume and specialized-service viability that a larger catchment population supports) increases. The health post or health center level, serving a population of roughly 3,000 to 40,000, functions purely as the first point of contact

-- trained in clinical breast examination and breast health awareness education, tasked with identifying suspicious findings and referring them onward, but performing no diagnostic imaging or tissue sampling at this level at all. The primary hospital (PHCU) level, serving a somewhat larger population of 60,000 to 100,000, adds clinical breast examination and risk factor counseling as a more formalized service, together with basic breast ultrasound where locally available and fine-needle aspiration cytology in selected cases, while still generally initiating referral onward for definitive imaging, biopsy, and staging rather than completing that full workup at this level. The general (secondary) hospital level, serving a substantially larger population of 500,000 to 1.5 million, is where full diagnostic imaging capability (ultrasound and, where available, mammography), core needle biopsy, complete staging, basic chemotherapy administration, and definitive surgery (both modified radical mastectomy and, where appropriate, breast-conserving surgery) become available -- this is precisely the tier identified in Slide 68 as the intended locus of Pillar 2's timely-diagnosis strategy. The specialized (tertiary) hospital level, serving the largest catchment population of 3.5 to 5 million, is where the full multidisciplinary team, complex reconstructive surgery, radiotherapy, the complete range of systemic therapies (including, where accessible, HER2-targeted agents), advanced diagnostic capability, and clinical research activity are concentrated -- and this tier is explicitly described as functioning as a hub that should provide outreach support back down to the secondary level, an important design detail indicating that the tiered system is intended to function as an integrated, mutually-supporting network rather than as a set of entirely siloed, independently operating service levels.

Multidisciplinary team-based decision-making is described as a required component of breast cancer management specifically at the tertiary level, with a team composition -- breast surgeon, medical oncologist, radiation oncologist, radiologist, pathologist, clinical nurse specialist, and social worker -- that closely mirrors internationally recognized MDT models, ensuring that treatment planning integrates surgical, medical, radiation, imaging, pathologic, nursing, and psychosocial perspectives together before a definitive management plan is finalized. The guideline specifically requires a second, follow-up consultation whenever cancer management plans are being discussed, reflecting a deliberate design choice to ensure patients and their families have adequate opportunity to process complex treatment information and ask questions, rather than being expected to absorb and consent to a complex multidisciplinary treatment plan within a single encounter.

Because Ethiopia's tiered referral structure inherently requires patients to move between different facilities and levels of care over the course of their diagnostic and treatment journey -- a structural feature that creates real opportunity for patients to be lost to follow-up at any transition point -- the guideline places specific emphasis on patient navigation as a dedicated strategy: a formal patient navigation program providing standardized follow-up contact is designed specifically to reduce loss to follow-up, with a designated patient navigator functioning as a continuous point of contact and support specifically across the transitions between referral levels, where the risk of a patient becoming lost to the system is judged to be highest. Supporting this navigation function, the guideline calls for clearly written, explicit referral pathways connecting the primary level all the way through to tertiary care, together with standardized appointment scheduling tools (referenced in the guideline's own Appendix D) intended to reduce the ad hoc, informal scheduling practices that can otherwise contribute to missed appointments and care discontinuity. Beyond pure logistics, the guideline explicitly incorporates psychosocial support as an integrated component of care, calling for a social worker to be embedded directly within the multidisciplinary team rather than functioning as a separate, optional referral, and for the establishment of patient support groups specifically designed to connect breast cancer survivors with newly diagnosed patients, leveraging peer support as a recognized component of comprehensive cancer care. Finally, the guideline names financial navigation as an explicit priority, identifying the high cost of care as a key, specifically named barrier to treatment completion -- this direct acknowledgment that financial barriers, not merely clinical or logistical ones, threaten the guideline's own Pillar 3 target of 80% treatment completion is an

important and candid feature of the document, correctly identifying cost as a structural determinant of outcome alongside the clinical and health-system factors detailed throughout the rest of this section.

SLIDE 70 -- Ethiopia versus International Standard -- Key Gaps and Priorities

Slide content recap:

- Screening: international annual mammography from age 40 (NCCN)/45 (ACS); Ethiopia CBE annually from age 30, mammography if available; priority: scale-up CBE, advocate mammography expansion
- Stage at diagnosis: international <30% Stage III-IV; Ethiopia 65-75% Stage III-IV; priority: shift to early detection (60:60:80 goal)
- Biomarker testing: international ER/PR/HER2/Ki-67 all specimens; Ethiopia ER/PR routine, HER2 expanding (inconsistent); priority: standardize HER2 testing nationally
- Genomic testing: international Oncotype DX RS for adjuvant decisions; Ethiopia not available; priority: not feasible short-term, future aspiration
- CDK4/6 inhibitors: international standard 1st-line MBC; Ethiopia not routinely available in public system; priority: WHO EML advocacy, access programs
- HER2-targeted: international trastuzumab standard, T-DXd 2nd line, pertuzumab; Ethiopia trastuzumab expanding, T-DXd not available; priority: secure trastuzumab supply chain first
- Immunotherapy: international pembrolizumab + chemo for TNBC Stage II-IV; Ethiopia not available; priority: long-term access program, WHO EML
- Radiotherapy: international available all centers, PMRT standard Stage III; Ethiopia limited centers (3); priority: expand RT centers (4 more planned)
- Systemic adjuvant: international capecitabine residual TNBC, T-DM1 residual HER2+; Ethiopia capecitabine increasingly available; priority: procure capecitabine for residual TNBC

Detailed Explanation and Source Context

[ETHIOPIA] This slide is the culminating, most direct expression of the entire Ethiopian-integration theme carried throughout this document -- a nine-row table placing the international standard of care and current Ethiopian practice side by side across every major domain of breast cancer management discussed in this document, with each row also naming a specific, actionable priority for closing that particular gap, rather than leaving the comparison as an abstract observation.

The screening gap restates, in its most condensed form, everything already established in Section 03 (Slide 11) and Section 06 (Slide 24): the international standard of annual mammography beginning at age 40 (per NCCN) or 45 (per ACS) is contrasted with Ethiopia's CBE-centered approach beginning at age 30, with mammography deployed only where available -- and the named priority action is explicitly twofold, calling both for continued CBE scale-up (the near-term, resource-appropriate lever) and for advocacy toward eventual mammography expansion (the longer-term, aspirational goal), rather than treating these as competing rather than sequential strategies. The stage-at-diagnosis gap restates the single starkest epidemiologic comparison in this entire document, already established in Section 03 (Slide 11): fewer than 30% of patients present at Stage III or IV internationally, compared with 65 to 75% in Ethiopia, with the named priority action being direct execution of the 60:60:80 target framework itself (Slide 68) -- this row functions as the outcome metric that every other row on this table is, in some sense, working to improve.

The biomarker testing gap restates the HER2-testing-consistency issue already flagged in Section 07 (Slide 29) and Section 18 (Slide 68): the international standard tests ER, PR, HER2, and Ki-67 on every specimen, while Ethiopian practice performs ER/PR testing routinely but HER2 testing only inconsistently, expanding but not yet standardized nationally -- the named priority is specifically to standardize HER2 testing at a national level, reflecting that this single biomarker gap has outsized downstream consequences, since it determines eligibility for trastuzumab and the broader anti-HER2 treatment cascade detailed in Section 13. The genomic testing gap

acknowledges that Oncotype DX and the related multigene assays detailed in Section 11 (Slide 41), used internationally to guide adjuvant chemotherapy decisions in hormone-receptor-positive disease, are simply not available in Ethiopia -- and the guideline is candid rather than aspirational-to-a-fault about the near-term feasibility of closing this specific gap, explicitly characterizing genomic testing access as "not feasible short-term" and instead framing it as a longer-term future aspiration, an honest acknowledgment that not every gap identified in this comparison can be prioritized or closed on the same timeline.

The CDK4/6 inhibitor gap and the HER2-targeted therapy gap both restate content already developed in full in Section 15 (Slide 60): CDK4/6 inhibitors, the international first-line standard for hormone-receptor-positive metastatic disease, are not yet routinely available in the Ethiopian public health system, with the named priority being advocacy through the WHO Essential Medicines List process together with the development of dedicated access programs; and while trastuzumab access is expanding, the newer HER2-targeted agent T-DXd remains entirely unavailable, with the named priority explicitly sequenced -- securing a reliable trastuzumab supply chain is named as the priority to establish FIRST, before pursuing access to newer, more advanced HER2-targeted agents, reflecting a pragmatic, foundational-before-advanced sequencing logic that recurs as a theme across several rows of this table. The immunotherapy gap likewise restates content from Section 13 (Slide 52) and Section 15: pembrolizumab combined with chemotherapy, the international standard for Stage II through IV triple-negative disease, is not available in Ethiopia, with the named priority being a long-term access program pursued again through the WHO Essential Medicines List mechanism, acknowledging this as a longer-horizon goal rather than a near-term achievable target.

The radiotherapy gap restates the specific facility count already established in Section 18 (Slide 68): radiotherapy is available at essentially every treating center internationally, with post-mastectomy radiation standard for Stage III disease as detailed in Section 12 (Slide 46) and Section 14 (Slide 56), while Ethiopia currently has only three radiotherapy centers nationally -- the named priority is direct and concrete: expand radiotherapy capacity, with four additional centers already specifically planned, restating the active capacity-building effort already noted in Slide 68 rather than treating this purely as a static, unaddressed deficiency. Finally, the systemic adjuvant therapy gap addresses the specific response-adaptive escalation strategies detailed in Section 13 (Slides 50 to 51) -- capecitabine for residual triple-negative disease per CREATE-X, and T-DM1 for residual HER2-positive disease per KATHERINE -- noting that capecitabine specifically is described as increasingly available in Ethiopia (representing genuine, incremental progress on this particular gap), with the named priority being to complete and secure capecitabine procurement specifically for the residual-TNBC indication, a comparatively achievable, cost-effective near-term target relative to some of the more resource-intensive gaps addressed elsewhere on this table.

Read as a whole, this table's structure -- pairing every identified gap with a specific, named, and appropriately time-horizoned priority action, from "achievable now" (capecitabine procurement, HER2 testing standardization) through "sequenced, foundational-first" (trastuzumab before T-DXd) to "long-term aspiration" (genomic testing, immunotherapy access) -- is itself the guideline's most important structural contribution: it transforms what could have been a purely descriptive list of deficiencies into an actionable, appropriately prioritized roadmap for progressively closing the gap between current Ethiopian practice and the 2026 international standard documented throughout the rest of this preparation note.

SECTION 19 -- Follow-Up and Quick Reference

ASCO-endorsed post-treatment surveillance guidelines and the deliberate evidence base for what surveillance testing should NOT be routinely performed, followed by a consolidated quick-reference comparison table, key numeric facts, and a drug mechanism-and-indication summary spanning the entire document.

SLIDE 71 -- Section Divider -- Follow-Up and Quick Reference

Detailed Explanation and Source Context

This final content section closes the clinical narrative of the document by addressing what happens after active treatment ends, and then serves a second, explicitly different purpose: consolidating the highest-yield numeric facts and comparisons from across all eighteen preceding sections into a single, rapid-review reference for exam preparation or quick clinical lookup.

SLIDE 72 -- Follow-Up Guidelines (ASCO Recommendations)

Slide content recap:

- H and P (History and Physical): every 3-6 months years 1-5, annually year 5+; focus on new symptoms, toxicity, recurrence signs
- Ipsilateral mammogram (after BCS): 6-12 months post-RT then annually years 1-5, annually year 5+; earlier if concerns; MRI in BRCA carriers
- Contralateral mammogram: annually both periods regardless of surgery type
- Pelvic exam: annually (tamoxifen -> endometrial risk); report abnormal bleeding immediately
- Bone density (on AIs): baseline then q1-2 years; calcium/Vit D + exercise; bisphosphonate if T-score <-2.0
- LFTs/CBC/tumor markers: NOT routinely recommended either period; only if clinically indicated
- CT/Bone scan/PET: NOT routine (asymptomatic) either period; only if symptomatic or signs of recurrence
- Genetic counseling: offer if not done at diagnosis; update testing criteria change (NCCN 2026 panels)
- KEY PRINCIPLE: no imaging or tumor markers (CA 15-3, CEA) in asymptomatic patients -- not proven to improve OS
- Symptoms drive investigations

Detailed Explanation and Source Context

ASCO's post-treatment surveillance guidelines are built around a single, deliberately evidence-based governing principle, stated explicitly as the "key principle" closing this slide: routine imaging and serum tumor marker testing (CA 15-3, CEA) in asymptomatic patients has NOT been shown to improve overall survival, and is therefore specifically NOT recommended as part of routine follow-up -- symptoms, not scheduled scans or blood tests, should drive further investigation. This principle may initially seem counterintuitive given how much of this document has emphasized early detection (Section 06), but the distinction is a critical one: population screening in an asymptomatic, previously-untreated woman aims to find a first cancer before it becomes symptomatic, a scenario with clearly demonstrated mortality benefit; surveillance for recurrence in an already-treated patient, by contrast, has not been shown to achieve an analogous survival benefit through routine imaging, because the trials that specifically tested intensive imaging-and-marker-based surveillance against symptom-directed follow-up alone found no difference in overall survival between the two strategies, reflecting that effective systemic therapy is generally initiated at the point recurrent disease becomes clinically apparent regardless of whether it was detected slightly earlier by a scheduled scan.

The recommended follow-up schedule accordingly focuses on a compact set of interventions with genuine demonstrated value. History and physical examination is recommended every three to six months for the first

five years after treatment, transitioning to annual examination thereafter, with the clinical focus specifically on eliciting new symptoms, assessing treatment-related toxicity, and examining for physical signs of recurrence -- this is, in essence, the ongoing clinical vigilance that makes the "symptom-driven investigation" principle actually workable in practice, since a thorough, regularly scheduled clinical encounter is what identifies the symptoms that should then trigger further, targeted investigation. Ipsilateral mammography, for patients who underwent breast-conserving surgery, begins 6 to 12 months after completion of radiation therapy and then continues annually, both during the first five years and beyond, with MRI substituted or added specifically in BRCA carriers given their already-established need for more intensive breast surveillance (Section 05, Slide 19) -- earlier or more frequent imaging is appropriate whenever a specific clinical concern arises. Contralateral mammography is recommended annually indefinitely, regardless of which type of surgery (breast-conserving surgery or mastectomy) was performed on the treated side, reflecting the ongoing contralateral breast cancer risk already quantified in Section 04 (Slide 14) and Section 17 (Slide 66) that persists throughout a patient's lifetime rather than resolving after some fixed interval. Annual pelvic examination is recommended specifically for patients on tamoxifen, given this drug's well-established association with increased endometrial cancer risk (already introduced in Section 06, Slide 23, and Section 13, Slide 49), with any abnormal vaginal bleeding requiring prompt reporting and evaluation rather than being deferred to the next scheduled annual visit. Bone density assessment via DEXA scan is recommended at baseline and then every one to two years specifically for patients receiving aromatase inhibitor therapy, given AIs' well-established accelerating effect on bone density loss (Section 13, Slide 49); general bone-health measures (adequate calcium and vitamin D intake, together with regular weight-bearing exercise) are recommended broadly, with pharmacologic bisphosphonate therapy added specifically when the bone density T-score falls below negative 2.0, following standard osteoporosis management thresholds applied in this AI-treated population.

Consistent with the governing principle stated above, liver function tests, complete blood count, and serum tumor markers are explicitly NOT recommended as routine surveillance tests during either the first five years or beyond, to be obtained only when a specific clinical indication (new symptoms, abnormal examination findings) prompts their use rather than as scheduled, routine screening. Likewise, CT scanning, bone scan, and PET imaging are NOT recommended routinely in an asymptomatic patient during either follow-up period, reserved specifically for patients who develop symptoms or examination findings raising genuine concern for recurrence -- this is the direct clinical application of the governing "symptom-driven investigation" principle to the specific imaging modalities that were used extensively during initial staging (Section 10, Slide 38) but are deliberately not carried forward into routine, asymptomatic surveillance. Finally, genetic counseling should be offered during follow-up to any patient for whom it was not already completed at the time of initial diagnosis, and testing criteria should be revisited and updated as guideline panels evolve -- directly reflecting the expanding, evolving gene panel already detailed in the NCCN 2026 update of Section 05 (Slide 20), since a patient tested some years earlier against an older, narrower gene panel may become newly eligible for expanded testing (PALB2, ATM, CHEK2, HOXB13) as guideline criteria are periodically updated.

SLIDE 73 -- Quick Reference -- Subtypes, Key Numbers, and High-Yield Distinctions

Slide content recap:

- LCIS vs DCIS -- Nature: risk marker (bilateral) vs obligate precursor (-> invasive if untreated)
- LCIS vs DCIS -- Laterality: bilateral risk vs unilateral (usually)
- LCIS vs DCIS -- Margins required: NO vs YES (≥ 2 mm)
- LCIS vs DCIS -- Radiation indicated: NO vs YES (after BCS)

- LCIS vs DCIS -- Axillary staging: NO (unless invasive) vs only if mastectomy or high suspicion of invasion
- LCIS vs DCIS -- Tamoxifen benefit: YES (BRCA2, ADH subgroup) vs YES (ER+ DCIS)
- Lifetime BC risk in US women: 12.5% (1 in 8)
- Mammography screening mortality reduction: ~15-25%
- Tamoxifen chemoprevention (P-1): 49% decreased in ER+ BC risk
- BCS = mastectomy for survival: 6 RCTs confirmed
- SLNB false-negative rate (dual tracer): ~5-8%
- Z0011: ALND vs SLNB alone (OS): no difference (N=891)
- Trastuzumab benefit (HERA): 50% decreased in recurrence
- pCR in TNBC with chemo: ~30-55%
- DCIS upgrade to invasive at excision: ~20%
- IMN location: intercostal spaces 1-6, within 3 cm of sternal edge
- Tamoxifen: SERM -- blocks ER in breast tissue; pre/post-menopausal ER+ adj; chemoprevention
- Letrozole/Anastrozole: non-steroidal AI -- blocks peripheral aromatization; post-menopausal ER+ adjuvant
- Trastuzumab: anti-HER2 mAb -- ADCC + signaling inhibition; HER2+ adjuvant + metastatic; 1 year duration
- Palbociclib/Ribociclib: CDK4/6 inhibitor -- arrests cell cycle G1; ER+/HER2- metastatic 1st line (+AI)
- Olaparib/Talazoparib: PARP inhibitor -- synthetic lethality in BRCA mut.; BRCA1/2-mutated HER2-neg metastatic BC
- T-DXd: ADC (T + deruxtecan); HER2+ 2nd line MBC; HER2-low MBC
- Sacituzumab govitecan: ADC (anti-TROP2 + SN-38); TNBC 2nd+ line; HR+ HER2- 3rd+ line

Detailed Explanation and Source Context

This quick-reference slide is deliberately structured as a rapid-review consolidation rather than as new content, and its value lies specifically in placing the document's most commonly confused, most frequently tested facts directly adjacent to one another for rapid comparison. The LCIS-versus-DCIS comparison table condenses the complete discussion of Section 08 (Slides 31 and 32) into six paired contrasts that should be internalized as a single mental unit: LCIS is fundamentally a bilateral RISK MARKER while DCIS is an obligate, unilateral anatomic PRECURSOR that will progress to invasive disease in a meaningful proportion of untreated cases; correspondingly, LCIS requires no specific margin clearance and no radiation, since there is no discrete anatomic lesion requiring complete surgical eradication, while DCIS requires a minimum 2-millimeter margin and standardly receives radiation after breast-conserving surgery, exactly as detailed in Section 08; axillary staging is not indicated for either lesion in its pure form, except that DCIS warrants sentinel node biopsy specifically when mastectomy is chosen or when invasion is highly suspected, following the forward-looking logic already explained in Section 14 (Slide 54); and tamoxifen provides a demonstrated risk-reduction benefit in both conditions, though through somewhat different populations and mechanisms -- in LCIS, benefit is most established in BRCA2 carriers and in the atypical-hyperplasia-associated risk context (Section 05, Slide 19), while in DCIS, the benefit applies specifically to estrogen-receptor-positive disease (Section 08, Slide 32).

The key numbers table assembles ten of the single most consequential, most frequently cited statistics from across this entire document, each traceable to its full discussion earlier: the 12.5% (one-in-eight) lifetime breast cancer risk in US women (Section 03, Slide 11); the 15 to 25% mortality reduction achieved by mammographic screening (Section 06, Slide 22); the 49% reduction in estrogen-receptor-positive breast cancer incidence achieved by tamoxifen chemoprevention in the NSABP P-1 trial (Section 05, Slide 19; Section 06, Slide 23); the six independent randomized controlled trials confirming breast-conserving surgery's survival equivalence to mastectomy (Section 01, Slide 4; Section 12, Slide 44); the 5 to 8% false-negative rate of sentinel lymph node biopsy using dual-tracer technique (Section 02, Slide 8; Section 12, Slide 45); the ACOSOG Z0011 trial's finding of no overall survival difference between axillary dissection and sentinel node biopsy alone in its specifically defined eligible population of 891 patients (Section 12, Slide 45); the approximately 50% reduction in recurrence risk achieved by adjuvant trastuzumab in the HERA trial (Section 01, Slide 4; Section 13, Slide 51); the 30 to 55% pathologic complete response rate achievable in triple-negative breast cancer with modern chemotherapy (Section 11, Slide 40; Section 13, Slide 52); the approximately 20% rate at which a DCIS diagnosis made on

core needle biopsy is upgraded to invasive carcinoma upon complete surgical excision (Section 08, Slide 32); and the precise anatomic location of the internal mammary nodal chain within intercostal spaces one through six, within three centimeters of the sternal edge (Section 02, Slide 8) -- together, these ten figures function as a compact numeric backbone spanning epidemiology, screening, chemoprevention, surgical technique, and systemic therapy across the entire document.

The drug mechanism-and-indication table performs the same consolidating function specifically for the systemic therapy agents most central to this document's treatment algorithms, pairing each drug with its precise mechanism of action and its core indication in a single line for rapid recall: tamoxifen, a selective estrogen receptor modulator that directly blocks estrogen receptor activity in breast tissue, used in both pre- and postmenopausal adjuvant and chemoprevention settings (Section 13, Slide 49); letrozole and anastrozole, non-steroidal aromatase inhibitors that block peripheral conversion of androgens to estrogen, restricted to postmenopausal adjuvant use and demonstrated superior to tamoxifen in this population (Section 13, Slide 49); trastuzumab, an anti-HER2 monoclonal antibody combining direct signaling inhibition with antibody-dependent cellular cytotoxicity, used across both adjuvant (fixed one-year duration) and metastatic HER2-positive settings (Section 13, Slide 51); palbociclib and ribociclib, CDK4/6 inhibitors that arrest the cell cycle at the G1 checkpoint, used as first-line therapy combined with an aromatase inhibitor in hormone-receptor-positive, HER2-negative metastatic disease (Section 14, Slide 57); olaparib and talazoparib, PARP inhibitors that exploit synthetic lethality specifically in BRCA-mutated tumor cells already deficient in homologous recombination repair, used in BRCA1/2-mutated, HER2-negative metastatic disease (Section 05, Slide 19; Section 14, Slide 57); [UPDATE] T-DXd, an antibody-drug conjugate combining trastuzumab with a topoisomerase I inhibitor payload, used as preferred second-line therapy in HER2-positive metastatic disease and as a newly established option in HER2-low metastatic disease across both hormone-receptor-positive and triple-negative biology (Section 11, Slide 42; Section 15, Slide 60); and [UPDATE] sacituzumab govitecan, an antibody-drug conjugate combining an anti-TROP2 antibody with an SN-38 cytotoxic payload, used as second-line-or-later therapy in triple-negative disease and as third-line-or-later therapy in hormone-receptor-positive, HER2-negative disease (Section 15, Slide 60) -- this final pair of entries deliberately closes the quick-reference table with the two agents most representative of the practice-changing, 2026-cycle antibody-drug conjugate advances that distinguish this revised document from what a pre-2026 edition of Schwartz or DeVita alone would contain.

SLIDE 74 -- Summary -- Key Take-Home Messages

Slide content recap:

- Breast cancer is biologically heterogeneous -- molecular subtyping (ER/PR/HER2/grade + genomics) drives modern management
- BCS + RT = mastectomy for survival; Z0011 supports SLNB alone in cT1-2N1 BCS patients -- ALND not needed
- ER+ BC: endocrine therapy backbone; Oncotype DX guides chemo; CDK4/6 inhibitors adjuvant for high-risk N+
- HER2+ BC: dual anti-HER2 (HP) neoadjuvant; residual disease -> T-DM1; T-DXd preferred 2nd-line MBC (ESMO 2026)
- TNBC: pembrolizumab + chemo neoadjuvant (KEYNOTE-522 pCR 65%); capecitabine for residual; SG/dato-DXd MBC
- [ETHIOPIA] CBE-based screening, 65-75% Stage III-IV, limited access to targeted therapies -- bridging the gap is the priority

Detailed Explanation and Source Context

This closing summary distills the six ideas that recur most persistently across all nineteen sections of this document into a final, single-page synthesis, and each one is worth restating precisely because of how many separate strands of evidence it draws together.

First, breast cancer's biological heterogeneity -- and the consequence that molecular subtyping, not anatomic stage alone, drives modern treatment selection -- is the organizing principle first fully established in Section 11

and applied throughout every subsequent treatment section; the AJCC 8th edition's "prognostic stage" innovation (Section 10, Slide 38), incorporating grade and receptor status directly into the staging system itself, is the single clearest formal acknowledgment of just how far this biology-over-anatomy principle has been institutionalized into standard practice.

Second, the equivalence of breast-conserving surgery plus radiation to mastectomy for survival, and the subsequent Z0011-driven finding that even sentinel-node-positive patients meeting specific criteria do not require completion axillary dissection, together trace the historical arc from Halsted's centrifugal, maximal-surgery paradigm (Section 01) to the modern, evidence-driven de-escalation approach detailed fully in Section 12 -- this progression from "more surgery is safer" to "the correct amount of surgery, no more, is safer" is arguably the single most important paradigm shift documented across this entire preparation note.

Third, fourth, and fifth, the subtype-specific systemic therapy backbones -- endocrine therapy with genomically-guided chemotherapy addition and CDK4/6 inhibition for hormone-receptor-positive disease; dual anti-HER2 blockade with response-adapted escalation to T-DM1 for residual disease, and T-DXd as the new preferred second-line metastatic agent, for HER2-positive disease; and pembrolizumab-based neoadjuvant immunotherapy with capecitabine escalation for residual disease, plus the newer TROP2-targeted antibody-drug conjugates in the metastatic setting, for triple-negative disease -- together represent the complete, current-as-of-2026 systemic treatment framework developed across Sections 11, 13, 14, and 15, and mastering the specific trial names and hazard ratios behind each (TAILORx, monarchE, KATHERINE, KEYNOTE-522, CREATE-X, DESTINY-Breast03/04, ASCENT) is what separates a superficial understanding of "which drug for which subtype" from a genuinely evidence-grounded one.

Sixth and finally, [ETHIOPIA] the Ethiopian guideline's own framing of its central challenge -- CBE-based rather than mammography-based screening, a 65 to 75% Stage III/IV presentation rate, and meaningfully limited access to the newest targeted and immunotherapy agents detailed in Section 15 -- is restated here as the document's closing thought specifically because it reframes everything preceding it: the biological and therapeutic sophistication documented throughout Sections 01 through 17 represents genuine, hard-won medical progress, but that progress delivers benefit only to the extent a health system can actually deliver it to patients, and the honest, specific, action-oriented gap analysis of Section 18 is this document's answer to the question of what "bridging that gap" concretely requires. The sources underlying this entire synthesis -- Schwartz's Principles of Surgery 11th Edition Chapter 17, DeVita's Cancer 10th Edition Chapter 106, the ESMO 2026 Metastatic Breast Cancer guideline, NCCN Breast Cancer version 6.2026, the JNCCN Genetic/Familial High-Risk Assessment version 2.2026, and the Ethiopian FMOH National Breast Cancer Guideline 2024-2028 -- are listed in full, with their specific role in this document, in the References section that follows.

References and Source Documents

This document synthesizes and elaborates the content of the 74-slide teaching deck "Breast Cancer Management -- Enhanced Presentation" using the following primary sources, cited throughout by short name.

Schwartz's Principles of Surgery

Brunicaardi FC (editor-in-chief), 11th Edition. Chapter 17: Diseases of the Breast. Parker CC, Damodaran S, Bland KI, Hunt KK. McGraw-Hill Education. Cited throughout as 'Schwartz Ch.17' for surgical anatomy, operative technique, classic trial data (NSABP B-06, Z0011), and the Halsted-Fisher historical paradigm.

DeVita, Hellman, and Rosenberg's Cancer: Principles and Practice of Oncology

10th Edition. Chapter 106: Cancer of the Breast. Burstein HJ, Harris JR, Morrow M. Wolters Kluwer. Cited throughout as 'DeVita Ch.106' for tumor biology, epidemiology, systemic therapy trial data, and molecular subtyping.

ESMO Clinical Practice Guideline -- Metastatic Breast Cancer

2026 Update. European Society for Medical Oncology, Annals of Oncology. Cited as 'ESMO 2026' for ESR1/PIK3CA-pathway agents, ESCAT and MCBS v2.0 scoring, and HER2-low/TNBC antibody-drug conjugate algorithms.

NCCN Clinical Practice Guidelines in Oncology -- Breast Cancer

Version 6.2026. National Comprehensive Cancer Network. Cited as 'NCCN v6.2026' for risk-adapted screening, adjuvant systemic therapy algorithms, and staging workup recommendations.

JNCCN -- NCCN Guidelines Insights: Genetic/Familial High-Risk Assessment

Version 2.2026: Breast, Ovarian, Pancreatic, and Prostate. Cited as 'JNCCN v2.2026' for BRCA1/2-associated prostate cancer risk, HOXB13, and multi-gene panel testing criteria.

Federal Ministry of Health, Ethiopia -- National Breast Cancer Guideline

2024-2028. Cited as 'Ethiopian FMOH 2024' for the three-pillar national framework, CBE-based screening protocol, health-system-tier scope of practice, and identified gaps versus international standard practice.

Key Clinical Trials Referenced Throughout This Document

Trial	Key Finding and Cross-Reference
NSABP B-06 / Milan I	BCS + RT vs mastectomy: equivalent overall survival (Section 01, 12)
ACOSOG Z0011	1-2 positive SLNs, cT1-2N0, BCS+WBRT: ALND omission non-inferior (Section 02, 12)
HERA / NSABP B-31	Adjuvant trastuzumab in HER2+ early BC: ~50% recurrence reduction (Section 01, 13)
SOFT / TEXT	Ovarian function suppression + endocrine therapy in high-risk premenopausal ER+ BC (Section 01, 13)
TAILORx / RxPONDER	Oncotype DX Recurrence Score-guided chemotherapy omission in ER+/HER2- BC (Section 11)
KEYNOTE-522	Neoadjuvant pembrolizumab + chemo in Stage II-III TNBC: pCR 64.8% vs 51.2% (Section 13)
KATHERINE	Adjuvant T-DM1 for residual HER2+ disease after neoadjuvant therapy (Section 13)

Trial	Key Finding and Cross-Reference
CREATE-X	Adjuvant capecitabine for residual TNBC after neoadjuvant chemotherapy (Section 13)
monarchE	Adjuvant abemaciclib in high-risk HR+/HER2- early BC (Section 13)
DESTINY-Breast03/04	T-DXd superior to T-DM1 (HER2+); T-DXd effective in HER2-low MBC (Section 11, 15)
EMERALD / EMBER-3	Oral SERDs (elacestrant, imlunestrant) for ESR1-mutated MBC after CDK4/6i (Section 15)
INAVO120	Inavolisib + palbociclib + fulvestrant in PIK3CA-mutated MBC (Section 15)
CAPitello-291	Capivasertib + fulvestrant in AKT1/PIK3CA/PTEN-altered MBC (Section 15)
OlympiAD / EMBRACA	PARP inhibitors (olaparib, talazoparib) in BRCA1/2-mutated MBC (Section 05, 15)
ASCENT / TROPION-Breast01	TROP2-targeted ADCs (sacituzumab govitecan, dato-DXd) in TNBC/HR+ MBC (Section 15)

End of Document

Compiled for GSR-3 Surgical Oncology Residency -- September 2026. Dr. Muluamlak -- EBS Unit. This preparation note is a companion to BreastCancer_Management_Enhanced.pptx and should be read alongside the presentation slides for full visual context (tables, flowcharts, and color-coded banners).